DEGREE PROJECT

Social Media Bot Detection

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Abstract

Social media platforms have revolutionized how people interact with each other and how people gain information. However, social media platforms such as Twitter and Facebook quickly became the platform for public manipulation and spreading or amplifying political or ideological misinformation. Although malicious content can be shared by individuals, today millions of individual and coordinated automated accounts exist, also called bots which share hate, spread misinformation and manipulate public opinion without any human intervention. To make things worse, the sophistication of these automated accounts is so high that they become unidentifiable by humans and difficult for algorithms to detect.

In the past decade, researchers have developed novel methods to detect harmful, machine-controlled accounts. Most of these methods are supervised and are based on classical machine learning approaches. Recently, unsupervised and adversarial methods have also been researched. What is usually common in all these methods is that the detection is based on account level meta-data and feature extraction.

The work presented in this thesis, aims at designing, implementing and evaluating bot detectors that are based on deep-learning models and which do not require account level meta-data nor feature extraction. The thesis work relies on the PAN 2019 Bots and Gender Profiling task dataset where Twitter accounts had to be classified as humans or bots. Furthermore, the thesis shows that deep-learning models can yield an accuracy of 0.89 on the PAN 2019 Bots and Gender Profiling dataset; thus, they can compete with classical machine learning methods. Moreover, the findings of this work also show that pre-trained models will be able to improve the accuracy of deep-learning models.

Contents

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Abbreviations

AB AdaBoost. 18, 21, 22

ANNs Artificial Neural Networks. 26–28, 31

ASHA Asynchronous SHA. 30, 31, 42, 44

AUC Area under the ROC Curve. 18, 22

BCE Binary Cross Entropy. 42

BERT Bidirectional Encoder Representations from Transformers. vi, 30, 31, 40, 43–45, 48–50

BN Bayesian network. 21

BRNNs bi-directional RNNs. 27

CC Camisani-Calzolari rule set. 17, 21, 22

DFS Depth-First Search. 17

DT Decision Tree. 18, 21, 22

ET Extra Trees. 18, 21, 22

GA Genetic Algorithm. 20, 21

GloVE Global Vectors for Word Representation. 19, 35, 38, 41

GRUs Gated Recurrent Unit networks. 27

HDBSCAN Hierarchical-based Clustering. 20

LCS Longest Common Substring. 20, 21

LR Logistic Regression. 18, 21, 22

LSTM Long Short Term Memory. v, vi, 19–21, 27, 28, 31, 38–44, 48–50

MCC Matthews correlation coefficient. 22

MLM masked LM. 30

MLPs multilayer perceptrons. 26

NAS neural architecture search. 8

NB Naïve Bayes. 18, 21, 22

NLP Natural Language Processing. 8, 13

NSP next sentence prediction. 31

OSNs Online Social Networks. 14

PCA Principal Component Analysis. 20

RF Random Forest. 18, 21, 22

RLE Run-length Encoding. 20

RNNs Recurrent neural networks. 27

ROC AUC Area Under the Receiver Operating Characteristic Curve. 18

RTBust Retweet-Buster. 20

SbN Socialbot Network. 13

SGD Stochastic Gradient Descent. 42

SHA Successive Halving. 30

SMOTE Synthetic Minority Oversampling Technique. 18

SVM Support Vector Machine. 18, 21–23

TICA Time Independent Component Analysis. 20

TPE Tree-structured Parzen Estimator Approach. 29, 31, 42, 44

VAEs Variational Autoencoders. 20

Chapter 1

Thesis Introduction

1.1 Background

Social media bot detection has a decade of history. According to Cresci et al. [15], the first social media bot detector was made by Yardi et al. [60] who tried to detect spammers on Twitter. In the early days, most detection methods were based on supervised machine learning algorithms, and the detection was based solely on the information of individual accounts. However, as time went by, bots became more sophisticated and coordinated. Distinguishing bots from real accounts became almost impossible for humans [19]. At this point, due to the unavailability of ground truth data and due to the coordinated bot behaviour, researchers turned to other methods, namely, to unsupervised machine learning algorithms. These detectors, focused on groups of accounts rather than individual ones and aimed at detecting accounts with similar behavioural patterns. The problem with the earlier described methods is that detectors are only created once new previously undetectable bots are discovered, leaving significant life span for bots to operate. In order to solve this problem, researchers [17] today develop adversarial methods where they not only create detectors but also try to predict the evolution of bots to find vulnerabilities in their existing methods before they are exploited.

In order to combat malicious social media bots several institutions and universities have organized bot detection competitions. DARPA was a bot detection competition organized in 2015 [54] in order to test the effectiveness of influence bot detection methods. Recently the PAN 2019 Bots and Gender Profiling task has taken place, with the intention of using only the textual characteristics (Tweet) of social media accounts [46]. This has the advantage of creating platform-independent solutions as these methods do not rely on specific social media characteristics. The thesis work presented here will be based on the PAN 2019 Bots and Gender Profiling task and will aim at improving the results of the existing solutions which will be discussed in more detail in the literature review.

1.2 Motivation

Social media has changed how people acquire information and these platforms became an essential source of information for many [25]. Today, platforms such as Twitter have become a platform for manipulating public opinion and spreading or amplifying political misinformation. However, large scale manipulation and sharing of politically polarized content is not only done by humans but social media bots [8].

Bessi et al. [8] investigated the role of bots during the 2016 U.S. Presidential election. The researchers developed bot detection methods and estimated that about 400 000 bots were actively participating in political discussions generating roughly 3.8 million tweets which were about one-fifth of the entire conversation. Although at the time it was not clear, today we know that social media bots played a key role in the election [8] by spreading divisive messages which may have contributed to Trump's victory[1].

In their study, Stella et al. [50] analyzed the role of social bots during the Catalan Referendum of October 1, 2017, by observing the behaviour of the two opposite groups of polarities, the pro-independence and anti-independence. The study found that bots exploited and promoted content with the same polarity as the group they targeted. Furthermore, the group of pro-independence interacted with bots almost 100 times more than the opposite group of anti-independence, and in this group, bots had a larger influence and were spreading negative content and hatred in order to exacerbate online social conflict [50].

Bots almost always play a key role in political elections. During the Brexit referendum in the United Kingdom, they manipulated public opinion in favor of the country to leave the European Union [26]. Bots also interfered with the French presidential elections in 2017 by spreading the MacronLeaks campaign against Emmanuel Macron [22]. As we can see bots actively shape public opinion, usually by negative campaigning [26] and influence individuals who may not be able to distinguish between human or bot-generated content. Therefore, countermeasures have to be taken in order to neutralize bots that exploit and spread misinformation by creating state-of-the-art bot detection methods.

1.3 Problem Definition

The project will be based on the PAN 2019 Bots and Gender Profiling task [46] where author profiling had to be solved by classifying Twitter feeds as bots or humans, based solely on the account's textual form of the tweets without any additional information, such as tweet time, name, followers, accounts followed or profile picture. Furthermore, in the event where the feed belonged to a human, the gender of the user also had to be classified. The dataset consisted of 6760 labeled English and 4800 Spanish users, each

[1] `https://www.bloomberg.com/news/articles/2018-05-21/twitter-bots-helped-trump-and-brexit-win-economic-study-says`

with 100 tweets [1]. Additionally, bots were classified into the following subcategories which can help the model evaluation [2]:

1. Template: Bots which follow a predefined template.
2. Feed: Bots which retweet or share news about a certain topic.
3. Quote: Bots which tweet quotes from famous people.
4. Advanced: Bots which use advanced machine learning techniques to generate human like language.

This thesis work aims to research current state of the art approaches for bot detection and implement a deep-learning-based model that can predict whether a Twitter account belongs to a bot or a human based, solely on their tweets without additional information. The project will not implement the gender classification part of the PAN 2019 Bots and Gender Profiling task and will focus solely on the English dataset.

Furthermore, the thesis aims at answering the following research questions:

1. Can deep-learning-based approaches compete with classical machine learning methods on the limited dataset which was provided?
2. How would increasing the available data with data augmentation influence the performance of the models?
3. What is the effect of using pre-trained models and language representations on the results?

1.4 Equality and Ethics

Although Twitter and other social media platforms do not restrict the use of bots, they are often created to steal data or limit free speech [2, 8]. These actions break laws; therefore, they are also unethical. Having said that, bots can also be unethical and still be lawful, for example, by hiding their identity or spreading fake news [2]. In 2016, Microsoft launched its Twitter chatbot called Tay, which was developed to entertain people with casual conversations. Unfortunately, after hours of its launch, the friendly chatbot became racist due to malicious human interactions[2]. The question of who is responsible for the behaviour of automated accounts is challenging. Is it the developer's responsibility or is it the responsibility of those who teach it to behave in an unethical way? According to Alves et al. [2], this is determined whether the developer(s) knowingly designed their algorithms to behave maliciously or not; if so they will be responsible for its actions. However, if the developers followed industry guidelines and good development practices

[2] https://www.theguardian.com/technology/2016/mar/24/tay-microsofts-ai-chatbot-gets-a-crash-course-in-racism-from-twitter

they cannot be held accountable.

Bot detectors also raise ethical questions. Social media platforms use different methods to determine the originality of accounts, and take actions against them. As most of these methods are based on machine learning algorithms, the results are accurate but not perfect; therefore, real human accounts can also be classified incorrectly as bots and may face restrictions to social media platforms [55]. To avoid incorrect classification, the algorithms and classification decision making have to be transparent. However, too much transparency may create more sophisticated bots [55], taking advantage of the transparency and spreading even more malicious information. Therefore, there is a trade off between accuracy and transparency and the correct balance has to be found during the development of detectors. Another ethical issue is the available resources and data which can be used for training detectors. For languages, such as English or Spanish, there are more data sets, more known bots and more resources to train unsupervised machine learning algorithms, whereas for other languages there are fewer. This means there could potentially be more undetected malicious automated accounts for languages that were not the primary focus of researchers.

1.5 Sustainability

Climate change is affecting every country, CO2 levels are at a record high[3] and electricity consumption is increasing year after year[4]. One of the goals of the United Nations is to combat global warming and to reduce carbon emissions[5]. Today, deep neural networks achieved impressive results in several Natural Language Processing (NLP) applications such as sentiment analysis, text classification or language translation mainly due to the available computational resources [53]. State-of-the-art machine learning algorithms are trained on specialized hardware such as GPUs or TPUs, sometimes for weeks or months, which require high electricity consumption. Even though clean energy is becoming more common, non-renewable energy still counts as the main source of energy[6]. This has a substantial cost to the environment. Strubbel et al. [53] analyzed the cost and energy consumption of several NLP models. The results show the computational costs are between $41-$3000000 and the CO2 emissions range from 27lbs to 626155lbs, where the lower end models are transformer models, whereas the upper end models are neural architecture search (NAS) models. In comparison, the carbon foot print of one passenger for a flight from New York to San Francisco is less than 2000lbs [53]. This comparison shows just how important it is to write efficient code and how important it is to develop

[3]https://news.un.org/en/story/2020/04/1062332

[4]https://www.statista.com/statistics/280704/world-power-consumption/\#:~:text=The\%20world\'s\%20electricity\%20consumption\%20amounted,seeing\%20a\%20more\%20pronounced\%20growth

[5]https://www.un.org/sustainabledevelopment/climate-change/

[6]https://ourworldindata.org/energy-mix\#:~:text=Globally\%20we\%20get\%20the\%20largest,than\%2080\%25\%20of\%20energy\%20consumption

methods that speed up the training phase.

Although the bot detectors which will be introduced later in the thesis fall on the lower end of the spectrum of energy consumption, training the models on servers using Intel Xeon CPUs or Nvidia Tesla P100-PCIE-16GB GPUs took around three and a half hours. With that being said, hundreds of computational hours have been spent on training and developing the bot detector, which according, to my estimations, has a carbon footprint of around 10lbs.

1.6 Thesis Structure

The thesis structure is organized as follows:

- Chapter 2: Relevant research on Twitter, bots and different bot detection methods are reviewed. In addition, the chapter also focuses on how datasets can be collected, what bot detection challenges researchers faced and the top 3 best performing solutions of the the PAN 2019 Bots and Gender Profiling task are discussed.
- Chapter 3: Introduces the reader to the theoretical background, algorithms and methods of deep-learning-based bot detectors.
- Chapter 4: Describes the architecture and design of the bot detectors created in this work.
- Chapter 5: Evaluates the proposed models on the test set.
- Chapter 6: Discusses the results of the models.
- Chapter 7: Summarizes the most important findings of the thesis and discusses future improvements.

Chapter 2

Literature Review

2.1 Twitter

Twitter is a popular social media networking service in San Fransisco, California, established by Jack Dorsey, Noah Glass, Biz Stone, and Evan Williams in 2006. Twitter is a microblogging service, which means that the information shared, i.e the tweets (small messages), are limited to a small size of 280 characters, making them easy to read and write [41]. Tweets are similar to emails or blogs; they allow users to share their experiences and ideas with any member on the platform. Users can post, like, retweet, follow other users; furthermore each user has their own timeline where their posts are collected. When users retweet, they forward a tweet to their own followers [37] and although one can retweet in order to disagree with an opinion[1] as Metaxas et al. [37], show a retweet usually indicates agreement, trust and interest in a message. When someone starts following a user, such as a friend, a company or a celebrity, they get a stream of the following users' tweets [41]. Moreover, users can mention and reply to others using the sign "@" followed by the target username, which creates a two-way communication channel. Users can also group posted tweets by a topic using hashtags, which are words prefixed by the hash symbol "#".

As of Q1 2019, Twitter was estimated to have 330 million monthly active users[2], and 187 million daily active users[3]. This is a substantial growth compared to its 30 million monthly active users in Q1[3] 2010 and making it one of the most visited websites with around 6 billion daily visits[4]. Furthermore, today more than a billion Tweets are created each week[5].

[1] https://www.thebalanceeveryday.com/retweet-definition-what-retweet-means-and-how-to-use-them-896699

[2] https://www.statista.com/statistics/282087/number-of-monthly-active-twitter-users/

[3] https://www.statista.com/statistics/242606/number-of-active-twitter-users-in-selected-countries

[4] https://www.similarweb.com/website/twitter.com/

[5] https://blog.twitter.com/official/en_us/a/2011/numbers.html

As of today, Twitter is not only a social media platform where individuals can share their opinions, but it has also become one of the key real time communication channels during natural disasters and political events [41]. In addition, Twitter became one of the most important platforms where politicians can communicate their campaign messages, mobilize their supporters and influence the public agenda [52]. For example, on the 2016 presidential Election Day, more than 75 million tweets were posted, which is far more than the 31 million during the 2012 elections[6].

The authors of Uddin et al. [56] conducted a study in which 716 randomly selected Twitter profiles were used to identify six classes of Twitter users:

1. Personal users: Users who use the platform for fun, learn or read news, do not promote any business or product, and their social interactions are considered to be low or mild.
2. Professional users: Home users who have many followers and follow many. They mainly focus on one area by sharing useful information about it.
3. Business users: Users who frequently tweet but have little interactions. Their behavior follows a similar pattern to each other, following a marketing and business strategy.
4. Spam users: Randomly follow users and spread malicious tweets, their behavior follows the same pattern to each other.
5. Feed/News: These accounts post news-related tweets. They are not interactive.
6. Viral/Marketing Services: Accounts that serve to increase sales, brand awareness or marketing objectives.

The first three of the above-mentioned user classes belong to human accounts, whereas the last three to machines also known as digital actors or bots. In the next section digital actors, or bots, will be discussed in more detail.

2.2 Bots

Bots, agents or actors, are applications that can execute predefined tasks and do so repetitively. Moreover, social media bots or spambots are a special type of bots which act as humans hidden behind social media accounts interacting with humans and influencing them with political, ideological, or commercial purposes [59]. In 2016, nearly 19% of all tweets related to the US election were generated by bots, and these tweets were retweeted almost at the same rate as if they had been originated by human accounts [8]. Today, bots are able to create human-like tweets and form botnets spreading malicious

[6]https://money.cnn.com/2016/11/09/technology/election-trump-social-media-records/index.html

information even faster.

In their paper, Stieglitz et al. [51] ask how social media bots can be distinguished from one another. Ferrara et al. [23] distinguish two types of bots, namely, benign bots and harmful bots. Benign bots aggregate content, automatically respond to inquiries, and in general, can be useful. On the other hand harmful or malicious bots are designed for harmful purposes [51]. As Ferrara et al. [23] highlight, benign bots can also be harmful, for example by spreading unverified information. Boshmaf et al. [9] distinguish social bots and other bots based on whether a bot is trying to hide its identity. Social media bots act and pretend to be real social media users by simulating human behavior and manipulating public opinion using artificial intelligence [9].

There is an intermediate class of bots called hybrid social bots (cyborgs) where once a human registers an account the user can automate specific tasks such as posting automatic tweets. In this case, hybrid social bots are either human-assisted bots or bot-assisted humans [14], where humans are responsible for content production and creation [1]. Moreover, hybrid bots are an effective and low-cost approach to gain influence on social media [1]. Bots can also form together and create botnets or armies where bots coordinate and follow each other [57]. Boshmaf et al. [9] define a Socialbot Network (SbN) as “a set of socialbots that are owned and maintained by a human controller called the botherder”. These networks have three components [9]:

1. Socialbots: Control a social media account and can execute commands sent by the botmaster such as to post a message or interact with other users.
2. Botmaster: A social media-independent software which the botherder uses to define commands to be executed by the bots.
3. Command & Control channel: A communication channel between the social bots and the botmaster.

While early bots were naïve and easy to spot, on platforms such as Twitter, bots today became increasingly sophisticated, making their detection very difficult [51]. Many of these bots create their own profiles by stealing users’ identities, making conversations using Natural Language Processing (NLP) and aiming to strengthen their influence by increasing the number of their followers [51].

2.3 Twitter Bots

A bot on Twitter is a special type of account controlled by a machine rather than a human. These accounts can automatically tweet, re-tweet, like, follow/unfollow and send direct messages. Twitter does not restrict the use of bots as long as they follow the automation rules[7]. Bots can broadcast helpful information, respond to users in direct

[7] `https://help.twitter.com/en/rules-and-policies/twitter-automation`

messages or try new solutions which help people[8]. On the other hand, bots are not supposed to spam, spread misinformation or violate any other policies [8].

However, Twitter bots are similar to regular bots, they can be benign, broadcasting news and helpful information, but they can also be harmful, spreading misinformation and promoting hate speech. There are different types of Twitter bots. The purpose of fake follower bots is to increase the number of followers of a target account [16]. While they may seem harmless, artificially increasing someone's followers by fake accounts makes them more influential and seem more trustworthy than they are, and as a result, people are more likely to follow them [16].

A study conducted by Varol et al. [57] shows that around 15% of Twitter accounts are bots. While this is a very high number, as demonstrated by Varol et al., this may still be an underestimation as increasing evidence suggests the presence of hybrid human-bot accounts, which need some human supervision [57]; therefore, the actual number of automated accounts could be higher.

2.4 Bot Detection Methods

This section discusses the different types of methods, their features, evaluation metrics and datasets used for detecting bots. Moreover, the evolution of bot detectors and bots will also be discussed in this chapter.

2.4.1 Evolution of Social Media Bots and Their Detectors

Cresci et al. [15] distinguish three major waves of different characteristics of bots. During the first wave, until around 2011, Online Social Networks (OSNs) had simplistic bots, which showed clear signs of automation with little social interactivity and their purpose was solely spam related. On the other hand, the second wave of social bots increased their social interactivity by following each other (creating bot nets) and had detailed profiles. Moreover, these bots were more sophisticated, not spamming the same messages over and over again. Lastly the third wave of bots mimic human behavior, have a large number of real followers as well as friends and typically share malicious messages with many neutral ones [15].

As bots were evolving, so were their detectors. According to Cresci et al. [15], the first social network bot detection method dates back to 2010, when detectors were based on supervised machine learning algorithms where the classification was based solely on the information of the account which was being detected. These detectors operated on the assumption that bots had individual characteristics which could be used to distinguish them from human accounts. As bots became more sophisticated with the third

[8] https://help.twitter.com/en/rules-and-policies/twitter-automation

wave of bots, detecting them with supervised methods assuming that bots are separable from real human accounts was no longer possible [15]. Rather than classifying accounts one by one, researchers targeted groups of accounts [15, 11], as these bots coordinate with one another and leave traces of synchronization behind [15]. Many of these bot detectors are unsupervised or semi-supervised in order to generalize the flaws of supervised methods which are limited by data availability [15].

When new types of bots that cannot be detected by available solutions are discovered, researchers start developing detection systems capable of detecting them. As Cresci et al. [15] describe, this results in a lag between the development of detectors and the deployment of them; thus bots have a large time span when they can operate. Moreover, "scholars and OSN administrators are constantly one step behind of malicious account developers"; therefore, the influence of bots do not seem to decrease on social media networks [15]. In order to mitigate these challenges in the future, adversarial machine learning techniques could be used [15]. As most machine learning models are incorrectly based on the assumption that the test data is drawn from the same distribution as the train data, the accuracy of the models can degrade in the case of adversaries [24]. Adversarial machine learning is a machine learning paradigm based on creating data that is first based on legitimate data, but through perturbation, the model is being fooled and incorrectly classifies the sample; thus, model vulnerabilities can be detected before they are exploited by adversaries [24].

2.4.2 Data Collection

While the performance of the models does depend on the model configurations, the datasets used for training the models are just as or even more important than the models themselves. In their study, Beskow et al. [7] demonstrate four tiers of Twitter data based on data availability which can be used for training machine learning models. As the study shows there is a tradeoff between data richness and computational time to classify accounts. While a Tier 1 model is suitable for classifying accounts in a large data stream where the account metadata with a tweet is available, Tier 3 is suitable for problems where high accuracy is expected, using all available information of an account [7]. As table 2.1 demonstrates, using a Tier 0 dataset, the classification is instant, whereas a model trained on a Tier 3 dataset takes around 20 hours.

Annotating Twitter accounts as bots or humans is a challenging task. In their study, Cresci et al. [19] performed a crowdsourcing campaign where 247 real-world trusted twitter users from 42 different countries were asked to classify an already labeled set of Twitter accounts. The dataset consisted of 4428 accounts of humans and bots, where each account had to be classified by at least 3 contributors; thus, the class was decided by majority voting [19]. The results show that the accuracy by human classification was less than 24% for social bots, classifying more than 1000 social bots as humans (False

Negative). Human and traditional bot accounts had been classified correctly with 91% and 92% accuracy. The study shows that humans may not be able to detect social bots and distinguish them from real human accounts [19]. Therefore, manually creating ground-truth datasets is not applicable for sophisticated social bots, so Cresci et al. [19] suggest the use of a new annotation method which takes into account the similarities and the behavior of the accounts.

In their study Almaatouq et al. [1] created a dataset based on the accounts which had been suspended by Twitter. The authors collected tweets and user information during a period of one month, using the Twitter public streaming API and labeled the accounts as spammers which were later suspended or removed. As a result, 7% of the dataset were classified as bots. As the authors highlight, the classified accounts may contain accounts which had been deactivated by the users themselves, but the vast majority, 93% of the suspensions, are a result of spamming behavior [1].

In their study, Lee et al. [31] deployed 60 honeypot accounts on Twitter which were designed to avoid interacting with regular users and could tweet normal text, mention other honeypot accounts, tweet a link or tweet one of Twitters top 10 topics. These honeypot accounts reported which accounts interacted with them, such as started following them. As the honeypots posted random messages and only interacted with other honeypots, a human would not find them interesting. Therefore, accounts which started following them or interacted with them should not be legitimate accounts. The study conducted by Lee et al. [31] concluded that Twitter suspended around 23% of the accounts which had been reported by the honeypots. Although 77% of the reported accounts were not suspended by Twitter, the researchers clustered the accounts and came to the conclusion that they were content polluters.

Other ways to collect data is by labeling accounts that participated in known bot attacks. Beskhow et al. [7] collected bot accounts during a Twitter bot attack against NATO and

Tier	Description	Focus	Collect/process Time per 250 Accounts	# of Data Entities (i.e tweets)
Tier 0	Tweet text only	Semantics	N/A**	1
Tier 1	Account + 1 Tweet	Account Metadata	1.9 seconds	2
Tier 2	Account + Timeline	Temporal patterns	3.7 minutes	200+
Tier 3	Account + Timeline + Friends Timeline	Network patterns	20 hours	50,000+

Table 2.1: Four tiers of Twitter data. Source: [7]

the DFR Lab in 2017. On average, these accounts have 130 and 43 interactions on a daily basis [7]. On the other hand, during the attack, they averaged 6000 interactions per hour. The authors [7] were able to label 99% of this activity as bot activity; by manually verifying them using random sampling. Moreover, 8 months later, Twitter had suspended 95.5% of the accounts [7].

Many studies used a variety of the above-mentioned data collection methods. Varol et al. [57] trained their models using manually labeled accounts, both manual and honeypot collected accounts and a mixture of these two with different ratios. Cresci et al. [16] created a bot dataset by buying 3000 fake accounts from different Twitter online markets and Chu et al. [14] manually analyzed the accounts which they have collected using the Twitter API using a Depth-First Search (DFS) approach.

2.4.3 Modeling Approaches

This section discusses the different types of modeling approaches used in the cited papers. The section is divided into five subsections which go into depth about rule-based, supervised, unsupervised and adversarial methods. Finally, these methods are discussed and compared with one another.

Rule-Based Methods

Cresci et al. [16] used the Camisani-Calzolari rule set (CC) [10], which assigns an account positive human/active and negative bot/inactive scores. Based on the two scores, the classification of an account can be given by calculating the distance between the two scores. If the sum of the scores is greater than 0, the account is considered human; if it is between 0 and -4, it is neutral; otherwise the account is classified as a bot. The algorithm uses a set of human criterion, such as whether the profile contains an image, physical address or if at least one of its tweets has been re-twitted by other accounts and if so, one human point is given. On the other hand, one bot point is given for each human criterion that cannot be verified and two bot points if the account only uses APIs. Other rule-based methods include the Socialbakers rule set, which determines the classification based on 8 rules, such as whether the account has ever tweeted, the percentage of links in tweets or if the account uses spam phrases in its tweets [16].

Supervised Methods

Machine learning algorithms can be characterized based on the desired outcome of the algorithm [43]. Supervised methods generate a function that maps the inputs into desired outputs based on the input-output pairs on which the model is trained. Therefore, human and bot related labeled data can be fed to a machine learning algorithm that can be trained to classifying unforeseen accounts [30].

Most papers reviewed and cited in this literature review utilize supervised machine learning methods in order to detect social media bots. Moreover, Cresci et al. [15] reviewed 236 bot detectors published since 2010, and the majority of the detectors are based on supervised algorithms. Most papers use classical machine learning methods [59, 45, 30, 56, 16, 57, 7, 31, 1, 14] such as Random Forest (RF), Logistic Regression (LR), AdaBoost (AB), Decision Tree (DT), Support Vector Machine (SVM) or Naïve Bayes (NB). However, deep-learning-based approaches are also researched [30].

Below, I discuss three papers in more detail, two of which use classical machine learning methods and one using deep learning. These papers are examined here as they represent different types of approaches and for many authors they have served as benchmarks and reference studies.

BotoMeter was developed at Indiana University and has been a publicly available service since 2014 to determine whether an English Twitter account is human or machine. The system can be accessed via REST call APIs or via a website. BotoMeter calculates a score called the bot-likelihood score of a given account and returns it to the user. The classification consists of several steps. First, more than 1000 features are extracted from a given account, such as network features (retweets or mentions), user features (user location, account creation date or language), features regarding friends, temporal features (tweet rate), content features and sentiment features. Next, a random forest classifier predicts whether the account is a machine or a human. The model has been trained on 31k accounts (15k accounts belonging to bots, 16k to humans) and 5.6 million tweets based on the dataset collected by Lee et al. [59]. The model achieved a 0.95 AUC, and every day more than a quarter-million requests are served [59].

Pozzana et al. [45] in their study used two datasets, one dataset which the researchers labeled with BotoMeter of the French Elections and another created by Cresci et al., which contains social spambots. The dataset was organized into sessions, which are the consecutive tweets posted by the same user and separated from the previous tweet by T minutes. The researchers extracted features, such as the fraction of retweets, fraction of replies, the number of mentions in a tweet or the length of the tweet itself. Four supervised machine learning algorithms were trained using 10-fold cross-validation with different combinations of features, namely Decision Tree (DT), Extra Trees (ET), Random Forest (RF) and AdaBoost (AB). All models reported an Area Under the Receiver Operating Characteristic Curve (ROC AUC) of 97% except for the AB, which scored 84%.

Kudugunta et al. [30] aimed at solving account-level and tweet-level bot detection, using deep learning and classical machine learning methods. For the account level classification, the researchers used user metadata to determine the classification, and AUC of 98.45% was achieved with RF. However, the researchers managed to increase the accuracy of account-level classification to 98.81% using AdaBoost and by balancing the dataset via oversampling techniques, namely Synthetic Minority Oversampling Technique (SMOTE)

[12]. Regarding the tweet-level detection, the researcher first preprocessed the tweets by tokenizing them, replacing hashtags, URLs, numbers, emojis and mentions with tags and converting all letters to lowercase. Then the tokenized tweets were transformed into embeddings using Global Vectors for Word Representation (GloVE), and the sequence was fed into Long Short Term Memory models. Moreover, the researchers also created Contextual LSTM models (which are a special type of LSTM models with multiple inputs and outputs), which were trained on the tweet's embedding along with user metadata. The LSTM model trained on tweets only, reached 95% accuracy, whereas the best contextual LSTM network achieved about 96%.

Unsupervised Methods

Due to the drawback of supervised machine learning approaches, such as the limited availability of ground-truth and reliable training datasets [36], researchers during the past years have shifted to unsupervised methods [15]. While supervised machine learning algorithms learn on labeled data, unsupervised methods learn patterns from untagged data, such as by clustering accounts. Moreover, most of the unsupervised approaches are group-based; that is, the analysis is based on groups of accounts rather than individual accounts; therefore, they are effective at identifying coordinated and synchronized accounts [15]. Although most approaches use connectivity patterns, such as directed graphs in order to identify bots with similar behavior [15, 28], other approaches identify bots by spotting anomalous tweeting and retweeting patterns [15, 11, 28].

An example of an unsupervised approach is DeBot [11], a social media bot detection method that correlates millions of users in near real-time to identify bot accounts. DeBot is based on a lag-sensitive hashing technique, which hashes the correlated users into the same buckets based on the users wrapping correlation. The system developed by Chavoshi et al. [11] calculates the correlation of the users based on user actions, such as posting, sharing, liking tweeting, retweeting or deleting. This sequence, including a time stamp, forms a time series; thus, the method aims to identify accounts with similar activity patterns. The architecture of DeBot consists of four components:

1. Collector: Collects tweets for a given set of keywords for T hours and creates the time series of the users with one second sampling time.

2. Indexer: Hashes each time series into buckets and reports suspicious users in a given bucket.

3. Listener: Collects and creates the activity time series of the suspicious users employing all user activities.

4. Validator: Calculates a wrapped correlation matrix of the suspicious users and clusters them. Users belonging to a cluster are reported as bots.

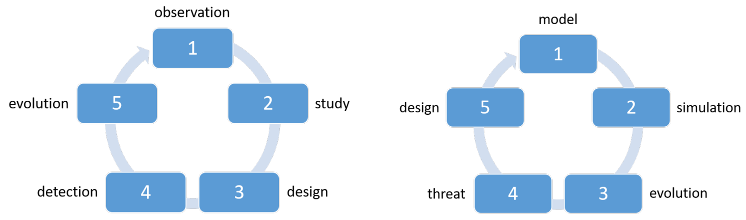
DeBot has been compared against Twitter's suspension process by identifying accounts as bots over a given period of time and every few days checking if the identified accounts has been suspended by Twitter. After 12 weeks, 45% of the accounts identified as bots by DeBot were also been suspended by Twitter. In addition, DeBot identifies bots at a higher rate than Twitter, which could mean Twitter does not suspend each identified bot.

Mazza et al. [36] created an unsupervised group based bot detection method called Retweet-Buster (RTBust). RTBust consists of 3 logical components. First, during the data preparation phase, retweet time series are created and compressed using Run-length Encoding (RLE) to minimize the amount of data to process and maximize the information of the time series. Next, a dimensionality reduction technique is used, namely Variational Autoencoders (VAEs), which learns the latent features of the data by using an LSTM based encoder and decoder. Finally, Hierarchical-based Clustering (HDBSCAN) is used to cluster the users. Large clusters may indicate synchronized accounts. Therefore, they are labeled as bots, where as human accounts are considered as noise. The authors experimented with different configurations and different dimensionality reduction methods, such as Principal Component Analysis (PCA) and Time Independent Component Analysis (TICA). The results show that the best model using VAE achieved an F1 score of 0.87.

Adversarial Methods

All bot detection methods mentioned so far has been created following a reactive schema where suspicious OSN behaviors are first detected and then studied, and finally, new detection methods are designed and deployed [17]. Therefore, as described by Cresci et al. [17], scholars and OSN administrators are always one step behind of bot developers. A proactive approach on the other hand would consist of two models. First, a simulation model would produce representations of malicious accounts that do not exist yet, but are similar to existing ones. Then, each representation would be evaluated, and those that are able to evade detection are considered threat and need to be taken into consideration during the development and design of new detectors [17]. This approach could have the advantage of foreseeing future bot evolutions and taking countermeasures before the actual evolution takes place. Figure 2.1 demonstrates the two different types of schemes described above.

Following the proactive scheme (Figure 2.1) Cresci et al. [17] have created a simulation model using a Genetic Algorithm (GA) based on digital DNA modeling. The Digital modeling technique creates a string representation of the user's online actions, where each action, such as a tweet, is represented as a character, such as a "T", and these actions encoded as characters form the digital DNA. Then the genetic algorithm using mutation and crossover generates new sequences of digital DNA of bots, which are evaluated against a bot detector. The authors have used DNA fingerprinting to detect bots which, given a list of digital DNAs, compares them and finds similarities of automated accounts by calculating the Longest Common Substring (LCS) of their DNA. Accounts sharing a long LCS or behavior pattern are likely to be similar as well. Moreover, Cresci et al. [17] show



(a) Reactive schema. (b) Proactive schema.

Figure 2.1: Source: From reaction to proaction: Unexplored ways to the detection of evolving spambots [17]

that bots have long LCS even with large numbers of accounts, whereas human accounts show little similarity. The authors have designed and launched multiple experiments and demonstrated that using the above-mentioned methods, newly evolved bots can avoid detection.

Discussion

Table 2.2 shows information of the inspected papers such as the type of approach the authors used and algorithms. Table 2.3 shows the results of the papers. As we can see, most researchers used supervised machine learning algorithms. Moreover, classical machine learning algorithms were the most common. Authors of [59, 45, 18] used unsupervised methods, whereas the author of [16] created detectors using a rule-based approach, while the rest of the authors created supervised bot detectors. It is important to highlight that the models had been trained and evaluated on different datasets, with different features and evaluation metrics; therefore, they cannot be directly compared to one another.

Researcher(s)	**Type**	**Algorithm(s)**
Cresci et al. [16]	Rule Based	CC
Yang et al.[59]	Supervised	RF
Pozzana et al. [45]	Supervised	RF, DT, ET, AB
Mazza et al. [36]	Unsupervised	RTBust (VAE)
Chavoshi et al. [11]	Unsupervised	DeBot
Varol et al. [57]	Supervised	RF, AB, LR, DT
Beskow et al. [7]	Supervised	NB, LR, SVM, DT, RF
Lee et al. [31]	Supervised	30 classifiers, best RF
Almaatouq et al. [1]	Supervised	ZeroR, BN, NB, LR, DT, RF
Chu et al. [14]	Supervised	RF
Kudugunta et al [30]	Supervised	LR, AB, LSTM
Cresci et al. [18]	Unsupervised	GA

Table 2.2: Summary of the cited papers.

Researcher(s)	Algorithm	Acc.	Prec.	Recall	F_1	MCC	AUC
Cresci et al. [16]	CC	0.55	0.98	0.06	0.12	0.18	
Yang et al.[59]	RF						0.95
Pozzana et al. [45]	RF						0.97
	DT						0.97
	ET						0.97
	AB						0.84
Mazza et al. [36]	RTBust (VAE)	0.87	0.93	0.81	0.87	0.76	
Chavoshi et al. [11]	DeBot		0.94				
Varol et al. [57]	RF						0.95
Beskow et al. [7]	NB	0.61	0.56	0.99			0.89
	LR	0.67	1.0	0.32			0.92
	SVM	0.90	0.85	0.97			0.95
	DT	0.96	0.96	0.97			0.96
	RF	0.98	0.99	0.97			0.99
Lee et al. [31]	RF	0.98			0.98		0.99
Chu et al. [14]	RF	0.96					

Table 2.3: Results of the papers

2.5 Overview of the Best Results of the Author Profiling Task

This section summarizes the top 3 highest performing approaches of the 7th International Author Profiling Shared Task at PAN 2019, focusing solely on the English bot detection challenge. Moreover, the results highlighted here will serve as a benchmark for the bot detection model proposed in the methodology chapter of the thesis.

Johanson et al. [29] have achieved the highest accuracy of the bot detection challenge [46]. As a first step, the authors extracted several tweet level features, such as tweet length, number of capital letters/URLs/mentions etc., and created account-based statistics of these features. The similarity of subsequent tweets of different accounts was also calculated, using the Damerau-Levenshtein distance, and the most common unigrams and bigrams were extracted by applying tf-idf to the concatenated tweets belonging to the same account. The authors used two classifiers; first, a logistic regression classifier was trained on the tf-idf features, and then the output of the LR along with the statistical features was fed to a random forest classifier. The authors achieved an accuracy of 0.95, the best result for the bot detection task.

Meanwhile, Fernquist [21] extracted three main features. First, features from tf-idf were extracted from the concatenated tweets of an account, just as in [29]. Then compression features were extracted by compressing the concatenated tweets of each user and extracting statistical features from them. And lastly, tweet features were extracted, such as the retweet ratio, Shannon entropy of all tweets concatenated or the number of hashtags used divided by the total number of hashtags. The researchers trained a CatBoost classifier with 4158 features and reached an accuracy of 0.949.

Bacciu et al. [3] extracted tweet level features such as the average number of emojis used in each tweet, links shared or the average number of hashtags used. Furthermore, cosine similarity was calculated, sentiment analysis was applied to each tweet and text distortion was used to emphasize the use of special characters. The classifier consists of two layers. The first layer consists of an SVM and an AdaBoost instance, whereas the second layer has a Soft-Voting classifier that ensemble the predictions of the first layer. The classifier described reached an accuracy of 0.94, making it the third-best performing classifier.

As can be seen, the best performing approaches all reached more than 0.94 accuracy. In addition they all used classical machine learning algorithms. The organizers of the PAN 2019 challenge [46] have evaluated all 46 submitted approaches, and the majority of them used classical machine learning approaches with only a few using deep learning methods.

Chapter 3

Theory

3.1 Introduction

This section introduces the fundamentals of natural language processing and the algorithms of the proposed bot detection models. Moreover, this section aims to explain design choices of the implementation phase, such as why certain algorithms are used instead of others. Furthermore, this section will also discuss the challenges that need to be overcome and the methods available to increase the accuracy of bot detectors.

As seen in the literature review, bot detection methods are usually based on supervised or unsupervised approaches sometimes combined with adversarial methods. Since this work is based on the PAN 2019 Bots and Gender Profiling task where labeled tweets were the only source of data, the work presented in this thesis work is based on supervised machine learning methods where the prediction whether an account is a human or a bot is based on the inspected account's tweets without additional information. Therefore, the task which is being solved is considered a text classification problem. In addition, as most submissions of the PAN 2019 Bots and Gender Profiling task are based on classical machine learning algorithms (see the literature review for more details) which are already well explored, or those based on deep learning methods did not achieve high accuracy, the proposed bot detectors in chapter 4 will be based solely on deep learning methods. Thus this chapter will also only focus on this research area.

3.2 Fundamentals of the proposed bot detector

This section describes the most important algorithms and methods that the proposed bot detectors build on.

3.2.1 Artificial neural networks

Artificial Neural Networks (ANNs) are computational units inspired by biological neurons, first researched during the 1940s [27]. ANNs are well suited for a variety of tasks, including classification problems. The simplest form of an ANN consists of one artificial neuron, also called a perceptron. A perceptron consists of n inputs $x_i \in X$, weights $w_i \in \mathbb{R}$ and a bias $b \in \mathbb{R}$ where w_i corresponds to the weight of x_i and determines the importance of the input (see figure 3.1). The neuron's output is calculated by adding the weighed sum of the inputs to the bias and applying the result to an activation function σ (see equation 3.1 [40]). The activation function determines how the inputs are transformed into outputs. There are several activation functions such as the sigmoid, relu or the step function [40].

$$Y = \sigma\left(\sum_{i=1}^{n} x_i w_i + b\right) \tag{3.1}$$

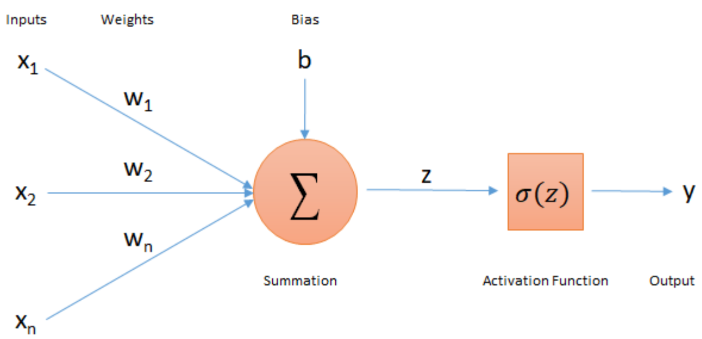


Figure 3.1: Architecture of a perceptron.

ANNs can consist of a single layer of neurons or multiple layers. Single-layer networks organize neurons into a single layer, whereas multi-layer networks have multiple layers of neurons and usually consist of an input layer, hidden layer and output layer. They are also called multilayer perceptrons (MLPs) (see figure 3.2).

ANNs learn by tuning their parameters using backpropagation algorithms. During the training phase, each training sample is fed into the network and an error value is calculated based on a cost function. The backpropagation algorithm sends back the error to the network, calculates the gradient of the loss function and updates the weights in order to reduce the error.

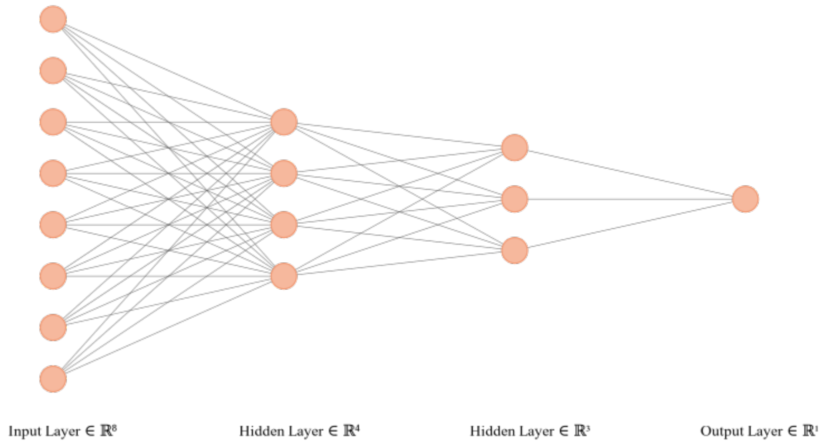


Figure 3.2: MLP with one input layer, two hidden layers and one output layer.

3.2.2 Recurrent neural networks

The problems with ANNs in regard to NLP applications is that they cannot handle and learn long dependencies. Moreover, ANNs need to use fixed lengths context which needs to be specified during the design phase of the model [38]. On the other hand, Recurrent neural networks (RNNs) use recurrent connections to allow information to cycle within the network and for information to persist [38, 42] while processing variable length inputs. This allows RNNs to learn correlations between data points that are close in the sequence [48].

Figure 3.3 shows the basic architecture of RNNs as well as unfolded in time. Instead of using a fixed number of inputs, RNNs are fed one input x_i at a time and the network predicts h_i for each input. As can be seen, the loop passes the information from one step to the next allowing the network to learn from previous states.

In order to use future information, Schuster et al. [48] proposed an RNN architecture consisting of two RNN networks. The two networks can be combined where each network learns from the input data for both time directions (forward/backward). These RNN networks are called bi-directional RNNs (BRNNs).

Unfortunately RNNs are unable to learn long dependencies due to the gradient vanishing and exploding problems [4]. In order to solve this issue researchers have developed novel approaches such as Long Short Term Memory (LSTM) networks or Gated Recurrent Unit networks (GRUs) which overcome the drawbacks of traditional RNNs.

3.2.3 LSTM

The key idea behind LSTMs is that at each time step the network passes a cell state and a hidden state to the next step. The cell state contains information for the entire

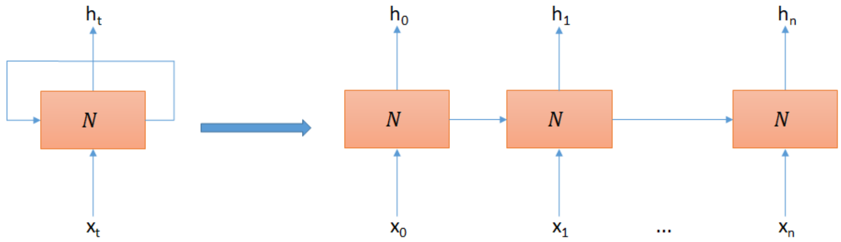


Figure 3.3: Architecture of ANNs

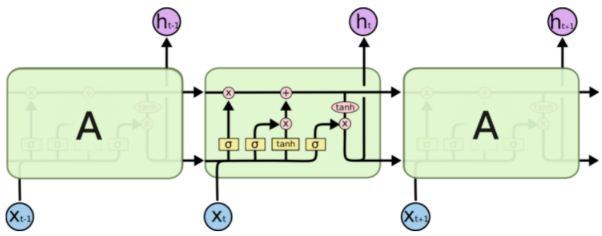


Figure 3.4: LSTM Architecture: [42]

sequence and can be considered as a long term memory. On the other hand, the hidden state contains information about the previous state. LSTMs consist of multiple gates which control the flow of information [42] (see Figure 3.4). LSTMs have three gates composed of a sigmoid neural net layer and a pointwise multiplication operation. The sigmoid layer determines how much information should be passed through. Folowing are the three gates:

1. Forget gate: Decides the information to keep from the previous hidden state and concatenates the current input to it.

2. Input gate: Decides what information to store in the cell state based on the concatenated previous hidden state and the current input.

3. Output gate: Computes the new hidden state based on the newly calculated cell state, the concatenated previous hidden state and current inputs.

3.2.4 Data augmentation

Data augmentation has shown promising results in computer vision, even by using simple techniques such as cropping, rotating and flipping images which became a great way to reduce overfitting of models [44]. However, augmenting textual data is a challenging task as there are neither generalized rules nor best practices for transforming this type of data. Yu et al.[61] generated new data by translating sentences from English to French and back to English. Other approaches include replacing words by their synonyms or using data noising as smoothing [58]. Wei et al. [58] developed a data augmentation technique to improve the performance of text classification models. The technique consists of synonym replacement, random insertion, random swap and random deletion. The results show an average of around 1% absolute increase in performance but on small datasets the method boosted the performance by even 3%.

3.2.5 Hyper-parameter optimization

Typically ANN-based models have between 10 to 50 hyper-parameters to be configured before training the model [6]. This is a difficult task and it is often considered an art rather than science since [6] there is no rule on how machine learning models should be configured as the parameters may change from one problem to another. Hyper-parameter optimization aims to answer the question of how optimal hyper-parameters can be chosen for a given machine learning algorithm. The most popular approaches that will be discussed here include grid search, Bayesian optimization and random search.

Grid Search: It is one of the most widely used automatic strategies for hyper parameter optimization [5]. During the parameter tuning, a subset of the hyper-parameter space has to be manually specified. The algorithm trains the model on each pair of the specified parameters, evaluates the model performance and outputs the settings which achieved the best results. As the algorithm is based on a brute-force, it is computationally expensive [34].

Random Search: The algorithm, rather than trying each parameter configuration (as grid search), randomly selects the parameters, evaluates them and selects the best configuration based on a given metric. Random search is able to find models which are just as good or even better than grid search but in a fraction of its computational time [5].

Bayesian optimization: In contrast to random and grid search, Bayesian optimization keeps track of previous optimization results and any prior information in order to help the exploration of the search space. The algorithm is based on the Bayes' theorem and aims to optimize an objective function that selects the hyper-parameters. Bayesian optimization has shown that it is able to find superior results to human experts or state-of-the-art approaches [49]

Tree-structured Parzen Estimator Approach (TPE): Similarly to the Bayesian optimization TPE also constructs probabilistic models based on previous evaluations in order to approximate the performance of hyper-parameters. The algorithm uses Parzen Estimators to create densities where samples are taken and evaluated [6].

Hyper-parameter optimization is a costly procedure requiring lots of computational resources. Another question arising from hyper-parameter optimization is not only how to select the parameters but how to allocate the resources for the randomly selected hyper-parameter configurations. Li et al. [33] developed a novel algorithm called Hyperband which early stops poorly performing configurations. Hyperband uses a method called Successive Halving (SHA) which for a given amount of resources evaluates all configurations, keeps the best ones and repeats the same step until only one configuration remains. Moreover, Hyperband eliminates incorrectly early stopping configurations by evaluating the configurations on a different number of resources. Other approaches use similar methods. Asynchronous SHA (ASHA) [32] also uses SHA but effectively paralyses early stooping and configuration promotion, making it suitable for large-scale hyper-parameter optimization problems.

3.2.6 Transfer Learning

The classical paradigm for supervised machine learning is to train a model based on a single dataset. This requires a large amount of data which is often not available. Moreover, for each task and problem, a new model is trained from scratch [47] which is time and resource consuming. Transfer learning aims to solve these problems and reduce over-fitting and to improve performance by transferring or adapting knowledge from one problem or dataset to another [39]. By sharing pre-trained models between different problems, the models also generalize better. Indeed, one of the most promising advances in NLP applications also came from transfer learning [47].

Pretrained models can be used in two ways for a given problem [47]:

1. Feature Extraction: The pretrained model is used as a feature extractor where the features are used as inputs for a separate model. In this case the weights of the pretrained model are frozen.

2. Fine-tuning: This approach consists of two steps. First, the pretrained model is frozen and only the model is trained. Next, the weights of the pretrained model are unfrozen and the entire network is trained together.

A common form of transfer learning in NLP applications is to use pre-trained word embeddings. Word embedding is the representation of words as real-valued vectors by encoding their meaning, such that words with similar meanings are closer in the vector space than dissimilar ones. One of the most widely used word embeddings are GloVE, ELMO or Word2Vec.

3.2.7 BERT

Bidirectional Encoder Representations from Transformers (BERT) is a language representation model which trains Transformer encoders [20] bidirectionally. As opposed to LSTMs which learn the input sequence left-to-right or right-to-left, Transformer encoders read the input sequence at once. The training of BERT consists of two steps. First some input tokens are masked and the model has to predict the original tokens. This is called masked LM (MLM). Then, in order for the model to learn relationships between two sentences, the model is trained to predict the next sentence from a corpus. This is called next sentence prediction (NSP). Results from BERT show that it achieves new state-of-the-art results in a variety of NLP tasks such as MultiNLI, SQuAD and pushes the GLUE score by 7.7% [20].

3.3 Conclusion

Several design choices have to be made for bot detectors which require the theoretical background introduced above. In addition, there are many solutions to the given problem, which means the design space is large. In the following lines the theoretical design choices such as why certain algorithms or ANN architectures are used instead of others will be discussed.

The biggest question is which neural network model to use as the base model of the detector. As described earlier, ANNs are not suitable for NLP problems; therefore, LSTMs will be used. Regarding hyper-parameter optimization, TPE will be applied using ASHA, as they have shown promising results. As for transfer learning, feature extraction will be used for word embeddings; fine-tuning will be used for constructing a BERT based bot detector; and for data augmentation, a new approach will be introduced in the next chapter.

Chapter 4

Implementation

4.1 Data Understanding

The dataset of the PAN 2019 Bots and Gender Profiling task is split into three sets, namely training (2800 accounts), validation (1240 accounts) and test (2640 accounts) sets. Each account has 100 tweets and an account can belong to a human or a bot. In this section, the data exploration of the training set will be discussed and explained.

In order to avoid bias for supervised machine learning algorithms the dataset has to be balanced. The training dataset contains 1400 humans and 1400 bots making it ideal for machine learning algorithms (see Figure 4.1).

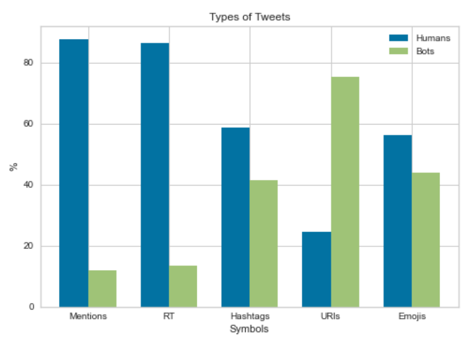


Figure 4.1: Symbol usage

Figure 4.2 shows the most frequent tokens for the entire corpus which also include stopwords (such as ’the’, ’to’ or ’and’) or URLs (https). These tokens will have to be preprocessed during the data preparation phase of the implementation in order to reduce the dimensionality of the corpus. It is also interesting to see how the most frequent tokens differ between humans and bots. As can be seen on Figure 4.3 humans and bots both share links (co, https, https) but bots use these tokens at a higher rate. Moreover, bots share content containing tokens such as hiring, developer, engineer and software which can be an indication of job promotion and spamming behaviour.

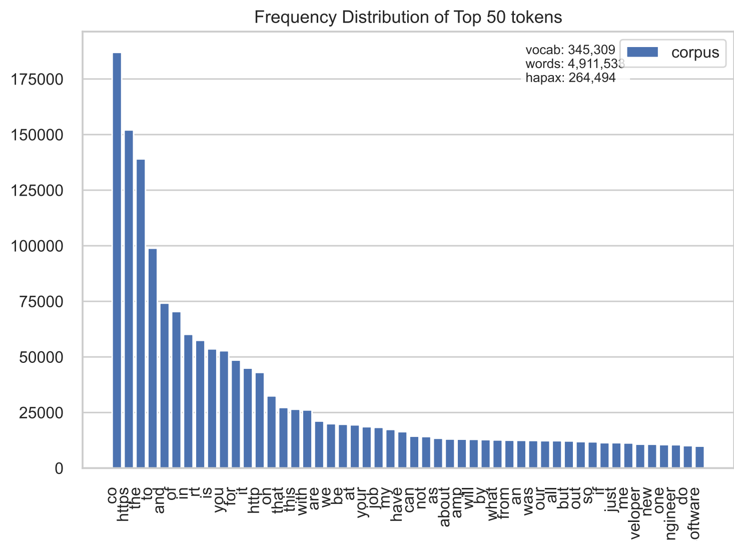
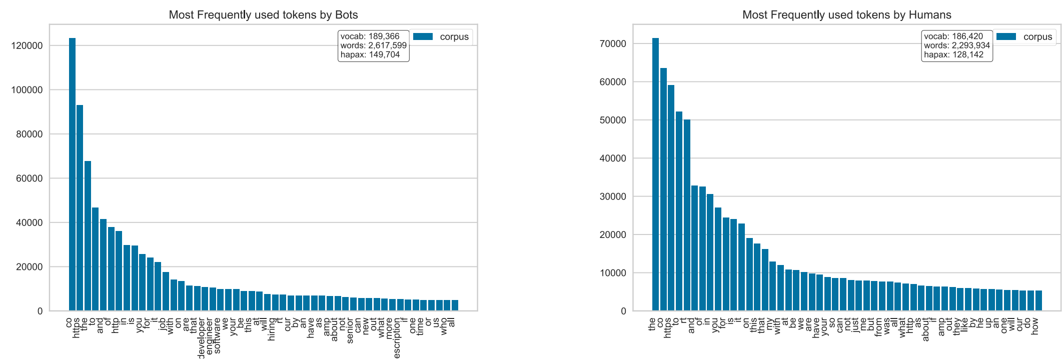


Figure 4.2: Most Frequently used tokens by Bots and Humans.



(a) Most Frequently used tokens: Bots.

(b) Most Frequently used tokens: Humans.

Figure 4.3: Most frequent tokens

As the predictive models are built on word embeddings, an important aspect is the sequence length of the tweets. Figure 4.4 demonstrates the distribution of the number of tokens within a tweet. Most tweets have less than 25 tokens, but some have more than 40, and one has almost 100 words. It is also worth looking into which type of account (bots or humans) prefer using which type of Twitter key words such as hashtags, mentions, emojis or URLs. Based on Figure 4.1, humans retweet and use hashtags and mentions more than bots. On the other hand, bots post more than three times as many URLs as humans, indicating malicious activity.

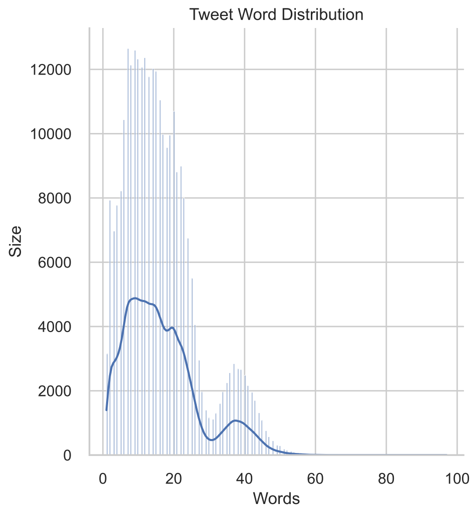


Figure 4.4: Tweet length distribution.

4.2 Data Preparation

The bot detectors which will be introduced shortly use GloVE as a pre-trained word embedding. During the data preparation phase an important aspect is to preprocess the data in a similar manner as the authors of the word embedding. This would allow the predictive model to learn the semantic and syntatic meaning of words, which could also boost the detectors' performance.

As word embeddings require a fixed input size, tweets also need to have a fixed length. There is a trade-off between the sequence length. If the input size is too small, valuable information may be neglected and the models may not learn important patterns. On the other hand, if the input is too large, the dimensionality of the corpus may permit the predictive models to learn. There are many common strategies for handling this, nevertheless, in this work, based on Figure 4.4, the maximum tweet length will be limited to Q3 + 1.5*IQR 40 tokens/tweet. This means that tweets which have more than 40 tokens will be truncated and tweets with a smaller size will be padded.

The data set was preprocessed as follows:

1. HTML tags were removed.
2. URLs were replaced with a <url>tag.
3. Mentions were replaced with a <user>tag.
4. Hashtags were replaced with a <hashtag>tag. Moreover, word segmentation was applied in order to split hashtags into their corresponding word(s). For example, #HashTag is represented as <hashtag>hash tag.
5. Numbers were replaced with a <number>tag.
6. Emojis were mapped to a tag, such as <smile>, <sadface><heart>etc.
7. Accented characters were removed.
8. Repeated characters were replaced with an <elong>tag. For example ... is represented as . <elong>.
9. Contractions were fixed.
10. Upper case tokens were replaced with <allcaps>and were lower cased.
11. Special characters such as ", !, ? etc. were surrounded by spaces.
12. Tweets were lower cased and extra spaces were removed.

Additional preprocessing steps had also been experimented such as using named entity recognition to replace words based on their entity, removing punctuations, stop words and lemmatizing the corpus. Unfortunately, these steps did not improve the accuracy; hence, they were omitted from the data preprocessing steps.

4.3 Modeling Approaches

This section introduces different modeling approaches, which were implemented during the modeling phase. 12 deep-learning-based methods will be discussed here, which will be evaluated in the Evaluation chapter of the thesis. The core of seven models are based on LSTM networks, four are based on BERT models and one a combination of the two. The detectors were written in Python and the deep-learning models are built using Pytorch.

4.3.1 Tweet Classification

All models introduced in this chapter build on a predictor which can classify individual tweets as humans or bots. The aim of this model is to classify tweets and based on the individual predictions new models can be trained which can classify the accounts. From here on, this model will be referenced as Tweet Classifier.

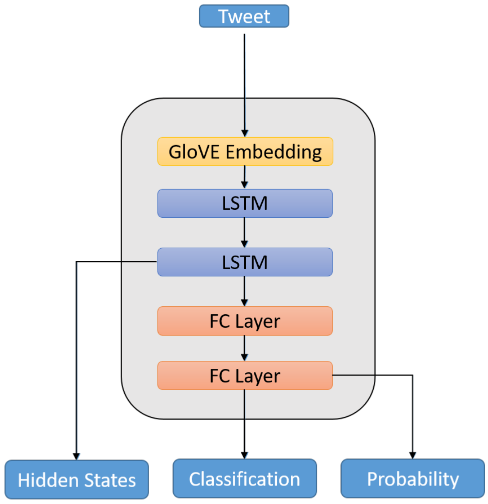


Figure 4.5: Tweet Classifier

Figure 4.5 shows the architecture of the Tweet Classifier. After preprocessing the data, the sequences of the token indexes are fed into the GloVE embedding, which creates the embedding representation of the tokens. Next, a two-layer bidirectional LSTM model learns the patterns of the tweets and finally, two fully connected layers output the prediction. The model outputs the classification, the probability of a tweet being bot and the hidden states of the last LSTM layer. These outputs will be used by models which classify the individual accounts.

Figure 4.6 gives an example of a prediction. The tweet is first preprocessed as described in the data preparation phase, next the tokens are converted to padded indexes and finally the Tweet Classifier makes the prediction.

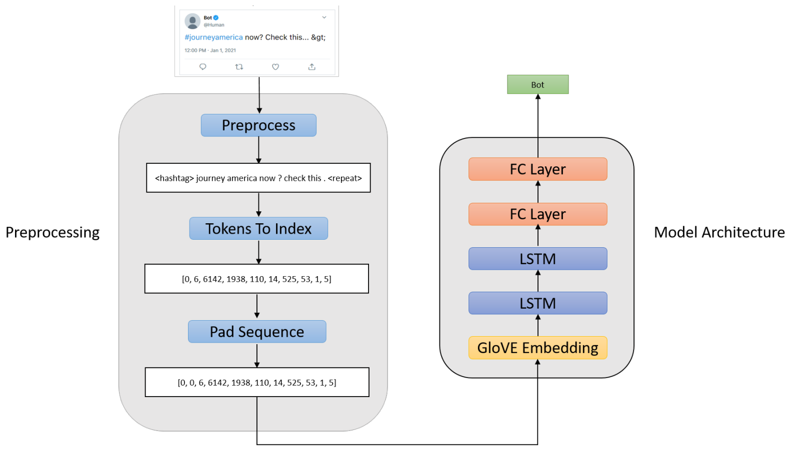


Figure 4.6: An example of a prediction

4.3.2 Majority Vote

In order to classify an account one could simply feed the individual tweets to the Tweet Classifier, collect the predictions and, based on a majority vote, classify the account. Figure 4.7 demonstrates the architecture of a bot detector using a majority vote. This approach does not require the implementation of additional models, which makes it fast to train and use.

4.3.3 Probability Based Prediction

A similar approach to the majority vote is to use tweet probabilities rather than classification predictions. In this scenario, the account's probability is calculated by averaging the sum of each of its tweets' probability. If the account probability is larger than 0.5, the account is considered a bot, otherwise a human. The advantage of this method over the majority based prediction is that certainty of the Twee Classifier is also taken in to

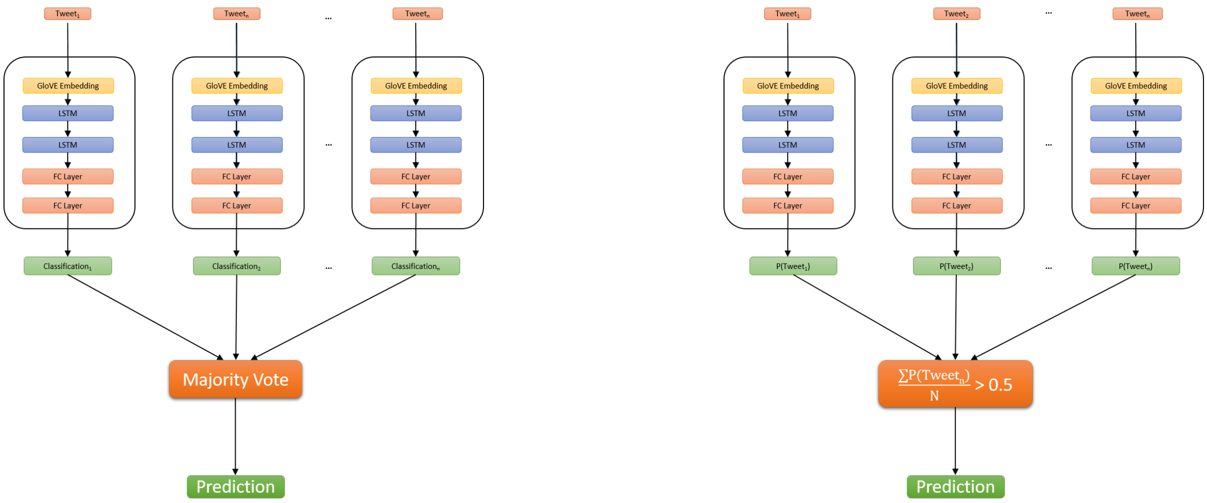


Figure 4.7: Majority Vote (left) and Probability based prediction (right).

consideration during the prediction. For example given 51 tweets of an account have a probability of 0.49 being a bot and 49 tweets with a probability of 1 being a bot, the majority vote based predictor would classify the account as a human, whereas the probability based account would classify it as a bot, as based on the probabilities the account has a higher likely-hood of being a bot than a human.

4.3.4 Combined Model

The output of the Tweet Classifier can also be the input of an account classifier model. For this approach, each tweet of an account is first fed into the Tweet Classifier model and the hidden representation of the tweet is used as inputs for a second model to classify accounts. The individual hidden states are first collected, then used as inputs for the second model.

In this model configuration, the Tweet classifier learns the features and most important characteristics of tweets regarding their originality. The hidden states represent this information; therefore the Tweet Classifier serves as an encoder where the model's output is disregarded.

The model consists of two LSTM layers and a fully connected layer which outputs the prediction for a given account (see Figure 4.8). Moreover, during the implementation, 5-fold cross validation was used, where 5 independent models were trained on different partitions of the training set. The prediction of the models is based on the majority vote of the 5 predictions or based on the average of the 5 probabilities, creating an ensembled configuration.

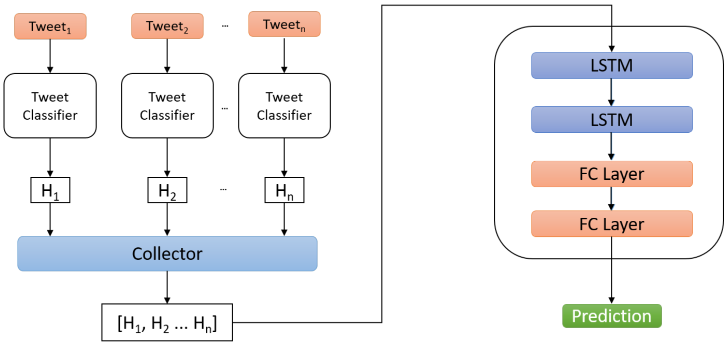


Figure 4.8: An example of a prediction

4.3.5 Fine-tuning BERT

BERT has also been fine-tuned to classify tweets as bots or humans. For this task, the pre-trained BertForSequenceClassification model was used from the hugging face library. The preprocessing steps described earlier are irrelevant for BERT as BERT requires special preprocessing. First, the input sequence was tokenized, all sentences were padded or truncated to a constant length, and special tags were added to the beginning and ending of each input [CLS]/[SEP]. Embeddings are also not required for BERT models as the pre-trained model already has one.

Regarding the account classification, the same steps had been tried as described for the LSTM models that is, majority vote, probability-based account prediction and representing each tweet as the last hidden state of BERT and feeding them to the same combined LSTM model. Figure 4.9 demonstrates the architecture of how a fine-tuned BERT model combined with an LSTM model can classify accounts. Moreover, the LSTM based account classifier was also trained on the hidden states of a BERT model wich was not fine-tuned.

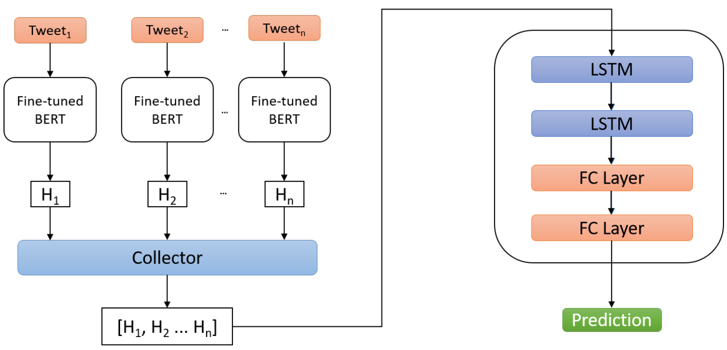


Figure 4.9: Fine-tuned BERT combined with an account classifier LSTM Model

4.3.6 Combining Hidden States

The hidden states of BERT and the LSTM Tweet Classifier can also be combined. As can be seen on Figure 4.10 the information of the BERT model, hidden size of 768 and the LSTM Tweet classifier, hidden size of 32 are concatenated and are fed into the LSTM Account Classifier. This approach allows the account classifier to learn from both the BERT and LSTM models which may supplement each other.

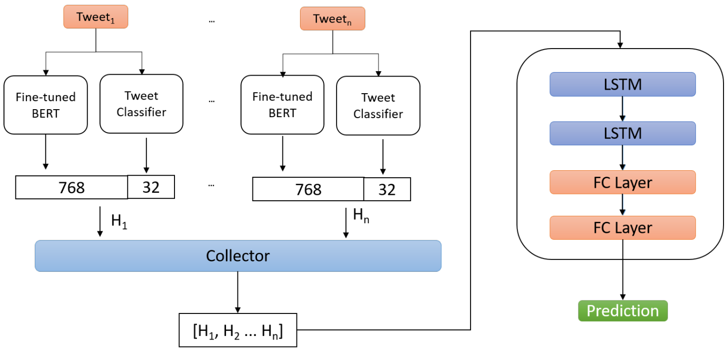


Figure 4.10: Hidden State Concatenation

4.4 Data Augmentation

As the training set has less than 3k accounts, training the LSTM based account classifier network is quite challenging. In order to overcome this issue data augmentation was used on the hidden states of the accounts. Let A be the set of {T1, T2...T_n} representing an account with n tweets and the set of the hidden states H={H1, H2...H_n}. We define an augmentation function f(H, l) where l $\leq$ n, which returns all n-l consecutive subsets of H of length l. This augmentation method significantly increased the availability of the training data, which should also increase the accuracy of the model.

In addition to the above, tweets were also augmented using an external library called nlpaug[1], which was used as a word embedding augmenter. The data augmentation was applied on the preprocessed data, as on the raw data URLs, hashtags and emojis were incorrectly segmented by the library. The augmenter used GloVE and substituted tokens based on embedding similarity.

[1]https://github.com/makcedward/nlpaug

4.5 Hyper-parameter Selection

This section discusses model configuration selection along with hyper-parameter optimization.

Ray-Tune [35] was used for hyper-parameter optimization using TPE as the search algorithm and ASHA to early terminate bad configurations. From here on, a training run with a fixed hyper-parameter configuration will be referred to as a trial.

4.5.1 Tweet Classifier

In order to find the best configuration 30 trials were executed running for a maximum of 20 epochs using Stochastic Gradient Descent (SGD) as the optimization algorithm and Binary Cross Entropy (BCE) as the loss function. The following hyper-parameter search space was used:

- Learning rate: 0.0001-0.2
- LSTM hidden size: 32, 64, 128, 356
- LSTM layers: 1,2
- Batch size: 32, 64
- Fully Connected Layer size: 32, 64, 128, 256
- LSTM dropout: 0.0-0.5
- Fully Connected Layer dropout: 0.0-0.5
- weight decay: 0-0.1

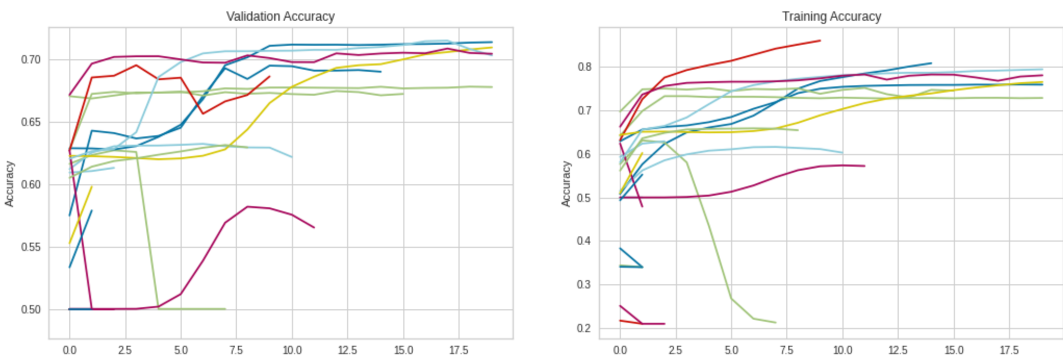


Figure 4.11: Training and Validation Accuracy of Trials

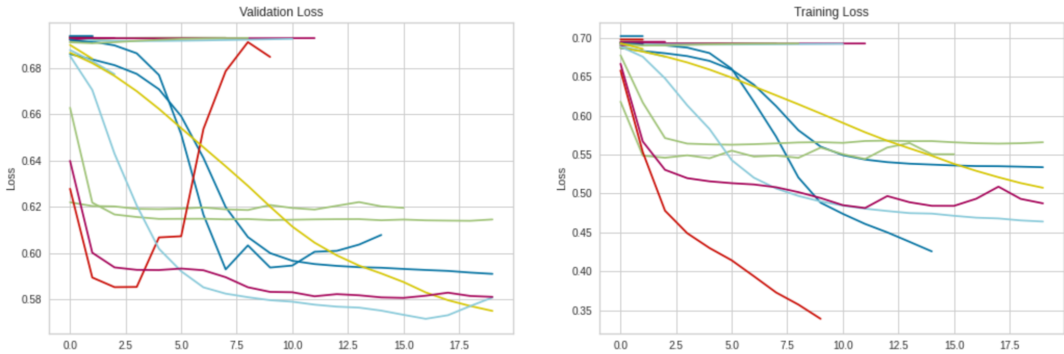


Figure 4.12: Training and Validation Loss of Trials

As can be seen in Figures 4.11 and 4.12, several bad trials early terminated, whereas the promising trials finished the 20 epochs. As the trials only ran for 20 epochs after the parameter optimization the best configuration was found manually by increasing the epochs for the most promising configurations.

4.5.2 BERT

The authors of BERT [20] recommend choosing the hyper-parameters from a pre-defined set of configurations. Therefore, exploring the search space can be done with a couple of iterations. For the tweet classification problem, the following parameters achieved the highest validation accuracy:

- Learning rate: 2e-5
- Batch size: 32
- epsilon: 1e-8
- max sequence length: 100
- epochs: 4

The sequence length was determined by the distribution of the number of tokens within the tweets after preprocessing since BERT also generates sub words of the tokens, which increases the dimensionality of the input data.

4.5.3 LSTM based Account Classifier

Three different models had been constructed for account classification, one for the BERT based hidden representation, one for the Tweet Classifier based tweet representation and one which combines the hidden states of BERT and the Tweet classifier. Moreover, different models had been evaluated using the original data and the augmented data. It is also important to highlight that an additional preprocessing step lies before the execution of the account classifier which serves to normalize the hidden states of the tweets.

Model	**LR**	**Hidden Size**	**Layers**	**Batch Size**	**FC Size**	**LSTM Dropout**	**FC Dropout**	**L2**
Tweet Classifier	0.00100	32	1	64	128	0.400	0.100	0.1
BERT	0.00100	768	-	64	-	-	-	-

Table 4.1: Best hyper-parameters for tweet classification.

Model	**LR**	**Hidden Size**	**Layers**	**Batch Size**	**FC Size**	**LSTM Dropout**	**FC Dropout**	**L2**
Account Classfier (LSTM, LSTM)	0.00800	32	2	64	128	0.499	0.900	9.1e-08
Account Classifier + Aug. (LSTM, LSTM)	0.00890	64	2	64	32	0.466	0.055	1.1e-06
Account Classifier (BERT, LSTM)	0.00040	128	2	64	512	0.088	0.100	6.7e-06
Account Classifier + Aug. (BERT, LSTM)	0.00329	512	2	64	32	0.198	0.500	2.8e-06
Account Classifier (BERT w/o fine-tuning, LSTM)	0.00017	256	1	32	32	0.402	0.176	3.9e-07
Account Classifier (BERT + LSTM, LSTM)	0.00171	128	2	64	64	0.4661	0.1055	9.9e-06

Table 4.2: Best hyper-parameters for account classification.

In order to find the best configurations for each model configuration, hyper-parameter optimization was used. 30 trials were executed and evaluated against the validation set for a maximum of 100 epochs. TPE and ASHA were used for exploring the search space and for early terminating bad trials. Table 4.1 and 4.2 shows the hyper-parameters of each model.

4.6 Software Architecture

Figure 4.13 demonstrates the software architecture of the thesis work. During the design phase, the main objective was to create classes that have only one responsibility and can be easily extended with new functionalities. Below is a summary of each class:

- DataUnderstander: Class used for data exploration and which creates several plots such as distributions of tweets or tokens, boxplots and bar charts of the input data.
- Preprocessor: Responsible for preprocessing the raw data as specified by the data preparation phase earlier.
- DataSetLoader: Reads the XML files and constructs the accounts, tweets and target labels for the training validation and test sets.
- Optimizer: Responsible for the hyper-parameter optimization of the models.
- Evaluator: Evaluates the models on the test set and creates metrics.
- Model: Abstract class for all modeling classes, holds the methods which all models have to implement.
- TweetClassifier: Implements the Tweet Classifier model, which was described in chapter 4.3.1.
- AccountClassifier: Implements the Account Classifier model which was described in chapter 4.3.4.
- BertDetector: Contains the fine-tuned BERT model which was previously described in more detail in chapter 4.3.5.
- TweetAugmenter: Generates new tweets as described in chapter 4.4.
- DataAugmenter: Augments the hidden states of the tweet classifier or of the hidden sequence representations from the fine-tuned BERT model.
- DatasetMaper: Overrides torch.utils.data.Dataset holds the input data and target attributes which is used by TweetClassifier.
- DatasetMaperList: Overrides torch.utils.data.Dataset holds the input data, sequence length and target attributes for the AccountClassifier.

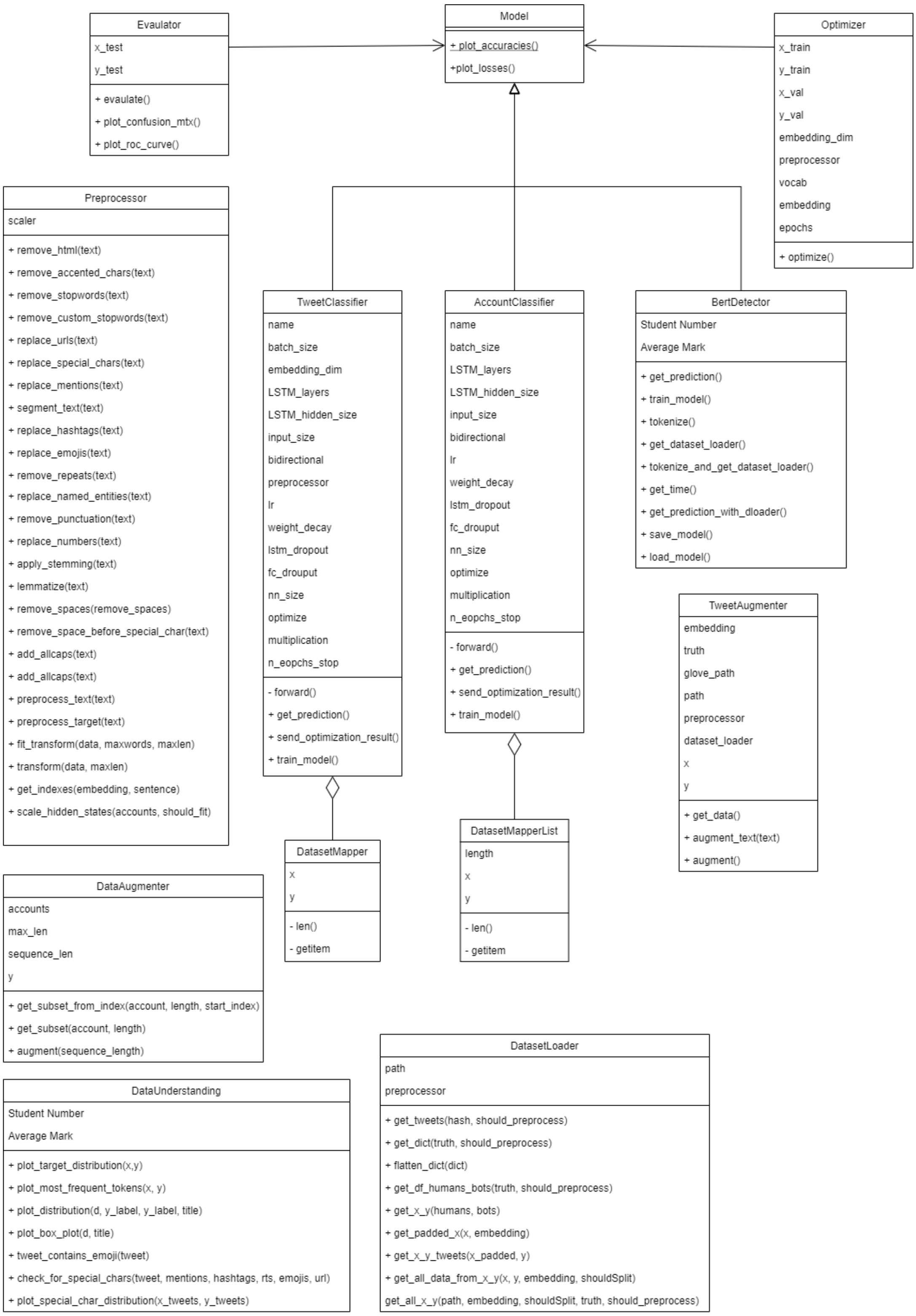


Figure 4.13: Class Diagram

Chapter 5

Evaluation

5.1 Introduction

This chapter evaluates and analyses the proposed models from chapter 4. The test set will be used while the models are configured employing the highest performing parameters described in more detail in the previous chapter. Four novel metrics will be used to evaluate the models, namely the accuracy, F1 score, precision and recall.

5.2 Tweet Classifier

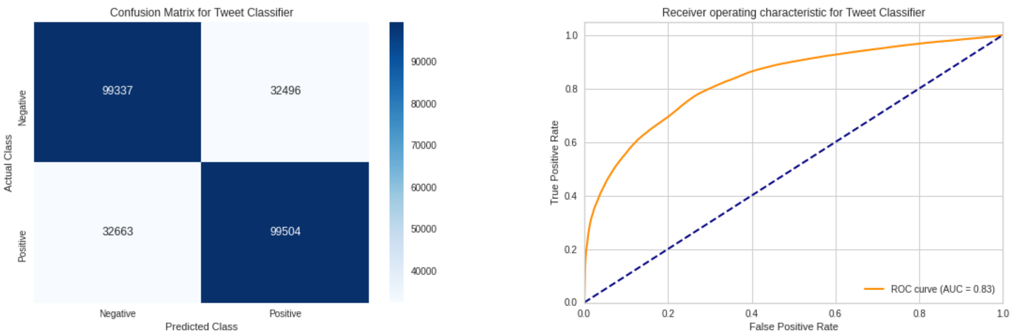


Figure 5.1: Tweet Classification - Left: Confusion Matrix. Right: ROC Curve

In this section, the Tweet Classifier model will be evaluated on the individual tweets. Moreover, the account predictability of the classifier will also be evaluated by the earlier mentioned majority vote and probability-based prediction.

The Tweet Classifier model reached an accuracy, f1 score, precision and recall of 0.75. This means the model is not biased towards any of the classes. Figure 5.1 shows the confusion matrix and the ROC curve where the AUC is 0.83.

Predicting accounts using majority vote achieves an accuracy of 0.87 and 0.86 for the F1 score. On the other hand, the probability-based account prediction reaches 0.879 accuracy and 0.87 F1 score beating the majority vote by 0.8%.

5.3 Fine-tuned BERT Model

The fine-tuned BERT model classifies individual tweets with an accuracy of 0.82, F1 score of 0.80 and AUC score of 0.91. In Figure 5.2, we can see the ROC curve and the confusion matrix. We can clearly see the model predicts almost 40k accounts as bots where as they were actually humans.

When predicting the accounts with a majority vote, the fine-tuned BERT model had an accuracy and F1 score of 0.849 while the probability-based prediction yielded an accuracy of 0.851 and 0.83 F1 score. This is almost 3% less than the LSTM based account prediction, while the BERT model predicts tweets 7% better.

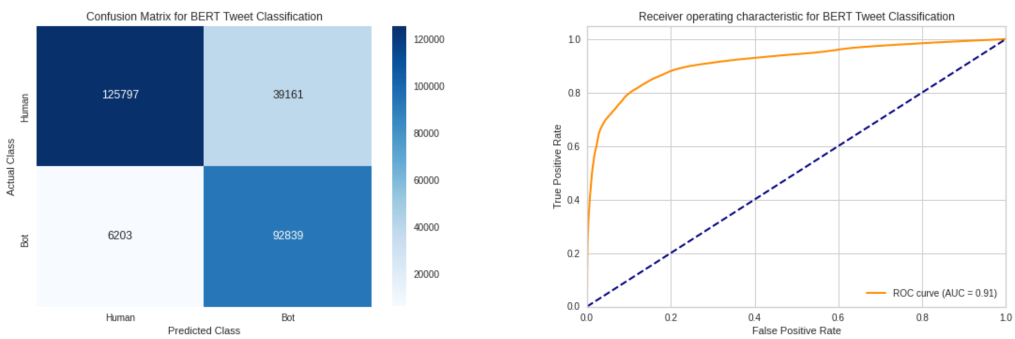


Figure 5.2: BERT Tweet Classification - Left: Confusion Matrix. Right: ROC Curve

5.4 Combined Model using Tweet Classifier

This section focuses on the combined LSTM models, namely with and without data augmentation. Furthermore, as described in the previous chapter, the results presented here are an average of 5 different models.

The model without data augmentation on average yielded an accuracy of 0.89 and F1 score of 0.89. On the other hand, with data augmentation, the model only reached an accuracy of 0.88 and F1 score of 0.881, which is 1% less than the model trained on the data that was not augmented.

5.5 Combined Model using BERT

Although the fine-tuned BERT model reached the highest accuracy for predicting individual tweets, the combined model using the hidden states from BERT did not achieve superior results.

Without data augmentation, the combined model reached an accuracy of 0.836 and an F1 score of 0.828. Similarly to the Tweet Classifier based combined approach, without data augmentation the model performed slightly better than with data augmentation. The model yielded an accuracy of 0.835 and F1 score of 0.821. Although the model without data augmentation performed slightly better, augmentation did not make a recognizable difference. It is also important to mention, the combined model with the BERT model which was not fine-tuned yielded an accuracy of 0.806 which is almost 3% less than if the fine-tuned BERT model was used.

5.6 Combining Hidden States

After combining the hidden states of both the fine-tuned BERT and LSTM models, the model yielded an accuracy of 0.838 and an F1 score of 0.82 on the test set. Although, the account classifier could use the information of both models it wasn't able to predict accounts better than the Account classifier based on BERT.

5.7 Summary of the results

Table 5.1, 5.2 and 5.3 demonstrates the results of all 13 models designed in this work. As can be seen the combined LSTM model yielded the highest accuracy of 0.892. Although the fine-tuned BERT model classifies individual tweets at the highest rate (0.828), all LSTM based models performed better during the account classification, including models based on majority vote and probability-based predictions.

The results are somewhat surprising as one would expect current state-of-the-art BERT

Model	**Accuracy**	**F1**	**Precision**	**Recall**
LSTM Tweet Classifier	0.753	0.753	0.753	0.754
BERT Tweet Classifier	**0.828**	**0.800**	**0.937**	**0.700**

Table 5.1: Tweet classification results.

Model	Accuracy	F1	Precision	Recall
Tweet Classifier - Majority Vote	0.873	0.864	0.930	0.806
Tweet Classifier - Probability based pred.	**0.878**	**0.870**	**0.936**	**0.811**
BERT - Majority Vote	0.849	0.826	0.975	0.717
BERT - Probability based pred.	0.852	0.830	0.976	0.721

Table 5.2: Account classification results based on single models.

Model	Accuracy	F1	Precision	Recall
Account Classifier - Majority Vote (LSTM, LSTM)	0.891	0.890	0.895	0.886
Account Classifier - Probability based pred. *(LSTM, LSTM)*	**0.892**	**0.893**	**0.881**	**0.906**
Account Classifier + Augmentation (LSTM, LSTM)	0.880	0.880	0.878	0.884
Account Classifier - Majority Vote (BERT, LSTM)	0.836	0.828	0.872	0.787
Account Classifier - Probability based pred. (BERT, LSTM)	0.838	0.830	0.874	0.790
Account Classifier + Augmentation (BERT, LSTM)	0.835	0.821	0.903	0.753
Account Classifier (BERT w/o Fine-tuning, LSTM)	0.806	0.793	0.853	0.740
Account Classifier (BERT + LSTM, LSTM)	0.838	0.820	0.916	0.743

Table 5.3: Account classification results based on multiple models.

based models to be superior to LSTM based approaches. In the next chapter, the results will be discussed, including the challenges faced during the implementation. Moreover, an explanation will be given of the under-performance of the BERT and data augmentation based models.

Chapter 6

Discussion

This chapter summarizes and discusses how the PAN 2019 Bots and Gender Profiling task was solved and how the results compare to the baseline models summarized in chapter 2. Furthermore, this chapter will also reflect on the research questions formalized in Chapter 1.

6.1 Discussion of the results

In order to solve the bot detection task of the PAN 2019 Bots and Gender Profiling task, 13 different deep-learning based models were designed, implemented and evaluated. Some models are basic in the sense that they can only classify individual tweets; others are more complex and combine different deep-learning models such as LSTMs with fine-tuned Transformer encoders to classify accounts as bots or humans.

The best result was achieved by combining two LSTM models, one to classify individual tweets as humans or bots, and another which takes the hidden states from the first model and classifies the inspected account. This approach yielded an accuracy of 0.89. Surprisingly, augmenting the input data for this model resulted in a 1% decrease in the accuracy. Although, data augmentation typically increases the accuracy of machine learning models, there can be several reasons why the model did not perform better. One reason is that during the hyper-parameter optimization, the search space was ill-defined and the optimization algorithm did not explore configurations that reach a high performance. On the other hand, maybe the augmented data was too complex and the model could not learn all the patterns. This can explain why the training accuracy was not improving while the validation accuracy was during training.

Another interesting point is that although the fine-tuned BERT model predicts the individual tweets better (+6%) than the LSTM based tweet classifier, the majority and probability-based account predictions are worse than the majority or probability predictions of the LSTM tweet classifier. This can be because of two factors. First, the

BERT model has a low recall which means many human accounts are classified as bots; second, the distribution of the incorrect tweets are spanned across more accounts than the LSTM based classifier, which can have a significant impact during the account classification. Moreover, the LSTM account classifier model performed poorer with the BERT hidden states. Again, this could be because of the incorrect hyper parameter configurations of the LSTM account classifier, or the hidden representation of the tweets are too similar; that is, the LSTM model cannot differentiate humans and bots.

It is also important to highlight that the LSTM based tweet classifier classifies accounts just 1% lower than the highest performing model introduced in this work. However, it can be trained in a fraction of the training time of a complex model such as fine-tuning BERT or the combined LSTM models.

6.2 Comparison of the best results of the PAN 2019 Author Profiling task

The best results and the majority of the PAN 2019 Bots and Gender Profiling task sumbmissions solved the task by classical machine learning algorithms (see Chapter 2 for more information). The top 3 submissions achieved more than 0.94 accuracy and these results serve as a benchmark for this work. The work presented here is based on deep-learning methods; moreover, only the textual data was used to train the models without additional metadata. On the other hand, the benchmarked solutions all extracted features from the data; therefore, it is hard to compare them. In summary it can be stated that the work presented is not superior to the benchmarked solutions and further improvements should be made to improve performance.

6.3 Reflection on the research questions

The thesis work aims at answering three research questions, which are described in Chapter 1. Referring to RQ1 of Chapter 1, whether deep-learning based approaches can compete with classical machine learning algorithms on the limited dataset: it can be argued that deep-learning based methods, especially LSTM based models, achieved good but not superior results.

For RQ2 in Chapter 1, data augmentation did not improve the accuracy because of the already mentioned reasons, such as by incorrectly specifying the hyper-parameter search space or incorrect implementation. Nevertheless, data augmentation in the NLP domain is challenging and an area that needs to be researched.

Regarding RQ3 of Chapter 1, it is clear that pre-trained models and language representations do improve the results. The fine-tuned BERT model beat the LSTM based tweet classification model by almost 7%.

Chapter 7

Conclusion and Future Work

7.1 Summary

The present thesis work gives the reader an overview of the area of automatized social media bots and, their detection methods. The work can be broken down into a theoretical and a practical part.

During the literature review, an overview is presented of the evolution of bots, their detectors and current state-of-the-art bot detection approaches, including several examples of supervised, unsupervised and adversarial methods. Challenges are also described, such as the difficulties of data collection. Chapter 3 focuses on the fundamentals of this work. The theoretical background of the proposed bot detection approaches is introduced, such as how RNNs work, how hyper-parameters can be optimized or how transfer learning can be used.

Chapter 4 and 5 focus on the design, implementation and evaluation of 13 deep-learning based detectors. A data augmentation approach is implemented and described for this task. The selection of the hyper-parameters is also discussed. Finally, in Chapter 6, the results are discussed and the research questions are answered.

7.2 Conclusion

The bot detectors introduced here successfully solve bot detection with an accuracy reaching almost 0.90. Although it can be argued, the presented deep-learning models are not superior to traditional classical machine learning methods. Having said that, the data set was quite limited in size and on large scale data the outcome may have been different. In addition, to the best of my knowledge, the presented architecture of the combining LSTM/BERT models with another LSTM model for classifying accounts have not been researched before. Therefore, this work can also serve as a baseline for future improvements for deep-learning based approaches.

7.3 Further development

Even though the work presented here solves the task, there are several improvements to be explored. In the following paragraphs, two future improvements will be described, namely, how data augmentation methods could be used to improve the data set and what other deep-learning architectures could be applied to improve the performance of the models.

7.3.1 Data Augmentation

In Chapter 4, it was mentioned that text augmentation was also experimented by randomly substituting tokens based on embedding similarity using GloVE. Although, at the time, it did not improve accuracy, due to time constraints this approach was not fully explored. In the future, text augmentation will be applied, such as the approach by Wei et al. [58], which combines synonym replacement, random insertion, random swap and random deletion.

7.3.2 Different Deep-learning approaches

One of the challenges faced during the thesis work was that the complex models did not improve the accuracy as much as it was expected. The reasons were described in the previous chapter and the improvements will be described in the next lines.

One way to improve the models is by making architectural changes to them, such as by adding more fully connected layers, using different loss functions or optimizers. One could also replace LSTMs by GRUs or use different word embeddings which all may improve the overall accuracy.

Another approach could be to improve the hidden representation of the models by using Siamese neural networks [13]. These networks are trained to predict whether two input samples are the same or not by calculating the similarity of the inputs (for example, images of two people). These networks have shown great results in computer vision in the area of face verification, but they could also be used to improve the bot detection models of this work. If a Siamese network were to be used, which is trained on tweets of humans and bots, the model would be forced to learn the characteristics of bots and humans in order to differentiate them. Therefore, the hidden representations of the tweets would also be more similar if they came from the same account or same type of accounts such as a bot or a human. These hidden states could then be used by the account classification model, which was introduced earlier.

And last, the best models from the PAN 2019 Bots and Gender Profiling task could also be combined with the approaches introduced in this thesis work as the solutions could be complementary to each other.

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ORIGINAL ARTICLE

Machine learning-based social media bot detection: a comprehensive literature review

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Abstract
In today's digitalized era, Online Social Networking platforms are growing to be a vital aspect of each individual's daily life. The availability of the vast amount of information and their open nature attracts the interest of cybercriminals to create malicious bots. Malicious bots in these platforms are automated or semi-automated entities used in nefarious ways while simulating human behavior. Moreover, such bots pose serious cyber threats and security concerns to society and public opinion. They are used to exploit vulnerabilities for illicit benefits such as spamming, fake profiles, spreading inappropriate/false content, click farming, hashtag hijacking, and much more. Cybercriminals and researchers are always engaged in an arms race as new and updated bots are created to thwart ever-evolving detection technologies. This literature review attempts to compile and compare the most recent advancements in Machine Learning-based techniques for the detection and classification of bots on five primary social media platforms namely Facebook, Instagram, LinkedIn, Twitter, and Weibo. We bring forth a concise overview of all the supervised, semi-supervised, and unsupervised methods, along with the details of the datasets provided by the researchers. Additionally, we provide a thorough breakdown of the extracted feature categories. Furthermore, this study also showcases a brief rundown of the challenges and opportunities encountered in this field, along with prospective research directions and promising angles to explore.

Keywords Social media security · Bot detection · Machine learning · Social bots · Feature engineering · Cybersecurity

1 Introduction

In this modern world, OSNs such as Twitter, Facebook, Instagram, LinkedIn have become a crucial part of each one's life (Albayati and Altamimi 2019). It radically impacts daily human social interactions where users and their communities are the base for online growth, commerce, and information sharing. Different social networks offer a unique value chain and target different user segments. For instance, Twitter is known for being the most famous microblogging

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social network for receiving rapid updates and breaking news. While Instagram usage is mainly by celebrities and businesses for marketing (Meshram et al. 2021). Whereas professional communities use LinkedIn. As social networks' popularity grows combined with the availability of vast personal information that users share makes the same valuable features of social platforms for ordinary people a tempting target for malicious entities (Adikari and Dutta 2020). The most prevalent form of malware on social media networks is thought to be bots (Aldayel and Magdy 2022; Cai, Li, and Zengi 2017b). Some bots are benign. However, the majority of bots are utilized to perform malicious activities such as fabricating accounts, faking engagements, social spamming, phishing, and spreading rumors to manipulate public opinion, such activities not only disturb the genuine users' experience but also lead to a negative effect on the public's and individual's security. As a result, in recent years, researchers have dedicated a significant amount of attention to social media bot detection (Ali and Syed 2022; Ferrara 2018; Rangel and Rosso 2019; Yang et al. 2012) and prevention (Thakur and Breslin 2021).

1.1 Social media platforms

OSNs have revolutionized communication technologies and are now an essential component of the modern web. The most popular social networks globally as of January 2022 are shown in Fig. 1, ordered by the number of monthly

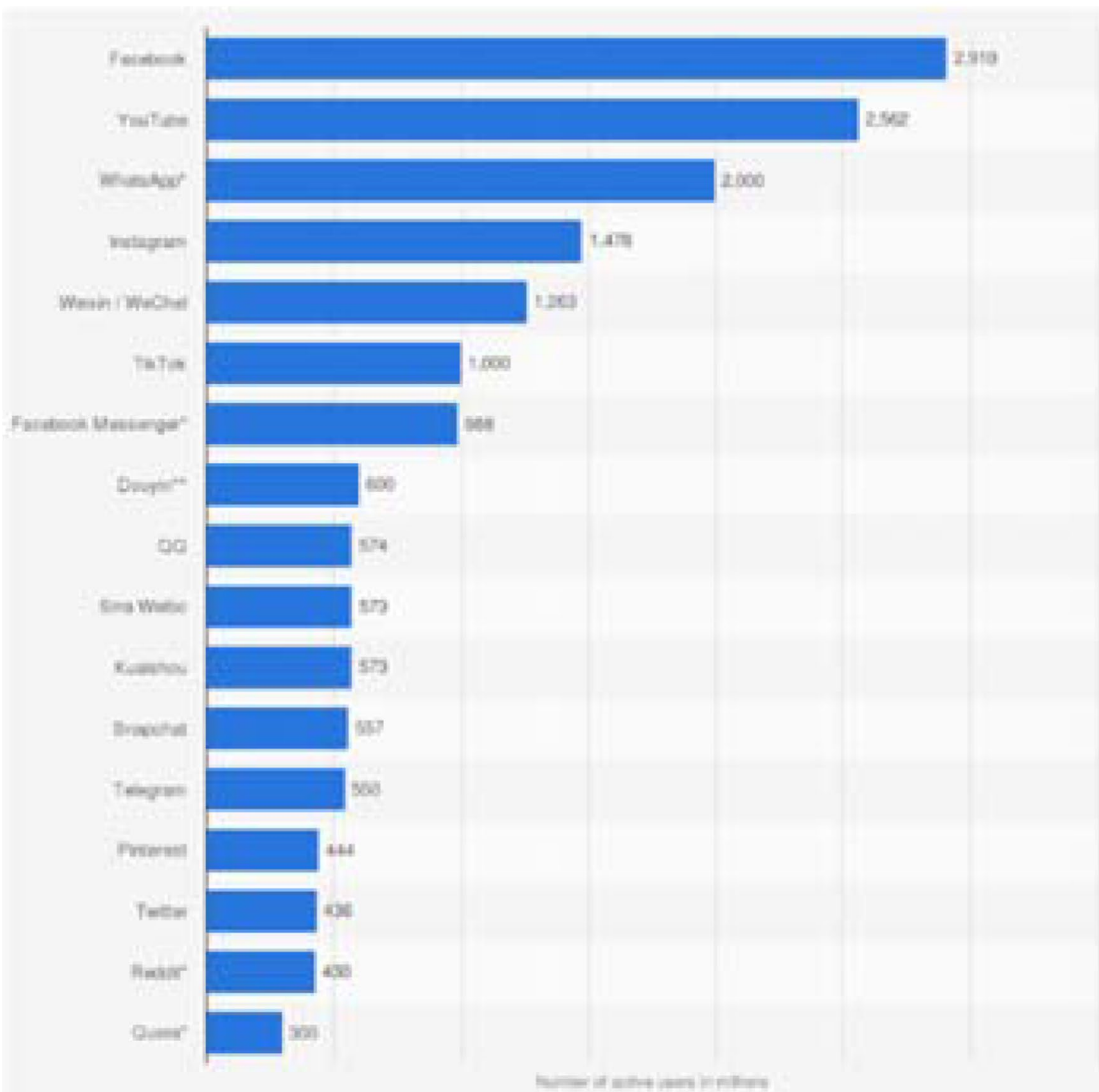

Fig. 1 Most popular social networks globally as of January 2022, ordered by no. of monthly active users.https://www.statista.com/statistics/272014/global-social-networks-ranked-by-number-of-users/

active users in millions. The social media platforms which are included in the scope of our study are namely Twitter, Facebook, Instagram, LinkedIn, and Weibo. On these platforms, user growth and popularity have been increasing at an exponential rate. These platforms enable users to produce and exchange user-generated content (Kaplan and Haenlein 2010). For instance, only 2.375 billion people were using Facebook in the first quarter of 2019 (Siddiqui 2019), thereby representing one-third of the world population (Caers et al. 2013). One of the most widespread and extensively used OSN by people from all walks of life is Twitter. Twitter allows individuals to express their sentiments on different topics such as entertainment, the stock market, politics, and sports. (Wald et al. 2013). It is one of the fastest means of circulating information as a result extremely affects people’s perspectives. Over the past few years, Twitter has become a replacement for mainstream media for obtaining news (Wald et al. 2013). On the other hand, Instagram is an OSN for sharing photos and videos and is accessible on both Android and iOS since 2012. Dated May 2019, there were more than a billion users registered on Instagram, according to collected data (Thejas et al. 2019). Moving on, Facebook is an online social networking site that makes it convenient for people to connect and share with family and friends. It was developed in 2004 initially for students by Mark Zuckerberg. With more than 1 billion users globally, Facebook is one of the biggest social networks in the current times (Santia et al. 2019). One of the most well-known professional social networks is LinkedIn, a platform that focuses on professional networking and career advancement (Dinath 2021). Sina-Weibo, also known as Chinese Twitter, was launched in 2009, and this microblogging website or application is one of China’s biggest social media platforms. It offers a plethora of features which include posting images, instant messaging, Weibo stories, using location-based hashtags, trending topics, etc. Furthermore, it also gives businesses the privilege to set up accounts for the purpose of advertisements and services (Tenba Group 2022).

1.2 Social media security

Security and trustworthiness among users, service providers, platform owners, and third-party supervisors are critical factors for social media platforms’ success and stable existence (Zhang and Gupta 2018). According to recent surveys (Shearer and Mitchell 2022), a considerable segment of the population prefers social networks to TV, newspapers, and other traditional media when looking for information. Trust in social networks as a source of information is predicted to rapidly grow (Kolomeets and Chechulin 2021). As a result, social bots can pose significant security risks by influencing public opinion and disseminating false information (Shao et al. 2017), spreading rumors and conspiracy theories (Ferrara 2020), creating fake reputations, and suppressing political competitors (Pierri et al. 2020; Benkler et al. 2017). Despite the fact that bots are extensively used, little research has been done to examine how they affect the social media environment. This indicates that nearly 48 million of the accounts on Twitter are bots (Sheehan 2018). It was also stated that Facebook acknowledges that 270 million of its accounts are fake (Sheehan 2018). Further, there is evidence that social media bots were utilized to attempt to influence political communication dates during the US midterm elections in 2010. There were also allegations that social bots on Twitter played a significant role in the 2016 US presidential election (Cresci et al. 2017; Mahesh 2020; Sedhai and Sun 2015). Bots can be employed to spread misinformation to promote a particular view of a public person, grow an account's following, and repost user-generated content. Bot detection on OSNs is therefore the most frequently requested security feature from businesses and law enforcement organizations (Kolomeets and Chechulin 2021). The dearth of publicly accessible datasets for OSNs such as Facebook, Instagram, and LinkedIn is one of the greatest obstacles in this research area. Unlike Twitter, this restriction results from some of these OSNs’ limited data collection policies.

1.3 Types of bots on social media

The term "bot" refers to a robot, a computer program that works more quickly than humans at recurring, automated tasks. More precise terminology can be used to define bots in OSNs "a computer software that generates content automatically and engages with users of social media to replicate and possibly modify their behavior" (Benkler et al. 2017). Bots can be used for useful or harmful reasons and often replicate human behavior to some degree (Fonseca Abreu et al. 2020). Good bots can significantly reduce the need for human customer service representatives for some businesses, such as chatbots and news bots that automatically upload new articles or news for journalists or bloggers. Bots can be employed for negative as well as positive purposes. According to (Gorwa and Guilbeault 2020), bots are responsible for a sizable portion of online activity, are used to manipulate algorithms and recommender systems, stifle or promote political speech, and can be crucial in the spread of hyper-partisan "fake news." According to (Benkler et al. 2017), there are four different categories of social media bots: spambots, social bots, sybil bots, and cyborgs. Promoter bots, URL spambots, and false followers are only a few examples of the various types of spambots that spread harmful links, uninvited messages, and hijack popular subjects on social networks (Meshram et al. 2021). On the other hand, social bots are algorithmically controlled user accounts that mimic the activity of human users but carry out their tasks at a considerably faster rate while successfully concealing their robotic identity (Ferrara 2018). While cyborgs bots are half-human, half-bot accounts that exist between people and bots, sybil bots are anonymous identities, i.e., user accounts, utilized for a significantly big effect (Gorwa and Guilbeault 2020). In this review, the collected papers were incorporating three categories of bots which are social bots spambots, and sybil bots.

1.4 The different machine learning-based techniques and algorithms

The development of algorithms that allow a computer to learn on its own from data and prior experiences is the core of ML, a subfield of artificial intelligence (AI). Arthur Samuel was the first to originate the term "Machine Learning" (Wiederhold and McCarthy 1992). ML system develops prediction models based on previous data and makes predictions up until new data are gathered. The amount of data used to create a model determines its accuracy (Saranya Shree et al. 2021). The various types of ML techniques include Supervised, Semi-supervised, Unsupervised, and Reinforcement. However, we have only included Supervised, Semi-supervised, and Unsupervised as a part of our study. Three types are present under supervised category: Classification, Regression, and Forecasting. Some of the most popular supervised algorithms include Random Forest (RF), Naïve Bayes (NB), Decision Trees (DT), Logistic Regression (LR), Support Vector Machine (SVM), Neural Networks (NN), and many more. Deep learning (DL) is a subset of supervised ML techniques that employs multiple layers to gradually extract higher-order features from the input data. In order to create patterns and process data, this AI technology mimics the actions and processes of the human brain (Gannarapu et al. 2020). Convolutional Neural Networks (CNNs), Long Short-Term Memory Networks (LSTMs), Generative Adversarial Networks (GANs), etc., are some of the well-known DL algorithms. Whereas unsupervised learning algorithms are namely categorized into Clustering and Association which mainly deal with unlabeled data. Some of the primarily used ones include K-nearest Neighbor (KNN), K-means clustering, Principal Component Analysis (PCA), etc. However, a small amount of labeled data and a large amount of unlabeled data are utilized in semi-supervised learning, which results in a hybrid of supervised and unsupervised learning (Mahesh 2020).

1.5 Machine learning implementation on social media security

The way people use social media is evolving as a result of the proliferation of ML techniques in social media and the increased sophistication of cyberattacks on computer information systems (Aljabri et al. 2021a, b). On social networking sites such as Facebook and Twitter, numerous ML techniques were employed. For instance, existing ML algorithms can determine the user's location, carry out sentiment analysis, (Aljabri et al. 2021a, b) offer recommendations, and much more. Diverse ML methods have been successfully deployed to address wide-ranging problems in cybersecurity which include detecting malicious URLS, classification of firewall log data, phishing attacks detection, etc. (Aljabri et al. 2022a, b, c; AAljabri and Mirza 2022). However, malicious software can be used to target social media platforms and carry out cyber-attacks. In terms of social media, several efforts have been undertaken to investigate ML techniques to detect such types of malware. For instance, (Alom et al. 2020) detected Twitter spammers using DL techniques. Moreover, the study conducted by (Kantartopoulos et al. 2020) addressed the effects of hostile attacks and utilized KNN as a measure to tackle the problem. The authors presented a methodology that uses SVM and Ensemble algorithms to effectively detect cyberbullying (Gupta and Kaushal 2017). Additionally, models have been developed for social media systems' access control (Carminati et al. 2011). Yet, bot and fake account detection on social media platforms are still one of the primary challenges for cyber security researchers (Thuraisingham 2020).

1.6 Key contributions

This section firstly puts forth the existing literature reviews done on different social media platforms as shown in Table 1. It also briefly discusses the previously used taxonomies along with the prevailing gaps. Starting with the literature review on Twitter, (Alothali et al. 2019) included literature concerning from 2010 to 2018 based on various techniques which include Graph-based, Crowdsourcing, and ML. They analyzed the common aspects such as datasets, classifiers, and the selected features employed. The challenges present in the domain were also addressed. (Derhab et al. 2021) discussed existing techniques and put forth a taxonomy that addressed the state-of-the-art tweet-based bot detection techniques in the timeline from 2010 to 2020. Based on tweet-based bot detection techniques, they provided the main features utilized. For tweet-based bot detection, they also described big data analytics shallow and DL techniques, in addition to their performance results. Finally, the challenges and open issues in the area of tweet-based bot detection were presented and discussed (Derhab et al. 2021). Furthermore, (Orabi et al. 2020) discussed the studies from 2010 to 2019 on Graph-based, ML-based, Crowdsourcing, and Anomaly-based. Their research revealed some gaps in the literature, such as the fact that studies discussed mainly Twitter, and that unsupervised ML is rarely used, in addition to the majority of publicly available datasets being either inaccurate or

Table 1 Summary of existing literature reviews

References	Range of papers reviewed	Taxonomy	Open issue/future discussion
Alothali et al. (2019)	2010–2018	Graph-based, crowdsourcing, and machine learning	✓
Derhab et al. (2021)	2010–2020	Shallow learning-based (supervised learning, semi-supervised learning, and unsupervised learning)	✓
		Deep learning-based-Deep learning-based	
Orabi et al. (2020)	2010–2019	Graph-based, machine-learning based, crowdsourcing, anomaly-based	✓
Gheewala and Patel (2018)	2010–2017	Clustering algorithms, classification algorithms, hybrid	✗
Ezarfelix et al. (2022)	2018–2021	Logistics regression, naive bayes, random forest, support vector machine	✓
Rao et al. (2021)	2015–2020	URL list-based spam filtering techniques, honeypot/honeynet-based techniques, machine learning and deep learning techniques	✓

insufficiently large. In (Gheewala and Patel 2018), contributed a review on ML twitter spam detection for the years 2010–2017 based on Clustering, Classification, and Hybrid algorithms. Some of the issues concluded were regarding the results being lowered as a result of concerns with feature fabrication, class imbalance, spam drift, etc., for spam detection. The study (Ezarfelix et al. 2022) performed was based only on the Instagram platform where a multitude of analyses, and evaluations have been performed on the studies from 2018 to 2021. It was concluded that in order to detect fake accounts, using NN is the most effective method. (Rao et al. 2021) presented a comprehensive review of the social spam detection techniques studied from 2015 to 2020 based on different social spam detection techniques which include Honeypot/Honeynet-based techniques, URL List-based spam filtering techniques, and ML and DL techniques. Numerous feature analysis and dimensionality reduction techniques used by different researchers were outlined. A thorough analysis was given, describing the datasets utilized, features used, ML/DL models used, performance measures used, and pros and cons of each model.

To the best of our knowledge, no study in the literature has carried out a comprehensive analysis of the existing studies in the time period (2015–2022) in the domain of applying ML-based techniques for social media bot detection (social bots, spambots, sybil bots). For this specific timeline, the existing reviews have studied either only ML or DL-based studies or only addressed a specific bot type. We perceived that there was a need for a recent literature review to be conducted so that researchers could identify the findings and gaps in this field and use that information as a roadmap for future research directions and further in-depth study. In response to this demand, in this study, we discuss what is currently known and being researched regarding the several concepts, theories, and techniques linked to bot detection on social media platforms.

This paper makes the following key contributions:

- Provide summaries and analysis of the used ML-based (supervised, semi-supervised, and unsupervised) classification techniques to detect various types of bots on some particular social media platforms.
- Provide a unique taxonomy based on the various ML-based techniques which has not been provided in the existing literature.
- Identify and analyze the most commonly extracted and used features on each social media platform.
- Study the most affected social media platform from malicious bots, the class of bots mostly found on these platforms. Additionally, highlight the most studied social platforms and analyze the gaps of research on other platforms.
- Examine and analyze the popular public datasets used for each platform and the methods used for the self-collected datasets.
- Highlight challenges and gaps in existing research thereby providing potential directions for further research.

The rest of this paper is structured as follows: Sect. 2 presents the methodology adopted for this paper. Section 3 puts forth a detailed analysis including tables and figures demonstrating the ML-based techniques used in the existing literature. In Sect. 4 based on all the reviewed studies, the datasets used, features extracted, and algorithms implemented are discussed thereby performing an extensive analysis. Section 5 sheds light on the insights gained and presents a discussion on the challenges and opportunities in existing research thereby providing future research directions. Section 6 provides a conclusion to summarize our literature review.

2 Methodology

The objective of this review is to study the existing literature from 2015 to 2022 in the domain of bot detection and classification using ML techniques on various social media platforms. We searched for social media bot detection-related papers on various well-known databases mainly Google Scholar, Mendeley, IEEE Xplore, ResearchGate, ScienceDirect, Elsevier, acm.org, arxiv.org, SpringerLink, MDPI, etc. The total number of papers reviewed were 105. All these 105 papers were summarized and elaborately discussed in this paper. Figure 2 demonstrates the range of the reviewed papers.

Figure 3 shows the created taxonomy for the paper. The first tier is based on ML-based techniques, followed by the second tier on the type of social media platform and lastly, the third tier is based on the type of social media bot which includes social bots, spambots, and sybil bots. The logic behind the taxonomy created in this literature review is mainly to identify the most effected social media platforms from bots, the class of bots mostly found on those platforms, and to highlight the most studied social platforms and analyze the gaps of research on other platforms. This is different from most existing literature reviews which focus on the ML techniques and algorithms used in research which can be inefficient to highlight findings and identify gaps since many studies use several techniques and algorithms applied on one platform.

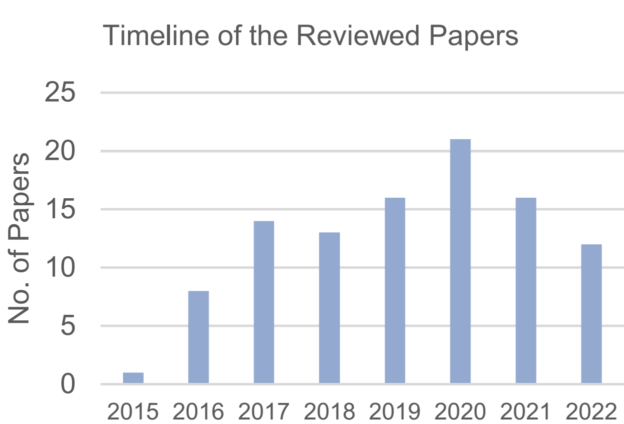

Fig. 2 Bar chart showcasing the range of the reviewed papers

3 Machine learning-based techniques for detecting bots on social media platforms

Numerous studies have been published addressing the use of ML-based techniques for bot detection. This section reviews the existing research studies on the subject by discussing previous studies and findings. The summaries are organized based on the three different ML types followed by different social media platforms and the affecting bot types.

3.1 Using supervised ML

Most of the studies we reviewed have implemented supervised ML and DL to detect social bots, spambots, and sybil bots which shall be discussed below.

3.1.1 Facebook—detecting social bots

Very few studies were found that used the supervised approach to detect social bots on Facebook. To improve classification accuracy, (Wanda et al. 2020) built a supervised learning architecture using a CNN model. To train and evaluate the model, the CNN used a Deep Neural Network (DNN) with a number of hidden layers. In order to minimize the objective function using the model's parameters, it also used a gradient descent. To optimize and accelerate training time in the NN, a pooling layer was used. The results with an optimizer Stochastic Gradient Descent (SGD) $m = 0.5$ showed a training loss of 0.5058 and a testing loss of 0.5060.

Secondly, 4.4 million publicly generated Facebook postings were collected and described in a dataset by (Dewan and Kumaraguru 2017). On their dataset of harmful posts, they used two different filtering techniques: one that used URL blacklists and another that used human annotations. They used NB, DT, RF, and SVM models, among other supervised learning methods. These models are based on a set of 44 publicly accessible attributes. After evaluation, RF was shown to have the highest accuracy of over 80%. Based on their findings, they proceed to develop Facebook Inspector (FBI), a browser plug-in that uses a Representational State Transfer (REST) API to identify harmful Facebook postings in real-time.

3.1.2 Facebook—detecting spambots

Some studies have identified spambots on Facebook using various data collection techniques. Due to the restrictive security policies on Facebook, accessing and acquiring relevant data is challenging.

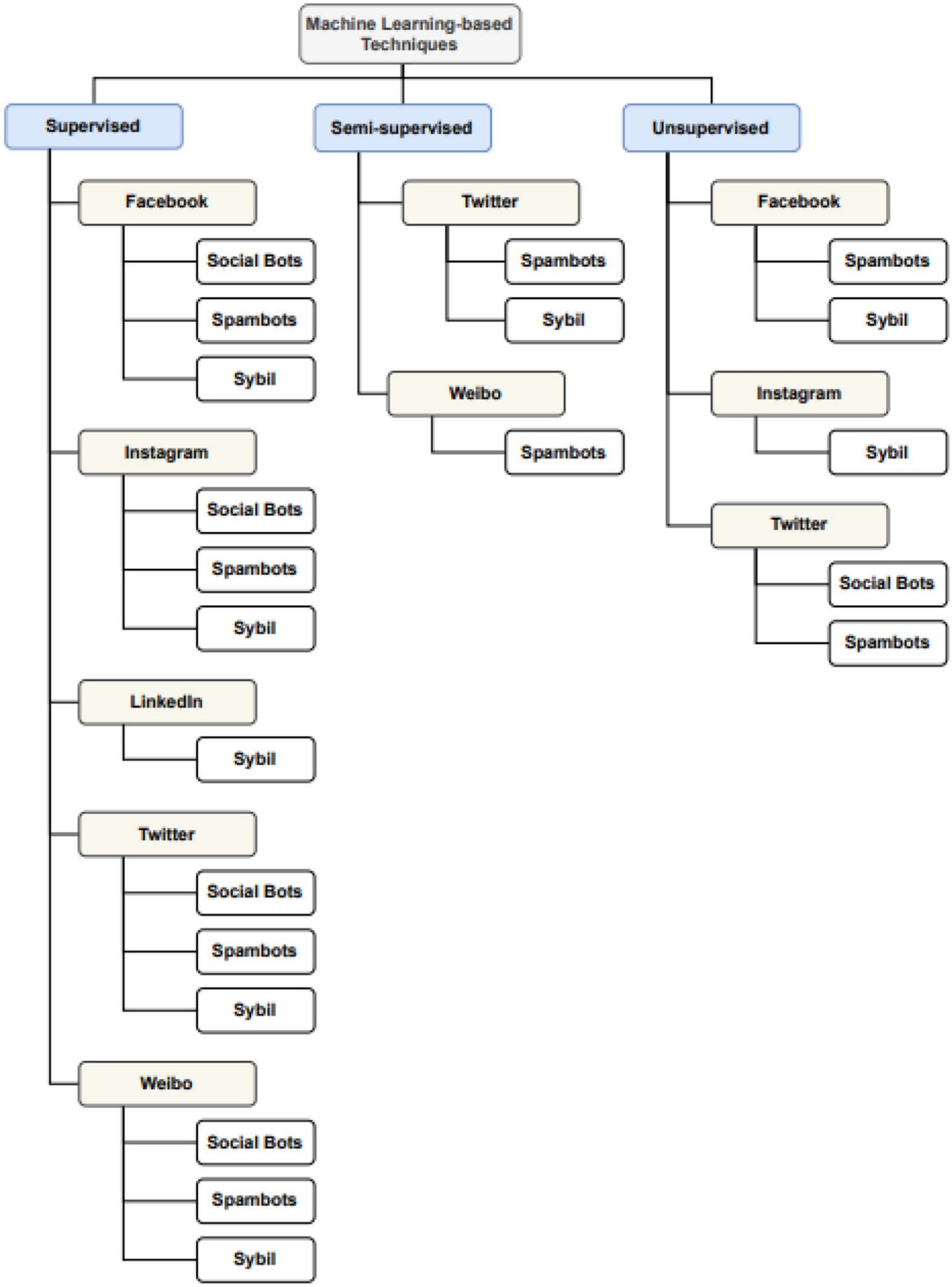

Fig. 3 Taxonomy of social media bot detection using ML-based techniques

Sahoo and Gupta (2020) implemented a spammer detection system on Facebook. The Particle Swarm Optimization (PSO) algorithm was used in this study to determine the popularity of the content and feature selection. The dataset included 1600 profile posts in total. Twelve profile- and content-based features were chosen after the generated content underwent data pre-processing. The PSO algorithm used these features as an input parameter to find fraudulent accounts. In this experiment, classifiers RF, RT, Bagging, JRip, J48, and AdaBoost were utilized. Using the classifier, the detection rate produced the best accuracy of 99.5%.

Followed by, (Rathore et al. 2018) who introduced an efficient spammer detection method called SpamSpotter that uses an Institute for Data, Systems, and Society (IDSS) to differentiate spammers from real Facebook users. A dataset made up of 1000 profiles was employed. The framework made use of features based on profiles and content. They used the Bayesian Network (BN), RF, Decorate (DE), J48, JRip, KNN, SVM, and LR as the eight supervised ML classifiers. The BN classifier outperformed all others with an accuracy of 0.984.

3.1.3 Facebook—detecting sybil bots

We found a reasonable number of studies that thrived in recognizing sybils (Fake profiles) on Facebook. This study proposed by (Albayati and Altamimi 2019) was about a smart system known as FBChecker that checks if a profile is fake. A set of behavioral and informational attributes were analyzed and classified by the system using the data mining approach. Four data mining algorithms which include KNN, DT, SVM, and NB were used. The RapidMiner data science platform was used to implement the selected classifiers. The dataset of 200 profiles was prepared by the authors. A Receiver Operating Characteristic Curve (ROC) graph comparison was created to check the accuracy and all classifiers showed a high accuracy rate, but SVM outperformed with an accuracy rate of 98%.

Subsequently, (Hakimi et al. 2019) proposed supervised ML techniques based on only five characteristics that play a key role in distinguishing fake and true users on Facebook. The important characteristics finalized were Average Post Likes Received, Average Post Comments, Average Post Comments Received, Average Post Liked, and Average Friends. A sample data of 800 users were generated by Mockaroo. The data were categorized into four clusters: Inactive User, Assume Fake account User, Fake account user, and Real User. Classifiers namely KNN, SVM, and NN were implemented. Results showed that KNN outperformed with an accuracy of 0.829. It was concluded that the features “likes”, and “remarks” add a significant value to the job of detection.

Moreover, (Singh and Banerjee 2019) created a dataset on Facebook using their graph API to be utilized for sybil accounts detection. Also, a comparative analysis of various algorithms over the dataset was performed. The dataset contained 995 both real and fake accounts. Twenty-nine features were extracted including textual, categorical, and numerical features. AdaBoost, Bagging, XGBoost, Gradient Boost (GB), RF, LR, Support Vector Classifier (LinearSVC), and ExtraTree algorithms were applied for evaluation. AdaBoost was the best-performing algorithm with a 99% F1-score.

However, (Saranya Shree et al. 2021) suggested Natural Language Processing (NLP) pre-processing techniques and ML algorithms such as SVM and NB to classify fake and genuine profiles on Facebook. A dataset of 516 profiles was used and trained until 30 epochs. It predicted 91.5% fake accounts and 90.2% genuine accounts correctly.

Another strategy for identifying sybils on Facebook was presented by (Babu et al. 2021). By using the Facebook graph API, they gathered a dataset of 500 users from a survey of 500 Facebook users in order to better understand the nature and distinguishing characteristics of sybil. The tested dataset was used to identify fake profiles using the NB classifier. Seven profile-based features were used in the model. Their suggested solution had a 98% efficiency rate. Moving on, (Gupta and Kaushal 2017) has described an approach to detect fake accounts. The key contributions of the authors’ work include a collection of a private dataset using the Facebook API through Python wrappers. After data collection, a set of 17 features was shortlisted which included likes, comments, shares, tag, apps usage, etc. A total of 12 supervised ML classification algorithms were used (from Weka), namely, k-Nearest Neighbor, Naive Bayes, Decision Tree classifiers (J48, C5.0, Reduced Error Pruning Trees Classification (REPT), Random Tree, Random Forest), etc. Two types of cross-validation were performed, namely, the holdout method, and tenfold cross-validation. A classification accuracy of 79% was achieved. The user activities contributed the maximum to the detection of fake accounts.

3.1.4 Instagram—detecting social bots

Only one study by (Sen et al. 2018) aimed to detect fake likes on Instagram thereby detecting social bots. A dataset of 151,117 likes of both fake and genuine likes was captured and labeled manually by the authors. A limitation of this study was the noisiness of the dataset. However, various types of features were extracted from the dataset, which were Network Effect, Internet Overlap, Liking Frequency, Influential Poster, Hashtag Features, and User-based features to be used with extensive analysis. LR, RF, SVM, AdaBoost, XGBoost, NN, and Multilayer Perceptron (MLP) algorithms were applied. MLP showed the best results with 83% Precision and 81% Recall (AUC of 89%). According to the authors, the model's high efficacy in capturing the parameters that influence genuine liking behavior is the model's main strength.

3.1.5 Instagram—detecting spambots

Two studies were found that used the ML approach for fake and automated accounts detection on Instagram. Firstly, (Akyon and Esat Kalfaoglu 2019) contributed by generating two labeled public datasets. A dataset for fake accounts (1203 accounts) and another for bots (1400 accounts). However, both datasets had problems. The fake accounts dataset had an uneven number of real and fake accounts. As a result, the Synthetic Minority Over-sampling Technique-for-Nominal and Continuous (SMOTE-NC) algorithm was implemented. While cost sensitive genetic algorithm was implemented to correct the automated accounts dataset unnatural bias. Profile-centric features were fed into NB, LR, SVM, and NN algorithm. SVM and NN provided promising F1-scores for both datasets. 94% with oversampling for fake accounts and 86% for automated accounts dataset.

Similarly, a method to identify spam posts was also presented by (Zhang and Sun 2017). 1983 user profiles and 953,808 media posts made up a manually labeled dataset. Profile-based, Color Difference Histogram-based, and Media Post-based feature vectors were extracted from user profiles and media postings. The near duplicate posts were grouped into the same clusters using two-pass clustering techniques, Minhash clustering and K-medoids clustering. The best pair has an accuracy of 96.27%: RF, (maxDepth: 8, numTrees: 20, impurity: entropy).

3.1.6 Instagram—detecting sybil bots

Many studies were able to detect sybil bots starting with (Meshram et al. 2021) proposed an automated methodology for fake profiles detection. The authors collected 1203 accounts including real and fake accounts using Instagram API. In addition, a list of eight content- and behavior-based features were extracted. Authors needed to oversample the dataset using SMOTE-NC before applying any algorithm due to the unevenness of the real-fake accounts ratio. Afterward, NN, SVM, and RF algorithms were applied. RF depicted the best-performing results with an accuracy of 97%.

Whereas, using the same records and features, (Sheikhi 2020) presented a bagging classifier and performed a comparative analysis with five well-known ML algorithms, which were RT, J48, SVM, Radial Basis Function (RBF), MLP, Hoeffding Tree, and NB with 10-cross-validation. The bagging classifier showed better performance by successfully classifying 98% of the accounts. Moreover, the author presented the best feature types for different sizes of datasets.

Additionally, (Dey et al. 2019) also assessed fake and real different Instagram accounts. A publicly labeled dataset of sixteen accounts was obtained from Kaggle. Twelve profile-based features were extracted from the sample dataset. Missing Value Treatment, OuSybiltlier Detection, and Bivariate Analysis were carried out as a part of the Exploratory Data Analysis. Median imputation was done to deal with the outliers. For the extent of this paper, LR, and RF—two supervised classification algorithms were used. Lastly, out of the two mentioned classifiers, RF showed the best performance with 92.5% accuracy.

Subsequently, the research of (Purba et al. 2020) aimed to identify fake users' behavior. Furthermore, different approaches of classification have been proposed. 2-class (authentic, fake) and 4-classes (authentic, spammer, active fake user, inactive fake user) classifications. The total number of fake and authentic users in the dataset was 32,460 users. They used seventeen features based on metadata, media info, media tags, media similarity, and engagement. Using these features with RF, MLP, LR, NB, and J48 algorithms showed promising results. RF showed an accuracy of up to 91.76% for 4-classes classification. Moreover, analysis outcomes showed that metadata and statistics results are the foremost predictors for classification.

Nevertheless, (Kesharwani et al. 2021) utilized a six-layered DL model NN to classify fake and genuine Instagram accounts. The designed model used 12 profile-based features. An open dataset of 696 Instagram users available on Kaggle was used for this experiment and was collected using a crawler. The dataset had 10 profile-based features. The model's training was done using 20 epochs and therefore giving an accuracy of 93.63%.

Quite interestingly, (Bazm and Asadpour 2020) proposed a behavioral-based model. A labeled dataset was collected by the authors including 2000 accounts of both fake and genuine users. Seven behavioral features were extracted from the dataset. KNN, DT, SVM, RF, and AdaBoost algorithms were tested and analyzed. AdaBoost showed the best-performing results with an accuracy of 95%. Additionally, the Max feature was identified as the most effective for classification followed by standard deviation, following count, and entropy. Three of the above-mentioned most effective features were behavioral.

Lastly, the work of (Thejas et al. 2019) also focused on detecting valid and fake likes of Instagram posts by applying automated single and ensembled learning models. A labeled dataset of 10,346 observations and 37 features has been composed. The authors used numeric features and text-based features to perform extensive analysis of fake likes related patterns. Various single classifiers have been used such as LR SVM, KNN, NB, and NN with different versions. Adjacent to ensembled-based classifiers as RF with multiple versions as well. Moreover, bot detection using an autoencoder has been experimented. RF showed the highest performance among all with 97% accuracy.

3.1.7 LinkedIn—detecting sybil bots

Only two studies were found on this platform to detect sybils, (Adikari and Dutta 2020) proposed a methodology for identifying bot-generated profiles based on limited publicly available data of profiles using data mining techniques. Many existing research assumes the availability of static and dynamic data of a profile, which is not the case with LinkedIn as it has more restrictive privacy policies that impede access to dynamic data. The profile features were extracted from a dataset of 74 profiles only. Thirty-four fake accounts were collected by searching blogs and websites for known LinkedIn fake accounts. The lack of verified fake accounts was a limitation of this research. NN, SVM, PCA, and Weighted Average algorithms were used in several combinations for detecting fake profiles. SVM showed the highest accuracy (87.34%) when employing PCA-selected features with a polynomial kernel.

Furthermore, (Xiao et al. 2015) proposed a scalable offline framework using the pipeline to identify clusters of fake accounts on LinkedIn. Cluster-level fake accounts are identified rather than account-level to detect fake accounts after registration rapidly. Statistical features generated by users at or after registration time, such as name, email address, company, were grouped into clusters. Cluster-level features were exclusively fed into the RF, LR, and SVM models. The authors have collected a set of labeled data for 260,644 LinkedIn accounts. RF algorithm's performance evidently provided the best results for all metrics; an AUC of 0.95 and a recall of 0.72 at 95% precision for out-sample test data.

3.1.8 Twitter—detecting social bots

Numerous studies were able to detect social bots on Twitter starting (Echeverrï£¡a et al. 2018) tested 20 unseen bot classes of varying sizes and characteristics using bot classifiers. Two datasets were collected using Twitter's API consisting of 2.5 million accounts. Twenty-nine Profile- and Content-based features were employed for classification. The classifiers used to test were GB Trees (XGBoost and LightGBoost Model (LGBM)), RF, DT, and AdaBoost. LGBM showed the highest accuracy rate of 97.84% on both the subsampling used—C30K and C500.

Moreover, (Fonseca Abreu et al. 2020) examined whether feature set reduction for Twitter bot detection yields comparable outcomes to large sets. Five Profile-based features were used for classification. The dataset used consisted of 4565 records of both social bots and genuine users. The ML algorithms tested namely were RF, SVM, NB, and one-class SVM. AUC's greater than 0.9 were obtained by all multi-class classifiers. However, RF exhibited the best results with an AUC of 0.9999.

Varol et al. (2017) used more than a thousand features which were based on metadata primarily based on friends, tweet content, sentiment, network patterns, and activity time series. A publicly accessible dataset of size 31 K that contains manually verified Twitter accounts as bots or real was used to train the model. The model's accuracy was evaluated using RF, AdaBoost, LR, and DT classifiers. The best performance was depicted by RF of 0.95 AUC. Furthermore, it was concluded that the most significant sources of data are user metadata and content features.

Twenty-eight features were extracted based on profile, tweets, and behavior (Knauth 2019). For easy future portability, language-agnostic features were mainly focused on. LR, SVM, RF, AdaBoost, and MLP classifiers were used for experiments. AdaBoost outperformed all competitors with an accuracy of 0.988. Smaller quantities of training data were analyzed, and it was shown that using a few, expressive characteristics provides good practical benefits for bot identification.

In this study, after a long process of feature extraction and data pre-processing, (Kantepe and Gañiz 2017) employed ML techniques. Thousand eight hundred accounts were used to get the data from Twitter API and Apache Spark, which was In this study, after a long process of feature extraction and data pre-processing, (Kantepe and Gañiz 2017) employed ML techniques. One thousand eight hundred accounts data was obtained with Twitter API and Apache Spark, which was then used to extract 62 different features. The features extracted were mainly profile-based features, Twitter features and periodic features. Four classifiers were used which include LR, Multinomial Naïve Bayes (MNB), SVM and GB. The highest accuracy result 86% was shown by the GB trees.

This research conducted by (Barhate et al. 2020) used two approaches for the detection of bots and analyzed their influence in trending a hashtag on Twitter. First, the bot probability of a user was calculated using a supervised ML technique and a new feature bot score. A total of 13 features were extracted for data pre-processing and Estimation of Distribution Algorithms (EDA). The data were trained using RF classifier, which produced an AUC result of 0.96. This study also came to the conclusion that bots had a high friend-to-follower ratio and a low follower growth rate.

The dataset that was acquired by (Pratama and Rakhmawati 2019) is from the supporters of the Indonesian presidential candidate on Twitter. The top five hashtags for each candidate were used to collect tweets, which were then manually labeled with the accounts' bot characteristics, resulting in a limit of about 4.000 tweets. SVM and RF, two ML models, are utilized for bot detection. These two models were trained with cross-validation ten-folds to improve the overall score. From these two models, RF has a higher overall score than SVM of 74% in F1-Score, Accuracy, and AUC. Comparing the 10 retrieved features from the dataset, they discovered that the account year creation had the biggest separation between humans and bots.

Davis et al. (2016) made use of RF classifier to evaluate and detect social bots by creating a system called BotOrNot. A public dataset of 31 K accounts was used to train the model. From six main groups of characteristics—network, user, friend, temporal, content, and sentiment features—the framework collected more than 1000 features. These various classifiers—one for each category of features and one for the overall score—were trained using extracted features. The system performance was assessed using ten-fold cross-validation, and an AUC value of 95% was obtained.

Likewise, a Twitter bot identification technique was also presented by (Shevtsov et al. 2022). 15.6 million tweets ‘total, including 3.2 million accounts sent during the US Elections, were included in their dataset from Twitter. The XGBoost algorithm was used to pick 229 features from approximately 337 user-extracted features. Their suggested ML pipeline involves training and validating many three ML models which are SVM, RF, and XGBoost. Performance was best for XGBoost where their findings indicate that it performs well on the collected dataset compared to the training data section because of its great generalization capabilities. Only 2% of the F1 score is going from 0.916 to 0.896, and 0.03% of the ROC-AUC indicates a decline in performance from 0.98 to 0. 977.

Additionally, SPY-BOT, a post-filtering method based on ML for social network behavior analysis, was introduced by (Rahman et al. 2021). Six hundred training samples were used to extract eleven characteristics. They contrast the two ML algorithms LR and SVM throughout the training phase. After comparing outcomes, tuned SVM was the best performing. On the validation dataset, their method achieves up to 92.7% accuracy while up to 90.1% accuracy was obtained on the testing dataset. As result, they suggest that the proposed approach able to classify the users’ behavior in Social Network-Integrated Industrial Internet of Things (SN-IIoT).

Also, a real-time streaming framework called Shot Boundary Determination (SBD) was also suggested by (Alothali, Alashwal, et al. 2021a) as a way to detect social bots before they launch an attack to protect users. To gather tweets and extract user profile features, the system uses the Twitter API. They used a publicly available Twitter dataset from Kaggle, which has a total of 37,438 records, as their offline dataset. Friends count, Followers count, Favorites count, Status count, Account age days, and Average tweets per day were the six features that were extracted and further used as input to their ML model. They use RF algorithm to differentiate between the bots and human accounts. The outcomes of their methodology demonstrated the effectiveness of retrieving, publishing the data, and monitoring the estimates.

Shukla et al. (2022) proposed a novel AI-driven multilayer condition-based social media bot detection framework called TweezBot. Moreover, the authors have performed a comparative analysis with several existing models and an extensive study of features, and exploratory data. The proposed method analyzed each Twitter-specific user profile features and activity-centric characteristics, such as profile name, location, description, verification status, and listed count. 2789 distinct user profiles were used to extract these features from a public labeled dataset from Kaggle. ML models used for comparative evaluation and analysis were RF, DT, Bernoulli Naïve Bayes (BNB), CNB, SVC, and MLP. TweezBot attained a maximum accuracy of 99.00049%.

Since bots are used to manipulate activities in politics as well (Fernquist et al. 2018) presented a study on political Twitter bots and their impact on the September 2018 Swedish general elections. To identify automatic behavior, an ML model that is independent of language was developed. The training data consist of both bots and genuine accounts. Three different datasets (Cresci et al. 2015; Gilani et al. 2017; Varol et al. 2017) were used to train the classification model. Furthermore, a list of 140 user metadata, Tweet and Time features were extracted. Various algorithms such as AdaBoost, LR, SVM, and NB were tested. RF outperformed with an accuracy of 0.957.

Similarly, (Beğenilmiş and Uskudarli 2018) made use of collective behavior features in hashtag-based tweet sets, which were compiled by searching for relevant hashtags. A dataset of 850 records was utilized to train the model using algorithms including RF, SVM, and LR. From tweets collected during the 2016 US presidential election, 299 features were retrieved. To capture the coordinated behavior, the features represent user and temporal synchronization characteristics. These models were developed to distinguish between organic and inorganic, political and non-political, and pro-Trump or pro-Hillary or neither tweet set behavior. The RF displayed the best outcomes, with an F-measure of 0.95. In conclusion, this study found that media utilization and tweets marked as favorites are the most dominant features and user-based features were the most valuable ones.

On the other hand, in this approach, (Rodríguez-Ruiz et al. 2020) one-class classification was suggested. One benefit of one-class classifiers is that they do not need examples of abnormal behavior, such as bot accounts. The public dataset (Cresci et al. 2017) was used. Bagging-TPMiner (BTPM), Bagging-RandomMine (BRM), One-Class K-means with Randomly projected features Algorithm (OCKRA), one-class SVM, and NB were the classifiers that were taken into consideration. For categorization, only 13 numerical features were extracted. With an average AUC value of 0.921, Bagging-TPMiner outperformed all other classifiers over a number of experiments.

Moreover, (Attia et al. 2022) proposed a new multi-input DNN technique-based content-based bot detection model. They used the 6760 records from the public PAN 2019 Bots and Gender Profiling Task (Rangel and Rosso 2019) dataset. The proposed multi-input model includes three phases. Their proposed Multi-input model includes 3 phases. The first phase represents the first input as an N-gram model of a 3D matrix of 100*8*300 as model input to two-dimensional CNN. On the other hand, the second phase input is one-dimensional CNN model that has a vector with M length (100 tweets) as model input. The final phase has the previous models with fully connected neural networks to combine them. Each model was trained using suitable hyper-parameters values. Their model achieved a detection accuracy of 93.25% and outperforms other newly proposed models in bot detection.

In the work of (Sayyadiharikandeh et al. 2020) for each class of bots, they recommended training specialized classifiers and combining their conclusions using the maximum rule. In the most recent version of Botometer, they also produced Ensemble Specialized Classifier (ESC). Additionally, the authors used 18 different public labeled datasets from Bot Repository, and over 1200 features were extracted. Features were divided into 6 categories: metadata, retweet/mention networks, temporal features, content information, and sentiment features. Accordingly, a cross-domain performance comparison and analysis was performed using all the 18 different datasets. The authors recommend considering the three types of bot class as in (Cresci et al. 2017) dataset. Moreover, the authors provided a list of the most informative features per bot classes in the used public dataset.

A comprehensive comparative analysis was conducted by (Shukla et al. 2021) to determine the optimal feature encoding, feature selection, and ensembling method. From the Kaggle repository, a total of 37,438 records comprising the training and testing dataset were acquired. Scaling of numerical attributes and encoding of categorical attributes were two steps in the pre-processing of the dataset. A total of 19 attributes were extracted. The model used the classifiers: RF, Adaboost, NN, SVM, and KNN. It was determined that employing RF for blending produced the best results and the highest AUC score of 93%. Since the proposed approach uses Twitter profile metadata, it can detect bots more quickly than a system that analyzes an account's behavior. However, the system's reliance on static analysis reduces its efficiency.

Ramalingaiah et al. (2021) represented an effective text-based bag of words (BoW) model. BoW produces a numerical vector that can be utilized as inputs in different ML algorithms. Using resulted features from feature selection process, different ML algorithms were implemented like DT, KNN, LR, and NB to calculate their accuracies and compare it with their classifier which uses the BoW model to detect Twitter bots from a given training data. The utilized dataset from Kaggle with 2792 training entries and 576 testing entries for evaluation of their models. As a result, the performance of the decision tree gives the highest accuracy which further uses a bag of bots' algorithm to increase accuracy in detecting bots. Their classifier performs the best as it uses a bag of words model with test data yields an accuracy of over 99%.

A ML method based on benchmarking was proposed by (Pramitha et al. 2021) to choose the best model for bot account detection. Dataset obtained from Kaggle with 24,631 records then scraping was performed using the Twitter API to obtain profile features. Furthermore, over-sampling using SMOTE is applied to overcome imbalanced data and improve the models' accuracy. Both RF and XGBoost algorithms were evaluated. XGBoost algorithm outperforms RF, with an accuracy of 0.8908. Additionally, after ranking fifteen different features, they discovered that three significant features—verified, network, and geo-enable—can identify between human and bot accounts.

Many studies implemented effective DL algorithms instead of ML, such as a Behavior-enhanced Deep Model (BeDM) proposed by (Cai, Li, and Zengi 2017b) for bot detection using a real-world public labeled dataset of size 5658 accounts and 5,122,000 tweets from Twitter, which have been collected with honeypots. The model fused tweets content as temporal text data and the user posting behavior information using DL by applying a DNN to detect bots. The DL frameworks used in the BeDM are CNN and LSTM. Compared to Boosting (Gilani et al. 2016; Lee et al. 2006; Morstatter et al. 2016) baselines, the BeDM attained the highest F1 score of 87.32%, which proved the efficacy of the model.

Later in the same year, (Cai, Li, and Zeng 2017a) proposed analogous work. Yet, the novel Deep Bot Detection Model (DBDM) avoids the laborious feature engineering and automatically learns both behavioral and content representations based on the user representation. Additionally, DBDM took into consideration endogenous and exogenous factors that have an impact on user behavior. DBDM achieved a better results with an F1-score of 88.30%.

Additionally, (Hayawi et al. 2022) also proposed a DL framework, DeeProBot used eleven user profile metadata-based features. Five training and five testing datasets were used from Bot Repository. Additionally, the text feature was embedded using GLoVe which aided in enhanced learning from the features. To detect bots, DeeProBot employed a hybrid Deep NN model. On the hold-out test set, DeeProBot gave an AUC of 0.97 for bot detection.

However, in a novel framework called GANBOT (Najari et al. 2022) modified the (Generative Adversarial Network) GAN concept. The generator and classifier were connected via an LSTM layer as a shared channel between them, reducing the convergence limitation on Twitter. By raising the likelihood of bot identification, the suggested framework outperformed the existing contextual LSTM technique. A total of 8386 from the Cresci2017 dataset were used. Results were assessed for four distinct vector dimensions: 25D, 50D, 100D, and 200D; the highest result was 949/0.951 for 200D.

A total of seventeen state-of-the-art methods for bot detection were described by (Kenyeres and Kovács 2022) together based on DL models. They classified Twitter feeds as bots or humans, based solely on the account's textual form of the tweets. PAN 2019 Bots and Gender Profiling task (Rangel and Rosso 2019) dataset was used which consisted of 11,560 labeled users. The core of seven models was based on LSTM networks, four based on Encoder Representations from Transformers (BERT) models, and one a combination of the two. For tweet classification, the best accuracy was obtained using fine-tuned BERT model of 0.828. While for account classification, the Adaboost model archived the best accuracy of 0.9. Their findings demonstrate that, even with a small dataset, DL models may compete with Classical Machine Learning (CML) methods.

Moreover, (Martin-Gutierrez et al. 2021) provide a multilingual method for detecting suspect Twitter accounts through DL. Dataset used in their work was collected using Twitter API of 37,438 Twitter accounts. Several experiments were conducted using different combinations of Word Embeddings to obtain a single vector regarding the text-based features of the user account. These features are later on concatenated with the rest of the metadata to build a potential input vector on top of a Dense Network denoted as Bot-DenseNet. The comparison of these experiments showed that the Bot-DenseNet when using the so-called RoBERTa Transformer as part of the input feature vector with an F1-score of 0.77, produces the best acceptable trade-off between performance and feasibility.

In this research, (Ping and Qin 2019) proposed a social bot detection model DeBD based on the DL algorithm CNN-LSTM for Twitter. CNN was used by DeBD to extract the joint features of the tweet content and their relationship. To carry out the experiments, a dataset of 5132 accounts was created. Secondly, the potential temporal features of the tweet metadata were extracted using LSTM. Finally, in order to achieve the purpose of detecting social bots, the temporal features were finally fused with the joint content features. The dataset used in this experiment was from (Cresci et al. 2017). All the experiments achieved a detection accuracy of more than 99%.

Daouadi et al. (2019) proved that a Deep Forest algorithm combined with thirteen metadata-based features is sufficient to accurately identify bot accounts on Twitter. Two datasets were used which were published by (Lee et al. 2006; Subrahmanian et al. 2016). The Twitter API was used to gather the dataset. The implementation was performed for more than 30 conventional algorithms, including Bagging, MLP, AdaBoost, RF, SL, etc. With an accuracy of 97.55%, the Deep Forest method surpassed the other conventional supervised learning techniques.

In this paper, (Cable and Hugh 2019) implemented the algorithms: NB, LR, Kernel SVM, RF, and LSTM-NN to identify political trolls across Twitter and compared their accuracies. A dataset of tweet ids related to the 2016 elections was used by scraping the Twitter API and obtaining a total of 142,560 unique tweets. The features were extracted using several methods: Word count, TF-IDF, and Word embeddings. The LSTM-NN obtained a test accuracy of 0.957.

Since it is important to determine the best features for enhancing the detection of social bots. To locate these ideal features, (Alothali, Hayawi, et al. 2021b) offer a hybrid feature selection (FS) technique. This method evaluates profile metadata features using random forest, naive Bayes, support vector machines, and neural networks. Using a public dataset made accessible by Kaggle that had a total of 18 profile metadata features, they investigated four feature selection approaches. In order to find the best feature subset, they employed filter and wrapper approaches. They discovered that, when compared to other FS methods, the cross-validation attribute evaluation performed the best. According to their findings, the random forest classifier has the best score using six optimal features: favorites count, verified, statuses count, average tweets per day, lang, and ID.

Lastly, (Sengar et al. 2020) proposed both ML and DL to distinguish bots from genuine users on Twitter. This was done by gathering user activity and profile-based features, then applying supervised ML and NLP to accomplish the goal. A labeled Twitter dataset which contains more than 5000 users and 200,000 tweets was used to train the classifiers. After analysis and feature engineering, eight features were extracted. Different learning models were compared and analyzed to determine the best-performing bot detection system namely KNN, DT, RF, AdaBoost, GB, Gaussian Naive Bayes (GNB), MNB, and MLP. Results showed that NN-based MLP algorithm gave the most accurate prediction with an accuracy of 95.08%. A CNN architecture was proposed for tweet level analysis by combining user and tweet metadata. The MIB Dataset (Cresci et al. 2017) was used.

The novel approach gave a staggering improvement. RF and GB gave the highest accuracy of 99.54%.

3.1.9 Twitter—detecting spambots

Some studies demonstrate the detection of spammers, starting with a hybrid method for identifying automated spammers based on their interactions with their followers was presented (Fazil and Abulaish 2018). Nineteen distinct features were retrieved, integrating community-based features with those from other categories like metadata-, content-, and interaction-based features. A real public dataset of 11,000 labeled users was used. The performance was analyzed using three supervised ML techniques namely RF, DT, and BN which were implemented in Weka. All three metrics—DR-0.976, FPR-0.017, and F-score 0.979, were found to be the best for RF. Lastly, it was determined that interaction- and community-based features are the most successful for spam identification in comparison after executing a feature ablation test and examining the discrimination capability of various features.

Oentaryo et al. (2016) categorized bots based on their behavior as broadcast, consumption, and spambots. A systematic profiling framework was developed which included a set of features and a classifier bank. Numeric, categorical, and series features were taken into consideration. The private manually labeled dataset used consisted of bots and non-bot 159 K accounts. Four supervised ML algorithms were employed which include: NB, RF, SVM, and LR. It was seen that LR outperforms the other classifiers by depicting an F1 score of 0.8228.

The research conducted by (Heidari et al. 2020) firstly, they created a new public data set containing profile-based features for more than 6900 Twitter accounts from the (Cresci et al. 2017) dataset where the input feature set consisted of age, gender, personality, and education from users' online posts. To build their system, they compare the following classifiers: RF, LR, AdaBoost, Feed-forward NN (FFNN), SGD. The results showed that the FFNN model with 97% accuracy provides the best results as compared with the other classifiers. Lastly, a new bot detection model was introduced which uses a contextualized representation of each tweet by using Embeddings from Language Model (ELMO) and Global Vectors for Word Representation (GloVe) in the word embedding phase to have a complete representation of each tweet's text. The model created multiple FFNN's models on top of multilayer bidirectional LSTM models to extract different aspects of a tweet's text. The model detected bots from human accounts, regardless of having the same user profile and achieved 94% prediction accuracy in two different testing datasets.

A spam detection AI approach for Twitter social networks was proposed by (Prabhu Kavin et al. 2022). The dataset (7973 accounts) was collected using Twitter Rest API and combined with the public dataset "The Fake Project" (Cresci et al. 2015). For pre-processing, dataset tokenization, stop word removal, and stemming were applied. User-based and content-based features were extracted from the dataset. To develop the model, a variety of ML methods, including SVM, ANN, and RF, were applied. With user-based features, the findings showed that SVM had the highest precision (97.45%), recall (98.19%), and F measure (97.32%).

In this research, (Eshraqi et al. 2016) determined a clustering algorithm that identified spam tweets (anomaly problem) on the basis of the data stream. The dataset consisted of 50,000 Twitter user accounts and 14 million tweets. The pre-processing was done by RapidMiner and then, transferred into Massive Online Analysis (MOA) for implementation. The features extracted were based on Graphs, Content, Time, and Keywords. When using the DenStream algorithm (Cao et al. 2006), regulating needed to be done properly. The model successfully identified 89% of available spam tweets. Furthermore, the results achieved by the model showed an accuracy of 99%.

Mateen et al. (2017) used 13 user-, content—as well as graph-based features to classify between human and spam profiles. The real public dataset used for this study was provided by (Gu 2022) which consisted of 11 K user accounts and 400 K tweets approximately. Three classifiers namely J48, DE, and NB were used for evaluation. J48 and DE outperformed the other classifiers using the hybrid technique of combined features by showing a 97.6% precision. Results showed that for the dataset employed, the hybrid technique significantly improved precision and recall. Additionally, compared to content- and graph-based features, which demonstrated 92% accuracy, user- and graph-based features correctly classified only 90% of cases.

Moreover, (Chen et al. 2017a, b) found that over time, the statistical characteristics of spam tweets in their labeled dataset changed, which impacted the effectiveness of the existing ML classifiers and is known as Twitter spam drift. Using Twitter's Streaming API, a public dataset of 2 million tweets was gathered. The Web Reputation Technology from Trend Micro was used to identify the tweets that were considered spam. The Lfun system, which was learned from unlabeled tweets, was proposed. Day 1 training and Day 2 to Day 9 testing results showed that RF only obtained DR ranging from 45 to 80%, whereas RF-Lfun increased to 90%. The Detection Rate of RF was roughly 85% from Day 2 training to Day 10 testing, but that of RF-Lfun was over 95%.

Kumar and Rishiwal (2020) explored and provided a framework for identifying spammers, content polluters, and bots using a ML approach based on NN usage. A data set consisting of 5572 tweets containing the text messages and their categorization labeling was used. Various algorithms were trained mainly MNB, Bernoulli, NB, SVM, and

Complementary NB. The most effective and best classification of spam account detection was shown by MNB with an accuracy of 99%.

In this study, (Güngör et al. 2020) used a dataset of 714 tweets that had been manually labeled and retrieved through the Twitter API. Eight profile-based features and five tweet-based features were extracted and analyzed. Additionally, a set of guidelines had been discovered via adding followers and friend FF rate, and spam accounts had been detected. For this experiment, the algorithms NB, J48, and LR were used. J48 performed the best, achieving an accuracy of 97.2%. In conclusion, the accuracy rate increased as a result of the usage of both tweet- and profile-based features.

By utilizing a dataset of 82 accounts of tweeters who use both Arabic and English, (Al-Zoubi et al. 2017) improved spam identification. J48, MLP, KNN, and NB were the algorithms used and compared in tenfold cross-validation with stratified sampling as a training/testing methodology. With an accuracy of 94.9, J48 demonstrated the best spam detection ability using the top seven features discovered by ReliefF.

For bot detection, (Heidari et al. 2021) analyzed the sentiment features of tweets' content for each account to measure their impact on the accuracy of ML algorithms. The authors have used (Cresci et al. 2017) dataset of the size of 12,736 accounts and 6,637,615 tweets. The bot detection methodology proposed by the authors is centered on the number of tweets that show a concentration on extreme opinions for an individual account. Whether the opinions are overly negative, positive, or neutral, it indicates the user is a bot. ML models such as RF, NN, SVM, and LR were examined using the proposed sentiment features. The highest result was achieved using Support Vector Regression (SVR) with an F1-score of 0.930.

The research work (Rodrigues et al. 2022) focused on identifying live tweets as spam or ham and performed sentiment analysis on both live and stored tweets to classify them as either positive, negative, or neutral. The proposed methodology used two different datasets from Kaggle. Vectorizers like TF-IDF and BoW models were used to extract sentiment features, which were then fed into a variety of ML and DL classifiers. The classifiers achieved the highest accuracy rate using LSTM in both spam detection with 98.74% and sentiment analysis with 73.81% accuracy.

The work (Andriotis and Takasu 2019) proposed a content-based approach to identify spambots. Technically, four public datasets were used in this study, which was (Cresci et al. 2017; Varol et al. 2017; Yang et al. 2012, 2013). Collectively, the datasets contain tweets of nearly up to 20 K accounts of both bots and genuine users. The methodology proposed employed metadata, content, and sentiment features. Furthermore, the performance of the KNN, DT, NB, SVM, RF, and AdaBoost algorithms was tested. AdaBoost showed the best result with a 0.95 F1-score. Additionally, the study depicted that sentiment features add value when combined with known features to bot detection algorithms.

Also, (Sadineni 2020) detect spam using a dataset from Kaggle that included 950 users and ten content-based attributes, demonstrating that SVM and RF outperform NB in terms of performance.

On the other hand, (Kudugunta and Ferrara 20182018) presented a contextual LSTM architecture based on a DNN that uses account metadata and tweet text to identify bots at the tweet level. The tweet text served as the primary input for the model. It was tokenized and converted into a series of GloVe vectors before being fed into the LSTM, which then fed the data into a 2-layer NN with ReLU activations. High classification accuracy can be attained using the suggested model. Additionally, the compared techniques for account-level bot identification that used synthetic minority oversampling reached over 99% AUC.

In this study, Arabic spam accounts were detected using text-based data with CNN models and metadata with NN models by (Alhassun and Rassam 2022) utilizing Twitter's premium API, and a dataset of 1.25 million tweets was collected. By flagging terminated accounts, data labeling was carried out. 13 features based on tweets, accounts, and graphs were retrieved. The findings demonstrated that the suggested combination framework used premium features to reach an accuracy of 94.27%. The performance of spam detection improved when premium features were compared to standard features when used with Twitter.

An efficient technique for spam identification was introduced by (Inuwa-Dutse et al. 2018). They suggested an SPD Optimized set of features that are apart from historical tweets. They focused on user-related attributes, user accounts, and paired user engagement. MaxEnt, Random Forest, ExtraTrees, SVM, GB, MLP, MLP+, and SVM were among the classification models that were utilized and evaluated based on three datasets, Honeypot (Lee et al. 2006), SPDautomated, and SPDmanual. The performance reached a peak of 99.93% when using GB on the SPD Optimized set. This technique can be used in real-time as the first step in a social media data gathering pipeline to increase the validity of research data.

Instead of employing the LCS method, (Sheeba et al. 2019) discovered spams using the RF classifier technique. The study used a dataset of 100,000 tweets. Latent Semantic Analysis was used to further identify the account after the RF classifier had identified it as a spambot using Latent Semantic Analysis (LSA). The proposed approach delivered benefits in terms of time consumption, high accuracy, and cost effectiveness.

An approach to spam identification based on DL methods was developed by (Alom et al. 2020). CNN architecture was utilized for the text-based classifier, while CNN and NN were merged for the combined classifier to classify tweet text and metadata, respectively. On two distinct real-world public datasets, Honeypot (Lee et al. 2006) and 1KS-10 K (Yang et al. 2013), the suggested approach's performance was compared to those of five ML-based and two DL-based state-of-the-art approaches. For the datasets Honeypot and 1KS-10KN, the accuracy of 99.68% and 93.12%, respectively, was attained.

In this research, (Reddy et al. 2021) implemented some supervised classification algorithms to detect spammers on Twitter. Information was obtained from tweepyAPI which comprised 2798 accounts in the training set and 578 accounts in the test set. Eighteen profile-base features were extracted. In terms of accuracy, Extreme Machine Learning (EML) obtained a better accuracy of 87.5.

3.1.10 Twitter—detecting sybil bots

Firstly, (Narayan 2021) used ML algorithms for the detection and successful identification of bogus Twitter accounts/bots. The algorithms used were DT, RF, and MNB. The dataset used included 447 Twitter accounts. Twitter API was used for the excavation of the data. DT has been found to be more accurate as compared to RF and MNB.

In their work, (Bindu et al. 2022) proposed three efficient methods to successfully detect fake accounts. The classification algorithms used were as follows: Linear and radial SVM, RF, and KNN. The data set used contained a total of 3964 records. RF gave more accurate prediction results accordingly overcoming the overfitting problem. The K-Fold Cross-Validation Scores for RF include a mean of 0.979812 and a standard deviation of 0.019682. On the other hand, in comparison Radial SVM did not perform well, and gave more False Negatives. However, using the Ensemble approach, higher accuracy was achieved.

Likewise, (Alarifi et al. 2016) studied the features used for detecting sybil accounts. Twitter4j was used to gather a manually labeled sample dataset of 2000 Twitter accounts (humans, bots, and hybrid-both human and bot tweets). Eight content-based features were selected. Four supervised ML algorithms which include J48 (C4.5), Logistic Model Tree, RF, Logitboost, BN, SMO-P, SMO-R, and multilayer NN were used. RF performed the best with a DR of 91.39 for two-class and 88.00 for three-class classification. Lastly, in order to maximize the use of the classifier, the authors developed an efficient browser plug-in.

David et al. (2017) leveraged a public labeled dataset from the project BoteDeTwitter to build half of their data set related to Spain politics. Using the Twitter API, a sample of 853 bot profiles and the most recent 1000 tweets from each user's timeline was collected. To create an initial feature set, 71 features based on profiles, metadata, and content were extracted. The following supervised ML methods were compared: RF, SVM, NB, DT, and NNET. Even though the increases were not significant after the first six features, RF managed to get the highest average accuracy of 94% by using 19 features.

In (van der Walt and Eloff 2018) paper, Twitter data were mined using the twitter4J API and a non-relational database yielding a total of 169,517 accounts. Engineered traits that had previously been used to successfully identify fraudulent accounts made by bots were added to a sample of human accounts. Without relying on behavioral data, these features were applied to several supervised ML models, enabling training on very little data. The results show that engineered traits, which were previously employed to identify fake accounts created by bots, could only reasonably predict fake accounts created by humans with an F1 score of 49.75%.

Kondeti et al. (2021) implemented ML to detect fake accounts on the Twitter platform. Different ML algorithms were used such as SVM, LR, RF, and KNN along with six account metadata features likes, Lang-code, sex-code, status-count, friends-count, followers-count, and favorites-count. Further to improve these algorithms' accuracy, they used two different normalization techniques such as Z-Score and Min–Max. Their approach achieved high accuracy of 98% for both RF and KNN models.

Khaled et al. (2019) suggested a new algorithm—SVM-NN to efficiently detect sybil bots. Four public labeled datasets were used by the authors. A total of 4456 accounts of both fake and human classes, result from combining them. Sixteen user-based numerical features were extracted from the datasets after applying features reduction, and they were then fed into the SVM, NN, and SVM-NN algorithms. The authors of the researchers assert that their novel SVM-NN uses fewer features than existing models. SVM-NN was the best-performing algorithm as it showed an accuracy of around 98%.

In the study, (Ersahin et al. 2017) collected their own dataset of fake and real accounts using Twitter API. The dataset consisted of 1000 accounts' data later pre-processed using Entropy Minimization Discretization (EMD) on sixteen user-based numerical features. NB with EMD showed the best result with 90.41% accuracy.

However, in order to predict sybil bots on Twitter using deep-regression learning, (Al-Qurishi et al. 2018) introduced a new model. The authors used two publicly available labeled datasets that had been generated during the 2016 US election and collected using Twitter API. The first dataset consisted of 39,467 profiles and 42,856,800 tweets. Whereas the second dataset consisted of 3140 profiles and 4,152,799 tweets. The authors extracted 80 online and offline features based on Profile-, Content- (Temporal, Topic, Quality, and Emotion-based), and Graph. Accordingly, the features were fed into the Deep Learning Component (DLC) FFNN. When fed with noisy and unclear data, the results depicted an accuracy of 86%. Categorical features showed clear segregation that all sybil bots disable their geographical location and have an unverified account. While numerical features showed that sybil bots have a noticeably young account age (recently created). Additionally, the number of re-post and mentions are significantly higher in the sybil's accounts.

Gao et al. (2020) proposed a content-based method to detect sybils. The proposed method included three main phases: CNN, bi-SN-LSTM, and the dense layer and softmax classifier stacked to output the classification results. The proposed bi-SN-LSTM network, in contrast to the bi-LSTM, employs SELU as the activation function of its recurrent step, enabling limitless modifications to the state value. The proposed model achieved a high F1-score of 99.31% on the "My Information Bubble" (Cresci et al. 2015) dataset.

3.1.11 Weibo—detecting social bots

Data collection, feature extraction, and detection modules were all included in the DL technique known as TPBot proposed by (Yang et al. 2022). To begin with, the data collection module used a web crawler to obtain user data from Sina Weibo using dataset collected by (Wu et al. 2021). Then, depending on each user's profile, the feature extraction module extracted temporal-semantic and temporal-metadata features. Finally, in the detection module, a detection model based on BiGRU was developed. TPBot outperformed baselines, by achieving an F1-score of 98.37%. Additionally, experiments were carried out on two Twitter datasets (Cresci et al. 2015, 2017) to assess the generalization capabilities of TPBot, and on both datasets, it outperformed the baselines.

Behavioral analysis and feature study were performed by (Dan and Jieqi 2017) to extract the effective features of Weibo accounts and build a supervised model to detect bots. A dataset of 5840 accounts from the Sina-Weibo data warehouse was used to discriminate between real and bot users. Eleven users' behavioral-based features were extracted and fed into DT, C4.5, and RF algorithms. The RF algorithm performed measurably better with a 0.944 F-measure.

Moreover, (Huang et al. 2016) built a classifier that combined NB and Genetic Algorithm on Weibo. The genetic algorithm was used to create an optimal threshold matrix which efficiently increased the precision of the model by improving the conditional probability matrix. Two models were built using two different datasets. One dataset (1000) was crawled by R and the other consisted of spammers purchased from the sales platform (600) and legitimate users crawled from friends and relatives (400). 9 profile-based features were set as attributes. In the comparison of the performance with LR, DT, and NB showed a higher precision of 0.92.

3.1.12 Weibo—detecting spambots

In this paper, for effective spammer detection, an EML-based supervised ML approach was proposed by (Zheng, Zhang, et al. 2016b). The study started by crawling Weibo data to create a labeled dataset. 1000 messages, both spam and normal, were chosen from the collected dataset. Message content and user behavior-based features were then extracted for a total of 18 features, which were then fed into the classification algorithm. With a TPR of spammers and non-spammers reaching 99 and 99.95%, respectively, the experiment and evaluation demonstrated that the suggested approach offers good performance.

Zheng, Wang, et al. (2016a) proposed a two-phase-based spambot detection approach. In the first phase, authors took existing work about user features. In the second phase, the authors introduced content mining for spambot detection. Using web crawlers, a dataset of 517 accounts and 381,139 tweets was collected. Eighteen behavioral and content-based features were extracted. The experiment results were compared with SVM, DT, NB, and BN algorithms. The proposed two-phased method performed better than the mentioned algorithms with an accuracy of 90.67%.

However, (Wu et al. 2021) used DNN and active learning (DABot) as a technique to detect bots. They classified bots into three types: spammers, bots that engage with accounts to increase impressions, and bots involve with politics. Thirty features were extracted and classified as metadata, interactions, contents, and time. A data collection of 20 K users and 214,506 posts from all users was produced as a consequence of the authors manually labeling the user accounts. Different stages made up the modeled architecture: data input for each user, ResNet block, BiGRU block, Attention layer, and Interference layer.

Another spam detection technique was put forth by (Xu et al. 2021) and relied on the self-attention Bi-LSTM NN model in conjunction with ALBERT. Two datasets were employed in the experiment: one self-collected (582 accounts) and the other microblogPCU (2000 accounts). They converted the text from social network sites into word vectors using ALBERT and then, input those word vectors into the Bi-LSTM layer. The final feature vector was created after feature extraction and combination with the information focus of the self-attention layer. To get the result, the SoftMax classifier performed classification.

3.1.13 Weibo—detecting sybil bots

In this research, (Bhattacharya et al. 2021) suggested a detection model that performed improved prediction of fake Weibo accounts using a variety of Ensemble ML algorithms. The 918 HTML pages that made up the public Weibo dataset were obtained from Kaggle. Data scraping was used to construct the fake accounts dataset. Content-based attributes were extracted. Five supervised models—RF, SVC, NB, LR, and GB—were taken into consideration. For determining the final result, the RF classifier's highest F1 score of 0.93, precision, and recall were taken into account. Finally, a plot confusion matrix revealed an inaccurate prediction for 44 accounts, providing the opportunity for additional research.

3.2 Using semi-supervised ML

Few studies on only two platforms have implemented semi-supervised ML to detect spambots and sybil bots which are discussed below.

3.2.1 Twitter—detecting spambots

Sedhai and Sun (2018) were the earliest that utilized a semi-supervised approach for spam detection. Their proposed S3D approach contains two main components which are spam detection components in real-time mode, and model update components in batch mode to periodically update the detection models. For spam detection, they apply four detectors which are a blacklisted domain detector using blacklisted URLs, a near-duplicate detector to label near-duplicate tweets using clustering, a reliable ham detector to label tweets that are posted by trusted users and that do not contain spammy words, and a multi-classifier using NB, LR, and RF models to labels the remaining tweets. Their approach achieved good accuracy results for spam detection on the public HSpam14 dataset along with four types of features to represent tweet and cluster Hashtag, in addition to being effective in detecting new spamming forms.

In this research work, (Alharthi et al. 2019) proposed a semi-supervised ML technique that classified Twitter accounts as spam or genuine accounts based on their behavior and profile information. A dataset consisting of (500) active Arab users was collected through a Twitter API and manually labeled. Label spreading and label propagation algorithms were implemented using 16 extracted features. The features (TweetsAverage), (Number of the accounts' followers to the number of his/her friends), (Tweet Source), and (is all the tweets have the same source?) were proven to be the most efficient features. The proposed model achieved the following results an F-measure of 0.89, an accuracy of 0.91, and an AUC of 0.90.

3.2.2 Twitter—detecting sybil bots

In this study, (Zeng et al. 2021) used semi-supervised self-training learning by utilizing a Kaggle data set of real and fake Twitter accounts. In this suggested technique, a self-training method was applied to automatically classify Twitter accounts. Further, to effectively reduce the impact of class imbalance on the identification effect, the resampling technique was incorporated into the self-training process. The proposed framework displayed good identification results on six different base classifiers, particularly for the initial batch of small-scaled labeled Twitter accounts.

3.2.3 Weibo—detecting spambots

Only a single study based on a semi-supervised approach by (Ren et al. 2018) detected spambots on Weibo. The authors have collected the dataset (31,147 users and 754,112 tweets) using a crawler. Behavioral and Content-based features were utilized to feed the model. Compared to NB, LR, SVM, and J48 algorithms, the proposed approach showed better results in all the evaluation metrics applied.

3.3 Using unsupervised ML

Few studies on only three platforms have implemented unsupervised ML to detect social bots, spambots and sybil bots which are discussed below.

3.3.1 Facebook—detecting spambots

Sohrabi and Karimi (2018) carried out the Facebook platform's spam filtering mechanism for posts and comments.

Different exploration techniques and optimization techniques, including PSO, simulated annealing, ant colony optimization, and Differential Evolution (DE) could be used with the suggested filtering strategy. Seven metadata features were recovered from the dataset, which was made up of 200,000 wall posts and comments on them. They examined the DB index and DE clustering method, SVM, and DT, three algorithms with PSO-based feature selection. The hybrid algorithm created by integrating SVM and clustering techniques produced the best outcomes.

3.3.2 Facebook—detecting sybil bots

Fake Facebook profiles Detection using a group of supervised and unsupervised mining algorithms was performed by (Albayati and Altamimi 2020). The main components were the Crawler and the analyzer modules. A dataset of 982 profiles and a set of 12 behavioral and profile-based features. In the analyzer module, using the mining tool RapidMiner Studio, they implemented two unsupervised algorithms, K-Means and K-Medoids, along with three supervised algorithms: ID3, KNN, and SVM. The findings of the performance evaluation method revealed that supervised algorithms outperformed unsupervised algorithms in terms of accuracy rates. With a 97.7% accuracy rate, ID3 surpasses other classifiers.

3.3.3 Instagram—detecting sybil bots

In this paper, (Munoz and Paul Guillen Pinto 2020) detected fake profiles on Instagram. Web scrapping techniques were used for data extraction on the third-party site to Instagram. A dataset of 1086 true and false profiles was designed. 17 features were extracted based on metadata and multimedia information. Various ML algorithms such as DT, LR, RF, MLP, AdaBoost, GNB, Quadratic Discriminant Analysis, Gaussian process classification, SVM, and NN were deployed. RF obtained the best accuracy of 0.96 as well as the best true and false prediction precision.

3.3.4 Twitter—detecting social bots

A bot detection technique was put forth by (Chen et al. 2017a, b) that used shortened URLs and tweeting almost duplicate content over an extended period of time to look for a particular class of malicious bots. This method automatically gathered bot groups from real-time Twitter streams as opposed to earlier work. The following nine URL shortening services were investigated: bit.ly, ift.tt, ow.ly, goo.gl, tinyurl.com, dlvr.it, dld.bz, viid.me, and ln.is. The model is made up of four sequentially operating parts: a crawler, a duplicate filter, a collector, and a bot detector. In order to conduct the experiment, 500,000 tweets were collected. According to the experiments, bot networks and accounts made up a mean of 10.5% of all accounts that employed shortened URLs.

Interestingly, (Mazza et al. 2019) presented a visualization technique named Retweet Tweet (RTT) for gaining insights into the retweeting behavior of Twitter accounts. For the purpose of identifying retweeting social bots, Retweet-Buster (RTBUST), an unsupervised group-analysis method, was employed. Using the Twitter Premium Search API, a dataset of 10 M Italian retweets shared by 1446, 250 unique users was compiled. RTBust was built around an LSTM variational autoencoder. Based on the results of the Hierarchical Density-Based Spatial Clustering (HDBSCAN) algorithm, it was decided whether the account was a bot or legitimate. In comparison with using it with PCA and TICA, the proposed RTBUST technique using the VAE produced the best detection performance, i.e., F1 = 0.87.

Anwar and Yaqub (2020) proposed a quick way to isolate bots from the Twitter discussion space. The dataset used was unlabeled data collected through Twitter Search API during the 2019 Canadian elections. It consisted of 103,791 accounts and 546,728 tweets. 13 metadata features were extracted using PCA implemented in K-means clustering. Results showed that bots have a higher rate of retweet percentage, daily tweets, and daily favorite count, which are incorporated with the known characteristics of bots.

In this paper, to enhance the detection accuracy of social bots, (Wu et al. 2020) proposed an improved conditional GAN to extend imbalanced data sets prior to applying training classifiers. The Gaussian kernel density peak clustering algorithm (GKDPCA), an unsupervised modified clustering algorithm, was put into practice. 2433 users' data was compiled into a dataset. On the basis of six different feature types—user meta-data, sentiment, friends, content, network, and timing, eleven different features were retrieved. With an F1 score of 97.56%, the enhanced CGAN performed better than the three popular oversampling methods.

Khalil et al. (2020) used two unsupervised clustering algorithms DBSCAN and K-Mean. Six publicly available datasets (2232, 3465, and 1969) were used mentioned in (Kantartopoulos et al. 2020). Eight profile-based features were extracted. It was concluded that DBSCAN performed better by achieving an accuracy of 97.7%.

The second contribution of (Barhate et al. 2020) is aimed at using an unsupervised ML approach. Hashtag data from the Twitter API was mined and a dataset of 140 K users was created. Using the PCA and K-means clustering algorithms, users were divided into four groups based on activity-related features. This enabled the analysis of each cluster's bot percentage. The age distribution of users in a trending hashtag was also plotted by the authors.

3.3.5 Twitter—detecting spambots

Some analyses were able to detect spammers successfully using unsupervised learning methods for instance, (Cresci et al. 2016) put forth a novel behavioral-based unsupervised approach for spambots accounts detection, inspired by biological DNA. The proposed methodology extracts and analyzes digital DNA sequences from users' actions. The authors manually created a dataset (4929 accounts) of verified spambot and genuine accounts. Each account got associated with a string that encodes its behavioral information. Compared to other benchmark work done, DNA fingerprinting model achieved the highest result with an MCC of 0.952.

Furthermore, (Koggalahewa et al. 2022) proposed an unsupervised spammer detection approach. In Stage 1, the clustering based on user interest distribution was performed. In Stage 2, spam detection was performed based on peer acceptance. Lastly, by assessing the user's peer acceptability against a threshold, a user was categorized as spam or genuine. Three datasets were used namely Social Honey Pot (Lee et al. 2006), HSpam14 million Tweets, and The Fake Project (Cresci et al. 2017). Detection accuracies pointed out that three features Local Outlier StandardScore (LOSS), Global Outlier Standard Score (GOSS), and Entropy when combined gave the best results. SMD performed the best with an accuracy of approximately 0.98 on the three datasets.

4 Discussion

To begin with, from all the reviewed studies we noticed that Twitter is the most researched platform with a total of 71 studies carried out, followed by 12 studies on Facebook, 11 studies on Instagram, 9 studies on Weibo, and lastly only 2 studies were conducted on LinkedIn. Appendix Table 2 summarizes all the reviewed ML-based studies focusing on the dataset used, feature's type, best-performing algorithm, and the highest result obtained, respectively. With respect to the most detected type of bot on each platform, Twitter had 36 studies on social bots, Facebook had 7 studies on sybil bots, Instagram had 8 studies on sybils, Weibo had 5 studies on spambots, and lastly, LinkedIn had only 2 studies which were on sybils.

Researchers in the reviewed papers used different datasets both publicly available and self-created to evaluate their models to classify bots from humans on the five addressed social media platforms. A summary of the 38 publicly available datasets has been provided in Appendix Table 3. From the Appendix Table 3, the most widely used datasets are MIB datasets which are the Cresci2017 and Cresci2015. However, the Cresci2017 dataset was the most used dataset by researchers because it includes five distinguished types of social media bots, namely genuine accounts, social spambots, traditional spambots, fake followers or Sybil, and a test set consisting of a mix between genuine and social spambots. Besides the variety of dataset's bot types, it is relatively a recent and large labeled dataset consisting of 12,736 accounts and 6,637,615 tweets in total, which may have attracted researchers to conduct their studies using the Cresci2017 dataset to detect spam and social bots in the Twitter platform. While Cresci2015 includes three fake follower's datasets, and two human accounts datasets making it more efficient in the detection of sybil bots on Twitter. The Fake Project dataset is one of the Cresci2015 which is much more used together with Honeypot dataset to detect spambots on Twitter. Different Kaggle's public datasets were used to detect different types of bots on Twitter. Due to the majority of papers related to Twitter compared to other platforms, more provided datasets are publicly available than the self-collected (private datasets) one. While other platforms such as Facebook and Instagram have more datasets that were self-created (private datasets). Weibo has almost equal types of datasets while LinkedIn has only self-created. Figure 4 illustrates public and collected datasets on each platform.

Despite the fact that there are numerous datasets available, some of them only contain human or bot IDs and labels. As a result, scraping is done using the appropriate collection API or method to obtain profile features or other information from an ID or account. For instance, the Twitter API is used to gather real-time datasets from publicly accessible Twitter data (Rodrigues et al. 2022). Many researchers have created their own datasets using these collection methods on different platforms as shown in Appendix Table 4. In

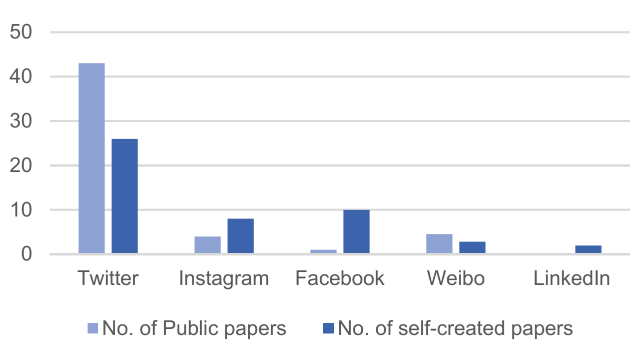

Fig. 4 Datasets distribution on each platform for the reviewed papers

Twitter, Twitter API was the most used collection method while methods like Twitter4j, Tweepy, ML, Twitter Premium Search API, and REST API were less used. Instagram datasets were collected using Instagram API, Selenium Web Driver tool, 3rd-party Instagram websites, and some manually. For Facebook, mostly used Facebook Graph API to collect data while web crawler was mostly used on the Weibo platform. Lastly, for LinkedIn in only two studies, the dataset was collected manually.

To distinguish between human and automated users on social media platforms, it's critical to identify an ideal collection of attributes (Alothali, Hayawi, et al. 2021b). A general observation was made that bots have a high friend-to-follower ratio and a low follower growth rate. This can be done by using a variety of features that have been reported in various studies. On the basis of the extracted features in all the reviewed papers, the features were classified into the following categories: Content/Language, User (Profile), Metadata, Behavioral, Network (Community/Interaction), Sentiment, Timing/Temporal, Graph, Numeric/Categorical/Textual/Series, Statistical¸ User Friends, Media and Engagement, Entity and Link¸ Keywords, Internet Overlap, hashtag features, and Periodic features. Content features are based on linguistic cues computed through NLP, mainly part-of-speech tagging. User features are based on properties of the users' accounts and users' relationships. User Meta Data features are information regarding the profile's characteristics. Locating an information source via metadata is known to be effective. Behavior features are calculated by statistical properties from the data. Different aspects of information diffusion patterns are captured by network features. General-purpose and Twitter-specific sentiment analysis algorithms are used to build sentiment features. Time features include statistics of time. Graph features are extracted by modelling the social media platform as a social graph model. Descriptive statistics relative to an account's social contacts are included in user friend features. Interest Overlap features include overlap between two users such as Topical affinity. Appendix Table 5 of the literature review provides examples of features from the reviewed studies as well as a summary of the features used in various social media platforms. According to the table, the most popular feature types from all the reviewed papers are content-based, profile-based, metadata-based, and behavioral-based features on essentially all types of platforms. Content-based features were utilized in 44 studies, followed by user/profile-based features in 42 studies, metadata-based in 27 studies, and behavioral-based in 16 studies. User friend, media, engagement, and keywords on various types of platforms are among the less popular feature types.

In regard to the Twitter platform, 34 studies used profile-based features followed by 32 studies that used content-based features and achieved high results. Meta-data-based features were used in 17 studies. Features based on Timing, Statistical, Keywords, Interaction, Periodic, Latent, Numeric, Categorical, and Series were used only once by single studies and achieved reasonable results. Four studies (Alhassun and Rassam 2022; Al-Qurishi et al. 2018; Mateen et al. 2017; Eshraqi et al. 2016) utilized graph-based features. It came to the notice that when (Eshraqi et al. 2016) combined graph-based features along with Content, Time, and Keywords, a very high accuracy of 0.99 was achieved. Only 5 studies (Wu et al. 2020; Davis et al. 2016; Inuwa-Dutse et al. 2018; Sayyadiharikandeh et al. 2020; Varol et al. 2017) made use of the network-based features. (Inuwa-Dutse et al. 2018) combined such network- and profile-based features and achieved the highest result AUC 99.93%. The number of features utilized in all 71 studies ranged from as less as 5 features to as high as 1000 features. (Varol et al. 2017) and (Davis et al. 2016) used approximately 1000 features and achieved an AUC of 0.95 whereas (Fonseca Abreu et al. 2020) used only 5 profile-based features and still obtained an AUC of 0.999. Regarding crucial features, interaction- and community-based features hold high value in spambot detection (Fazil and Abulaish 2018).

In regard to the Facebook platform, 7 out of 12 studies utilized profile-based features followed by content-based being used by 5 studies. Moreover, by examining the results of this platform's studies, it can be concluded that the highest results were achieved when profile-based and content-based features were combined hence showing a high accuracy of 0.984 in the research conducted by (Rathore et al. 2018). Noteworthy, textual, Categorial, and Numerical-based features were used only in 1 study.

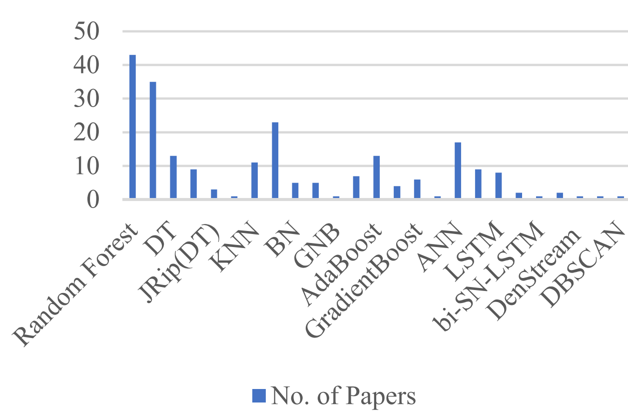

Fig. 5 Bar chart for ML algorithms

Singh and Banerjee (2019) but gave promising results (F1-score 0.99). Features such as “likes”, “remarks”, “user activities” contributed the maximum for the detection of sybils. Moving on to the third-most researched platform, Instagram, behavior-, content, and profile-based were used in 4 out of 11 studies. The combination of behavior- and content-based features showed the highest performance with an accuracy of 0.9845. In Weibo-based studies, content-based were the most widely used in addition to behavior-based features. Timing as well as semantic features were the least used but, on the contrary, gave the highest results. Since only 2 studies were found on LinkedIn, they made use of profile and statistical features. This platform needs to be extensively explored using other feature types which include content-, metadata-, behavioral-based, etc.

In terms of the different ML-based (supervised, semi-supervised, and unsupervised) techniques utilized in the reviewed papers which were built to compare and detect different types of social bots, Appendix Table 6 presents a list of all the respective papers that utilized the different algorithms. This Table 6 highlights the classifier bot type with the highest performance achieved for each algorithm. Figure 5 shows the number of papers that utilized each ML algorithm. As shown, RF is the best-performing and most applied algorithm among all algorithms in research and SVM is the second most applied algorithm in research followed by NB, DT, and AdaBoost. The least applied algorithms were GNB, ELM, bi-SN-LSTM, and clustering algorithms were the least applied algorithm. For supervised ML, the best performing algorithms of classical ML algorithms, the best performing were RF, JRip, AdaBoost, with their accuracy reaching up to 99.5%, and the least utilized algorithms were ID3 and GNB. As for DL algorithms, the best performing algorithm was CNN with the highest accuracy of 99.68% achieved though most perform well. The least utilized and least popular algorithm was ELM, even though it is considered simple with less training time. Moreover, it was noticed that ML classifiers work well with small-size datasets and DL algorithms with large-size datasets. However, no algorithm can be considered good or bad as it depends on a number of factors such as the dataset size, data preprocessing, and the number and type of features.

Further in terms of semi-supervised learning, despite them being powerful techniques in terms of discovering patterns in big data only four studies were found: three on Twitter and one on Weibo (Sedhai and Sun 2018; Alharthi et al. 2019; Zeng et al. 2021; Zeng et al. 2021). Since large datasets are derived from the Twitter platform which makes labeling an expensive and time-consuming process, semi-supervised techniques such as label propagation and label spreading show the ability to be applied more often. Moreover, integrating resampling along with self-training is a helpful way to reduce the impact of class imbalance when using semi-supervised learning. For the unsupervised learning approach, the DenStream unsupervised clustering algorithm achieved the highest result as compared with other used clustering algorithms like K-Mean and DBSCAN. Though this approach has less popularity and performance compared with the supervised approach (Albayati and Altamimi 2020), this method offers the benefit of not requiring a labeled dataset. Since, there was only one paper that applied this approach, hence additional investigation into this particular algorithm is not possible.

To conclude, not much evidence could be drawn from this analysis that the most researched bot type or the most researched social platforms are necessarily the ones most affected by social bots.

5 Challenges and opportunities

In this section, we shall put forth an elaborate discussion on challenges and future research directions based on our study and analysis. The findings showcased that social bot detection is challenging and this challenge is aggravated as the social network volume increases. To begin with, the most detected and researched bot types are social bots (42 studies), followed by spambots (34 studies), and lastly sybil bots (29 studies). Evidentially, Twitter is the most studied social network with a large number of bots of all types, especially social bots, mainly because of how easy it is to collect data through their API and the vast collection of accessible public datasets. However, social networks such as Instagram, LinkedIn, and Weibo need further in-depth study. Specifically, there is a dearth of studies on Facebook and LinkedIn due to the immense difficulty in obtaining publicly available datasets, which is caused by certain strict privacy policies on those networks. LinkedIn, in particular, does not have much of recent studies conducted on it. Furthermore, only sybil bots were found in the publicly available LinkedIn datasets. Moreover, with slight modifications, the ML techniques used for Instagram will have the potential to be applied to LinkedIn. From our studies, we conclude that Cresci2017 is the most used dataset in social media bot research due to its classification of bots based on their types. Whereas, Instagram has a greater number of sybil bots and two studies based on fake engagements. User and content-based features are the most frequently used for Instagram thereby showing high-accuracy results. Nevertheless, there is a scope for more research on this platform. In terms of features, on twitter (Fonseca Abreu et al. 2020) showed that even with 5 significant features, high results can be achieved. Therefore, new studies can be carried out by using as less features as possible. On Facebook, since profile-, content-, textual-, categorial-, and numerical-based features contributed high value in various studies, a new research direction can be explored by combining all the above-mentioned five feature types. LinkedIn needs to be extensively explored by using feature types which include content-, metadata-, behavioral-based, etc.

In terms of the reviewed algorithms, it is seen that RF is the best performing in terms of accuracy and the most applied algorithm among all algorithms on all social media platforms in the conducted study. SVM is the second most applied algorithm in research followed by NB, DT, and AdaBoost. DenStream unsupervised clustering algorithm achieved the highest result compared with other used clustering algorithms like K-mean and DBSCAN. Though this approach has less popularity, it has added the advantage of not requiring a labeled dataset. Different algorithms based on Bayes Theorem were used to classify Spam and social bots like NB, MNB, GNB, and NB. However, MNB overperforms the others. Different types of algorithms were used to build Decision Trees like basic DT, J48, JRip and ID3. DT, and J48 were the most applied forms. Yet, the JRip algorithm achieved the best performance among them on spam detection. Different types of boosting algorithms were applied such as AdaBoost, XGBoost, and GradientBoost. AdaBoost was the most applied, whereas GradientBoost performed the best amongst them on social bots detection. The DL approach was mostly applied to detect social bots type on the Twitter platform. Among the different DL algorithms, CNN and LSTM were the highest-performing and most promising algorithms in terms of accuracy.

Comparing algorithms on different platforms, RF achieved the best accuracy result on Weibo and Instagram platform. While AdaBoost achieved highest Detection Rate on Facebook platform. On the other hand, CNN and ANN achieved highest accuracy on twitter platform.

Moving on, future enthusiastic researchers are encouraged to investigate and conduct studies on unstudied social media platforms such as TikTok, Telegram which are known to have bots. As seen above, only four studies employed semi-supervised learning techniques, and a few used unsupervised technique; therefore, these fields need more exploration and contribution. The semi-supervised approach gives unlabeled instances the same weight as labeled ones while also minimizing the cost of labeling the data. More importantly, it is advised that researchers make their datasets available to the scientific community. This will support the training of new models, their testing, and the evaluation of the existing models. Additionally, new public datasets that contain the most recent type of bots are needed. The main gap was observed in the collected research. Only a few papers proposed a ML-technique to detect bots at registration or creation in real-time. As none of the existing research are designed to catch bots and act before they make connections with real users. Whereas in practice, it is desired to detect bots as soon as possible after registration in order to prevent them from interacting with real users. However, this has its own challenges as the bot's detection needs to be done from the basic information provided during the registration time.

Lastly, as the great novelist, Patricia Briggs quotes "Knowledge is a better weapon than a sword". The users on various social media platforms need to gain cybersecurity awareness in order to not get deceived and be able to distinguish between bots and benign accounts and be responsible in situations if a malicious bot was recognized to immediately report it to the platform.

6 Conclusion

This paper made an effort to provide a comprehensive review of the existing studies in the area of utilizing ML for bots detection on social media platforms which are affected by three types of bots—social, spam, and sybils to provide a starting point for researchers to identify the knowledge gaps in this field and conduct future in-depth research.

Furthermore, the usage of supervised, semi-supervised, and unsupervised ML-based approaches was also summarized. Numerous ML and DL methods were analyzed for bots detection, including KNN, RF, DT, NB, SVM, KNN, LSTM, ANN, etc. Visual aids were created to analyze the reviewed papers based on the nature of their datasets, the various categories of features, as well as the performance of employed algorithms. From the analysis for bots detection, we discovered that RF exhibited the highest performance in terms of accuracy and is the most frequently used ML algorithm. Whereas CNN and LSTM were the highest-performing and promising DL algorithms in terms of accuracy. Last but not the least, we addressed and listed some of the challenges, limitations as well as recommended suggestions that can be utilized by enthusiastic future researchers for adding more value and thereby contributing to the field of cybersecurity.

Appendix

See Tables 2, 3, 4, 5 and 6.

Table 2 Summary of Social Media Bot Detection using ML Techniques

References	Technique	Platform	Bot type	Dataset	Feature's type	Best-performing algorithm	Highest result
Adikari and Dutta (2020)	Supervised ML	LinkedIn	Sybil	Shalinda2020	Profile	SVM	Accuracy (87.34%)
Akyon and Esat Kalfaoglu (2019)	Supervised ML	Instagram	Spam, sybil	Fatih2019	Behavioral, content	SVM, ANN	F1-score with oversampling (94%) without (86%)
Alarifi et al. (2016)	Supervised ML	Twitter	Sybil	Twitter4j	Content	RF	DR-91.39, 88.00
Alharthi et al. (2019)	Semi-supervised	Twitter	Spam	Alharthi2019	Behavioral, profile	Label spreading, Label propagation	F-measure (0.89) Accuracy (0.91) AUC (0.90)
Alhassun and Rassam (2022)	Supervised ML	Twitter	Spam	Atheer2022	Graph, profile, content	CNN	Accuracy (94.27%)
Alom et al. (2020)	DL	Twitter	Spam	Honeypot 1KS-10KN	Tweet text, users' meta-data (content + profile)	CNN + ANN	Accuracy (99.68%)
Alothali, Alashwal, et al. (2021a)	Supervised ML	Twitter	Social	Kaggle	Profile	RF	–
Alothali, Hayawi, et al. (2021b)	Supervised ML	Twitter	Social	Twitter bots accounts (Kaggle)	Metadata	RF + CVAE method	AUC (94.3%)
Al-Qurishi et al. (2018)	DL	Twitter	Sybil	2016 US elections	Profile-, content-, graph-based	FFNN	Accuracy (86%)
Al-Zoubi et al. (2017)	Supervised ML	Twitter	Spam	Ala2017	Binary	J48 DT	Accuracy (94.9)
Andriotis and Takasu (2019)	Supervised ML	Twitter	Spam	Onur2017 Chao2011 Chao2012 Cersi2017	Metadata, content, senti-ment	AdaBoost	F1-score (0.95)
Anwar and Yaqub (2020)	Unsupervised ML	Twitter	Social	2019 Canadian elections	Metadata	K-means clustering	–
Attia et al. (2022)	DL	Twitter	Social	CLEF 2019	Content	2D CNN	Accuracy (0.933)
Wu et al. (2020)	Unsupervised ML	Twitter	Social	Varol2017, Clark2016	Metadata, sentiment, friends, content, network, timing	GKDPCA	F1 score (97.56%)
Babu et al. (2021)	Supervised ML	Facebook	Social	Prateek2017	Profile	Bayesian	Efficiency (98%)
Barhate et al. (2020)	Supervised ML	Twitter	Social	Bot repository	Profile based (bot score)	RF	AUC (0.96)
Bazm and Asadpour (2020)	Supervised ML	Instagram	Sybil	Muhammad2020	Behavioral	AdaBoost	Accuracy (95%)
Beğenilmiş and Uskudarli (2018)	Supervised ML	Twitter	Social	Bgenilmis2017	Profile	RF	AUC (0.95)
Bhattacharya et al. (2021)	Supervised ML	Weibo	Sybil	Kaggle	Content	RF	F1-score (0.93)
Bindu et al. (2022)	Supervised ML	Twitter	Sybil	MIB projects	Profile	RF	Mean (0.979812)
Chen et al. (2017a, b)	Supervised ML	Twitter	Spam	Chao2017	Statistical	RF-Lfun	Detection rate (95%)
Cable and Hugh (2019)	DL	Twitter	Social	Littman r2018	Profile	LSTM	Accuracy (0.981)
Cai, Li, and Zeng (2017a)	DL	Twitter	Social	Fred2016	Behavior, content	DBDM	F1-score (88.30%)
Cai, Li, and Zengi (2017b)	DL	Twitter	Social	Fred2016	Behavioral, content	BeDM	F1-score (87.32%,)
Cresci et al. (2016)	Unsupervised ML	Twitter	Spam	StefanoCresci2016	Behavioral	DNA fingerprint	MCC (0.952)
Dan and Jieqi (2017)	Supervised ML	Weibo	Social	Sina Weibo data warehouse	Behavioral	RF	F-measure (0.944)

Table 2 (continued)

References	Technique	Platform	Bot type	Dataset	Feature's type	Best-performing algorithm	Highest result
Daouadi et al. (2019)	Supervised ML	Twitter	Social	Subrahmanian2016, Lee2011	Metadata, profile	Deep forest	Accuracy (97.55%)
David et al. (2017)	Supervised ML	Twitter	Sybil	BotsDeTwitter2016	Content	RF	Accuracy (94%)
Davis et al. (2016)	Supervised ML	Twitter	Social	Clayton2016	Network, user friends, temporal, content, sentiment	RF	AUC (95%)
Dewan and Kumaraguru (2017)	Supervised ML	Facebook	Social	Prateek2017	Entity, content, metadata, link	RF	Accuracy (80%)
Dey et al. (2019)	Supervised ML	Instagram	Sybil	Free4ever1	Profile	RF	Accuracy (92.5%)
Echeverrï£¡a et al. (2018)	Supervised ML	Twitter	Social, Spam	Echeverria2017 Besel2018 Chavoshi2016 Cresci2017, 2015 Yang2012 Subrahmanian2016 Z. Gilani2017	Content, profile	LGBM	Accuracy (97.84%)
Ersahin et al. (2017)	Supervised ML	Twitter	Sybil	Buket2017	User-based	NB w/EMD	Accuracy (90.41%)
Eshraqi et al. (2016)	Supervised ML	Twitter	Spam	Chao2011	Graph, content, time, keywords	DenSream	Accuracy (99%)
Fazil and Abulaish (2018)	Supervised ML	Twitter	Spam	GuofeiGu2017	Community, metadata, content, interaction	RF	F-score (0.979)
Fernquist et al. (2018)	Supervised ML	Twitter	Social	Cresci2015 Z. Gilani2017 O. Varol2017	Metadata, content, time	RF	Accuracy (0.957)
Fonseca Abreu et al. (2020)	Supervised ML	Twitter	Social, Spam	Cresci2017	Profile	RF	AUC (0.9999)
Gao et al. (2020)	DL	Twitter	Sybil	Cersi2015	Content based	(CNN + SELU) + bi-SN-LSTM	F1 score (99.31%)
Güngör et al. (2020)	Supervised ML	Twitter	Spam	Kubra2020	Content, profile	J48	Accuracy (97.2%)
Gupta and Kaushal (2017)	Supervised ML	Facebook	Sybil	Aditi2017	Profile, timeline	RF	Accuracy (94%)
Shukla et al. (2021)	Supervised ML	Twitter	Social	Kaggle	Metadata	RF	AUC (93%)
Hakimi et al. (2019)	Supervised ML	Facebook	Sybil	Ahmad2019	Content-based	KNN	Accuracy (0.829)
Hayawi et al. (2022)	DL	Twitter	Social	Varol 2017 Cresci2017a, b Yang2019–Yang2020b Mazza2019 Gilani2017	Metadata	GLoVe + LSTM	AUC (0.97)
Heidari et al. (2020)	Supervised ML & DL	Twitter	Spam	Cresci2017	Profile	-FFNN -Glove + ELMO(LSTM) + FFNN	Accuracy (0.971) Accuracy (0.981)
Heidari et al. (2021)	Supervised ML	Twitter	Spam	Cresci2017	Sentiment	SVM	F1-score (0.930)
Huang et al. (2016)	Supervised ML	Weibo	Social	Yingying2016	Profile-based	DT Bayesian algorithm	Precision (0.92)

Table 2 (continued)

References	Technique	Platform	Bot type	Dataset	Feature's type	Best-performing algorithm	Highest result
Inuwa-Dutse et al. (2018)	Supervised ML	Twitter	Spam	Honeypot, SPDautomated, SPDmanual	Account, network, user, optimised	GB	AUC (99.93%)
Kantepe and Gañiz (2017)	Supervised ML	Twitter	Social	Mücahit2017	Profile, content, periodic	GBT	Accuracy (86%)
Kenyeres and Kovács (2022)	Supervised ML & DL	Twitter	Social	PAN 2019 bots and gender profiling task	Tweet text, content, non-content	GLoVe + LSTM + Adaboost	Accuracy (0.9)
Kesharwani et al. (2021)	DL	Instagram	Sybil	Kaggle	Profile-based	ANN	Accuracy (93.63%)
Khaled et al. (2019)	Supervised ML	Twitter	Sybil	Cersi2015	User-based	SVM-NN	Accuracy (98%)
Khalil et al. (2020)	Unsupervised ML	Twitter	Social	Mohammad2020	Profile	DBSCAN	Accuracy (97.7%)
Knauth (2019)	Supervised ML	Twitter	Social	Cresci2017	Profile, content, behavior	AdaBoost	Accuracy (0.988)
Koggalahewa et al. (2022)	Unsupervised ML	Twitter	Spam	The fake project Social honey pot HSpam14	Content	SMD	Accuracy (0.98)
Kondeti et al. (2021)	Supervised ML	Twitter	Sybil	–	Account metadata	RF, KNN	Accuracy (98%)
Kudugunta and Ferrara (2018)	DL	Twitter	Spam	Cresci2017	Tweet text, account metadata	Contextual LSTM	AUC (> 96%)
Kumar and Rishiwal (2020)	Supervised ML	Twitter	Spam	Vinay2020	Content	MNB	Accuracy (99%)
Albayati and Altamimi (2020)	Supervised & unsupervised ML	Facebook	Sybil	Mohammed2020	Profile	ID3 DT	Accuracy (97.7%)
Albayati and Altamimi (2019)	Supervised ML	Facebook	Sybil	Mohammed2019	Profile	SVM	Accuracy (98%)
Martin-Gutierrez et al. (2021)	DL	Twitter	Social	David2021	Profile information (metadata), text	Bot-DenseNet + RoBERTa transformer	F1-score (0.77)
Mateen et al. (2017)	Supervised ML	Twitter	Spam	GuofeiGu2017	Profile-based, content-based, graph-based	J48, Decorate	Precision (97.6%)
Mazza et al. (2019)	Unsupervised ML	Twitter	Social	Michele2019	Latent	LSTM	F1 (0.87)
Meshram et al. (2021)	Supervised ML	Instagram	Sybil	Pranay2021	Behavioral. content	RF	Accuracy (98.45%)
Munoz and Paul Guillen Pinto (2020)	Unsupervised ML	Instagram	Sybil	Samuel2020	Metadata	RF	Accuracy (0.96)
Najari et al. (2022)	Supervised ML	Twitter	Social, Spam	Cresci2017	–	GLoVe + LSTM	949/.951—200D
Narayan (2021)	Supervised ML	Twitter	Sybil	Nirdhum2021	Profile	DT	Accuracy (93%)
Oentaryo et al. (2016)	Supervised ML	Twitter	Spam	Richard2016	Numeric, categorical, series	LR	F1-score (0.8228)
Ping and Qin (2019)	DL	Twitter	Social	Cersi2017	Temporal, content	CNN, LSTM	Accuracy (99%)
Prabhu Kavin et al. (2022)	Supervised ML	Twitter	Spam	Collected "the fake project"	User, content	SVM	F-score (0.973)
Pramitha et al. (2021)	Supervised ML	Twitter	Social	Kaggle	Profile	XGBoost	Accuracy (0.891)
Pratama and Rakhmawati (2019)	Supervised ML	Twitter	Social	Pandu2019	Twitter accounts	RF	F-score (0.74)

Table 2 (continued)

References	Technique	Platform	Bot type	Dataset	Feature's type	Best-performing algorithm	Highest result
Purba et al. (2020)	Supervised ML	Instagram	Sybil	Kristo2020	Metadata, media info, engagement, media tags, media similarity	RF	4-classes accuracy (91.76%)
Shukla et al. (2022)	Supervised ML	Twitter	Social	Kaggle	Profile	TweezBot	Accuracy (99.00049%)
Rahman et al. (2021)	Supervised ML	Twitter	Social	Md2021	Behavioral	SPY-BOT	Accuracy (0.901)
Ramalingaiah et al. (2021)	Supervised ML	Twitter	Social	Kaggle	Account	DT + BoW	Accuracy (> 0.99)
Rathore et al. (2018)	Supervised ML	Facebook	Spam	Shailendra2017	Profile-based, content-based	BN	Accuracy (0.984)
Reddy et al. (2021)	Supervised ML	Twitter	Spam	P. Muthi2021	Message content, behavior	EML	Accuracy (87.5)
Ren et al. (2018)	Semi-supervised	Weibo	Spam	Honglin2018	Behavior, content	–	Accuracy (0.936)
Rodrigues et al. (2022)	Supervised ML & DL	Twitter	Spam	Kaggle SMS	Sentiment	LSTM	Spam detection-accuracy (98.74%) Sentiment analysis-accuracy (73.81%)
Rodríguez-Ruiz et al. (2020)	Supervised ML	Twitter	Social Spam	Cresci2017	Content	Bagging-TPMiner	AUC (0.921)
Sadineni (2020)	Supervised ML	Twitter	Spam	Kaggle	Content based	RF	F1 score (0.83%) of spammers
Sahoo and Gupta (2020)	Supervised ML	Facebook	Spam	Somya2020	Profile, content	JRip	Detection rate (99.5%)
Saranya Shree et al. (2021)	Supervised ML	Facebook	Sybil	Instagram influencer dataset	Text, behavioral, graph	SVM + Naïve Bay	Accuracy (91.5%)
Sayyadiharikandeh et al. (2020)	Supervised ML	Twitter	Social	Botometer repository	Metadata, retweet/mention networks, temporal, content information, sentiment	ESC	Accuracy (0.99)
Sedhai and Sun (2018)	Semi-supervised	Twitter	spam	HSpam14	Hashtag, content user, domain	Clustring, NB, LR, RF	Best F1 scores
Sen et al. (2018)	Supervised ML	Instagram	Social	Indira2018	Network effect, internet overlap, liking frequency, influential poster, hashtag features, user-based features	MLP	Precision (83%)
Sengar et al. (2020)	Supervised ML & DL	Twitter	Social	Sandeep2017	User, Tweet metadata	RF, GB	Accuracy (99.54%)
Sheeba et al. (2019)	Supervised ML	Twitter	Spam	Sheeba2019	Content	RF	F1-score (66%)
Sheikhi (2020)	Supervised ML	Instagram	Sybil	Saeid2021	Behavioral, content	Bagging	98.45% accurate classification
Shevtsov et al. (2022)	Supervised ML	Twitter	Social	US 2020 Elections	User	XGBoost	F-score (0.896)
Singh and Banerjee (2019)	Supervised ML	Facebook	Sybil	Yeshwant2019	Textual, categorical, numerical	AdaBoost	F1-score (99%)

Table 2 (continued)

References	Technique	Platform	Bot type	Dataset	Feature's type	Best-performing algorithm	Highest result
Sohrabi and Karimi (2018)	Supervised & Unsupervised ML	Facebook	Spam	Mohammad2017	Metadata	(DB index, DE) + SVM	Accuracy (96.3%)
Thejas et al. (2019)	Supervised ML	Instagram	Sybil	Thejas2019	Text, numeric	RF	Accuracy (97%)
van der Walt and Eloff (2018)	Supervised ML	Twitter	Sybil	Estée2018	Account metadata	RF	F1 score (49.75%)
Varol et al. (2017)	Supervised ML	Twitter	Social	Onur2017	Metadata, content, network	RF	AUC (0.95)
Zhang and Sun (2017)	Supervised ML	Instagram	Spam	Wuxain2017	Content, user profile	RF	Accuracy (96.27%)
Wanda et al. (2020)	DL	Facebook	Social	VirusTotal, PhishTank	Profile	NN	Training loss (0.5058)
Xiao et al. (2015)	Supervised ML	LinkedIn	Sybil	Cao2015	Statistical	RF	AUC (0.95), recall (0.75) at 95% precision
Xu et al. (2021)	DL	Weibo	Spam	MicroblogPCU and weibo-Data (self-collected)	Semantic features from text	ALBERT + Bi-LSTM + self-attention	High accuracy
Wu et al. (2021)	DL	Weibo	Spam	SWLD-20 K	Metadata, interactions, contents, timing	ResNet + BiGRU	Accuracy (0.989)
Chen et al. (2017a, b)	Unsupervised ML	Twitter	Social	Zhouhan2017	–		
Yang et al. (2022)	DL	Weibo	Social	SWLD-20 K	Semantic, metadata, profile	TPBot	F-score (0.984)
Zheng, Wang, et al. (2016a)	Supervised ML	Weibo	Spam	Zinhu2015	Behavior, content	Two-phased Method	Accuracy (90.67%)
Zheng, Zhang, et al. (2016b)	Supervised ML	Weibo	Spam	Xianghan2015	Message content, user behavior	ELM	TRP (99%)

Table 3 Summary of Public Datasets Information

Platform	Datasets	Papers that used it	No. of papers	Bot type
Twitter	Cresci2017	Andriotis and Takasu (2019), Echeverrï£¡a et al. (2018), Fonseca Abreu et al. (2020), Hayawi et al. (2022), Heidari et al. (2020), Heidari et al. (2021), Knauth (2019), Sengar et al. (2020), Kudugunta and Ferrara (2018), Najari et al. (2022), Ping and Qin (2019), Rodríguez-Ruiz et al. (2020)	13	Spam, social
	Cresci2015	Bindu et al. (2022), Echeverrï£¡a et al. (2018), Fernquist et al. (2018), Gao et al. (2020), Khaled et al. (2019), Prabhu Kavin et al. (2022)	5	Sybil, social
	Kaggle	Alothali, Alashwal, et al. (2021a), Alothali, Hayawi et al. (2021b), Shukla et al. (2021), Knauth (2019), Pramitha et al. (2021), Shukla et al. (2022), Ramalingaiah et al. (2021), Rodrigues et al. (2022), Rodríguez-Ruiz et al. (2020), Sadineni (2020)	10	Spam, sybil, social
	Honeypot	Alom et al. (2020), Cai, Li, and Zeng (2017a), Inuwa-Dutse et al. (2018), Koggalahewa et al. (2022)	4	Spam
	The fake project	Koggalahewa et al. (2022), Prabhu Kavin et al. (2022)	2	Spam
	HSpam14	Koggalahewa et al. (2022), Sedhai and Sun (2018)	2	Spam
	Chao2011, Chao2012	Andriotis and Takasu (2019), Eshraqi et al. (2016), Fazil and Abulaish (2018)	3	Spam
	NSCLab	Chen et al. (2017a, b)	1	Spam
	O. Varol2017	Wu et al. (2020), Fernquist et al. (2018), Hayawi et al. (2022)	3	Social
	Z. Gilani2017	Echeverrï£¡a et al. (2018), Fernquist et al. (2018), Hayawi et al. (2022)	3	Social
	Onur2017	Andriotis and Takasu (2019), Varol et al. (2017)	2	Spam, social
	1KS-10KN	Alom et al. (2020)	1	Spam
	Sheeba2019	Sheeba et al. (2019)	1	Spam
	SPDautomate, SPDmanual	Inuwa-Dutse et al. (2018)	1	Spam
	PAN 2019 bots and gender profiling task	Attia et al. (2022), Kenyeres and Kovács (2022)	2	Social
	Yang2019/2020b/2012	Echeverrï£¡a et al. (2018), Hayawi et al. (2022)	2	Social
	GuofeiGu2017	Fazil and Abulaish (2018) Mateen et al. (2017)	2	Spam
	US (2020 elections	Shevtsov et al. (2022)	1	Social
	Littman2018	Cable and Hugh (2019)	1	Social
	Roeder2018	Cable and Hugh (2019)	1	Social
	Botometer repository	Barhate et al. (2020), Khalil et al. (2020), Sayyadiharikandeh et al. (2020)	3	Social
	Subrahmanian2016	Daouadi et al. (2019), Hayawi et al. (2022)	2	Social
	Echeverria2017, Besel2018, Chavoshi2016	Hayawi et al. (2022)	1	Social
	Lee2011	Daouadi et al. (2019)	1	Social
	Mohammad2019	Khalil et al. (2020)	1	Social
	Clark2016	Wu et al. (2020)	1	Social
	Mazza2019	Echeverrï£¡a et al. (2018), Hayawi et al. (2022)	2	Spam, social
	Clayton2016	Davis et al. (2016)	1	Social
	BotsDeTwitter	David et al. (2017)	1	Spam
	Begenilmi2017	Beğenilmiş and Uskudarli (2018)	1	Social
	Abdulrahman2016	Alarifi et al. (2016)	1	Sybil
	Morstatter2016	Cai, Li, and Zengi (2017b)		Social

Table 3 (continued)

Platform	Datasets	Papers that used it	No. of papers	Bot type
Instagram	Kaggle	Kesharwani et al. 2021)	1	Sybil
	Free4ever1	Dey et al. (2019)	1	Sybil
	Fatih2019	Akyon and Esat Kalfaoglu (2019)	1	Spam, sybil
	Instagram influencer dataset	Saranya Shree et al. (2021)	1	Sybil
Facebook	VirusTotal and PhishTank	Wanda et al. (2020)	1	Social
Weibo	MicroblogPCU	Xu et al. (2021)	1	Spam
	Sina Weibo data warehouse	Dan and Jieqi (2017)	1	Social
	SWLD-20 K	Wu et al. (2021), Yang et al. (2022)	2	Spam, social
	Kaggle	Bhattacharya et al. (2021)	1	Sybil

Table 4 Self-collected (private) Datasets and their Collection Methods

Platform	Papers	Data collection method	No. of papers
Twitter	Alarifi et al. (2016), van der Walt and Eloff (2018)	Twitter4j	2
	Reddy et al. (2021)	Tweepy	1
	Alhassun and Rassam (2022), Alothali, Alashwal, et al. (2021a), Al-Qurishi et al. (2018), Al-Zoubi et al. (2017), Anwar and Yaqub (2020), Wu et al. (2020), Barhate et al. (2020), Beğenilmiş and Uskudarli (2018), Bindu et al. (2022), Chen et al. (2017a, b), Cable and Hugh (2019), Daouadi et al. (2019), David et al. (2017), Echeverrï£¡a et al. (2018), Fazil and Abulaish (2018), Fonseca Abreu et al. (2020), Güngör et al. (2020), Kantepe and Gañiz (2017), Martin-Gutierrez et al. (2021), Narayan (2021), Oentaryo et al. (2016), Shukla et al. (2022), Rodrigues et al. (2022), Shevtsov et al. (2022), Pramitha et al. (2021), Varol et al. (2017), Chen et al. (2017a, b)	Twitter API	27
	Cresci et al. (2016), Shukla et al. (2021), Khaled et al. (2019), Pratama and Rakhmawati (2019)	Manually	4
	Ersahin et al. (2017)	Machine learning	1
	Cai, Li, and Zengi (2017b)	Honeypots	2
	Mazza et al. (2019)	Twitter premium search API	1
	Alharthali et al. (2019)	Twitter API	1
	Prabhu Kavin et al. (2022)	REST API	1
Instagram	Meshram et al. (2021), Sheikhi (2020)	Instagram API	2
	Akyon and Esat Kalfaoglu (2019), Bazm and Asadpour (2020), Sen et al. (2018)	Manually	3
	Munoz and Paul Guillen Pinto (2020)	Python	1
	Munoz and Paul Guillen Pinto (2020), Thejas et al. (2019)	Selenium web driver tool	2
	Purba et al. (2020)	3rd-party Instagram websites	1
Facebook	Wanda et al. (2020)	VirusTotal, PhishTank	1
	Babu et al. 2021), Dewan and Kumaraguru (2017), Gupta and Kaushal (2017), Rathore et al. (2018), Singh and Banerjee (2019), Sohrabi and Karimi (2018)	Facebook graph API	6
	Hakimi et al. (2019)	Mockaroo	1
	Albayati and Altamimi (2020), Rathore et al. (2018)	CRAWLER	2
Weibo	Bhattacharya et al. (2021), Ren et al. (2018), Wu et al. (2021), Yang et al. (2022), Zheng, Wang, et al. (2016a)	Web crawler and data scraping	5
	Huang et al. (2016)	Manually	1
LinkedIn	Adikari and Dutta (2020), Xiao et al. (2015)	Manually	2

Table 5 Summary of Features in Social Media Bot Detection using ML

Feature's type	Papers that used it	No. of papers	Platform	Description
Content/language	Akyon and Esat Kalfaoglu (2019), Alarifi et al. (2016), Alhassun and Rassam (2022), Al-Qurishi et al. (2018), Andriotis and Takasu (2019), Attia et al. (2022), Wu et al. (2020), Bhattacharya et al. (2021), Cai, Li, and Zeng (2017a), Cai, Li, and Zengi (2017b), David et al. (2017), Davis et al. (2016), Dewan and Kumaraguru (2017), Echeverrï£¡a et al. (2018), Eshraqi et al. (2016), Fazil and Abulaish (2018), Gao et al. (2020), Güngör et al. (2020), Hakimi et al. (2019), Kenyeres and Kovács (2022), Knauth (2019), Koggalahewa et al. (2022), Kumar and Rishiwal (2020), Mateen et al. (2017), Meshram et al. (2021), Ping and Qin (2019), Prabhu Kavin et al. (2022), Rathore et al. (2018), Reddy et al. (2021), Ren et al. (2018), Rodríguez-Ruiz et al. (2020), Sadineni (2020), Sahoo and Gupta (2020), Sayyadiharikandeh et al. (2020), Sheeba et al. (2019), Sheikhi (2020), Fernquist et al. (2018), Sedhai and Sun (2018), Shevtsov et al. (2022), Varol et al. (2017), Zhang and Sun (2017), Wu et al. (2021), Zheng, Wang, et al. (2016a), Zheng, Zhang, et al. (2016 b)	44	Facebook, Instagram, Twitter, Weibo	Content features are based on linguistic cues computed through NLP, mainly part-of-speech tagging
User (profile)	Adikari and Dutta (2020), Alharthi et al. (2019), Alhassun and Rassam (2022), Al-Qurishi et al. (2018), Al-Zoubi et al. (2017), Babu et al. 2021), Barhate et al. (2020), Beğenilmiş and Uskudarli (2018), Bindu et al. (2022), Cable and Hugh (2019), Daouadi et al. (2019), David et al. (2017), Dey et al. (2019), Echeverrï£¡a et al. (2018), Ersahin et al. (2017), Fonseca Abreu et al. (2020), Güngör et al. (2020), Gupta and Kaushal (2017), Heidari et al. (2020), Huang et al. (2016), Inuwa-Dutse et al. (2018), Kantepe and Gañiz (2017), Kesharwani et al. 2021), Khaled et al. (2019), Khalil et al. (2020), Knauth (2019), Albayati and Altamimi (2020), Albayati and Altamimi (2019), Mateen et al. (2017), Narayan (2021), Prabhu Kavin et al. (2022), Shukla et al. (2022), Rahman et al. (2021), Ramalingaiah et al. (2021), Rathore et al. (2018), Sahoo and Gupta (2020), Sedhai and Sun (2018), Sen et al. (2018), Sengar et al. (2020), Zhang and Sun (2017), Wanda et al. (2020), Yang et al. (2022)	42	Facebook, Instagram, Twitter, LinkedIn, Weibo	User features are based on properties of user account and users relationships

Table 5 (continued)

Feature's type	Papers that used it	No. of papers	Platform	Description
Metadata	Alom et al. (2020), Alothali, Alashwal, et al. (2021a), Alothali, Hayawi, et al. (2021b), Andriotis and Takasu (2019), Anwar and Yaqub (2020), Wu et al. (2020), Daouadi et al. (2019), David et al. (2017), Dewan and Kumaraguru (2017), Fazil and Abulaish (2018), Fernquist et al. (2018), Shukla et al. (2021), Hayawi et al. (2022), Kondeti et al. (2021), Kudugunta and Ferrara (2018), Martin-Gutierrez et al. (2021), Meshram et al. (2021), Munoz and Paul Guillen Pinto (2020), Pramitha et al. (2021), Pratama and Rakhmawati (2019), Purba et al. (2020), Sayyadiharikandeh et al. (2020), Sohrabi and Karimi (2018), van der Walt and Eloff (2018), Varol et al. (2017), Wu et al. (2021), Yang et al. (2022)	27	Instagram, Twitter, Weibo	User meta data features is information regarding the profile's characteristics. Locating an information source via metadata is known to be effective
Behavioural	Alharthi et al. (2019), Cai, Li, and Zengi (2017b), Meshram et al. (2021), Rahman et al. (2021), Akyon and Esat Kalfaoglu (2019), Sheikhi (2020), Knauth (2019), Zheng, Zhang, et al. (2016b), Saranya Shree et al. (2021), Cai, Li, and Zeng (2017a), Bazm and Asadpour (2020), Cresci et al. (2016), Dan and Jieqi (2017), Ren et al. (2018), Zheng, Wang, et al. (2016a), Reddy et al. (2021)	16	Facebook, Instagram, Twitter, Weibo	Behavior features are calculated by statistical properties from the data
Network (community/interaction)	Fazil and Abulaish (2018), Varol et al. (2017), Davis et al. (2016), Inuwa-Dutse et al. (2018), Sen et al. (2018), Sayyadiharikandeh et al. (2020), Wu et al. (2021), Wu et al. (2020)	8	Instagram, Twitter, Weibo	Different aspects of information diffusion patterns are captured by network features. They are built on retweets, mentions, hashtag occurrences etc On the basis of the frequency, all networks are weighed
Sentiment	Davis et al. (2016), Yang et al. (2022), Heidari et al. (2021), Rodrigues et al. (2022), Andriotis and Takasu (2019), Sayyadiharikandeh et al. (2020), Xu et al. (2021), Wu et al. (2020)	8	Twitter, Weibo	General-purpose and Twitter specific sentiment analysis algorithms are used to build sentiment features
Timing/temporal	Davis et al. (2016), Gupta and Kaushal (2017), Fernquist et al. (2018), Eshraqi et al. (2016), Wu et al. (2021), Sayyadiharikandeh et al. (2020), Ping and Qin (2019)	7	Facebook, Twitter, Weibo	Time features includes statistics of time such as retweets, consecutive tweets, statistics for the time between post etc Temporal capture timing patterns of content generation and consumption, such as tweet rate and inter-tweet time distribution
Graph	Eshraqi et al. (2016), Mateen et al. (2017), Saranya Shree et al. (2021), Al-Qurishi et al. (2018), Alhassun and Rassam (2022)	5	Instagram, Twitter	Graph features are extracted by modelling the social media platform as a social graph model
Numeric/categorical/textual/series	Oentaryo et al. (2016), Thejas et al. (2019), Saranya Shree et al. (2021), Singh and Banerjee (2019)	4	Facebook, Twitter	Examples: account ID, post ID

Table 5 (continued)

Feature's type	Papers that used it	No. of papers	Platform	Description
Statistical	Davis et al. (2016), Xiao et al. (2015), Chen et al. (2017a, b)	3	LinkedIn, Twitter	Examples: min, max, quartiles, Mean and variance, median, moments, and entropy etc
User friends	Davis et al. (2016), Wu et al. (2020)	2	Twitter	Descriptive statistics relative to an account's social contacts are included in user friend features
Media and engagement	Purba et al. (2020)	1	Instagram	Examples: average caption length, non-image percentage, engagement rate (like/comm.), location tag percentage, promotional keywords, followers' keywords, post interval
Entity and link	Dewan and Kumaraguru (2017)	1	Facebook	Examples: gender, parameter length, page category, has username, hyphen count username length, no. of subdomains, name length, no. of words in name, locale, likes on page, has HTTP/HTTPS, parameters count, path length etc
Keywords	Eshraqi et al. (2016)	1	Twitter	Examples: presence of the words "chat" and "naughty" in biography have a direct relationship with spam
Internet overlap, hashtag features	Sedhai and Sun (2018), Sen et al. (2018)	2	Instagram, Twitter	Interest overlap features include overlap between two users such as topical affinity
Periodic	Kantepe and Gañiz (2017)	1	Twitter	Examples: screen name, location, is protected, profile image

Table 6 ML Algorithms in Reviewed Papers

Algorithm	Papers that applied it	No. of papers	Highest classifier bot type detected	Performance measure (accuracy) (%)
RF	Alarifi et al. (2016), Alothali, Alashwal, et al. (2021a), Alothali, Hayawi, et al. (2021b), Andriotis and Takasu (2019), Barhate et al. (2020), Bazm and Asadpour (2020), Beğenilmiş and Uskudarli (2018), Bhattacharya et al. (2021), Bindu et al. (2022), Chen et al. (2017a, b), Dan and Jieqi (2017), David et al. (2017), Davis et al. (2016), Dewan and Kumaraguru (2017), Dey et al. (2019), Echeverrï£¡a et al. (2018), Fazil and Abulaish (2018), Fonseca Abreu et al. (2020), Gupta and Kaushal (2017), Shukla et al. (2021), Heidari et al. (2020), Heidari et al. (2021), Knauth (2019), Kondeti et al. (2021), Meshram et al. (2021), Munoz and Paul Guillen Pinto (2020), Narayan (2021), Oentaryo et al. (2016), Pratama and Rakhmawati (2019), Purba et al. (2020), Shukla et al. (2022), Rathore et al. (2018), Rodrigues et al. (2022), Sadineni (2020), Sedhai and Sun (2018), Sen et al. (2018), Sengar et al. (2020), Sheeba et al. (2019), Singh and Banerjee (2019), Thejas et al. (2019), Varol et al. (2017), van der Walt and Eloff (2018), Zhang and Sun (2017), Xiao et al. (2015)	44	Social	99.545
SVM	Adikari and Dutta (2020), Akyon and Esat Kalfaoglu (2019), Andriotis and Takasu (2019), Bazm and Asadpour (2020), Beğenilmiş and Uskudarli (2018), Bindu et al. (2022), David et al. (2017), Fernquist et al. (2018), Fonseca Abreu et al. (2020), Gupta and Kaushal (2017), Hakimi et al. (2019), Heidari et al. (2021), Khaled et al. (2019), Kantepe and Gañiz (2017), Knauth (2019), Kumar and Rishiwal (2020), Albayati and Altamimi (2019), Meshram et al. (2021), Munoz and Paul Guillen Pinto (2020), Oentaryo et al. (2016), Prabhu Kavin et al. (2022), Shukla et al. (2022), Rahman et al. (2021), Rathore et al. (2018), Ren et al. (2018), Rodrigues et al. (2022), Rodríguez-Ruiz et al. (2020), Sadineni (2020), Saranya Shree et al. (2021), Sen et al. (2018), Sheikhi (2020), Sohrabi and Karimi (2018), Thejas et al. (2019), Xiao et al. (2015), Zheng, Wang, et al. (2016a)	35	Sybil	98
NB	Akyon and Esat Kalfaoglu (2019), Andriotis and Takasu (2019), Babu et al. 2021), Bhattacharya et al. (2021), David et al. (2017), Ersahin et al. (2017), Fernquist et al. (2018), Fonseca Abreu et al. (2020), Güngör et al. (2020), Gupta and Kaushal (2017), Huang et al. (2016), Albayati and Altamimi (2019), Mateen et al. (2017), Munoz and Paul Guillen Pinto (2020), Oentaryo et al. (2016), Ren et al. (2018), Rodrigues et al. (2022), Rodríguez-Ruiz et al. (2020), Saranya Shree et al. (2021), Sedhai and Sun (2018), Sheikhi (2020), Purba et al. (2020), Thejas et al. (2019), Zheng, Wang, et al. (2016a)	24	Sybil	90.4
ANN	Adikari and Dutta (2020), Akyon and Esat Kalfaoglu (2019), Alarifi et al. (2016), Alom et al. (2020), Al-Qurishi et al. (2018), Hakimi et al. (2019), Heidari et al. (2020), Heidari et al. (2021), Kesharwani et al. 2021), Khaled et al. (2019), Meshram et al. (2021), Munoz and Paul Guillen Pinto (2020), Shukla et al. (2022), Sen et al. (2018), Thejas et al. (2019), Yang et al. (2022)	17	Sybil, Spam	94
DT	Bazm and Asadpour (2020), David et al. (2017), Echeverrï£¡a et al. (2018), Fazil and Abulaish (2018), Albayati and Altamimi (2019), Munoz and Paul Guillen Pinto (2020), Narayan (2021), Shukla et al. (2022), Ramalingaiah et al. (2021), Rodrigues et al. (2022), Sengar et al. (2020), Varol et al. (2017), Zheng, Wang, et al. (2016a)	13	Social	99
AdaBoost	Andriotis and Takasu (2019), Bazm and Asadpour (2020), Echeverrï£¡a et al. (2018), Fernquist et al. (2018), Heidari et al. (2020), Kenyeres and Kovács (2022), Knauth (2019), Munoz and Paul Guillen Pinto (2020), Sahoo and Gupta (2020), Sen et al. (2018), Sengar et al. (2020), Singh and Banerjee (2019), Varol et al. (2017)	13	Social	98.8

Table 6 (continued)

Algorithm	Papers that applied it	No. of papers	Highest classifier bot type detected	Performance measure (accuracy) (%)
KNN	Al-Zoubi et al. (2017), Andriotis and Takasu (2019), Bazm and Asadpour (2020), Fonseca Abreu et al. (2020), Gupta and Kaushal (2017), Hakimi et al. (2019), Kondeti et al. (2021), Albayati and Altamimi (2019), Rathore et al. (2018), Sengar et al. (2020), Thejas et al. (2019)	11	Sybil	98
J48 (DT)	Al-Zoubi et al. (2017), Güngör et al. (2020), Gupta and Kaushal (2017), Mateen et al. (2017), Rathore et al. (2018), Ren et al. (2018), Sahoo and Gupta (2020), Sheikhi (2020), Purba et al. (2020)	9	Spam	97.6
MLP	Al-Zoubi et al. (2017), Knauth (2019), Munoz and Paul Guillen Pinto (2020), Purba et al. (2020), Sen et al. (2018), Sengar et al. (2020), Sheikhi (2020)	7	Social	83
CNN	Alhassun and Rassam (2022), Alom et al. (2020), Attia et al. (2022), Cai, Li, and Zeng (2017a), Cai, Li, and Zengi (2017b), Gao et al. (2020), Martin-Gutierrez et al. (2021), Ping and Qin (2019), Wu et al. (2021)	9	Spam	99.68
Gradient boost	Bhattacharya et al. (2021), Echeverrï£¡a et al. (2018), Inuwa-Dutse et al. (2018), Kantepe and Gañiz (2017), Sengar et al. (2020), Singh and Banerjee (2019)	6	Social	99.54
LSTM (RNN)	Cai, Li, and Zeng (2017a), Cai, Li, and Zengi (2017b), Hayawi et al. (2022), Kenyeres and Kovács (2022), Mazza et al. (2019), Kudugunta and Ferrara (2018), Ping and Qin (2019), Wanda et al. (2020)	8	Sybil	99.31
BN	Al-Zoubi et al. (2017), Fazil and Abulaish (2018), Gupta and Kaushal (2017), Rathore et al. (2018), Zheng, Wang, et al. (2016a)	5	Spam	98.4
MNB	Kantepe and Gañiz (2017), Rodrigues et al. (2022), Kumar and Rishiwal (2020), Narayan (2021), Sengar et al. (2020)	5	Spam	99
XGBoost	Pramitha et al. (2021), Sen et al. (2018), Shevtsov et al. (2022), Singh and Banerjee (2019)	4	Social	89.6
JRip (DT)	Gupta and Kaushal (2017), Rathore et al. (2018), Sahoo and Gupta (2020)	3	Spam	99.5
Bi-LSTM (RNN)	Heidari et al. (2020), Xu et al. (2021)	2	Spam	98.1
BiGRU (RNN)	Wu et al. (2021), Yang et al. (2022)	2	Social	98.87
ID3 (DT)	Albayati and Altamimi (2020)	1	Sybil	97.7
GNB	Kantepe and Gañiz (2017)	1	Social	86
ELM	Zheng, Zhang, et al. (2016b)	1	Spam	99
bi-SN-LSTM	Gao et al. (2020)	1	Spam	F1(99.31)
DenStream	Eshraqi et al. (2016)	1	Spam	99
K-means	Koggalahewa et al. (2022)	1	Spam	96.9
DBSCAN	Mazza et al. (2019)	1	Social	97.7

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Declarations

Conflict of interest The authors declare that there is no conflict of interest.

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REVIEW

Social media bot detection with deep learning methods: a systematic review

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Abstract

Social bots are automated social media accounts governed by software and controlled by humans at the backend. Some bots have good purposes, such as automatically posting information about news and even to provide help during emergencies. Nevertheless, bots have also been used for malicious purposes, such as for posting fake news or rumour spreading or manipulating political campaigns. There are existing mechanisms that allow for detection and removal of malicious bots automatically. However, the bot landscape changes as the bot creators use more sophisticated methods to avoid being detected. Therefore, new mechanisms for discerning between legitimate and bot accounts are much needed. Over the past few years, a few review studies contributed to the social media bot detection research by presenting a comprehensive survey on various detection methods including cutting-edge solutions like machine learning (ML)/deep learning (DL) techniques. This paper, to the best of our knowledge, is the first one to only highlight the DL techniques and compare the motivation/effectiveness of these techniques among themselves and over other methods, especially the traditional ML ones. We present here a refined taxonomy of the features used in DL studies and details about the associated pre-processing strategies required to make suitable training data for a DL model. We summarize the gaps addressed by the review papers that mentioned about DL/ML studies to provide future directions in this field. Overall, DL techniques turn out to be computation and time efficient techniques for social bot detection with better or compatible performance as traditional ML techniques.

Keywords Social media · Bots · Deep learning · Machine learning · Bot detection · Systematic review

Abbreviations

DL	Deep learning
ML	Machine learning
SVM	Support vector machine
SMOTE	Synthetic minority oversampling
ENN	Edited Nearest Neighbours
CGAN	Conditional generative adversarial network
ADASYN	Adaptive synthetic
GloVe	Global Vectors for Word Representation
USE	Universal Sentence Encoder
LIWC	Linguistic Inquiry and Word Count
MTF	Markov transition field
GAF	Gramian Angular Field
SST2	Stanford Sentiment Treebank
BERT	Bidirectional Encoder Representations from Transformers
GPT	Generative pre-trained transformer
LSTM	Long short-term memory
CNN	Convolutional neural network
RNN	Recurrent neural network
GRU	Gated recurrent unit
BiGRU	Bidirectional gated recurrent unit
BeDM	Behaviour enhanced deep model
Bi-SN-LSTM	Bidirectional self-normalizing LSTM network
ResNet	Residual network
MLP	Multilayer Perceptron

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GCNN	Graph convolutional neural network
PSO	Particle swarm optimization
FFNN	Feed-forward neural network
RDNN	Regularized deep neural network
C-DRL	Content-based deep reinforcement learning
SNA-DRL	Social network analysis-based deep reinforcement learning
HAN	Hierarchical attention networks
RF	Random forest
LR	Logistic regression
SL	Simple logistic
DNN	Deep neural networks

1 Introduction

Nowadays, social media platforms are expanding at a fast pace in terms of number of users, data size and applications. They are basically internet-based applications that facilitate exchange of user-generated contents. A high rate of usage of such platforms creates a revolution in human communication. For instance, Facebook had almost one-third of the world population[1] as its users in the first quarter of 2019, and by 2015, the estimated number of users had grown to 1.3 billion in Twitter [1]. In general, people use social media platforms for interacting with others via various type of posts, by following them and being followed. In some platforms like Twitter, trending topics are discussed on a daily basis [2]. Malicious individuals and organizations exploit the flexibility and power of social media to gain influence by creating fake automated accounts, often called social bots or sybil accounts and, in this study, social media bots. These accounts can exploit the regular services for malicious purposes by manipulating the discussion and public opinion, spreading rumours and fake news, promoting harmful products/services, defaming other people or being fake followers of a user to handcraft a fake popularity and spamming/social phishing/profile cloning/collusion attacks [3, 4]. These attacks can be catastrophic. Some of the vicious examples of bot infiltration are attacks during US presidential election, Russiagate hoax[2] attack and rumour spread during Boston Marathon blasts [5] in Twitter. A very high percentage of bot accounts in social media, such as between 9 and 15% accounts (equivalent to 48 million accounts) in Twitter according to a recent study [6], makes these platforms highly vulnerable.

Hence, social bot detection research got a huge interest as a defence mechanism against such threats. Moreover, the human-like characteristic of these bots and a continuous evolution [7, 8] in their strategy make it very difficult to distinguish them from legitimate users and thus invoke the need of more sophisticated and dynamic countermeasures.

Machine learning can be a smart approach for learning the pattern of bot behaviour and thus for efficient detection, especially by making an effective use of the mammoth streaming data generated from these online platforms. Several machine learning techniques, including supervised, unsupervised and reinforcement learning, have been proposed to detect bots in Twitter and other platforms [9–11]. In general, ML models show good performance measures for social bot detection with easy implementation; however, they exhibit time and computation expensive feature extraction, slower learning time and less performance in case of large features. Deep learning is a special branch of machine learning which is distinguishable from traditional ML approaches by its layered architecture and ability to process and extract features from complex data such as images, text and speech. Many DL models are shown to outperform the traditional, shallow and ML classifiers for the bot detection task. In addition, to overcome the challenge of cyborgs with human-like behavioural attributes, DL techniques especially generative adversarial networks [12] can be really effective.

To the best of our knowledge, six review articles have been published so far in this field highlighting social bot detection techniques and taxonomy (see details in supplementary Table S1). Hence, the existing surveys cover the broader field of social bot detection, highlighting all of the technical approaches such as structure-based, crowd-sourced, hybrid, graph-based, machine learning and other techniques such as dynamic time warping, digital DNA-based or natural language processing (NLP) approaches [8, 13–17]. We have found at least one review article that surveyed machine learning algorithms in general for different categories of social media analysis including bot detection [14].

The motivation behind the current systematic review on only the deep learning articles for social media bot detection is mainly based on two reasons. First, the DL approaches showed high potential to detect the benign/malicious bots and to keep pace with their fast-evolving and highly variable characteristics, which is a pressing need right now. Second, none of the previous systematic reviews attempted to synthesize the efficacy, failure and challenges of exclusively these techniques for social media bot detection. It is important to figure out the status of the deep learning research in comparison with other techniques, including the classical machine learning

[1] https://www.statista.com/statistics/264810/number-of-monthly-active-facebook-users-worldwide/.

[2] https://www.nbcnews.com/tech/tech-news/after-mueller-report-twitter-bots-pushed-russiagate-hoax-narrative-n997441.

algorithms, to guide the future research in social media bot detection to the right direction and to maximize the usability of current data sources. Third, while a previous review article presented machine learning research in general for social media, narrowing it down to social media bot detection and going deeper from both technical and application points of view is crucial for ensuring future success in this field.

Here, we presented a comprehensive analysis on the model architecture, processing workflows, effectiveness and limitations of these cutting-edge approaches. Figure 1 shows a generalized overview of the end-to-end process for the DL-based social bot detection task.

The main contribution of our work can be summarized as follows:

- We provide a systematic review of various DL approaches which have been used particularly in social media bot detection, including all the related literature found between 2000 and 2021, from pre-defined resources, and based on pre-defined inclusion/exclusion criteria. Out of 1496 obtained publications using the search term, 40 are finally selected for this review. Up to the best of our knowledge, it is the first systematic review conducted on this topic.
- We present a comparative study among DL algorithms and between DL and traditional ML approaches for the bot detection task. It is worth mentioning here that only the comparisons with ML, as reported in the DL articles, were presented and analysed in this review.
- We provide the summary of the datasets, taxonomy of the features and their extraction mechanism that are extensively used for DL approaches in this field.
- We provide the future research direction by informing the researchers of the high potential of these approaches as well as of the current gaps and challenges.

These contributions can enrich DL-based research for the bot detection, especially by directing researchers to the most effective algorithms and their associated features and pre-processing strategies and to the gaps and future potentials of this research field.

This paper is structured as follows: Sect. 2 defines the social bots; Sect. 3 explains the methodologies of this paper; Sect. 4 explains the datasets, features and other pre-processing strategies used for DL methods; Sect. 5 analyses and compares the architecture and performance of DL models for bot detection; Sect. 6 presents a comparative study between DL and traditional ML approaches; Sect. 7 discusses the research gaps and challenges regarding the DL-based social bot detection and, finally, Sect. 8 concludes the paper with directions to future work.

2 Social media bots

Before going into detail studies on ‘Social media bots’, it is very important to define what they are. The word ‘bot’ generally means automated software agents that are designed to hold human conversation or to use in command-and-control networks to launch attacks [18, 19]. Still, the concept of bots in social media can be really intricate. By following the very recent systematic review by [17], we will use the term ‘social media bots’ to clearly differentiate them from more general categories such as social bots or sybils that are defined with the condition of mimicking human behaviour [20].

Now, automated accounts or bots can be benign or malicious. Also, the automated malicious accounts may not mimic human behaviour, known as traditional spam bots. Some authors proposed to define them as semi-automatic agents [21] as they are designed to fulfil a specific purpose and generally controlled by a botmaster or human who manages their activities. Some categorized malicious bots such as spam bots (disseminating spam), political bots (get involved in political discussion) and sybils (fake accounts) use to gain undeserved influence [17]. In our review, we emphasize on the word ‘bot’ or ‘sybil’ that represents automated accounts only. Hence, we excluded all those papers which mentioned fake profile or malicious account detection in mixed terms (human + automated) and included those reporting the automated accounts only. In some articles, although it is not clear which kind of bots

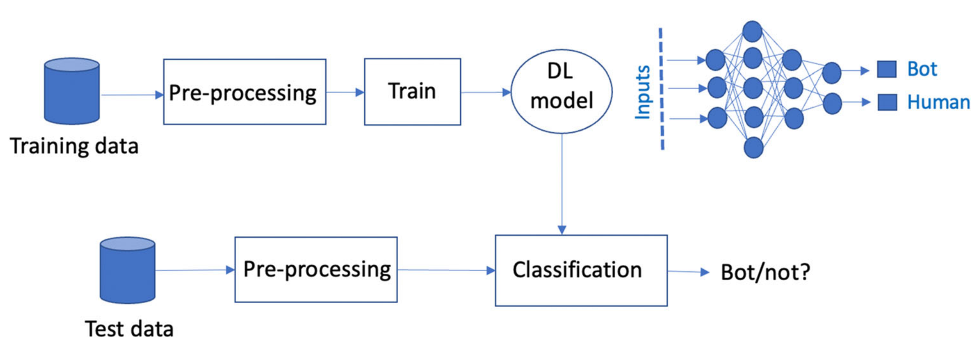


Fig. 1 A generalized workflow diagram of the DL-based social bot detection. The collected data are pre-processed first through labelling, data augmentation and feature extraction and then used to train a DL model. The trained model is used for the classification of human/normal users or social media bots on the unseen test data

they attempted to detect (benign or malicious), we did not exclude any paper based on that criteria. The papers used different terms such as normal or human users or legitimate accounts to define the opposite class of bots; however, we did not set any filtering criteria for that category either. Our search strategies and inclusion/exclusion criteria are distinctly defined in the next section.

3 Methodology

3.1 Research questions

For this review, the authors followed the guidelines suggested by [22] for conducting a systematic literature review (SLR). After identifying the need for such a review as depicted in 'Introduction', we specified the research questions. This work has been conducted precisely to report about the deep learning research in the field of social media bot detection. Hence, the study aimed to answer the following questions:

RQ1 What are the deep learning algorithms that are used for social bot detection or prediction?
RQ2 What are the pre-processing mechanisms needed for the above algorithms?
RQ3 What is the effectiveness (success or failure) of different DL models in association with combinatory feature inputs (multimodal inputs, e.g. user data, posts, etc.) for social bot detection?
RQ4 How is the performance of deep learning algorithms in comparison with traditional machine learning models?
RQ5 What are the gaps and future research directions in this area?

3.2 Search strategy

The search strategy was determined based on its objectives and research questions. It was derived through the consultation among the authors with relevant experience and through trial searches using various combinations of search terms to find the already known primary studies. The search is conducted in several stages: primary term-based search in three of the largest databases including Google Scholar/Scopus/ScienceDirect, crawling in published literature/systematic reviews for any missing papers (journal articles and conference proceedings), manual screening over the title and abstracts and finally applying exclusion and inclusion criteria to generate the targeted set of publications for review. We also informally searched over other databases such as IEEE/ACM to find any missing ones and did not find anything additional. The abstract screening and the outcome of search using scoping criteria were assessed based on the agreement among the researchers on each paper.

We used Rayyan Intelligent Systematic Review software[3] for an organized screening of the papers with the above-mentioned steps.

Figure 2 shows our selection steps from the initial search to finalizing the articles to be analysed and reported in this study.

3.3 Search terms

We generated a very wide variety of search terms and searched over three significant databases to include every published article that used DL for social media bot detection. Therefore, we included different bot terms ('bot', 'fake news', 'botnet' and 'sybil') and also both 'detection' and 'prediction' terms. However, the terms 'detection' and 'prediction' represent the same task, i.e. to classify bots vs. human. The search term was finalized through initial trials and manual inspection on the outcome.

Google Scholar:With all of the words: social bot detection ['in title' box]and articles dated between 2000 and until now (the time range was set considering social media bot detection with DL approaches as a relatively new field of research)Time stamp: 31 March 2021, 9:27 pm: count: 46

Scopus:

TITLE-ABS ('Review' AND 'social bot detection') OR TITLE-ABS (('detection' OR 'detecting' OR 'prediction' OR 'predicting') AND ('Bot' OR 'bots' OR 'fake news' OR 'botnet community' OR 'botnet' OR 'Social network polluting contents' OR 'sybil') AND ('deep' OR 'machine learning' OR 'neural network')) AND PUBYEAR > 2000Time stamp: 31 March 2021, 12:15 am: count: 1232

ScienceDirect:

In advanced search: 'Find articles with these terms: social bot detection, deep learning, machine learning and neural network'.

Year: 2000–2021Time stamp: 31 March 2021, 12:09 am: count: 311

4 Results

4.1 Overview of input features and pre-processing

This section presents the detailed information on the taxonomy of the features and pre-processing strategies including data labelling, balancing and feature extraction for the reviewed studies.

[3] https://www.rayyan.ai/.

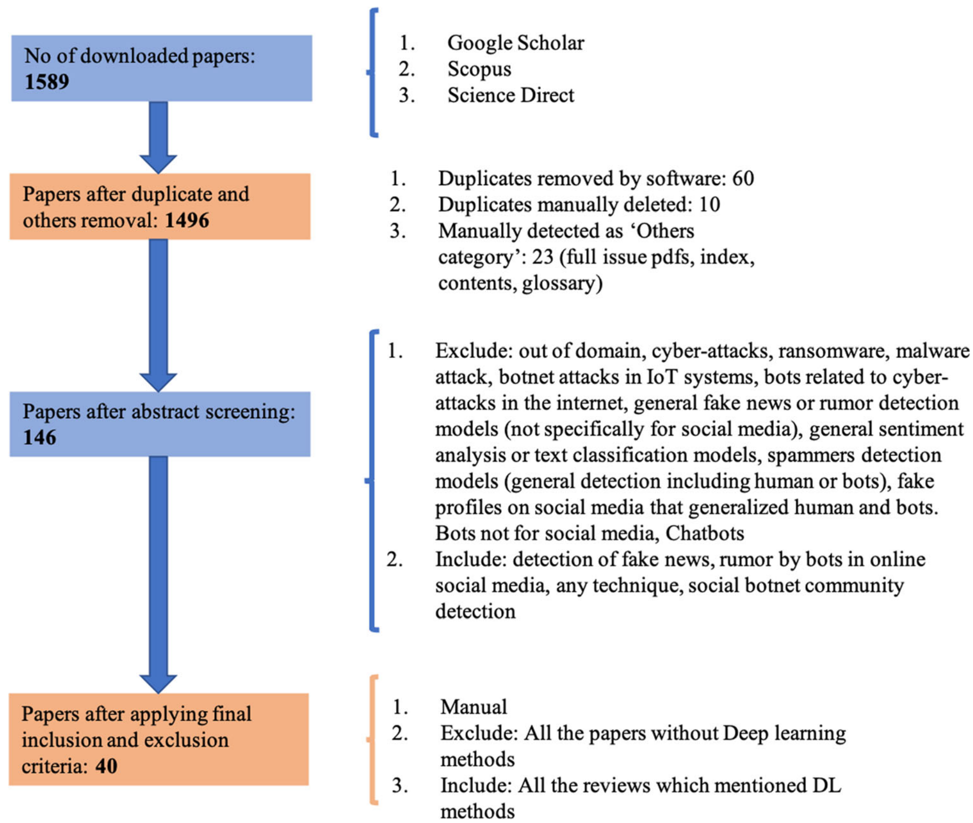


Fig. 2 Systematic steps for the selection of the articles for our final review

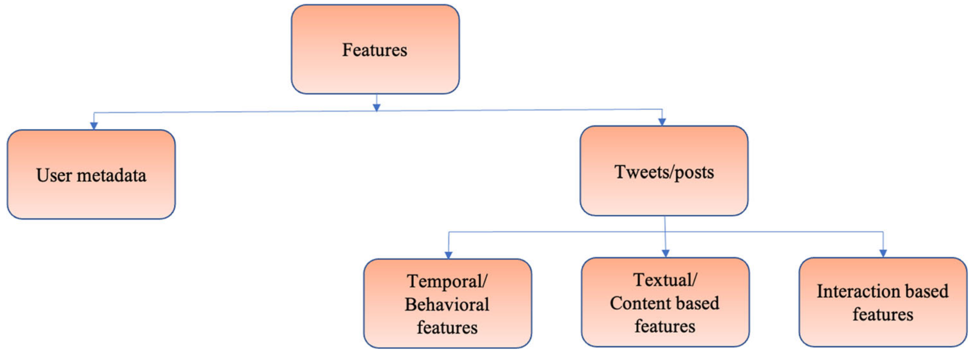
4.1.1 Types of features

Based on our detail analysis, the features used for DL models can be classified into two main groups as shown in Fig. 3.—user metadata and tweets/posts. These two main features with their subclasses are described below.**User metadata:** This group generally represents information that come from the user profile information, such as nick name, introduction, location, follower count, friend count, listed count, favourites count and statuses count, as used by several studies [23, 24].

Tweet/posts data: The following features are shown to be extracted from users' posts for classification.

- *Temporal/behavioural features* This group represents behaviour information including posting behaviour/connecting behaviour. Studies like [25] regarded user's history tweets as temporal text data instead of plain text in existing methods, by using two features, timestamps (posting inter-arrival time pattern) and posting type (original/retweet). In addition, they focussed on extracting endogenous (circadian rhythm) and exogenous features (cultural or environment influences) that represent social behaviour of a user. Posting times have been used by other studies too [26]. From the time interval between posts, another study [27] also derived burstiness parameters or information entropy to use them as features. Following a different approach, Lingam and colleagues represented the descriptive features of user's behavioural patterns as states, as inputs to the learning agent for the deep Q-learning model [28].
- *Textual/content-based features* These features can be directly extracted without any intermediary steps such as num_hashtags, num_urls, num_mentions, num_punctuations and num_interjections. In addition, the high-level representation of the textual data such as abuses, angry terms and named entities from the tweets can be derived by ConvNets [25, 29]. Al-Qurishi et al. [26] used quality of the generated content as features and derived emotional reactions from twitter posts to predict bots.
- *Interaction-based features* The previous studies [27] used interaction-based features such as number of comments/reposts/likes in the posts, repost ratio and diversity of sources of posts for the classification. Al-Qurishi et al. [26] mentioned the interactions as graph-based features. [30] also proposed to extract the bipartite graph of the following relationship and the retweet relationship between users to address the challenge of differentiating between anomaly and legitimate accounts.

Fig. 3 Taxonomy of features used in the DL research for bot detection



4.1.2 Data collection, labelling and augmentation

Most of the reviewed studies used existing or popular datasets for their model training and testing (see supplementary; Table S2). For example, Katarya, Mehta [29] used the popular MIB datasets that included manually annotated genuine and fake twitter accounts along with tweet data corresponding to these accounts. Cai, Li [25] used a public dataset [31] which was collected with honeypot method. Heidari and Jones [32] used the advantage of BERT's transfer learning approach and used labelled data of movie reviews to detect bots in social media, and thus proposed a new way to use labelled data from other online platforms to detect bots in Twitter.

Wu et al. [27] manually labelled a small portion of user data, collected from Sina Weiboo, based on their proposed six discrimination metrics. They introduced an active learning module composed of a query algorithm, a machine learning classifier and a supervisor, for building a large-scale experimental dataset. Kudugunta and Ferrara [33] proposed an effective technique to generate a large, labelled dataset from a minimal amount of labelled data based on a combination of synthetic minority oversampling (SMOTE) and two under sampling techniques (Edited Nearest Neighbours (ENN) and Tomek Links). Wu et al. [34] proposed another data-augmentation approach by adopting improved CGAN generative methods to extend social media bot samples with three types of oversampling methods: SMOTE, the adaptive synthetic (ADASYN) algorithm and the random oversampling methods.

4.1.3 Feature extraction

Some of the key feature extraction or use strategies are described as follows:

A couple of studies [23, 29] used a number of data processing steps to improve model efficiency, namely null value cleaning and reducing dimensionality, and thus ended up with an appropriate subset of the features for training. In contrary, Wu et al. [27] used the combination of metadata-, interaction-, content- and timing-based features to distinguish between social media bots and normal users and showed that all detection approaches performed best on the feature set that contained all the features.

Ping and Qin [35] used temporal features, content features and relationship between them as inputs to multiple DL models. Cai, Li [25] concatenated text vector, timestamp vector and posting type vector into a single sequence vector as the input to the bot detection classifier. Halvani and Marquardt [36] also performed some low-level preprocessing steps on their dataset including concatenation of all tweets in each XML file into a one long document, lowercasing of the entire text and substitution of noisy elements with a dummy token, for example twitter handles.

Most of the studies used appropriate language modelling technique to convert the textual content to suitable inputs for classification. A number of studies [25, 29, 32, 37, 38] used GloVe word embedding [39] for language modelling. Kudugunta and Ferrara [33] used an additional step by forming a string of tokens from each tweet by a python library that is called ekphrasis, which performed tokenization, word normalization, word segmentation (for splitting hashtags) and spell correction. Then, the tokenized tweets were transformed into an embedding using the GloVE model. Finally, the mean of the embeddings for each tweet had been calculated, and thus, each tweet was represented as a vector. Onose, Nedelcu [40] used pre-trained word2vec word embeddings for text representation. Mou and Lee [41] proposed a novel framework JntL that incorporated both handcrafted features and automatically learnt features by DL methods from various perspectives. They extracted Linguistic Inquiry and Word Count (LIWC), a dictionary that captured the general statistics of each user's profile, activities and their preferences in terms of word usage. Automatic feature learning was divided into two components: text embeddings and temporal behaviour embeddings. They used Universal Sentence Encoder (USE) for sentence-level embedding and BERT-Base for word-level embeddings. Given the sequences of each user's posting timestamp, they applied two methods for mapping those sequences into 3D images: GAFMTF and II Map for informative pattern recognition. (Heidari & Jones, 2020) used Google Bert for sentiment classification of tweets. BERT and RoBERTa (an adaptive version of BERT) embeddings were also used for bot detection by [42, 43]. While BERT facilitates multilingual embeddings, RoBERTa improves the performance in longer sequences, or when there are vast volumes of data.

4.2 Deep learning methods for social media bot detection: evaluation

Table 1 summarizes all the DL algorithms that the social media bot detection studies have used until now. From the list, we have explained here some groups of high performing or novel DL models and scrutinized them according to their pre-processing strategies, training/test datasets and prediction performance.

The holistic analysis will validate the information that the cutting-edge artificial intelligence techniques are generally working well in this field, and in addition to traditional ML algorithms, the future researchers might want to consider deep learning models for their study. This will help the researchers to find out the best suitable approach in terms of the model architecture, feature extraction, preprocessing mechanism and sample size for their particular dataset and objective of the study. If any model performance is worse or the approach has limitations, we have identified the reason and, in contrast, emphasized on their unique benefits. The key approach here is the same, i.e. the use of deep learning to deal with the highly heterogenous and evolving social media bots.

Here, we present a taxonomy for the most commonly used DL models for the bot detection task in Fig. 4.

Some of the highly performing approaches are as follows:

CNN/LSTM/Bi-LSTM/RNN/GRU: max accuracy 100%

A number of social media bot detection studies used different combinations of deep convolutional neural network (CNN), LSTM, recurrent neural network (RNN) and gated recurrent units (GRU). Ping and Qin [35] achieved the highest performance with 100% accuracy among all. The main characteristic of their proposed model was the use of tweet content and temporal features in a fused way. In general, CNNs work better to classify textual content and identify abuses, angry terms, named entities, etc.; however, representing the textual data in a vectorized form through language models such as Word2Vec was essential. In contrast, LSTMs and RNNS are both highly useful for finding the relationship between the consecutive time point data, where sequence of sentence is important. Their proposed DeBD model included the following layers:

- Joint content feature extraction layer for generating features of tweet contents and the relationship between them through CNNs from the concatenated tweets posted by social bots.
- Temporal feature extraction layer for the extraction of temporal features of tweet metadata using LSTM.
- Fusion layer to fuse the above-mentioned features with certain rules.
- Finally, a fully connected layer and a softmax layer to generate the binary labels: ‘social bot or not’.

Although they achieved a very high precision and nearly perfect accuracy on different datasets, it should be noted that it required a large amount of tweet information from the user and removal of the non-English users and their tweets from the dataset.

In comparison, the behaviour enhanced deep model (BeDM) model by [25] fused the tweet content information and behavioural information using deep learning method. The order of a specific tweet was determined by its posting time, and a high-level representation of each tweet was obtained using the shared CNNs. The posting behaviour information in terms of timestamp and posting types was concatenated with the text vector output of CNN. Finally, the sequence of concatenated vectors for all history tweets was fed into an LSTM network to finally label each input as bot or human. Although the accuracy measure was not available, it turned out that this strategy was not as successful as other studies with only 88.41% precision for detecting bots, even with a very large sample size of around 2 million in each of the bot and human class.

Sengar, Kumar [23] used an interesting model architecture with a machine learning classifier for the final prediction labels, from the multimodal feature inputs such as first level predictions by CNN from GloVe word embeddings fused with user account and tweet metadata. Their approach on an extremely large dataset with more than eight thousand user accounts and around 7 million tweets, achieved as good as or better performance when compared to only LSTM/Bi-LSTM model-based studies [33, 39] and the combinatory CNN-LSTM studies as mentioned above. The study by Mou and Lee [41] attained 91.8% precision and 90.1% accuracy with around 37,000 accounts in each class, with 84 handcrafted and automatically learnt features such as text embeddings and temporal behaviour embeddings and using CNN for final decision-making. In a less complicated way, another study [44] showed that bot accounts could be detected with a very high accuracy with a combination of CNN and ANN using a single post on the social media; however, it requires a huge amount of tweets (about 1 million) in each class. Gao et al. [45] used a self-normalizing CNN, adopted to extract lower features from the multi-dimensional input data, that

Table 1 Comparison among DL algorithms: architecture and performance and comparison with traditional ML algorithms [model performance measures are shown as the (min–max) range of values reported by combining the results from the cited studies in each row. Please note that some measures have single values as only one study reported that metric, and a range is not available. A range of values for a single study may represent measures for different test datasets or different combinations of features/models.]

Deep learning methods	Social media	Model performance	Comparison with other ML/DL methods	Dataset	Input features	Used in/mentioned in
CNN + Multilayer Perceptron/ other ML models	Twitter	Accuracy: 90.31–99.54% Precision: 90.76–99.54% Recall: 90.31–99.54% F1-score: 96.73–99.54%	When CNN predictions added with other features for a final classification, performance improved than KNN, support vector, decision tree, RF, AdaBoost, gradient boosting, Gaussian NB, Linear Discriminant and Multilayer Perceptron	Cresci	Profile details, tweets and tweet metadata	[23, 29]
CNN + ANN	Twitter	Accuracy: 93.69–99.43% Precision: 94–99% Recall: 94–99% F-measure: 94–99%	Performance improved over only CNN (tweet features based) and only ANN (profile features based) models CNNs can learn the difference between the text contents of tweets by bots and humans better than ANN	Cresci	Profile details, tweets and tweet metadata	[44]
CNN + LSTM/only Bi-LSTM/CNN + self-normalizing Bi-LSTM + dense layers	Twitter	Accuracy: 96.1–100% Precision: 87.58–99.99% Recall: 86.26–100% Specificity: 93.8–99.8% F1-score: 87.32–99.99% AUC: 99.32% MCC (Mathews correlation coefficient): 0.92–0.998	(CNN + LSTM) performance improved over Stweeler [54], Boosting [55], BoostOR [31] Performance of bi-SN-LSTM improved over commonly utilized RNN methods including vanilla RNN, bidirectional RNN (bi-RNN), gated recurrent unit (GRU), bidirectional GRU (bi-GRU), LSTM and bidirectional LSTM (bi-LSTM) Only Bi-LSTM application on Cresci'17 dataset [56] was either compatible or a little lower in performance in comparison with unsupervised DNA-inspired online behavioural model [57] CNN + LSTM models performed better than Twitter, human annotators, BotOrNot (supervised RF Classifier), decision tree, BayesNet and Decorate classifiers [7] unsupervised stream clustering algorithms (StreamKM + + and DenStream)[58], generic statistical approaches in [59] Unsupervised DNA-inspired online behavioural model proposed in Cresci'16 performed a little better in terms of precision and specificity in one of the test sets but all other measures	Social Honeypot, Cresci, Lee, Gilani, Varol, Midterm, Botometer, Botwiki, Celebrity, Pronbots, Vendor and Verified	Tweet, tweet metadata, timestamps, user profiles, structure of users' local subgraph, LIWC features and word/sentence/ temporal embeddings	[25, 35, 38, 41, 45]

Table 1 (continued)

Deep learning methods	Social media	Model performance	Comparison with other ML/DL methods	Dataset	Input features	Used in/mentioned in
			were lower than the DeBD model (CNN + LSTM) as in [35]			
CNN	Twitter	Accuracy: 85.65% Precision: 97.02% Recall: 94.35% F1-score: 83.67% AUC: 85.65%	Simpler models with less trainable weights work better than complicated ones	Pan19	Tweets	[54]
Bidirectional Encoder Representations from Transformers (Google Bert) + fast forward neural network	Twitter	Accuracy: 94.8–97.9% F1-score: 94.2–97.6%. MCC (Mathews correlation coefficient): 89.1–96.2%	Accuracy and overall performance improved over BotOrNot [61], RF/decision tree/BayesNet/Decorate [7], StreamKM + + (based on K-means) and DenStream (based on DBSCAN) [58], Naïve Bayes, Jrip, decision tree [59] and unsupervised sequence clustering [57]	Cresci	BERT embeddings	[32]
A novel deep neural network model called RGA = a residual network (ResNet) + a bidirectional gated recurrent unit (BiGRU) + attention mechanism	Twitter	Accuracy: 98.87%. Precision: 98.40% Recall: 99.33% F1-score: 98.86%	Performance improved over LR/SVM/ELM/RF/MLP/LSTM/ComNN/CNN and performance improved over only Resnet, only BiGRU, Resnet + BiGRU, Resnet + attention, BiGRU + attention models	Data collected from Sina Weibo	Metadata-, interaction-, content- and timing-based features	[27]
Contextual LSTM/Bi-LSTM/LSTM	Twitter	For tweet-level bot prediction: Accuracy: 79.64–98.88% Precision: 96% Recall: 96% F1-score: 96% ROC/AUC: 87.04%–96.43%	Contextual LSTM performance improved over only metadata-based LR/SGD/RF/AdaBoost classifiers Bi-LSTM performance improved over MLP, AdaBoost, LSTM, RNN, Bi-GRU and GRU classifiers	Cresci and pan19	User profile features, tweet metadata and temporal features	[33, 37, 46]
Embedding layer + two Bi-LSTM layers + attention mechanism at the top of the second Bi-LSTM layer + three dense layers	Twitter	Accuracy: 82.61–99.06% Precision: 83.65% Recall: 89.41% F1-score: 86.43% ROC/AUC: 88.70%	Performance improved over MLP/CNN/SVM/RF/LR/BNB/KNN/decision tree on the same dataset	Social Honeypot and Cresci	User profile features, tweet metadata and word embeddings from tweets	[47]

Table 1 (continued)

Deep learning methods	Social media	Model performance	Comparison with other ML/DL methods	Dataset	Input features	Used in/mentioned in
Feed-forward neural network/fully connected based neural network/MLP	Twitter, VKontakte	Precision: 92.59% Recall: 50% F1-score: 64.94% Accuracy: 86%–92% AUC: 73% (in VKontakte)	Accuracy is better for English tweet inputs in comparison with Spanish tweets Bot-DenseNet [43] was either compatible or a little lower in performance when compared to previous supervised (RF, one-class classifier, AdaBoost, logistic regression and decision tree classifiers) and unsupervised algorithms (LSTM autoencoder for feature extraction + hierarchical density-based algorithm)	Verified, Botwiki, Midterm, Cresci, Botometer, Vendor, Celebrity, Pronbots, Gilani, Varol, pan19 and collected datasets from Twitter, VKontakte	Profile-based, content-based features, tweet metadata, BERT contextualized text embeddings and emoji embeddings	[24, 26, 42, 43, 55, 56]
RNN	Twitter	88%	Performance improved over GRU/MLP/AdaBoost classifier	Cresci	Temporal features from tweets	[46]
Gated recurrent unit (GRU)/bidirectional GRU	Twitter	Accuracy: 93.1%	Performance improved over MLP/AdaBoost Classifier; Bi-GRU improved over RNN	Cresci	Temporal features from tweets	[46]
Deep Q-learning based on particle swarm optimization (DQL-PSO)	Twitter	Precision: ~ 95% Recall: > 90% F-measure: > 90% G-measure: > 90%	Performance improved over adaptive deep Q-learning, FFNN, RDNN, C-DRL and SNA-DRL	Social Honeypot and the Fake project	User behavioural patterns as states	[28, 57]
Improved conditional generative adversarial network (improved CGAN)	Twitter	Accuracy: 97% Precision: 97% Recall: 97% F1-score: 97% G-mean: 93%	Performance improved over the traditional CGAN algorithm	Cresci and Varol	User metadata, sentiment, friends, content, network and timing	[34]
Deep forest algorithm	Twitter	Accuracy with SMOTE: 97.55% Accuracy without SMOTE: 94.26%	Performance improved over RF algorithms	Varol and Social Honeypot	Metadata of user profile and posts	[58]
Hierarchical attention networks (HAN)	Twitter	Accuracy: 89.43%	Accuracy is better for English tweet inputs in comparison with Spanish tweets	Pan19	Tweets	[40]

were generated from user profiles and the structure of users' local subgraph and a bidirectional self-normalizing LSTM network (bi-SN-LSTM) to extract higher features from the compressed feature map sequence. Their unique technical contribution lied in proposing bi-SN-LSTM network with SELU as the activation function of its recurrent step, which provided unbounded changes to the state value and using alpha dropout to prevent overfitting while training neural networks. With 1950 sybils and 1950 human users' data, the model achieved 99.99% precision.

Thus, overall, it implies that the efficient fusion of textual, temporal and user account features with ensembled modelling is the key to the higher success for the bot detection.

By using GloVe word embeddings of tweets and Stanford CoreNLP toolkit for sentence splitting and tokenization to a bidirectional LSTM model, Wei and Nguyen [38] achieved lower performance (up to 96.1% accuracy and 94% precision) than the highest CNN + LSTM measures. However, their model required no prior knowledge or assumption about users' profiles, friendship networks or historical behaviour on the target account and thus no handcrafted features. On the contrary, using GloVe embeddings and a Bi-LSTM network, Luo et al. [37] achieved only 79.64% accuracy which could be rendered to their small sample size with 412,000 annotated tweets posted by 2060 bots and 2060 humans, respectively. Rajendran et al. [46] were able to distinguish the tweeting rate and frequency of bot accounts from the genuine accounts with a good classification accuracy of 98.88% using a Bi-LSTM model only. Although this is also lower than the maximum CNN + LSTM limit, they showed an important comparison among Bi-LSTM, Multilayer Perceptron (MLP), RNN, LSTM, GRU and bidirectional GRU models which revealed superiority of Bi-LSTM models with the highest accuracy among all. These results implied that the failure of RNN models to collect information from a long sequence can be overcome by replacing them with LSTM and GRU units.

Interestingly, the study by Kudugunta and Ferrara [33] proposed a contextual long short-term memory (LSTM) architecture that exploited both tweet content and user metadata to detect bots from a single tweet. Using a minimal amount of labelled data (roughly 3000 examples of sophisticated Twitter bots), they demonstrated a high classification accuracy (AUC > 96%), which could limit the model generalizability and usability. However, the study also supported the idea that utilizing tweet metadata in addition to tweet content can improve the model performance. Another work by Ilias and Roussaki [47] also got successful using a variant of LSTM model (with embedding layer, two Bi-LSTM layers and attention mechanism at the top of the second Bi-LSTM layer and finally three dense layers). This model outperformed earlier works with 99.06% accuracy on Cresci'17 dataset (legitimate tweets vs. malicious ones), however, failed on Social Honeypot dataset with 82.615% accuracy (humans and content polluters). Future studies may investigate more hybrids of LSTMs like sentence-state LSTMs in association with other architecture as described above.

Multilayer perceptron/feed-forward neural network: max accuracy: 99.92%

A novel but much simpler Bot-DenseNet model has been proposed by Martín-Gutiérrez, Hernández-Peñaloza [43] to address the bot identification task by a multilingual approach using BERT and RoBERTa to generate tweet embeddings. They are later on concatenated with the rest of the metadata to build a potential input vector on top of a dense network. The model achieved a lower F1-score than the previous LSTM autoencoder performance [48]. However, certainly, the Bot-DenseNet model had a unique capability for analysing text features from multiple languages and due to its low-dimensional representation, was suitable for use in any application within the information retrieval (IR) framework. In contrast, Al-Qurishi, Alrubaian [26] achieved a high accuracy of 99.92% on cleaned (outliers removed) by using a data harvesting module, a (profile-based, graph-based and content-based) feature extracting mechanism and a deep feed-forward neural network to detect sybil vs. honest accounts. Again, the model accuracy was only 86% on noisy/unclean datasets, thus limiting its real-time applicability.

(ResNet) + a bidirectional gated recurrent unit (BiGRU) + attention mechanism: max accuracy: 98.87%

Using a novel deep neural network model called RGA, consisted of a residual network (ResNet), a bidirectional gated recurrent unit (BiGRU) and an attention mechanism, and using an active learning module to efficiently expand the labelled data, Wu, Fang [27] achieved 98.87% accuracy for bot detection from the data from Sina Weibo, one of the most popular Chinese OSNs in the world. This model was successfully shown to be more effective than the state-of-the-art DL solutions (MLP [49]/LSTM [50]/ComNN [51]) and thus could be more explored in future studies, especially with limited datasets.

Graph convolutional neural network: F1-score: 67–91%

New types of solutions are a necessity to face the open challenge of identifying more advanced bots. Some of the studies addressed this issue by developing bot detection algorithms based on the social network graph in addition to the user profile features. Alhosseini et al. [52] proposed a model based on graph convolutional neural networks (GCNN) for spam bot detection, as the first attempt of this

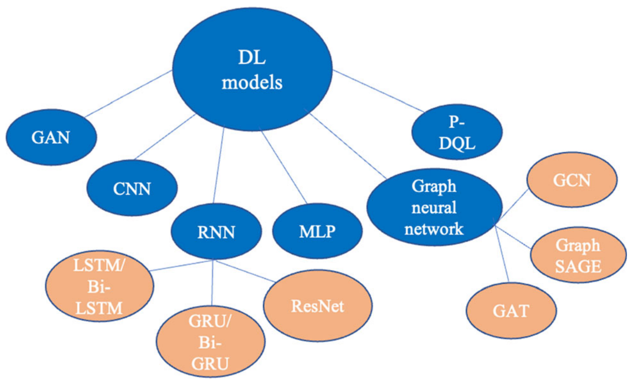


Fig. 4 Taxonomy of the DL models used for social media bot detection

kind. Their inductive representation learning approach included both the features of a node and a node's neighbourhood to better detect bots. They achieved 89% precision, 80% recall and 84% F1-score for detecting malicious accounts and social media bots. This approach outperformed (94% AUC) other classification algorithms (86% AUC with MLP and 79% with BP) with at least $\sim$ 8% improvement. Another study by Aljohani, Fayoumi [53] achieved only 71% classification accuracy by employing a GCNN, combining Twitter's social network properties in novel altmetrics data, with other user profile features.

In the same domain, Zhao et al. [30] presented a new semi-supervised graph embedding model based on a graph attention network for spam bot detection. By learning a complex method to integrate the different neighbourhood relationships between nodes to operate the social graph, the model achieved 88% recall (15% higher than MLP), 93% precision, 91% accuracy and outperformed other graph neural networks such as GCNN, GraphSAGE and GAT. This model was shown to be stable and robust even on the imbalanced dataset. However, this approach needs to be further nourished to be compatible with the more popular user metadata, tweet content and timestamp-based CNN studies for the bot detection.

Deep Q-learning: F1-score $>$ 90%

We found that Lingam, Rout [28] proposed a significantly different approach to combat the slower convergence of deep reinforcement models for the spambot detection task. They developed a particle swarm optimization (PSO)-based deep Q-learning algorithm (P-DQL) through generating an optimal sequence of actions associated with a goal state. The features such as spam content in the tweet and average number of tweets posted per day were represented as set of states. The model achieved around 95% precision and more than 90% F1-score, recall and G-measure. More importantly, they showed that the performance improved over adaptive deep Q-Learning, feed-forward neural network (FFNN), regularized deep neural network (RDNN), content-based deep reinforcement learning (C-DRL) and social network analysis-based deep reinforcement learning (SNA-DRL), thus could prove superiority over some of the deep learning approaches.

4.3 Comparison between DL and traditional ML techniques

Deep learning is basically a specialized subset of machine learning; hence, we can categorize the algorithms as traditional ML and DL. A traditional ML can be as simple as linear regression in which the algorithm learns to make a prediction based on patterns and inference. Deep learning algorithms can be regarded as more sophisticated and mathematically complex evolution of traditional ML algorithms. These algorithms can analyse data with a logical structure similar to how human brain makes intelligent decisions, by using a layer-by-layer processing strategy.

In the past few years, DL approaches begin to be more wildly used in social media bot detection and came up with some advantages over traditional ML approaches with better generalization performance, especially in case of big data.

Here, we followed up some of the DL-based social media bot detection studies that showed a comparison with the ML methods, either by applying the ML models on their data or by comparing with the previous studies, wherever appropriate, as also shown in Table 1. We, ourselves, did not search for related ML studies for such comparison as this was out of scope for the review.

By concatenating handcrafted and automatically learnt features from posting contents and temporal behaviours, the CNN + Bi-LSTM models ($\sim$ 90% accuracy) as proposed in [41] outperformed RF models ($\sim$ 87% accuracy) proposed by [55, 61], with a sufficiently large dataset of 37,497 malicious bots and 37,941 legitimate accounts. Wei and Nguyen, 2019, also showed that CNN + LSTM models performed better than Twitter, human annotators, BotOrNot model with supervised RF classifier [60], decision tree, BayesNet and Decorate classifiers [7], unsupervised stream clustering algorithms (StreamKM + + and DenStream) [58] and generic statistical approaches in [59]. Kudugunta and Ferrara [33] showed a detailed comparison of their proposed contextual LSTM architecture performance for social media bot detection at tweet level, with a number of ML methods. For account level bot detection with user metadata, RF and AdaBoost classifier with data enhancement techniques achieved 6–7% higher performance than a 2-layer neural network (NN) model. For tweet-level bot detection with textual content and tweet metadata, the LSTM model achieved $\sim$ 4–20% higher accuracy in comparison with metadata-based LR/SGD/RF/AdaBoost classifiers.

Ilias and Roussaki [47] showed that, a DL architecture that integrated an attention mechanism at the top of the Bi-LSTM layer, outperformed (in terms of all of the recall, F-measure, accuracy and AUROC scores) the existing approaches (SVM, RF, LR, DNA inspired behavioural modelling or digital DNA compression methods), to classify tweets. More specifically, recall score increased by at least 12.6% and up to 223.5%, F-measure score by 8.5–53.5%, accuracy by 7.7% and AUROC by at least 12.1%. A similar study by Rajendran et al. [46] showed that a Bi-LSTM network performed the best with 98.88% accuracy among RNN, LSTM, GRU, Bi-GRU and MLP models and outperformed the AdaBoost classifier by $\sim$ 21%. A different approach by [23, 29] using ML methods for the final classification of human vs. bots showed that by combining user data with tweet metadata and CNN predictions from tweet contents, RF and gradient boosting classifier outperformed KNN, support vector, decision tree, AdaBoost, Gaussian NB, Linear Discriminant and Multi-layer Perceptron models.

Bot-DenseNet, suggested by [43], was either compatible to previous supervised (RF, one-class classifier, AdaBoost, LR and decision tree classifiers) [6, 61, 62] or showed slightly lower performance when compared to LSTM autoencoder-based feature extraction followed by hierarchical density-based algorithm[47]. Using BERT and neural network, Heidari and Jones, 2020, showed that accuracy and overall performance improved over BotOrNot [60], RF/decision tree/BayesNet/Decorate [7], StreamKM + + (based on K-means) and DenStream (based on DBSCAN) [58], Naïve Bayes, Jrip, decision tree [59] and unsupervised sequence clustering [57].

The study by Wu et al. [34] proposed a DL framework combining a residual network (ResNet), a bidirectional gated recurrent unit (BiGRU) and an attention mechanism and the performance improved over RF, LR and SVM by $\sim$ 1–2% accuracy measures, from metadata-based, interaction-based, content-based and timing-based features.

A comparison of the semi-supervised graph embedding model for spam bot detection, as proposed by C. Zhao et al., (2020), with the classical machine learning algorithms showed an improvement of 15% in F1-score than RF models, an improvement of 15% in recall than MLP models and more than 60% improvement in F1-score, recall and precision than the BP models. The authors rendered this higher performance of such models with combined features and graph structure inputs to two reasons, 1. ML models considered either the feature set or graph structure and 2. The influence of imbalanced dataset was increased when traditional ML algorithms were used.

The deep Q-network architecture designed by Lingam et al. [28] showed approximately 5–20% improvement of precision, recall, F1-score and G-measures over the baseline algorithms (FFNN, RDNN, SNA-DRL, C-DRL and ADQL), from a combination of tweet-based, user profile-based and social graph-based features. The deep forest algorithm proposed by Daouadi et al. [58] yielded an accuracy of 97.55% information from the metadata of user profiles and posts, which was $\sim$ 2% higher than the RF model and outperformed other traditional ML algorithms such as bagging (B), AdaBoost (AB), random forest (RF) and simple logistic (SL), in terms of AUC measures.

We found only one study by Dukic et al. [42] that reported that LR models (weighted F1-score of 83%) outperformed deep neural networks (DNN) (81.78%) by using contextualized BERT embedding from tweets with some additional features that represent emojis, retweets, URLs, mentions and hashtags. They also demonstrated better interpretability of LR models over DNNs.

Considering all these facts together, we concluded that DL models outperformed ML models in almost all of the cases we reviewed that attempted to do such a comparison, with $\sim$ 5% exceptions. Data suggest that this high performance can be mainly rendered to the higher capability of processing complex data such as textual content and large volume and imbalanced datasets by DL algorithms.

5 Discussion

The objective of the current review study was to figure out the efficacy of deep learning techniques to mitigate current and future challenges in social media bot detection, considering this field is going through a very fast pace of evolution. Our review study showed that deep learning techniques in general are very successful in discriminating human and bot accounts in highly heterogeneous real datasets, in most cases over traditional machine learning models. We detected some special DL subfields/models/architecture that should direct the future bot research, such as ensembled models with fusion of various handcrafted and automatically learnt features, graph neural networks for bot cluster detection over feature-based model, latest transformer models, generative adversarial networks with an anomaly detection approach and combination of real and synthetic data-based models. We provided an accumulated information of all the datasets available for this type of study and analysed the significance and efficacy of currently used features over one another. Finally, we identified the significant challenges and improvement areas in DL-based bot detection research, irrespective of the social media platforms.

The fusion of features extracted by different types of DL models (e.g. CNN and LSTM extracted features from tweets) for the final classification has been proved effective in a number of studies [29, 35]. Therefore, future studies should explore more ensembled models and new strategies for the fusion of two or more model predictions. In contrast, the study by Halvani and Marquardt [36] suggested from their preliminary experiments that tweet-based simple feed-forward neural network outperformed advanced approaches such as CNN and LSTM models. However, generation of useful input vectors from tweets with the fast space of change in bot strategies can be highly challenging. Combination of handcrafted and automatically learnt features (content, behavioural and temporal) worked well in the previous studies [25, 41]. In addition, new type of features should be investigated to boost up the accuracy for real-world implementation and ensure generalizability. Graph neural network-based methods should be explored more in addition to individual feature-based model, to facilitate the detection of social media bot cluster or co-ordinated attack or campaigns [45].

Latest NLP models have been highly successful to generate more meaningful input vectors from tweets for a DL model [36, 43, 45]. Fine tuning of such transformer parameters and exploring the improvement scopes may further boost up these models performance [42, 52].

Most of the studies showed the superior performance of DL techniques over ML ones. This means that layered network has the property of understanding the pattern in the input data, as the data traverse through the hidden layers. This basic characteristic is missing in traditional ML models which rely on human-fed features. Still, at least one study by [42] found that deep neural network models are only redundant to shallow ML models by a BERT-based bot detection approach along with exploratory data analysis of tweets. These findings should be investigated more with advanced or stacked DL models and by cross-validation through different datasets or platforms. In addition, future research should invest on the explainability of DL model outputs in general, to make it as advantageous and interpretable as ML models. In addition, the current review particularly reported the benefits and pitfalls of deep learning approaches and presented a comparison with ML algorithms, if reported on the DL articles. Hence, future reviews may want to inform the researchers of the overall scenario of the artificial intelligence research including all the classical machine learning and deep learning studies available in social bot detection, identify their respective potential and gain deeper insights.

Generating large dataset from a minimal amount of labelled data by leveraging state-of-the-art oversampling techniques can potentially flourish this research field, by reducing overfitting and increasing bot detection accuracy. Transfer learning from another domain also may assist to overcome the limitation of labelled data.

Finally, one of the major challenges in the social media bot detection research is that the attributes and behaviour pattern of the social media bots are continuously evolving and are growing more complex, sophisticated and optimized for particular activities. Thus, new and adaptable approaches to the evolution of social media bots are highly required. Real-time processing by leveraging GANs or other anomaly detection approaches, unsupervised techniques or novel learning modules, can be keys to combat the evolution of bots. Considerable efforts are required to develop generalized models among social media platforms such as Twitter Instagram, Facebook and electronic mail services such as Gmail or so. Application of bot detection techniques to other areas such as fake news detection or enriching the quality of news events is a potential area to explore. Future research should be directed towards bot-type classification in addition to bot classification, which is expected to increase the overall efficacy.

From the methodical perspective, for a systematic review, we could possibly start with a literature review in this field; however, there is risk of not being thorough or fair. Therefore, we decided to do a systematic literature review in the first place that followed a pre-defined and fair search strategy. Also, other than the topic level inclusion and exclusion criteria, we did not introduce a study quality assessment criterion in this review, which could be a limitation. However, due to limited number of studies in this area so far and for this might be more appropriate in clinical research reviews, we did not put an extra ‘quality’ filter.

We could do a quantitative study, where the data from the reviewed studies using meta-analytic techniques are combined, and more real and unknown effects are possibly detected that individual smaller studies are unable to detect. Hence, the heterogeneity of the input data, features and the modelling approaches made it difficult to present a useful and concrete statistical analysis outcome on the results. Our study presents a baseline quantitative synthesis on the results though, which should enable the future researchers to present the next level of analysis with more statistical power, with the inevitable additional publications in this field. There is a concept of tertiary review which is a systematic review of systematic reviews, in order to answer wider research questions. As there are only six review articles we found in this field, we did not go forward in that direction either. However, any future study directed to tertiary review might figure out some interesting aspects of the social bot detection evolution path. and how the technical approaches are changing with the evolution.

6 Conclusions and future direction

In this study, we followed a systematic approach to review the DL applications, as a state-of-the-art and highly advanced technology, in the social media bot detection research to assess the current status and critical challenges. Our review shows that DL-based techniques can be much effective and may potentially outperform traditional ML approaches, with few exceptions that definitely represent a great room for further research. In general, DL methods can make better use of textual features than other ML methods and in many cases fusion of different DL models produced a better performance such as hybrids of CNN and LSTMs exhibited a consistently reliable performance across studies. Sophisticated NLP models such as Google Transformers and combination of multiple types of features in the intermediary pre-processing steps may further boost up the classification performance. However, research suggests that feature-based models may not be always the appropriate one, for example to detect co-ordinated attacks, and graph neural network models could be an effective and alternative solution in that regard.

Deep learning algorithms especially generative adversarial networks or semi-supervised techniques may play an important role to leverage anomaly detection approach to address the major challenge of continuous change in the overall social media environment and rapid evolution of social media bots. Retrainable models through real-time processing would be another solution to this issue. Finally, most of the models are confined on twitter now. Therefore, leveraging the DL solutions to overcome similar issues in other platforms may potentially increase the usability and impact of this research to a great extent.

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article and agree to the publishing of its content.

Data availability All data generated or analysed during this study are included in this published article (and its supplementary information files).

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Systematic Literature Review of Social Media Bots Detection Systems

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ABSTRACT

Online social networks (OSNs) are vital to people's daily lives. They offer free services that allow people to connect and interact with family and friends, post comments and images, express views on sports and politics, and influence other users on OSN. The significant risks to OSN security are malicious SMBs, and numerous studies have been done to identify them. This article, a Systematic Literature Review (SLR), aims to determine best practices in SMB recognition. This SLR covers research published between 2008 and 2022. As a result of this study, we classified OSN profiles into real, verified, and fake accounts. The malicious SMBs types are SMBs, spam bots, Sybil and cyborgs, stegobots, political bots, and game bots. We also proposed a classification of SMBs detection techniques, which are ML-based, DL-based, graph-based, anomaly-based, topic-modeling-based, DNA-inspired, genetic-based, and hybrid approaches. Additionally, our study revealed that most public datasets used for SMB detection only include some types of SMB, dependent on Twittersphere, include limited data, and needed to be more extensive, up-to-date and accurate. We studied challenges of SMB detection, like data labeling, features engineering, imbalanced data, etc. Finally, we proposed the opening axis for future works.

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1. Introduction

The emergence of OSNs, including Twitter, Facebook, and Instagram, has occurred in online communication. OSNs are now an essential component of communication for individuals, institutions, organizations, and businesses to stay in touch with friends and family and to maintain business relationships with clients and customers. According to "https://www.statista.com", Facebook continues to be the largest and most popular OSN in history, with over 2895 million (M) members, followed by YouTube (2514 M), WhatsApp (2000 M), Instagram (2000 M), TikTok (1051 M), Facebook Messenger (931 M), and Twitter (556 M). Parallel to that, these networks have evolved into performing additional tasks overseen by an autonomous agent known as a bot. A bot is a software that mimics human behavior and operates autonomously and automatically. His actions can be used for good or bad. An SMB is a social media bot that works. "Statista.com" estimates that 71% of Twitter accounts discussing U.S. stocks are social bots, whereas Twitter has only identified 37% of them. A recent study (Cresci et al., 2018) exposed the actions of spam bots in Twitter stock microblogs. The research revealed that 71% of Twitter accounts that retweeted stock posts were bots, and 37% of these accounts had their accounts banned. 2019 saw the identification of 11% of Facebook accounts as bots (Zago, et al., 2019), and a study found that social bots contributed to the "infodemics" by disseminating false information on the COVID-19 epidemic (Gallotti et al., 2020).

Bots can operate within networks, which are referred to as OSN botnets. It is a group of SMBs that collaborate to perform specific actions on OSN and are generated and remotely managed by a bot master via a command and control (C&C) channel. To avoid detection by bot detection systems, social botnets mimic the interactions of OSN users. (Ellaky et al., 2021). The SMB's creation is effortless today by utilizing automation and bot-building tools like "Seek Socially" and "UseViral". The most developed bots use cloud services like Heroku or Amazon Web Services (AWS). Additionally, cloud service companies offer Bot-as-a-Service (BaaS) services, which have intelligence features like producing sophisticated content or disseminating spam content. An SMB comprises two main components: the program that manages the account and the exposed account on OSN. An SMB usually allows two different types of generic operations. The first is social engagement, such as reading, writing, and responding to information uploaded by other users. The second is social networking capabilities like adding friends or following other users that are used to change the social network. Social media can be used for various purposes, including brand marketing, teamwork, networking, and communication. People and organizations use SMBs to increase their effect, distribute ideas and materials, and widen their reach. Social bots are capable of both malicious and good deeds (Orabi et al., 2020). Real accounts and verified profiles owned by companies, celebrities, and other entities manage legal operations. They were created to supervise business operations or to assist in the relationship between a client and a company, including exchanging goods and services, publishing earthquake alerts on Twitter, disseminating news, etc. On the other hand, malicious bots can also harass or burden users by engaging in unethical activities, such as stealing the identities of actual users, swaying the electorate to support specific candidates, and propagating hate speech and other polarizing content. The use of social bots in OSNs has been investigated in several fields, including politics (Ferrara, 2017); (Luceri et al., 2019), stock market manipulation (Ferrara, 2015), fake news (Grinberg et al., 2016), the propagation of extremist ideologies and hate speech (Shao et al., 2018); (Stella et al., 2018), health misinformation propaganda involving vaccines (Sutton, 2018); (Broniatowski, et al., 2018).

Comprehending, detecting, and categorizing social bots and botnets has taken much work. Therefore, the development of precise and scalable bot detection systems has been facilitated by breakthroughs in artificial intelligence, particularly those in machine learning (ML) and deep learning (DL) approaches. Even though there are more publications every year to stay up with social bot evolution, these latter continue to pose new risks and difficulties for detection systems.

This study aims to conduct an SLR of social bot detection technologies. To the best of our knowledge, only a few state-of-the-art articles (Orabi et al., 2020); (Alothali et al., 2018); (Ferrara et al., 2016); (Alothali et al., 2018) have addressed social bot identification. We acknowledge the value of the existing reviews. Still, we highlight that they were conducted a few years ago and may need to capture the most recent developments in social bot detection. It is crucial to conduct a review on social bot detection despite the existing reviews because the field of SMB detection is constantly evolving, and new methods and techniques are being developed. This SLR covers research on social bot detection techniques published between 2008 and 2022 and addresses various social bot kinds, including sybils, cyborgs, spam bots, stegobots, and political bots, on OSNs including Facebook, Twitter, Instagram, YouTube, Renren, and Sina Weibo. This SLR study offers a concise and clear summary of the methods currently used to categorize various types of SMBs, suggests doing a quantitative and analytical analysis, and gives future research directions. Furthermore, in our work, we studied all of a model's phases and life cycle, from data preparation to feature engineering to feature selection to dataset labeling, etc., which was not done in the previous state of the art.

The significant contributions of this research work are as follows:

1) Review of the most recent research works produced to detect SMBs.
2) Classification and categorization of detection techniques.
3) Identification of data sources, datasets, feature types, feature engineering techniques, detection approaches, algorithms, and performance evaluation.
4) Analysis of detection techniques for each type of SMB.

This article is presented as follows. Section II introduces the research methodology of our SLR. Next, Section III provides a literature review of works related to SMB detection. Section IV provides a complete overview of the research's findings. Section V gives a general discussion of the results. Finally, Section VI provides the conclusion and prospects.

2. Research methodology

A SLR is a type of literature review in which a collection of articles on a specific topic is gathered and critically analyzed in order to identify, select, synthesize, and assess relationships, limitations, and key findings in order to summarize both quantitative and qualitative studies. (Xiao and Watson, 2019). The SLR is a crucial step for researchers to investigate the current research topics regarding the detection technique of SMBs, highlight the weaknesses of the proposed approaches, and give new trends and orientations for researchers. We used the PRISMA approach, a widely recognized and rigorous method for conducting SLRs. It involves a structured approach to searching for and selecting relevant literature, as well as assessing the quality of the studies included in the review. The PRISMA approach consists of several key steps, including: Defining the research question and scope of the review; conducting a comprehensive search of relevant databases and other sources of literature; screening the search results based on predefined inclusion and exclusion criteria; extracting data from the selected studies and synthesizing the findings; and assessing the quality of the studies included in the review.

2.1. Search strategy

We used the most popular research databases because the search strategy directly impacts the relevance and completeness of the studies retrieved: ScienceDirect, the IEEE Xplore digital library, SpringerLink, ACM, ResearchGate, arXiv, and Google Scholar.

Table 1
Inclusion and exclusion criteria.

Inclusion Criteria	Exclusion Criteria
Paper-indexed Scopus or Web of Science.	Studies that are not related to computer science or information technology.
The paper is redacted in English.	Articles on social bots that do not propose detection systems.
The articles are from journals or conferences.	Absence of Digital Object Identifier (DOI).

2.2. Research questions

To understand the problem of social bot detection in OSNs and its challenges and to highlight the different aspects of this problem, we established the questions that this SLR must address as follows:

RQ1: What are the existing types of OSN profiles and SMBs? This question will help determine the different accounts and bots operating on OSNs.

RQ2: What are the approaches used to detect SMBs? This question aims to determine the methods used to detect and classify SMBs and to study their performance.

RQ3: Which features are used to identify SMBs effectively, and which are the most relevant? This question determines the account characteristics used in the detection process and shows the most decisive ones.

RQ4: Which feature engineering and preprocessing techniques are used for social bot detection? This question identifies the preprocessing techniques applied to data to improve their quality.

RQ5: What are the dataset's characteristics used for social bot detection? This question aims to identify the sources of data, the different problems related to datasets, and how data can be labeled.

2.3. Query terms

We used snowballing method, which involves starting with a small set of identified studies, often the most relevant or seminal papers in a field, and then expanding the search by identifying and examining the references cited in those papers. The process is repeated, with each new set of papers examined for additional references to include in the review. This iterative process continues until no further relevant studies are identified. The steps conducted for our snowballing research are described as follows:

- Employ a text mining process to find search keywords and terms pertinent to SMBs detection.
- Conduct a broad initial search using a search string constructed from what academics deem pertinent search terms.
- Indicate whether the resulting studies are pertinent (P) or not (NP) for the current review.
- Use text mining techniques on both P and NP texts to uncover and extract new key phrases and identify old essential terms that are no longer relevant.
- Update the search term and go back to step one.
- Then, using Boolean operators, a search process was carried out by ScienceDirect, the IEEE Xplore digital library, SpringerLink, ACM, ResearchGate, arXiv, and Google Scholar to find the pertinent articles that addressed the search questions based on predetermined keywords.

The aforementioned search methodology and screens have been merged with additional phrases based on the knowledge discovered during the searches to construct a suitable search string for the SLR now under development. So, the last search term we used for the search queries was:

Query 1: (‘‘Social bots" OR ‘‘Social media bot*" OR ‘‘OSN bots" OR ‘‘digital platforms bot*").

Query 2: (‘‘Sybil" OR ‘‘Spam bots" OR ‘‘Spammer* Social network" OR ‘‘Stegobot" OR ‘‘Cyborg" OR ‘‘Fake accounts" OR ‘‘Malicious social media bots").

Query 3: (‘‘Automated account*" OR ‘‘Trending Topics" OR ‘‘generating posts" OR" like farms" OR ‘‘automated repost*").

Query 4: (‘‘Social bots' detection" OR ‘‘Technique* to detect social bots" OR ‘‘Detection of social media bot*").

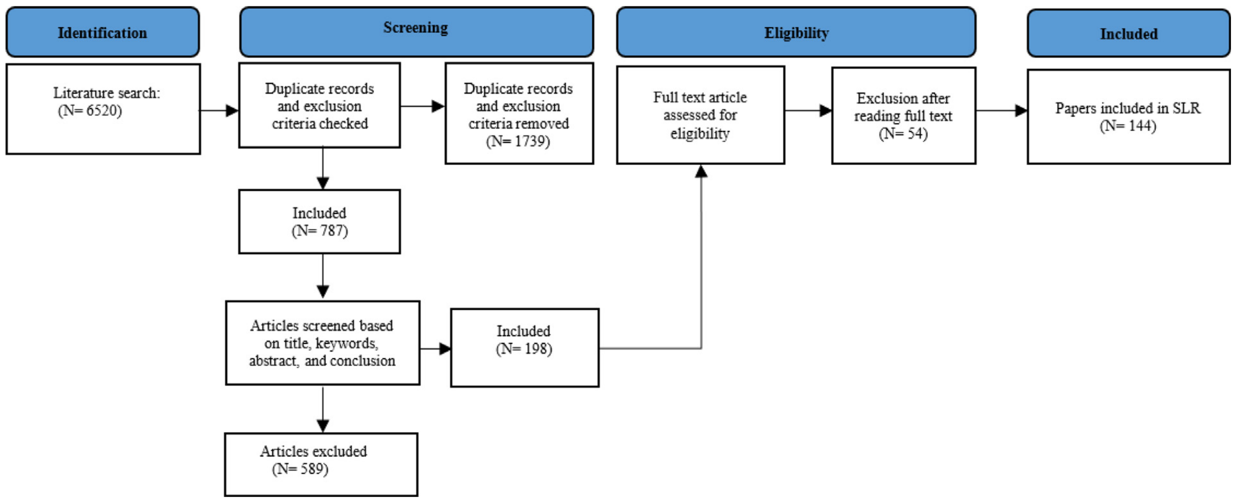

Fig. 1. Flowchart of the search process.

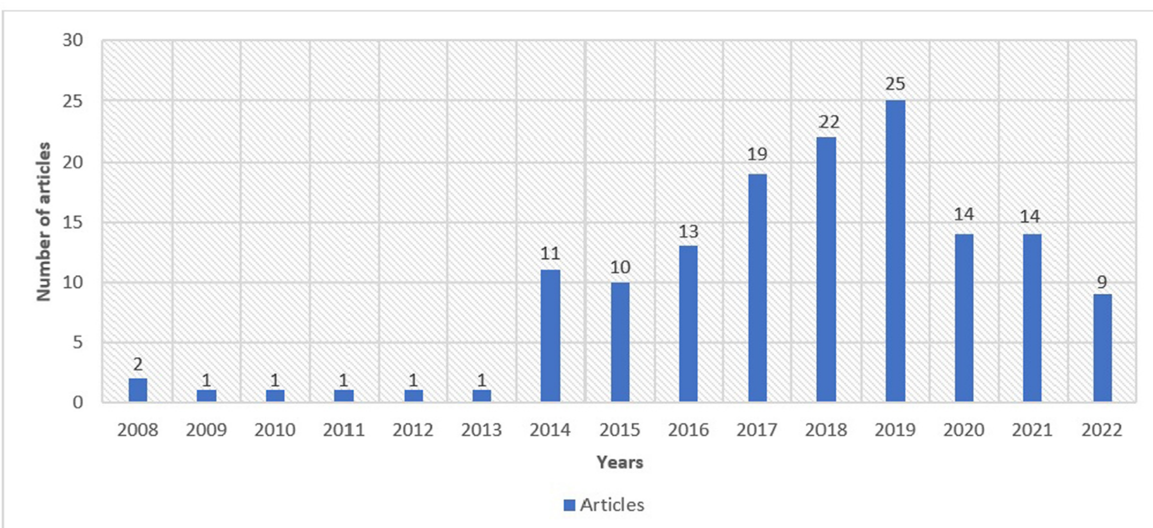

Fig. 2. Distribution of the state-of-the-art works over the years.

2.4. Study selection and quality assessment

The paper selection step was conducted based on a list of inclusion and exclusion criteria to ensure the pertinence and quality of papers. Table 1 presents the requirement of the article to be included in the SLR in column one and the exclusion criteria in the second column.

Specific quality assessment guidelines are used to determine the most pertinent works to evaluate the articles' quality. The first parts of the selected papers assessed were the title, abstract, conclusion, and keywords. Articles that do not correspond with our study were eliminated. The remaining articles were then subjected to the quality evaluation criteria for their dependability, integrity, and relevancy. For both systematic reviews and research syntheses, we developed a checklist of 12 guidelines with this objective in mind. If the work partially complies with the rule is merited in each situation:

- Satisfy the rules score = 1
- Satisfy some of the rules score = 0.5
- Does not satisfy the rule score = 0

Then The ultimate score of the article is determined by adding up the points awarded for each of the twelve quality evaluation criteria. As a result, only articles with a rating of 8 to 14 are included in the selected papers.

The rules are defined as follows:

1. Do the chosen articles refer to the research query?
2. Is the paper well-structured?
3. Do the research aims in the literature appear appropriately stated?
4. Does the literature provide sufficient baseline knowledge regarding SMBs?
5. Does the literature explain the terminology they used?

Table 2
Publication venues.

Document Type	Source Title	References
Conference Articles	International Conference on Digital Age and Technological Advances for Sustainable Development	(Ellaky et al., 2021)
	AAAI Conference on Artificial Intelligence	(Yang et al., 2020)
	ACM Asia Conference on Computer and Communications Security	(Lingam et al., 2020)
	ACM Conference on Computer and Communications Security	(Song et al., 2015)
	ACM Conference on Web Science	(Cresci et al., juin 2019) (Mazza et al., 2019)
	ACM International Conference on Information & Knowledge Management	(Feng et al., 2021) (Feng et al., 2021) (Sayyadiharikandeh et al., 2020) (Cai et al., 2017)
	Multidisciplinary International Social Networks Conference	(El-Mawass et al., 2018)
	ACM Transactions on Privacy and Security	(Ikram, et al., 2017)
	Artificial Intelligence Methods in Intelligent Algorithms	(Daouadi et al., 2019)
	Australasian Conference on Information Systems	(Stieglitz et al., 2017)
	Applications and Techniques in Information Security	(Ma et al., 2014)
	Bio-Inspired Computing - Theories and Applications	(Yang, et al., 2014)
	Companion of the World Wide Web Conference	(Davis et al., 2016) (Zannettou et al., 2019) (Ali Alhosseini et al., 2019)
	Communications of the ACM	(Ferrara et al., 2016)
	European Network Intelligence Conference	(Teljstedt et al., 2015)
	IEEE Conference on Communications and Network Security	(Feng et al., juin 2019)
	IEEE International Conference on Big Data	(Khaled et al., 2018)
	IEEE International Conference on Data Mining	(Chavoshi et al., 2016)
	IEEE International Conference on Intelligence and Security Informatics	(Fernquist et al., 2018)
	IEEE International Conference on Communications	(Chen et al., 2015)
	IEEE Symposium on Security and Privacy	(Yu et al., 2008)
	IEEE Transactions on Information Forensics and Security	(Dutta et al., 2020)
	IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining	(Dickerson et al., 2014) (Morstatter et al., 2016) (Minnich et al., 2017) (Feng et al., 2021) (Beskow and Carley, 2018) (Liu et al., 2014)
	IFIP Advances in Information and Communication Technology	(Mulamba et al., 2016)
	Information Retrieval Technology	(Almerekhi and Elsayed, 2015)
	Proceedings of the Fourth International Workshop on Social Sensing	(Hurtado et al., 2019)
	International AAAI Conference on Web and Social Media	(Cresci et al., 2018)
	International Bhurban Conference on Applied Sciences and Technology	(Mateen et al., 2017)
	International Conference for Convergence in Technology	(Kiran et al., 2019)
	International Conference on Social Informatics	(Chen et al., 2017)
	International Conference on Communication Technology Proceedings	(H. Ping et al., 2018)
	International Conference on Computer Science and Engineering	(M. Kantepe et al., 2017)
	International Conference on Computer, Communication, and Signal Processing	(Kumar et al., mai 2021)
	International Conference on Data Intelligence and Security	(Shaabani et al., 2018)
	International Conference on Data Science and Advanced Analytics	(Pizarro, 2020)
	International Conference on Dependable, Autonomic, and Secure Computing	(Dorri et al., 2018)
	International Conference on Pervasive Intelligence and Computing	
	International Conference on Big Data Intelligence and Computing	
	IEEE Cyber Science and Technology Congress	
	International Conference on Industrial and Information Systems	(Lingam et al., 2018)
	International Conference on Intelligent Systems Design and Applications	(De Meo et al., 2011)
	IEEE Conference on Computer Communications	(Wang et al., 2017)
	International Conference on Innovations in Information Technology	(Alothali et al., 2018)
	International Conference on Social Networks Analysis, Management, and Security	(Bello et al., 2018)
	International Conference on Intelligent Computing in Data Sciences	(Ellaky et al., 2021)
	International Conference on Web Intelligence, Mining, and Semantics	(Beğenilmiş and Uskudarli, 2018)
	International Conference on Machine Learning and Applications	(Alsaleh et al., 2014)
	International Conference on Services Systems and Services Management	(Jin Dan and Teng Jieqi, 2017)
	International Conference on ICT Systems Security and Privacy Protection	(Mulamba et al., 2016)
	International Conference on Trust, Privacy, and Security in Intelligent Systems and Applications	(Wei and Nguyen, 2019)
	International Conference on Web and Social Media	(Varol et al., 2017)
	International Workshop on Information Forensics and Security	(Andriotis and Takasu, 2018)
	International Conference on World Wide Web Companion	(Cresci et al., 2017)
	International Conference on Social Computing, Behavioral-Cultural Modeling and Prediction and Behavior Representation in Modeling and Simulation	(Chew et al., 2018)
	International Conference on Human-Computer Interaction	(Castillo, et al., 2017)
	Procedia Computer Science	(Fernandes et al., 2015) (Pratama and Rakhmawati, 2019)
	Proceedings of the World Wide Web Conference	(Abu-El-Rub and Mueen, 2019)
	Proceedings of the Annual Computer Security Applications Conference	(Chu et al., 2010)
	Proceedings of the 9th USENIX conference on Networked Systems Design and Implementation	(Cao et al., 2012)
	Proceedings of the Network and Distributed System Security Symposium	(Danezis and Mittal, 2009)
	World Conference on Smart Trends in Systems, Security, and Sustainability	(Mohammad et al., 2019)
	World Symposium on Web Applications and Networking	(AlRubaian et al., 2015)

(*continued on next page*)

Table 2 (*continued*)

Document Type	Source Title	References
Journal Articles	American Journal of Public Health	(Sutton, 2018) (Broniatowski, et al., 2018)
	ACM Transactions on Knowledge Discovery from Data	(Costa et al., nov. 2017) (Yang et al., févr. 2014)
	ACM Transactions on Multimedia Computing, Communications, and Applications	(Jr et al., avr. 2018)
	ACM Transactions on the Web	(Gilani et al., févr. 2019)
	ACM SIGWEB Newsletter	(Ferrara, 2015)
	Applied Network Science	(Luceri et al., 2019)
	Applied Sciences (Switzerland)	(Zhao et al., 2020)
	Automatic Control and Computer Sciences	(Zegzhda et al., 2017)
	Computational and Mathematical Organization Theory	(Beskow and Carley, mars 2019)
	Computer Communications	(Lee and Kim, 2014)
	Computer Journal	(Natarajan et al., 2015)
	Cluster Computing	(Murugan and Devi, 2019)
	Computers and Security	(Ji et al., 2016) (García et al., sept. 2014) (Rodríguez-Ruiz et al., 2020) (Boshmaf, août 2016) (Liu et al., août 2017) (Akiyama et al., août 2017)
	Decision Support Systems	(Cresci et al., déc. 2015)
	EPJ Data Science	(Bruno et al., 2022)
	First Monday	(Ferrara, 2017)
	Future Generation Computer Systems	(Al-Qurishi et al., 2018)
	Human Behavior and Emerging Technologies	(Yang et al., 2019)
	IEEE Access	(Martin-Gutierrez et al., 2021) (Wu et al., 2020) (Loyola-Gonzalez et al., 2019) (M. S. Al-Rakhami et A. M. Al-Amri, 2020)
	IEEE/ACM Transactions on Networking	(Yu et al., 2008)
	IEEE Communications Magazine	(Zago, et al., 2019)
	IEEE Intelligent Systems	(Cresci et al., 2016)
	IEEE Systems Journal	(Lingam et al., 2021)
	IEEE Transactions on Computational Social Systems	(Rout et al., 2020)
	IEEE Transactions on Control of Network Systems	(Wang and Paschalidis, 2017)
	IEEE Transactions on Industrial Informatics	(Al-Qurishi et al., 2018)
	IEEE Transactions on Dependable and Secure Computing	(Yang et al., 2016) (Cresci et al., 2017) (Chu et al., 2012)
	Information Processing and Management	(Orabi et al., 2020)
	Information Sciences	(Kudugunta and Ferrara, 2018) (Igawa, et al., 2016) (Miller et al., 2014) (Jeong et al., 2016)
	Information Systems	(Pham et al., 2022)
	Inteligencia Artificial	(Safaa et al., 2022)
	International Journal of Electrical and Computer Engineering	(Benabbou et al., 2022)
	Journal of Big Data	(Koggalahewa et al., 2022)
	Journal of Planning education and Research	(Xiao and Watson, 2019)
	Journal of Cyber Security and Mobility	(Shetty et al., 2021)
	Journal of Information Security and Applications	(Wanda and Jie, 2020)
	Journal of Computational Social Science	(Yang et al., 2022)
	Knowledge-Based Systems	(Wu et al., 2021)
	Multimedia Tools and Applications	(Venkatachalam and Anitha, 2017)
	Nature Communications	(Shao et al., 2018)
	Nature Human Behaviour	(Gallotti et al., 2020)
	Neurocomputing	(Inuwa-Dutse et al., 2018) (Wu et al., 2016)
	Online Social Networks and Media	(Cresci et al., 2019)
	PLoS ONE	(Rauchfleisch and Kaiser, 2020)
	Proceedings of the National Academy of Sciences of the United States of America	(Stella et al., 2018)
	Science	(Grinberg et al., 2016)
	Social Network Analysis and Mining	(Wanda and Jie, 2021) (Alothali et al., 2021)
	Social and Information Networks	(Gao et al., 2018) (Kharaji and Rizi, 2014) (Balestrucci et al., 2021)
	Social Science Computer Review	(Hagen et al., 2022)
	SpringerPlus	(Kang et al., 2016)

6. Are the detected SMBs types precisely defined?
7. Does the literature mention the source of the data used?
8. Does the literature mention the size and characteristics of the used datasets?
9. Is the detection experiment that was done appropriate and acceptable?
10. Is the research that was done on the ground legitimate?
11. Can you readily identify the dataset used?
12. Are the detection experiment results identified and published in detail?
13. Are the techniques utilized to evaluate the outcomes of the detection experiments suitable?
14. Is the method suggested by the literature considered valuable?

2.5. Study selection and flowchart of search process

Our study selection and research process followed four stages. In the first stage, 6420 papers were selected and got through the SLR process. At the screening stage, we excluded duplicate papers, applied exclusion criteria, and kept the papers on the research topic based on title, abstract, keywords, and findings. The unrelated articles were eliminated from the review. The quality assessment criteria were then used to evaluate the rest of the articles for trustworthiness, integrity, and relevance. Only 144 papers were identified as relevant. Fig. 1 represents the flowchart conducted for our search process.

2.6. Data analysis

We identified 144 studies published in the field of social bot detection during the period between 2008 and 2022. Fig. 2 displays the distribution of the state-of-the-art articles by year.

So, we can see that most of the investigations on social bots were conducted between 2017 and 2019. Research studies and publications related to this concept also increased significantly in 2014. The number of research work about social media bots has risen in recent years due to the proliferation of social media platforms and their impact on politics. Bots are often used to spread misinformation or propaganda, and their ability to amplify messages quickly and widely has raised concerns about their influence on public opinion. Researchers are interested in understanding how bots influence conversations on social media platforms and developing methods for detecting and preventing their use. Overall, the increasing use of social media and the impact of bots on public opinion and politics have made them an important area of research in recent years. Additionally, advancements in technology, like ML, DL, and NLP techniques and data availability, have made it easier and more feasible to conduct research on social media detection in 2019. Table 2 presents the publication venues of the retrieved articles, and the result shows that 50% are conference articles and 50% are journal articles.

2.7. Data extraction strategy

Once the cutting-edge papers have passed through the preceding stages, additional information is retrieved from the article based on the research objectives and study questions to thoroughly understand research issues and each work's contribution to social bot detection. The most important extracted information is the reference, article title, authors' names, publication year, type of detection method (classification or clustering or hybrid, etc.), type of SMB, OSN platform, dataset, labeling techniques, feature categories, feature extraction techniques, feature selection techniques, pre-processing techniques, models and algorithms, optimization techniques, performance metrics, and the state-of-the-art techniques compared to. The information gathered serves as the foundation for this study. The following sections will present and discuss it using graphs, figures, and comparative tables to conduct a methodical investigation to help readers understand SMBs and their detection challenges.

3. Related work

The scope of this section is to give the readers a summary of different proposals for SMBs detection techniques. In several research, the goal was to detect whether the OSN user is a human or a bot (normal /social bots, malicious/legitime) without specifying the type of bot. In this case, the paper aims to detect a generic SMB. The other research papers specify the type of bot targeted, such as Sybil and cyborgs, Stegobot, Game bots, political bots, and spam bots. As a result, our literature review organizes detection techniques by the type of SMB studied: SMB in general, Sybil and Cyborg, Stegobot, Political bot, and Game bot.

3.1. Generic SMBs detection techniques

The authors of this paper (Gilani et al., 2019) scraped a massive dataset from Twitter that included user information and content features. Then proceeded, an analysis of behavioral features such as the number of likes per tweet, the number of retweets per tweet, the number of replies, the mentions, etc. The findings of this study showed that Twitter bots tweet and re-tweet more than humans, post five times more URLs in their tweets than human posts, and receive more likes to their content.

- **ML-based detection systems**

The authors in (Dickerson et al., 2014) proposed a social bot detection system based on the AdaBoost model. They used network features, including a subgraph of the Twitter follower network of 40 M edges, content, and sentiment features. Several ML algorithms have been applied to detect SMBs, and the AdaBoost model gave the best AUC score of 73%. In addition, the authors showed that sentiment features are essential to identify bots. In (Lee and Kim, 2014), a new detection scheme was proposed to identify groups of malicious users once created. This method is employed to discrepancy between automatically produced usernames and ones created by effective human users. A clustering strategy was applied to group the accounts with similar names, and Support Vector Machine (SVM) algorithm categorized the groups as malicious or not. The study analyzed approximately 4.7 million Twitter accounts collected using an API. In (Yang, et al., 2014), the authors presented a new bot detection technique using an interaction graph based on user interaction features and a Back Propagation Neural Network algorithm (BPNN) model. BPNN algorithm with the interaction features performed an accuracy of 98.6% and outperformed the existing techniques. BotOrNot (Davis et al., 2016) is a system that evaluates the level of the menace of a Twitter user. Therefore, the data were extracted from the user profiles and classified into the following feature categories: network, user, friends, temporal, content, and sentiment features. BotOrNot is based on Random Forest (RF), the machine learning algorithm which gave the best AUC rate of 95%. A host-side system (Ji et al., 2016) can identify social botnets. Based on the analysis of evasion strategies employed by social botnets, the authors proposed new features classified as life cycle features, temporal, and spatial correlations. The RF model gave good results compared to the other ML algorithms, with a False Positive (FP) of 0.3%, a False Negative (FN) of 4.7%, and an F-measure of 96.3%. This work (Ikram, et al., 2017) provided a honeypot-based detection system using SVM and the co-clustering method. For this aim, the authors extracted data from Facebook users' timelines and classified features into lexical and non-lexical classes using the SVM algorithm. The non-lexical features included comments, likes, and shared content words per post, and the lexical features included grammatical and semantic correctness of the posted content. As a result, the model achieved a precision of 99% and a recall of 93%. In this study (Kantepe and Ganiz, 2017), the authors proposed SMB system detection based on behavioral features and the Gradient boosting Trees algorithm (GBT), which achieved an accuracy of 86%. Therefore, they applied Information Gain (IG), Mutual Information (MI), and Chi-Square as feature selection techniques. In this paper (Jin Dan and Teng Jieqi, 2017), behavioral analysis was conducted based on RF, which reached a precision of 94.4% using data from Sina-Weibo. The analysis showed that user information, content, and behavioral features are the most relevant for bots' identification. The experiments confirmed that the suggested attributes positively impacted identifying the bots. SocialBotHunter (Dorri et al., 2018) is a semi-supervised collective classification based on the One-Class SVM algorithm. This approach integrated network features, user information, and content features to identify Twitter botnets. The results showed that SocialBotHunter successfully detects spam bots with a precision of 99.95 % and an acceptable recognition time. To detect SMBs (Jr et al., 2018), the authors implemented a wavelet-based technique that classified social accounts as humans, legitimate or malicious bots based on content features. For feature extraction, the authors created a vector by applying discrete wavelet transformation and a weighting scheme called lexicon-based coefficient attenuation. The RF classifier was trained on datasets scraped using the Twitter API, and an accuracy score of 94.47% was achieved. To detect SMBs using their social names, the authors (Beskow and Carley, 2019) applied an SVM-based model and text-based content, user metadata, temporal, and network features. To extract textual features, they used the TF/IDF of a bigram, the number of lower-case words, the number of upper-case words, and the string entropy based on Shannon entropy. As a result, SVM performed an accuracy of 99.6% and a recall of 99.8 %. This paper (Fernandes et al., 2015) proposed a bot detection system based on an SVM classifier and a DBSCAN clustering model trained using content, user information, and network features. DBSCAN achieved an F1-score of 91% but struggled to nuance human and bot behavior. After that, SVM is applied to classify bot users into three classes: brands, celebrities, and promoters, with an F1-score of 74%. In this article (Ellaky et al., 2021), the authors studied the impact of feature presentation on SMB detection using Bert and Doc2Vec Word Embedding. They applied RF and DT algorithms as classifiers, and the experiment results showed a precision of 99.96% for RF and DT with Bert and outperformed a precision of 100% with Doc2Vec. This article (Alothali et al., 2021) presented a hybrid feature selection technique for SMB detection based on RF algorithms and user information features. Four feature selection techniques were experimented with: wrappers, filters, embedded, and cross-validation attribute evaluation (CVAE) techniques. As a result, RF achieved an accuracy of 89% using CVAE. They also found that the best-performing features are favorites_count, verified, statuses_count, and average_tweets_per_day. The authors of this article (Almerekhi and Elsayed, 2015) proposed a J48-based model for detecting Arabic social bots on Twitter using content and temporal features. The system was able to distinguish human accounts from bots with an accuracy of 92% using a dataset manually labeled with the help of the CrowdFlower platform. In this paper (Igawa, et al., 2016), the authors proposed a system for SMB classification using content features based on Lexicon Based Coefficient Attenuation (LBCA) and Wavelets. The RF model was trained on a dataset crawled from the 2014 FIFA World Cup, which performed an accuracy rate between 94% and 100%. This work (Rodríguez-Ruiz et al., 2020) presented an anomaly detection system implemented using RF as one-class ML classifiers and behavior features extracted from the Cresci-17 dataset. As a result, the proposed model achieved an AUC that exceeded 89% and allowed the detection of several types of bots. The authors of (Rout et al., 2020) proposed a social bot detection method based on content features. They added new features such as URL redirection, the relative position of the URL, frequency of shared URLs, spam content in the URL, anomalous URLs, etc. In addition, they evaluated the trustworthiness of tweets posted by each user using Bayesian learning and Dempster-Shafer theory (DST). Then they designed a learning automata-based malicious social bot detection LA-MSBD algorithm by incorporating a trust model with a set of URL-based features. As a result, the proposed system achieved a precision of 95.37% in detecting social bots. The authors in (Chen et al., 2017) proposed a system to detect active malicious botnets on Twitter by clustering tweets that contain embedded and short URLs. To train the model, they used real data extracted using the domain name of a URL shortening platform. Then picked a set of tweets that contained more than 20 duplicate tweets. The approach considered a cluster as a group of accounts sharing 200 tweets. In this study (Shaabani et al., 2018), the authors proposed an approach to detect social bots without using network features, user information, or content features. The suggested approach works based on collective causality metrics that provide a way to identify significant causal relationships. The results showed that the trained model, utilizing a causality-based clustering, could recognize groups of SMBs with an accuracy of 75%. TwiBot-20 (Feng et al., 2021) is a massive benchmark for Twitter bots. It includes 229,573 accounts, 33.48 M tweets, 8.72 M user ownership items, and 455,958 following connections and includes all the features that could be scraped using the Twitter API. Authors compared different studies on this dataset and others as Cresci-17 and PAN-19. As a result, all the state-of-the-art detection systems fail to perform well on the TwiBot-20 dataset.

- Topic **modeling based detection systems**

The authors of (Morstatter et al., 2016) proposed a bot recognition model with an 82.48% recall using the AdaBoost algorithm. The study is based on content features from two real-world Twitter datasets: the Libya dataset, which included the Arab spring tweets in Libya, and the Arabic Honeypot dataset. The authors suggested new features, which are a fraction of the time of retweets, the mean time between tweets, the average length of tweets, number of URLs, and used the Latent Dirichlet allocation (LDA) for topic modeling. DeBot (Chavoshi et al., 2016) was developed to aggregate user accounts into a cluster in real time using only content features (comments) without needing a labeled dataset. The proposed clustering model used the warped correlation to detect thousands of SMBs per day with an accuracy of 94% and reported daily activities of the detected accounts. Based on a comparative study conducted by the authors of this work, they observed that certain bots escaped Twitter's suspension process and remained active for months.

- **Anomaly-**based **systems**

BotWalk (Minnich et al., 2017) is an anomaly-based system that identifies SMBs by detecting new behavior patterns using user information, content, temporal, and network features. BotWalk can detect new bots by performing anomaly detection using a clustering algorithm based on behavioral characteristics. Twitter accounts previously labeled as bots are randomly selected, and the follower's attributes of each account are extracted in real-time using the Twitter API and Tweepy. When new bots are detected, they are added to the seed bank of collected bots. Therefore, this system recognized 6000 candidate bots per day with a precision of 90%. In this paper (Costa et al., 2017), an activity model (ActM) was developed to accurately fit the distribution of OSN of inter-arrival times of a user's posts (IATs). The suggested model is an anomaly-based model built using an ActM. This mathematic-based algorithm uses a stochastic process to detect SMBs based on user posts in the timeline with an accuracy of 93%.

- **DL-based** detection **systems**

In this paper (Cai et al., 2017), the authors proposed an SMBs detection approach based on Convolutional Neural Network (CNN) model. The model learned the representation of user behavior by capturing the external and internal factors influencing the OSN user's actions. Then a Long Short-Term Memory (LSTM) algorithm was used to build a content-based representation as temporal data rather than just text. As a result, the model achieved an F-measure of 88.3%. In the study (Kudugunta and Ferrara, 2018), the authors proposed a system based on the LSTM algorithm and used content features and user information to recognize SMBs at the posting scale. Because of the need for a large dataset to train the DL model, the authors used oversampling to balance a labeled dataset. The experiments showed that the model could detect social bots from a single tweet with more than 96% accuracy. DeBD (Beğenilmiş and Uskudarli, 2018) is a bot detection system based on the LSTM algorithm composed of three layers. The first one extracts the content features and the links between them. The second layer extracts the time-based features from the metadata of the tweets using the LSTM algorithm. The third layer mixed the extracted content features with the temporal features to identify the social bots with an accuracy of 100%. In this study (Mohammad et al., 2019), the authors implemented CNN and Artificial Neural Network (ANN) models to detect SMBs based on a single post. For the experiment, they used content features and user features. The trained model achieved an accuracy score that exceeded 90%. BotFlowMon (Feng et al., 2019) is a bot detection system based on Multilayer Perceptron (MLP) and CNN algorithms that classify SMB traffic using NetFlow and transaction data. The method classified social bots, including chatbot, post bot, crawler bot, and hybrid bot, with a precision of 98.87%. This research (Kiran et al., 2019) presented a methodology to identify Twitter SMBs based on behavioral features and an ANN model that achieved a best accuracy of 97.79%. The authors of this article (Wu et al., 2021) developed a system using Deep Neural Networks (DNN) and Active learning to identify SMBs in Sina-Weibo. The authors used user information, content, temporal, and interaction features for this aim. Therefore, to build a large-scale dataset, they used active learning to augment the size of the labeled dataset, which achieved a balanced dataset of up to 300,000 samples. Then they designed a model called RGA (ResNet & bidirectional Gated recurrent unit, & Attention), which included ResNet, Bidirectional Gated Recurrent Unit (BiGRU), and an attention layer. The proposed model reached an accuracy score of 98.87%. SATAR (Feng et al., 2021) is a self-supervised technique used for the representation learning of user information, network features, and the semantic information of tweets. Therefore, at the tweet level Word2Vec embedding was used to make the vector representation of each tweet, and the LSTM model was used as a classifier. As a result, SATAR outperformed other cutting-edge systems in detecting social bots, achieving an accuracy of 98.6% and the ability to generalize this approach to all Twitter bots. Bot-DenseNet (Martin-Gutierrez et al., 2021) is a bots detection approach based on RoBERTa embedding layer. The proposed system received an F1-score of 77% based on profile metadata and content features extracted from a Kaggle dataset. In this work (Lingam et al., 2020), the authors studied Twitter botnets and proposed Social Botnet Community Detection (SBCD) system based on graph theory and Deep Autoencoder. For this aim, they compared various similarity-based graphs of users' behaviors, including content, URL, interest, and social interaction similarity. As a result, SBCD achieved a precision of 90%. In (Fazil et al., 2021); a combination of Bi-LSTM and CNN models with an attention layer was proposed for social bot detection. The models were trained with five datasets, and the results showed that the approach outperformed the existing methods with an accuracy of 99.83%. In this work (Beskow and Carley, 2018), the authors studied the automated communication in OSN and proposed a detection system using snowball sampling and RF algorithms. The network features provided relevant information that influenced the model's precision, which reached 92%. The authors of the study (Kumar et al., 2021) used CNN, LSTM algorithms, and BERT for feature extraction to implement a DL-based model to detect SMBs. Based on users' posted content on Twitter, the suggested system achieved an accuracy of 97%. To address the problem of imbalance and the lack of large datasets needed to train the DL algorithm, the authors of (Wu et al., 2020) combined the enhanced Conditional Generative Adversarial Networks (CGANs) with the Gaussian Kernel Density Peak Clustering algorithm (GKDPCA) to prevent the production of noise during data augmentation. They used Wasserstein distance with a gradient penalty to overcome the model collapse and gradient vanishing of the CGAN model. As a result, CGANs achieved good precision with a score of 97.29% compared to other oversampling techniques such as ADASYN, SMOTE, and Random. Using one-class classifiers does not require examples of outlier behavior to distinguish between bots and human accounts.

- **Hybrid detection systems**

Toward understanding the behavior of malicious SMBs (Bello et al., 2018), the authors suggested a sentiment analysis based on network, content, and sentiment features. Then they analyzed services that create social bots like labnol, botlibre, cheapbotsdonequick, jetbots, and botize. Thus, for the experiments, they mined the data using the Twitter API to obtain a maximum of 3200 content posts from every account. Then, they extracted the content features set using bi-gram and Latent Semantic Analysis (LSA). To reduce the number of features to 11 features, linear SVM was used as a feature selection method. As a result, the DT achieved a precision of 98%. In this research, the authors presented an SMBs detection system based on Deep Forest (DF). They used user information, content, and statistical features. The trained model reached an accuracy score of 97.55%. Bot2Vec is a bot detection approach based on network representation learning (Pham et al., 2022). The authors used the skip-gram method to preserve the user's local neighborhood relationships and the intra-node structure of the user node while learning the mapping of OSNs. This technique used only the user's profile without additional features, and as a result, Bot2vec reached a ROC score of 97.68%. Deep-Friend (Wanda and Jie, 2021) is an approach that classifies the malevolent nodes in OSN using Dynamic Deep Learning (DDL) and link features. The system detected malicious nodes with an AUC of 99.70% using the Flickr dataset and 99.56% using a Facebook-based dataset. In this study (Mazza et al., 2019), the authors suggested a system to detect a botnet called Retweet Buster (RTbust) based on temporal patterns extracted from content features. They did this by extracting the LSTM algorithm from the retweets time series and then clustering the group of accounts using the hierarchical DBSCAN clustering algorithm. Therefore, the RTbust approach achieved an F1-score of 87%.

- **Graph-based detection systems**

BotRGCN (Feng et al., 2021) is a detection system based on the Relational Graph Convolutional Network (GRCN) model. The approach used user information, numerical metadata (like the number of followers, the number of followings, the number of active days, etc.), and content features. The experiment was conducted using the TwiBot-20 dataset (Feng et al., 2021), and BotRGCN achieved an accuracy of 84.62%. This work (Wang and Paschalidis, 2017) is a community-based approach that uses both a flow-based approximation that estimates the histogram of quantized flows and a graph-based method that approximates the degree distribution of interaction graphs between nodes, including Erdos-Renyi graphs and scale-free graphs. For the experiment, the authors used the CTU-13 botnet dataset (García et al., 2014), and the proposed system could detect social botnets with a precision of 86%.

3.2. Sybil and cyborg detection techniques

Malicious SMBs create profiles on OSNs under different identities to operate malicious activities. The number of accounts they create depends on the purpose of the attackers and the OSN community they target (AlRubaian et al., 2015). Fake or Sybil accounts are both fictive identities.

- **ML-based detection systems**

In this paper (Alsaleh et al., 2014), the authors compared the existing Sybil detection systems and explored the Sybil features for an accurate classifier. Data was collected from Twitter accounts using the Twitter REST API and Twitter4j, then manually labeled as human, bot, or hybrid accounts. The proposed MLP model with a gradient descent optimization algorithm achieved an F-measure of 95.1%. Íntegro (Boshmaf, août 2016) is an extension of Syrank that distinguishes between real and fake accounts by ranking the real ones higher than the fakes. Integro initially begins by spotting candidate victims through user activity via the RF algorithm, and then it assigns a low weight to edges linked to victims. They used real data with more than 15 million active users to validate the model. The proposed approach achieved higher performance than Syrank, with an AUC score of 76%. In this paper (Yang et al., 2014), the authors studied the characteristics of fake accounts to understand the behaviors of Sybils in Renren. Therefore, they built a dataset of 650,000 Renren Sybils and applied the SVM model to achieve an accuracy of 99%. The study's findings showed that Sybil accounts without explicit connections operate in combination to mount attack campaigns. In this study (Al-Qurishi et al., 2018); the authors proposed a Sybil identification technique based on Feed Forward Neural Network algorithm (FFNN) and using user information, content, and network features. In addition, they used a network-based mechanism that leveraged a connection matrix and a behavior-based mechanism that collected the user activities on the OSN. All these components operated systematically to investigate and evaluate the OSN accounts. The FFNN-based model reached an accuracy of 86%. This research (Al-Qurishi et al., 2018) presented a system based on the RF model to identify malicious behaviors of Sybil accounts based on content features, user information, temporal and network features, and user activities. The system provided three levels of analysis: network level, content level, and user level. Indeed, the authors collected data related to the US elections from YouTube and Twitter. Then, applied random sampling techniques to the training set. The RF-based model achieved an accuracy of 96%.

- **Graph-based detection systems**

SybilGuard (Yu et al., 2008) performs lengthy random walks O ($\sqrt{n}$ log n) that must intersect with the node to accept a suspect node as honest. But the Sybil admittance is nonetheless constrained by Sybillimit (Yu et al., 2008) by O(log n) per attack edge. SybilGuard and Sybillimit's limitations are: related to the number of attack edges, which had a high false negative rate; when the labeled nodes are severely imbalanced, the number of accepted sybils significantly rises. SybilInfer (Danezis and Mittal, 2009) is a threshold based on a probability system that provides strong assurances only on low attack edges and is independent of the topology of the adversary region. It also permits detecting a more significant number of attacker nodes than SybilLimit. In this research work (De Meo et al., 2011), the authors suggested community detection in large networks based on the Louvain method and computing the network's pairwise distance between nodes. Edges are weighted using a global feature that measures their ability to spread information throughout the network. As a result, the proposed system outperformed the state of art approaches. SybilRank (Cao et al., 2012) is an optimum Sybil detection method implemented using Barabasi's scale-free model. It exceeded the previous works by building a graph with fewer weights, requiring a longer mixing time for the Sybil region. And assuming that trustworthy users will not engage with unknown accounts. SybilRank detected Sybils with a TP over the score of 85%. SybilBelief (Gong et al., 2014) is a semi-supervised system implemented using Markov random fields (MRFs) and loopy belief propagation (LBP) to recognize Sybil nodes. A sample of benign and Sybil nodes was used as input, and it propagated the label from the input data to other nodes in the network. SybilBelief is resilient to noise and can identify Sybil nodes with an accuracy of 100%. SybilSCAR (Wang et al., 2017) is Local rule-based propagation used in OSN for Sybil detection. It combines loopy belief propagation with random walks. To train SybilSCAR, a sizable training dataset was necessarily required. As a result of the experimentation, the SybilSCAR overperformed the SybilRank with an AUC score of 100%. This article (Ma et al., 2014) provided an overview of the current online defense methods in OSN security. It proposed Sybil-Resist, a random-walk-based technique that can identify the Sybil nodes and the Sybil region. The achieved detection rate of 99.2%, which outperformed the state-of-the-art approaches in both detection accuracy and time of execution. SybilFrame (Gao et al., 2018) is a system that detects Sybil attacks by using Pairwise Markov Random Field. The authors validated the proposed approach using a synthetic and real network topology with 20 million nodes and 265 million edges. They showed that SybilFrame could detect Sybils better than structure-based approaches, with an accuracy rate of 95.4%. SybilRada (Mulamba et al., 2016) is a Sybil detection system based on the graph structure of OSN. To train the model, the authors utilized both the weights of edges and similarity values of the nodes on real and synthetic datasets. The finding is that SybilRadar detected Sybils with an AUC rate of 95% even when the attack edges between Sybils and benign are counted in tens of thousands (Cresci et al., 2015). This study (Zegzhda et al., 2017) suggested a system to identify active cyborgs in the OSN VKontakte. By implementing an ANN-based model trained using the user information. The experiment reached an AUC score of 73%. In this paper (Kharaji and Rizi, 2014), the authors proposed a Markov clustering technique to detect cloned profiles based on the network structure of OSN. Similar profiles are gathered into a mutual community, then the relationship strength of all real profiles is rated, and the profiles with fewer values are considered cloned profiles. The finding of this work showed that the proposed system could detect cloned profiles with a TP of 92%. Vote Trust is a graph structure-based technique for Sybils detection. Particular users in a group vote for the confidence of a new user that joined the group and then transmitted their voting to the newly joined user. Here, an interest group refers to a group of smartphones that have one or more common sensing assignments. Trust is then calculated based on the user's acceptance and rejection of their friend requests and their influence. Trust is also computed globally from the votes of the whole network's nodes. In this paper (Caruccio et al., 2018), the authors introduced a mechanism to discriminate real accounts from Sybil accounts on OSN by extracting Relaxed Functional Dependencies (RDF) and big data to characterize the typical patterns of fake accounts. For this aim, they used three datasets: fake accounts, verified accounts, and real accounts.

- **DL-based detection systems**

DeepProfile (Wanda and Jie, 2020) is a Dynamic CNN (DCNN) based system that identifies fake accounts using user metadata, content, and temporal features. In addition, they suggested a new pooling layer to enhance the effectiveness of the DCNN model in the training process. The proposed algorithm achieved a precision of 94%. This research work (Benabbou et al., 2022) proposed a system to recognize fake accounts using the BiGRU model and GloVe for word embedding presentation. They trained the model on the BiGRU algorithm based on content features. Therefore, the model outperformed the LSTM and the CNN algorithms and achieved an accuracy of 99.44%.

- **Hybrid detection systems**

The classification system presented in (Chu et al., 2010) consisted of the following four elements: (1) an entropy-based component, (2) an ML-based component, (3) a user's information component, and (4) a component to make decisions. Through experimentation, the proposed system could categorize tweets with a TP rate of 82.80%. In this paper (Chu et al., 2012), a probability-based method was developed to classify Twitter accounts as human, bot, or cyborg based on user information, behavior, and content features. For the experiment, they scraped 50,000 accounts from Twitter to implement the system, which was able to classify the Twitter accounts using the Bayes theorem and RF with a TP rate of 96%. In this work (Shetty et al., 2021), the authors proposed a hybrid system to detect fake accounts using SybilGuard, engagement rate, and profile statistics. This approach achieved an accuracy of 95.34%. This work (Khaled et al., 2018) aims to detect Twitter's Sybils based on the SVM-Neural Network (SVM-NN) algorithm and used user information. They applied feature selection techniques and dimension reduction techniques to select effective features. The SVM-NN detected fake accounts with an accuracy of 98.3%.

- **Topic modeling detection systems**

HawkesEye (Dutta et al., 2020) detects Sybils based on temporal features, the time series behavior of the retweeting, and the topic modeling of content features using LDA. However, the proposed model outperformed the state-of-the-art approaches with a precision of 96.4%.

3.3. Stegobot detection techniques

The stegobots are malicious SMBs that use steganographic images shared by OSN users to build social botnets (Venkatachalam and Anitha, 2017). This paper (Natarajan et al., 2015) proposed a methodology based on the Adaboost algorithm to recognize stegobot hosts in OSN using stegobot communication analysis. The study showed that stegobots have a unique design and implementation. The traffic patterns of stegobots and benign accounts can be detected using a multilevel analysis of the profile information and infected pictures posted on Flickr. The approach achieved an accuracy score of over 97% and an FP rate under 3%. The authors of this work (Venkatachalam and Anitha, 2017) introduced a new host-based solution to detect stegobot accounts and stegobot botnet activities based on the NB algorithm. They used user information and content features, including malicious and steganographic images extracted from Flickr, Google+, and social network features. Therefore, experimental results showed that combining content features with network features increased the results of the NB algorithm and achieved an accuracy of 96.24%.

3.4. Political bots detection techniques

A political bot is a software used and controlled by governments and political parties to manipulate public opinion and amplify polarization and sentiment by using fake likes, posting content, tweeting, and retweeting (Hagen et al., 2022); (Fernquist et al., 2018) . Their activity revolves around the most important political events and spreads misinformation, viral content, and fake news on OSN using memes and deep fake videos (Zannettou et al., 2019). In this research study (Bruno et al., 2022), the authors analyzed social bots' behavior during the United Kingdom (UK) elections of 2019 by analyzing online talks and tweets and then examining both the temporal pattern and statistics of the content posted about Brexit. The finding is that social bots participated in the social debate on Twitter, especially on the days near the UK elections, in which the discussion of bots increased; it jumped from [100000, 300000] in regular days to 900,000 during the election and then returned to the normal values observed before the elections. The authors of this paper (Hurtado et al., 2019) analyzed the political discussion on Reddit by building the network structures of this OSN. Unlike other OSNs, such as Facebook and Twitter, Reddit users register for a topic and then participate in the discussion by posting comments. This analysis showed that a group of fully connected members whose names contain the word "bot" have bot-like behavior. In this work (Hagen et al., 2022), the authors examined the influence of SMBs on the political discussion in OSNs. They used content and sentiment features to analyze the network structure of the active accounts in the political debate. As a result of the study, SMB influenced political debate using fake likes and network sentiment amplification.

- **ML-based detection systems**

In this study (Castillo, et al., 2017), the authors analyzed the activity performed by electoral candidates in the Chilean presidential campaign during the 2017 election for political bot detection. To do so, they extracted from Twitter user metadata, metadata of friends' accounts, network, temporal, tweet, and sentiment features to train the SVM model. SVM achieved over 80% for the training set and decreased to less than 60% for the test set. This work proposed (Beğenilmiş and Uskudarli, 2018) a system based on the RF model to detect the coordinated fake news activities promoted by human and social bots during the United States (US) presidential election of 2016. For the experiments, they collected data based on trending hashtags, and a set of data included user information and temporal features. RF achieved an average accuracy of 95%. To detect political bots, this work (Pratama and Rakhmawati, 2019) used an RF-based model trained using the comments posted by the supporters of Indonesian presidential candidates on Twitter. The data were extracted from Twitter using filters such as Joko Widodo (@jokowi) and Prabowo Subianto (@prabowo). The RF reached an accuracy score of 74%. BotCamp (Abu-El-Rub and Mueen, 2019) is a framework based on the Boosted DT algorithm for political bot detection based on temporal, content, and network features. They automatically collected data from the trending hashtags and applied a graph clustering of retweets, mentions, hashtags, and temporal characteristics. BotCamp detected political bots with an accuracy of 87.5% and a precision of 98%. This study revealed that the studied political SMBs were similar in temporal correlation, sentiment, and topic correspondence. This work (Sayyadiharikandeh et al., 2020) suggested ensembles of specialized classifiers (ESC) to identify political bots using user information, behavior, friends, networks, temporal, and content features. The proposed system is an update of the Botometer, which achieved an AUC rate of 99%. In this study (Teljstedt et al., 2015), the authors suggested a political bot detection approach that copes with the complexity of manual annotation of massive datasets, which requires additional effort and resources. To achieve this goal, the authors developed a semi-automatic annotation of datasets based on the RF model. For the experiment, the author used a Russia-Ukraine conflict dataset extracted from Twitter and proposed new features like "inter-tweet-content similarity" (ITCS), "Pearson's Chi-Square Statistics of Uniformity" (PCU), "Rao's Spacing Statistics of Uniformity" (RSU), "inter-tweet-content regularity" (ITC-R), "inter-tweet-delay shape" (ITD-S) and "inter-tweet-delay regularity" (ITD-R" The RF model achieved a precision of 97.4%.

3.5. Spam bots detection techniques

Spam bots are malicious SMBs that disseminate malicious Uniform Resource Locator (URL), spam messages, and comments, including misinformation and fake news (Ellaky et al., 2021). Spam bots routinely abuse OSN users and perform a variety of online crimes, including trolling, spreading rumors, stalking, and cyberbullying.

- **ML-based detection systems**

In this study (Miller et al., 2014), the authors dealt with detecting spam bots as an anomaly problem. They extracted 95 one-gram attributes of tweets and user information to do so. Therefore, to optimally handle the continuum of tweets, two kinds of stream clustering techniques were applied, StreamKM++ and DenStream, which were both modified to improve spam recognition. The two algorithms clustered benign users and considered the outliers as spam bots. Using these algorithms together gave a recall of 100% and a false positive of 2.2%. This research (Inuwa-Dutse et al., 2018) proposed a system based on Gradient Boosting (GB) model for spam bot detection that aimed to understand spam behavior. To collect data, they applied LSA on a honeypot dataset to extract concepts related to spam bots' posting. The Recursive Feature Elimination (RFE) method selected the most significative characteristics, excluding historical content. GB algorithm performed the best accuracy of 98.97%. In this research work (Andriotis and Takasu, 2018), the authors used AdaBoost to classify Twitter spam bots based on content features, user information, and sentiment features. They used four known datasets ICWSM17 (Varol et al., 2017), RAID11 (Liu et al., 2017), WWW17 (Yang et al., 2013), and Cresci-17 (Cresci et al., 2017). As a result, AdaBoost achieved a precision of 95%. The Social Spammers and Spam Messages in Microblogging Detection (S3MCD) (Wu et al., 2016) is a model based on the Alternating Direction Method of Multipliers (ADMM) algorithm. The approach used content and user information features and three relation matrices: user-message links, user–user links, and message–message links. The system exploited the posting links between the OSN accounts and the content posted to combine SMBs detection with spam content detection. The Term Frequency Inverse Document Frequency (TF-IDF) vector was employed to extract the feature vector. Logistic regression (LR) was used to assign scores to microblogging accounts and the posted content. As a result, S3MCD reached a precision score of 92.34% in spam bots detection. The authors of (Liu et al., 2017) studied the imbalanced spam and legitimate accounts. Therefore dataset, a fuzzy oversampling technique (FOS) was used to fix the problem of unbalanced data. The experiment results showed that this system improved the detection of spam bots using imbalanced datasets and an RF-based model, which achieved a TP score of 78%. Authors in (Loyola-Gonzalez et al., 2019) proposed a system based on a contrast pattern model to detect spam bots. The suggested framework conducts the classification in four steps: 1) data extraction from Twitter that includes user information, content, timing, and sentiment features (sub, irony, agreement, confidence, feeling) extracted using the Meaning Cloud API; 2) Multiple different DT generation from the proposed feature representation using different quality measures to evaluate the splits, where the contrast patterns are extracted from each DT; 3) identification of the best contrast patterns in features in the training set; 4) accounts classification. The RF performed higher than the reviewed techniques, with an AUC of over 99.56%. The authors in (Murugan and Devi, 2019) proposed a system for Twitter spammers detection based on RF algorithms. To extract the features, they used a hybrid technique by employing both LR with Principal Component Analysis (PCA) to reduce the dimension of the feature set. The overall performance of the selected features from the user profile showed an increase in detection rate, reaching an F-1 measure score of 92%.

- **DNA-inspired detection systems**

To deal with the fact that bots can evade detection by randomizing their actions, the authors of this work (Cresci et al., 2016) proposed a DNA-inspired model to detect spam bots. This approach encoded the behavior of OSN users in a sequence of characters that represented the behavioral lifetime. Based on the Longest Common Substring (LCS) and a similarity measure applied to each digital DNA sequence to cluster the group of spam bots. They considered that a cluster of users that have a similar series of actions are spam bots.

- **Hybrid detection systems**

This study (Fazil and Abulaish, 2018) proposed a hybrid approach to detect spam bots on Twitter based on tweets, network attributes, and user information. In addition, they introduced a new feature to conduct hybrid detection: community and interaction features. For the experiment, they used the social spammer dataset proposed in (Cresci et al., 2016) to train the RF algorithm, which exceeded the baseline models with an accuracy of 97.9%. The Spammer Detection Hybrid Model (SDHM) (Liu et al., 2014) is a hybrid model for social spammers' detection in Weibo based on topic modeling. They used LDA for topic extraction from content features, and content similarity was computed using the Longest Common Subsequence (LCS). In addition, user information and network features were investigated. The authors support that SDHM can detect spam bots with only one user's posting behavior and an F-measure of 91.8%. The experiment results showed that the proposed system achieved an accuracy score of 92.9% without requiring historical data or user information. In this paper (El-Mawass et al., 2018), the authors employed the Markov random field (MRF) technique to fix the evasion-induced errors in spam bot detection. They used the graph model in conjunction with the SVM model, which outperformed traditional classifiers with a precision of 95.2%.

- **DL-based detection system**

The results revealed that 79% of evolved spam bots evade state-of-the-art detection systems like in (Cresci et al., 2017) and (Miller et al., 2014). In this work (Lingam et al., 2021); the authors used Deep Q-learning (DQL) and Particle Swarm Optimization (PSO) to detect spam bots on Twitter. The features used are user information, content, and temporal features. Moreover, the study proposed an algorithm called Spam-Influential User Identification Integrated into Influential Community Detection (SIU-ICD). The SIU-ICD reduced the dissemination of spam influential communities on OSNs and achieved a precision of 94%. The authors in (Ali Alhosseini et al., 2019) presented a spam bots detection system using inductive representation learning and user information. The Graph Convolutional Neural Networks algorithm (GCNN) exhibited the best performance compared to MLP and Belief Propagation with a precision of 89%. This paper (Wei and Nguyen, 2019) aims to detect Twitter spam bots based on Bi-LSTM and content features. They used Word Embedding for feature extraction from users' tweets. This work (Mateen et al., 2017) presented an approach for spam bots detection based on the J48 algorithm to detect Twitter spam bots using user metadata, content, and network features. The J48 algorithm was able to detect spam bots with a precision of 97.6%. The spam bot detection approach proposed in (Zhao et al., 2020) used a heterogeneous stacking-based ensemble learning method. To adjust the weights of the classifiers' prediction results, they used cost-sensitive learning in the training stage of the DNN-based model that achieved a precision of 70%.

- **Graph-based detection systems**

This research work (Lingam et al., 2018) presented a social spam-botnet detection technique based on content and network features and the trust concept. The trust value between OSN users is calculated based on Bayesian theory, and the value of indirect trust among neighboring users is calculated based on the Dempster-Shafer theory. The incorporation of both parameters improved the accuracy of the detection by a score of 85%. Based on Triad Significance Profile (TSP), Social Status, and Cascaded filtering, this study (Jeong et al., 2016) proposed spam bot detection mechanisms. These approaches analyzed and exploited social network properties. This method's main advantage is scalability. Instead of analyzing the entire network, it examines user-oriented "two-hop" social networks. This approach detected spam bots with a TP score of 96.3%.

- **Topic modeling based detection systems**

In this paper (Koggalahewa et al., 2022), the authors implemented an unsupervised technique to detect spam bots using LDA and similarity-based clustering. They calculated the matching acceptance between OSN users based on frequent interests in specific topics between two users. Therefore, they used LDA for topic modeling and then trained the model, which achieved an accuracy of 96.9%.

- **Genetic-based detection systems**

GenBot (Cresci et al., 2019) is a genetically-based system that generated synthetic spam bots by generating digital DNA sequences of user information and content features by encoding user posts into chronological segments of action using a digital DNA technique. Identifying behavioral evolutions and mutations of spam bots in OSNs was made possible by converting user information and content features to digital DNA sequences. Transforming user information and content features to digital DNA sequences permitted the identification of behavioral evolutions and mutation of spam bots in OSNs. In (Cresci et al., 2019), authors applied an ad-hoc genetic algorithm to detect social spammers based on digital DNA. To represent the digital DNA of an OSN user, they embedded the behavior of each OSN account as a numerical DNA sequence. The behavior of clusters of spam bots was modeled via their numeric DNA and compared to a set of benign accounts. The finding, it is possible to examine the ability of spam bots to escape the existing behavior-based detection techniques by switching parameters of the genetic model. In this work (Akiyama et al., 2017), the authors studied the URLs included in spam content that redirects users to malicious websites based on a URL tracking system using Domain Genetic Algorithms (DGA) to blocklist the malicious links. This approach is a honeypot-based system that prevents URL redirection with an accuracy of 96.50%.

3.6. Game bots detection techniques

A game bot is an SMB that can play OSN games instead of human users to gain more money and prizes by playing continuously and winning more items that human users cannot. This study (Kang et al., 2016) aims to propose a system based on the RF model to detect social game bots active in Massive Multiplayer Online Role Playing Games (MMORPG). First, the authors investigated the game bots' behaviors and discovered they were engaged in repetitive trading for real-time money. They used different user data from MMORPG enterprises and focused on behavior-based data, including activity logs, chat contents, user information, and network features. The approach achieved an accuracy of 96.06%.

In this chapter, we have reviewed the SMB detection techniques proposed by researchers, including all existing types of SMBs, such as social bots, Sybils, cyborgs, stegobots, political bots, spam bots, and game bots. Based on these state-of-the-art works, we will address the research questions of this SLR and outline open questions for future work.

4. Review analysis

This section discusses the key findings of our systematic literature review, which were derived from the most important studies published between 2008 and 2022. One hundred forty-four articles were regarded as primary because they included several experiments in SMB detection. Therefore, it investigates the studied approaches by analyzing the techniques used by the researchers to build SMB detection systems from five perspectives: (1) The type of bots targeted by the SMB detection techniques; (2) The in-depth study of the algorithms and the technique used for the SMBs detection, performance comparison, discriminative features, and the dataset that gave better results for SMBs detection. (3) The classification of the features used by the SMBs detection techniques; (4) The influence of feature engineering techniques on the accuracy of the models; and (5) The analysis of the data sources used for SMB detection, including the type of OSNs, the presentation of dataset details, and dataset issues such as dataset size and labeling problems. The 5th research question raised at the outset of this study will be addressed by conducting a quantitative and qualitative analysis of the overall experiment results. This research will go over each stage of the lifecycle of an SMB detection model and provide insights for new research studies to improve the detection of SMBs.

4.1. Profiles and social media bot types (RQ1)

According to the literature, there are different types of profiles on the OSN; the legitimate profile is the normal profile of accounts belonging to real people conducting legitimate activities. This category of profiles also includes the verified profile, which is an

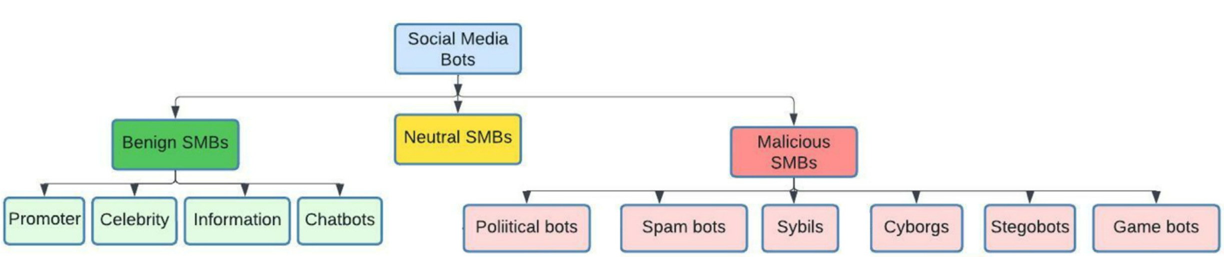

Fig. 3. Taxonomy of social media bots.

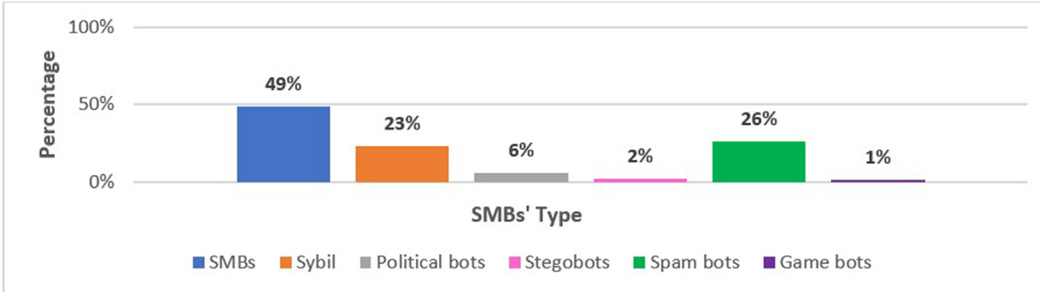

Fig. 4. Type of bots studied in the state-of-the-art works.

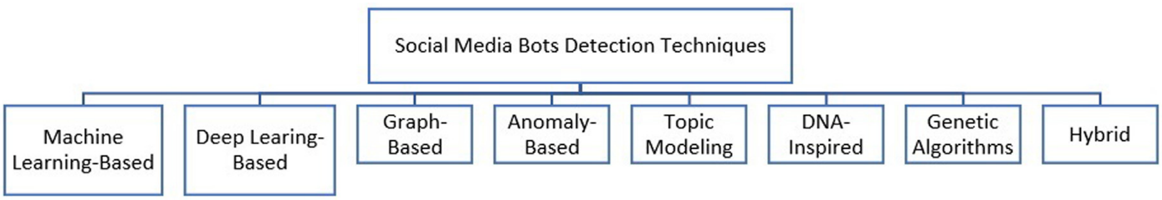

Fig. 5. Taxonomy of SMBs detection approaches.

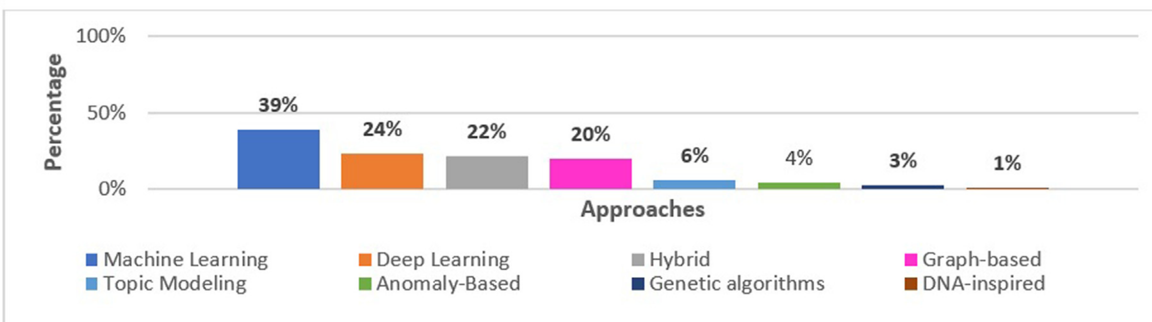

Fig. 6. SMBs Detection Techniques.

account belonging to companies, celebrities, or public figures whose profile authenticity has been confirmed by OSN. The cloned profile is the account whose identity and metadata have been stolen from a legitimate profile to conduct malicious activities. Fake accounts are pseudo-identities used to infiltrate the OSNs; they can be managed manually by a person or automatically by a bot master. Sock puppets are manually controlled OSN profiles, and they are designated to support something or someone on the OSN. Therefore, troll profiles are the politically engaged Sock Puppets controlled by government proxies or political parties to spread misinformation, viral content, and fake news to influence OSN users during political events and elections.

An SMB can mimic human behavior on OSNs and perform automated activities, including posting textual content, photos and interacting with other users. In general, an SMB supports two kinds of generic operations in OSN. The first one is social interactions, which are the operations of writing and reacting to content posted by other OSN users. The second is social networking operations that are used to alter the social structure of a group of users by adding or following other accounts. Therefore, each social bot requires a profile in an OSN to launch these activities. An SMB can be classified according to its intention and behaviors, whether benign, neutral, or malicious (Stieglitz et al., 2017). Benign bots automatically post alert tweets, disseminate news (such as news bots), interact with users, and respond to their needs (such as chatbots). Neutral bots post or repost jokes, such as "nonsense bots" that share nonsense content. Malicious SMBs are known as "zombies" or "slaves", which a bot master controls to perform automatic operations and coordinate attacks. The most known malicious SMBs are spam bots, political bots, Sybils, cyborgs, stegobots, and game bots. Based on the literature review, we classified malicious SMBs into six types according to their severity level, as shown in Fig. 3. In this taxonomy, we defined four categories for benign SMBs and six main categories for malicious SMBs.

Despite the variety of bots operating in different OSNs, many works have been devoted to SMB detection and classification. Most of these works are based on a binary supervised learning algorithm that classifies the OSN accounts as (human or non-human) or (Human legitimate or malicious bots). These works proposed generic SMB detection without specifying the type of social bot. Fig. 4 depicts the percentage of papers dedicated to each category of SMBs.

From Fig. 4, we observe that generic SMB detection is the most common approach carried out in the literature, followed by spam bots, Sybils, and political bots detection approaches. Few studies were conducted to detect stegobots and game bots. This can be explained by the fact that the researchers were initially more concerned with figuring out whether a particular account was legitimate or not, as well as whether it was a real account or a bot account. The growth of this field of study has made it possible to

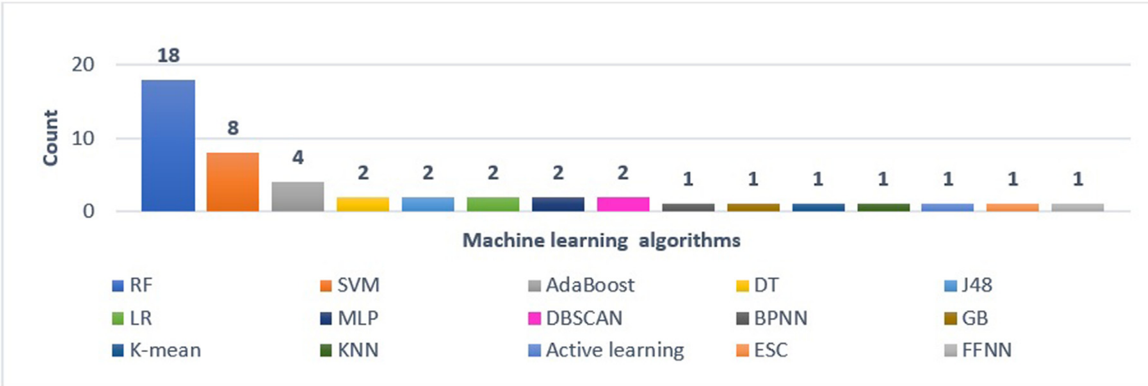

Fig. 7. ML Algorithms used for SMBs detection.

Table 3
ML-based detection systems.

Algorithm	References	Max Performance	Dataset of Max Performance	Features of Max Performance
AdaBoost	(Dickerson et al., 2014) (Morstatter et al., 2016) (Natarajan et al., 2015) (Andriotis and Takasu, 2018)	Accuracy: 93.89%, F1-score: 95.06%	ICWSM17, RAID11, WWW17, Cresci-17	User information, sentiment, and content features
SVM	(Ikram, et al., 2017) (Dorri et al., 2018) (Beskow and Carley, 2019) (Fernandes et al., 2015) (Bello et al., 2018) (Yang et al., 2014) (Castillo, et al., 2017) (El-Mawass et al., 2018)	AUC: 100% Recall: 99.8%	Scraped Dataset from RenRen	User information, content, network, and temporal features
DT	(Bello et al., 2018) (Ellaky et al., 2021)	Precision: 100%	Cresci-17	Content features
RF	(Davis et al., 2016) (Jr et al., 2018) (Ellaky et al., 2021) (Beğenilmiş and Uskudarli, 2018) (Ji et al., 2016) (Alothali et al., 2021) (Igawa, et al., 2016) (Rodríguez-Ruiz et al., 2020) (Pratama and Rakhmawati, 2019) (Beskow and Carley, 2018) (Boshmaf, août 2016) (Al-Qurishi et al., 2018) (Teljstedt et al., 2015) (Chu et al., 2012) (Loyola-Gonzalez et al., 2019) (Fazil and Abulaish, 2018) (Kang et al., 2016) (Murugan and Devi, , 2019) (Jin Dan and Teng Jieqi, 2017)	Precision: 100%	Cresci-17	Content features
BPNN	(Mazza et al., 2019)	Accuracy: 98.6%	_	Network features
J48	(De Meo et al., 2011) (Yu et al., 2008)	FP: 1.7% F-measure: 97.9%		User information, content, and network features
GB	(Inuwa-Dutse et al., 2018)	Accuracy: 98.97%.	Honeypot dataset	User information, content features
GBT ensemble learning algorithms	(Zannettou et al., 2019)	Recall: 85% Accuracy: 86% F1-score: 83%	Scraped dataset from Twitter	User information, content, and temporal features
KNN	(Mohammad et al., 2019)	Recall: 96%	Scraped dataset from Twitter	User information, content, and temporal features
MLP	(Ali Alhosseini et al., 2019) (Alsaleh et al., 2014)	Precision: 98.87%	NetFlow and transaction dataset	User information, content features
ESC	(Sayyadiharikandeh et al., 2020)	AUC: 99%	Pron-bots, Cresci-17 Fake followers	User information, content features
FFNN	(Al-Qurishi et al., 2018)	TP: 67.7%	Scraped dataset from Twitter	User information, content, temporal, linguistic, and network features

analyze social bots in-depth and comprehend their behaviors related to each type of social media attack and illegal behavior that SMBs engage in.

4.2. Detection approaches and performance (RQ2)

4.2.1. Artificial intelligence methods

This section discusses the different SMB detection approaches and the dataset features used to train models. Researchers have used many methods to build a system to detect malicious SMBs accurately and prevent their spread. These approaches are ML-based, DL-based, graph-based anomaly-based, topic modeling based, DNA-inspired, genetic-based, and hybrid. To better understand the method academics have suggested for detecting SMBs, we categorize the current SMBs detection methodologies. The taxonomy of these approaches is shown in Fig. 5.

Fig. 5. ML-based methods applied traditional ML algorithms such as SVM, RF, DT, KNN, LR, etc. Therefore, the DL-based methods utilized Deep learning algorithms like CNN, LSTM, MLP, etc. The graph-based approaches are implemented using graph theory

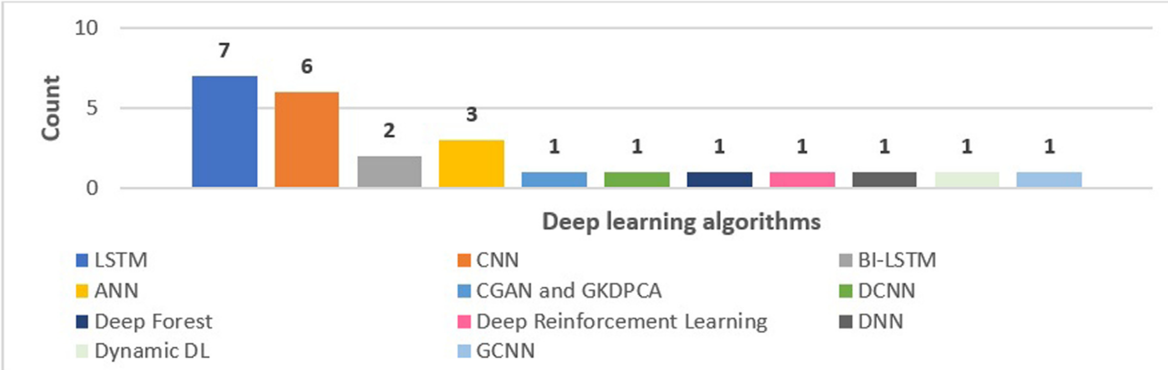

Fig. 8. Deep learning algorithms used for SMBs detection.

Table 4
DL-based detection system.

Algorithm	References	Max Performance	Dataset of Max Performance	Features of Max Performance
LSTM	(Mazza et al., 2019) (Kudugunta and Ferrara, 2018) (Beğenilmiş and 2018) (Feng et al., 2021) (Cai et al., 2017) (Kumar et al., 2021) (Benabbou et al., 2022)	Recall: 100% Accuracy: 100% Precision: 99.8%	Crisci-17	User information, content features
BI-LSTM	(Beğenilmiş and Uskudarli, 2018) (Loyola-Gonzalez et al., 2019)	Precision: 94% Recall: 97.6% Accuracy: 96.1%	Cresci-17	Content features, deep linguistic features
CNN	(Cai et al., 2017) (Mohammad et al., 2019) (Feng et al., 2019) (Fazil et al., 2021) (Kumar et al., 2021) (Benabbou et al., 2022)	Recall: 100% Accuracy: 100% Precision: 99,8%	Crisci-17	User information, content features
DCNN	(Wanda and Jie, 2020)	AUC: 95,47% Precision: 94% Recall: 93.21%	Scraped dataset from Twitter	User information, content, and temporal features
GCNN	(Ali Alhosseini et al., 2019)	AUC: 94% Recall: 80% Precision: 89%	Content polluters dataset Yang et al. 2013	User information
DNN	(Cresci et al., 2016)	Precision: 70%	The dataset used in (Chen et al., 2015) Cresci-2015	User information, content features
ANN	(Mohammad et al., 2019) (Kiran et al., 2019) (Zegzhda et al., déc. 2017)	Accuracy: 97.79%	Scraped dataset from Twitter	User information, content features
Deep Forest	(Daouadi et al., 2019)	Accuracy: 97.55%	_	User information, content features
CGAN and GKDPCA	(Kantepe and Ganiz, 2017)	Precision: 97.29% Recall:97.93% Accuracy:96.55%	Scraped dataset from Twitter	User information, content, and network features
RGA: ResNet, BIGRU, Attention	(Wu et al., 2021)	Precision: 99.33%	Scraped dataset from Sina Weibo	User information, content, network, and temporal features
Dynamic DL	(Wanda and Jie, 2021)	AUC: 99.70%	Flickr, Facebook	User information
DRL	(Akiyama et al., 2017)	Recall: 90% Precision: 94%	Social Honeypot, Cresci-2015	User information, content, and temporal features
BiGRU, LSTM, and CNN	(Benabbou et al., 2022)	Accuracy: 99.44% Precision: 99.25%	Cresci-17	Content features

algorithms like The Random Walk, Loopy Belief propagation, Pairwise Markov Random Field, etc. The aim is to build the network structure of a group of accounts to detect the Sybil nodes and to prevent the Sybils attacks. The anomaly-based approaches are used to identify OSNs intrusions and attacks conducted by SMBs. This approach is often used during the first stage of SMB detection by clustering the group of accounts with abnormal behaviors. The topic modeling techniques allow the detection of emerging topics in the microblogs of OSNs. These techniques, like LCS, LDA, etc., enable the detection of propaganda, manipulation, and disinformation campaigns carried out by malicious SMBs. The approaches based on DNA-inspired and Genetic algorithms used the notion of a sequence of interaction and timeline posting and action to detect abnormal behavior of SMBs. So, the hybrid approaches combine several algorithms to operate in a complementary and mutual way to increase the efficiency of malicious SMBs recognition.

Fig. 6 provides a statistical overview of the works based on the different categories of approaches. Therefore, it is noticeable that 39% of these detection systems are based on ML algorithms, then DL algorithms with 24%, and hybrid methods that combine different algorithms with 21%.

The supervised technique based on ML and DL algorithms can overcome the issue of scalability and the high cost by reducing

Table 5
Graph-based detection systems.

Algorithm	References	Max Performance	Dataset of Max Performance	Features of Max Performance
SybilGuard	(Yu et al., 2008)	_	1 M node from 45,000 Web documents	Network features
SybilLimit	(Yu et al., 2008)	_	Data crawled from: https://www.friendster.com https://www.livejournal.com https://www.dblp.uni-trier.de	Network features
SybilInfer	(Danezis and Mittal, 2009)	FP < 5%	LiveJournal and power law network dataset: 300,000 nodes	Network features
SybilRank:	(Cao et al., 2012)	TP > 85%	Facebook, CA-GrQc, CA-HepTh, CA-AstroPh, CA-CondMat, wiki-Vote, soc-Epinions, soc-Slashdot, and email-Enron datasets	User information, network, and behavioral features
Louvain method	(De Meo et al., 2011)	Normalized mutual information: 81.6%	Facebook dataset CA-GrQc, CA-HepTh, CA-AstroPh, and CA-CondMat datasets	Network features
SybilSCAR	(Wang et al., 2017)	AUC: 100% AUC: 99% AUC: 82%	Facebook dataset Small Twitter dataset Large Twitter dataset	Network features
SybilBelief	(Gong et al., 2014)	FP < 0.3% Accuracy: 100%	Facebook dataset Slashdot dataset Email dataset	Network features, User information
Sybil-Resist	(Ma et al., 2014)	Detection rate: 99.2%	Facebook dataset	Network features
SybilFrame	(Gao et al., 2018)	Accuracy: 95.4%	Massive Twitter dataset	Network features
SybilRadar	(Mulamba et al., 2016)	AUC < 95%	The Cresci-2015 dataset The Elezioni2013 dataset TWT dataset The INT dataset	Network features
Íntegro	(Boshmaf, 2016)	AUC: 76%	Pro-Facebook, Pro-Tuenti, Gra-facebookts Gra-facebookrd, Gra-hepth, Gra-astroph Graph	User information Network features
Similarities-based and relationships-based	(Kharaji and Rizi, 2014)	_	Facebook interactions dataset	User information Network features
Trust vote	(Yang et al., 2016)	Recall $\geq$ 80%	RenRen dataset Peking University (PKU) network: friend invitation graph for PKU network 5.01 M complete friend request records, 230 K PKU users	User information Network features
Erdos-Renyi graphs and scale-free graphs	(Wang and Paschalidis, 2017)	Precision: 86%	CTU-13 botnet dataset	Network features
Sybil Guard, ML, Loop Belief Propagation	(Shetty et al., 2021)	Accuracy: 95.34%	Scraped dataset from Twitter Kaggle and Machine Learning Detecting-Twitter-Bots dataset	User information Network features Statistical features
Node2Vec and intra-community random walk	(Pham et al., 2022)	ROC: 97.68%	Cresci-2015 Social spammer	User information

human intervention during the detection process. From the Fig. 6, we observe that 39% of the proposed approaches were implemented using ML algorithms, 24% using DL algorithms, 22% using hybrid, 20% using graph-based algorithms, 6% using topic modeling techniques, 4% using anomaly-based techniques, and then the genetic-based and the DNA-inspired algorithms. The count distribution of the employment of ML techniques in detecting SMBs is shown in Fig. 7.

RF, SVM, and AdaBoost are the three most well-known ML algorithms for social bot detection, as illustrated in Fig. 7. Also, we observe that supervised ML methods are applied to SMB identification more frequently than unsupervised methods. Table 3 presents the ML algorithm, references, greatest performance recorded, training dataset, and features used to obtain maximum performance. The aim of Table 3, evaluate the ML-based techniques used in SMB identification.

The RF algorithm is the most widely used model classifier, as shown in Table 3, and it successfully identified spammers, social media bots, and political bots. For the best accuracy, the RF algorithm reached 100% and a recall of 100% with models trained using the Cresci-17 dataset and content features (Ellaky et al., 2021); (Igawa, et al., 2016). In the second range, the SVM attained an AUC of 100% (Ikram, et al., 2017) and a recall of 99.8% with a model trained on a dataset containing user information, content, network, and temporal variables collected from RenRen. The performance findings of the additional ML methods, mainly when applied to the Cresci-17 dataset, are encouraging. Supervised ML algorithms have effectively identified SMBs, spam bots, Sybils, cyborgs, and political bots. They only need a few features to identify SMBs. From the state-of-the-art studies examined in the preceding sections and comparing the results presented in Table 3, we discovered that user information and content features are the most frequently used attributes for ML-based detection systems. To identify botnets and stegobots, which require more intricate information like steganographic images and videos that cannot be preprocessed appropriately or trained using ML algorithms. SMBs keep changing their attack strategies to evade detectors. Moreover, the complex experiments involve more significant amounts of data, for example, to reconstruct the structure of the social network or to extract the more advanced characteristics, which increase the complexity of the ML-based model and the complexity of the learning algorithm. Things are very different when it comes to DL-based models. Unlike traditional machine learning, deep learning does not require any feature engineering phase. It can learn the representation from the input data with no preconceived structure. Fig. 8 illustrates the count distribution of the employment of DL techniques to detect SMBs.

Table 6
Other detection systems.

Approach	Type of Approach	References	Max Performance	Dataset of Max Performance	Features of Max Performance
ActM	Anomaly-based	(Davis et al., 2016)	Recall: 92% Precision: 93%	Scraped datasets from Twitter, Reddit, Stack-Overflow	Temporal and content features
Bayes theorem and RF	Hybrid	(Chu et al., 2012)	TP: 96%	Scraped dataset from Twitter	User information, behavioral and content features
BotWalk	Anomaly-based	(Minnich et al., 2017)	Precision: 90%	Scraped dataset from Twitter	User information, content features, temporal, network, and behavioral features
Causality-based Clustering, label propagation algorithm	Hybrid	(Shaabani et al., 2018)	Precision: 75%	ISIS dataset from Twitter	User information, content, and sentiment features
Debot and Graph-based clustering	Hybrid	(Abu-El-Rub and Mueen, 2019)	Recall: 83.6% Accuracy: 87.5% Precision: 98%	U.S. election dataset 2016 from Twitter	Content, temporal, network, and sentiment features
One-class RF	Anomaly-based	(Rodríguez-Ruiz et al., 2020)	AUC: 89%	Cresci-17 dataset	Behavioral features
Trust Model & Dempster-Shafer theory	Hybrid	(Lingam et al., 2018)	Accuracy: 85%	Cresci-2015 dataset	User information, content, and network features
DGA	Genetic algorithms	(Akiyama et al., 2017)	Accuracy: 96.50%	Dataset scraped from Twitter of content injected with URL redirection	Content features
flow-based approximation & Graph modeling	Hybrid	(Kiran et al., 2019)	Precision: 86%	CTU-13	Botnet traffic
GenBot	Genetic algorithms	(Liu et al., 2017)	Improved results of state-of-the-art techniques	Cresci-17	User information and content features.
LA-SMBD & trust model & Bayesian learning & Dempster-Shafer	Hybrid	(Rout et al., 2020)	Precision:95.37%	Cresci-2015 Social Honeypot	User information, content, network, and temporal features
S3MCD: ADMM & LR	Hybrid	(Wu et al., 2016)	Recall: 93,08% Precision: 92.34% F1-score: 92.71%	Scraped dataset from Twitter	User information, content, and friend features
Similarity-based MRF and RF	Hybrid	(Al-Rakhami and Al-Amri, 2020)	Accuracy 95.2%	Scraped dataset from Twitter	User information, content network, and behavioral features
Social Fingerprinting: Digital DNA model	DNA-inspired	(Cresci et al., 2017)	Precision: 98.2% Accuracy: 97.7% F-measure: 97.7%	Retweeters of the political candidate Spammers of Amazon.com products Cresci-2015	Behavioral features
DEBOT & Warped Correlation		(Wei and Nguyen, 2019)	Precision: 94%	Scraped dataset from Twitter	Content features
Contrast model, DT, RF	Hybrid	(Loyola-Gonzalez et al., 2019)	AUC: 99.56%	Cresci-17 dataset	User information, content, temporal, and sentiment features
Big data and RDF		(Caruccio et al., 2018)	_	Cresci-2015 Verified and Real account dataset	User information
KNN & HawkesEye & Topic modeling	Hybrid	(Dutta et al., 2020)	Recall: 96%	Scraped dataset from Twitter	User information, content, and temporal features
StreamKM ++ & DenStream	Anomaly-based	(Miller et al., 2014)	Recall: 100% TF: 2.2%	Scraped dataset from Twitter	User information, content features
SVM-NN	Hybrid	(Khaled et al., 2018)	Accuracy: 98.3%	The Cresci-2015 dataset; Dataset, human accounts: human of politicians, journalists and bloggers, Fastfollowerz dataset., InterTwitter dataset, Twitter technology dataset	User information
Adaboost and Topic Modeling	Hybrid	(Morstatter et al., 2016)	Recall: 82.48% Precision: 75.41%	Libya dataset, Arab spring activity in Libya. Arabic Honeypot	Content features
random walk & Deep Autoencoder	Hybrid	(Lingam et al., 2020)	Precision: 90%	Scraped dataset from Twitter	Content, behavioral features
Entropy, LDA, Bayesian algorithm	Hybrid	(Chu et al., 2010)	TP: 82.80%	Scraped dataset from Twitter	User information, content features
LDA	Topic modeling	(Koggalahewa et al., 2022)	Accuracy: 96.9%	Dataset of Twitter user accounts posting about fashion	Content features
BotCamp: boosted TD, graph clustering	Hybrid	(Abu-El-Rub and Mueen, 2019)	Accuracy: 87.5% Precision:98%	Scraped dataset from Twitter	Content, temporal, and network features
LDA & LCS	Topic modeling	(Liu et al., 2014)	Precision: 99%	real spam dataset scraped from Sina Weibo	User information, content, and network features

Table 7
Online Social Networks Features.

Features	Description
User information	features and *meta*-data extracted from users' profiles, including username or screen name, language, bio, etc.
Friend	Include statistics and numerical attributes related to the user's friends or followers, such as the median, the number of followers, the number of accounts followed, the follower ratio, etc.
Temporal features	They are employed to discover the existence of synchronized activities on OSN. Examples are the average time between two successive publications, the time between two consecutive retweets, etc.
Content	The bot-generated contents typically use simple terminology and short clauses.
Sentiment	Features computed to capture the polarity of the content posted in OSN and the posture of an integral conversation based on emotions score, polarity score, n-grams features, etc.
Network features	Features that rely on users' social interactions and activities and capture patterns of the social network of a community or a group of accounts.
Contextual features or deep linguistic	Features that capture the aspects of the construction of tweets and conserve the semantic or syntactic form of tweets using Deep learning and NLP techniques.
Behavioral features	Features that describe the action and interactions of social bots in OSN. So, the behavioral similarity can be examined by comparing content similitude, URL co-occurrence in posts, social activity, and reaction similarity.

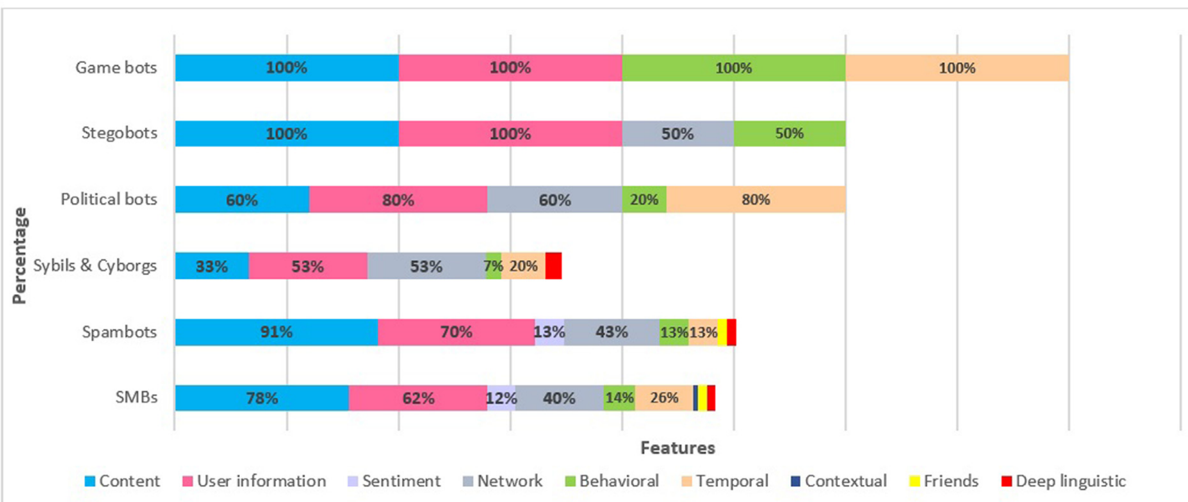

Fig. 9. Features classes used to detect each type of SMB.

For the DL algorithms, we observe that they are less used compared to traditional ML algorithms. Therefore, as shown in Fig. 8, the most commonly used algorithms are CNN, LSTM, ANN, and BI-LSTM. These algorithms were applied for the generic detection of SMBs, spam bots, and Sybil. To assess the performance of the SMB detection approaches based on DL techniques, we summarize in Table 4 the DL algorithms carried out and their highest performance related to the dataset and the features used to train the model.

To develop models to find SMBs, researchers have used a variety of DL algorithms. As shown in Fig. 8, the LSTM, CNN, and ANN models are the most commonly used algorithms. Based on the results presented in Table 4, we note that LSTM and CNN are the best-performing algorithms. Both used content features and user information to achieve 100% recall and 100% accuracy on the Cresci-17 dataset. In addition, we observed that user information, content, network, and temporal features are the most common characteristics used for DL-based detection systems. The methods based on graph models and graph theory enable building the graph structure of the OSN based on the connections and relationships among the OSN nodes. Generally, it uses graph theory to rank OSN accounts, which provides a robust mechanism for identifying SMBs and detecting trustworthy users. In other works, they used DL algorithms to detect the botnet and SMB community. Bots often form clusters or networks and may engage in coordinated behavior. Analyzing the network structure can help identify bot accounts. Table 5 gives a global view of the results of graph-based algorithms used in the studied works.

The majority of approaches for the detection of fake accounts are based on mathematical algorithms from graph theory. It is possible to build the social graph of the studied accounts through this algorithm. In this case, the nodes are the accounts, and the edges are the social links between the accounts. To build the structure of the social graph of the studied accounts, it is necessary to have additional information such as the network features, the list of friends, the set of interactions between users, etc. Combining user information and network features, the SybilBelief approach (Gong et al., 2014) achieved 100% accuracy in detecting Sybil nodes. Then comes Sybil-Resist (Ma et al., 2014); which earned a detection rate of 99.2% to detect the Sybil region and nodes based on the Random Walk algorithm.

Many other techniques were used for SMB detection based on genetic-based algorithms, digital DNA-based algorithms, anomaly-based algorithms, topic modeling-based approaches, and hybrid approaches. Table 6 presents the findings of each of these algorithms:

The anomaly-based approaches (Davis et al., 2016); (Miller et al., 2014); (Rodríguez-Ruiz et al., 2020) used Clustering algorithms to identify OSNs intrusions and attacks. Therefore, the topic modeling techniques were employed to detect active SMBs in emerging topics in OSNs. The highest precision achieved by the topic-modeling-based technique is 99% (Liu et al., 2014). The approaches based on DNA-inspired and Genetic algorithms were used to detect spam bots, reaching a precision of over 95%. The hybrid approaches proved that combining several algorithms increased the efficiency of the malicious SMBs recognition, and the precision is up to 95% which is very promising. To improve the performance of the SMB detectors, it's crucial to continually update the models with new and different types of content to ensure that they can recognize and block the latest tactics used by SMBs and other spammers. This requires ongoing training and testing of the models to ensure they are up-to-date and effective at detecting and blocking SMBs.

4.3. Feature types in social bot detection systems (RQ3)

Characterizing the relevant features of public social data is a critical step in detecting social bots. Several feature categories have been investigated, but six primary feature classes have been identified as important in distinguishing between human users and SMBs. We propose a new categorization of OSN features in this paper (Ellaky et al., 2021): user information features, friend features, temporal features, content features, sentiment features, network/interaction features, deep linguistic features, and behavioral features, as defined in Table 7.

All the SMB detection techniques under study were built using a set of OSN account attributes. Based on the feature mentioned above categories, we evaluated the feature utilization rate to determine each type of SMB. It is challenging to compare how frequently each feature type is employed in identifying each type of bot because we have a different number of articles in each category. Yet, it demonstrates the most common characteristics used to classify various social bots. Nonetheless, it would be intriguing to investigate whether the effectiveness of the detection systems examined in this paper correlates with the employed features. Fig. 9 illustrates the OSN features used to detect each type of SMB.

From Fig. 9, we can observe that all research studies on social bots' detection used content features, user information, network features, and behavioral features. We also noticed that the deep linguistic and contextual features had not been carried out enough to detect social bots. The same feature was the most used for Sybil and cyborgs' detection. All techniques investigated in the context of stegobots focused solely on content features and user information. Therefore, the content feature, user information, and network features were also the most commonly used for the spam bot. Indeed, for political bots' detection systems, the most used features are content, user information, temporal features, network features, friend features, and sentiment features.

4.4. Feature engineering and preprocessing techniques in social bot detection systems (RQ4)

OSN metadata includes numerous information associated with contents and events, such as the time and location from where a post was sent. To train a model using OSN-based datasets that contain misspellings, informal language, and symbolic phrases, preprocessing is necessary to process and analyze the data, as well as feature extraction and selection.

4.4.1. Preprocessing techniques

The content feature represents the comments posted by OSN users. The comments are written unconventionally and include text, hashtags, numbers, emoticons, etc. Natural Language (NLP) techniques are required to clean and preprocess text data. Preprocessing text data helps train a model with relevant data to keep the pattern required to build accurate ML models (Ellaky et al., 2021).

- **Cleaning**

Cleaning the data is necessary to make the dataset available in a computational format without changing its meaning or content. This includes removing elongations, diacritics, punctuation, digits,

Table 9
Feature selection techniques.

Techniques	References
Manual	(Pratama and Rakhmawati, 2019) (Rodríguez-Ruiz et al., 2020)
Correlation measure	(Natarajan et al., 2015) (Khaled et al., 2018) (Lingam et al., 2018) (Fernandes et al., 2015)
Entropy values	(Ji et al., 2016)
IG	(Kang et al., 2016) (Kantepe and Ganiz, 2017)
l1- regularized linear SVM	(Bello et al., 2018)
MI, Chi Square	(Kantepe and Ganiz, 2017)
PCA	(Alsaleh et al., 2014) (Beğenilmiş and Uskudarli, 2018) (Al-Qurishi et al., 2018) (Dickerson et al., 2014) (Gilani et al., 2019) (Khaled et al., 2018) (Lee and Kim, 2014)
SVM	(Khaled et al., 2018)
CVAE	(Alothali et al., 2021)
Wrapper subset evaluation (WSE)	(Alothali et al., 2021)
Filter	(Alothali et al., 2021)
Embedded	(Alothali et al., 2021)

Table 8
Feature extraction techniques.

Techniques	References
N-gram	(Miller et al., 2014) (Beskow and Carley, 2019) (Bello et al., 2018) (Inuwa-Dutse et al., 2018) (Lee and Kim, 2014) (Almerekhi and Elsayed, 2015) (Ikram, et al., 2017) (Ellaky et al., 2021)
TF-IDF	(Beskow and Carley, 2019) (Ferrara, 2017) (Luceri et al., 2019) (Ellaky et al., 2021) (Wu et al., 2016)
BOW	(Ellaky et al., 2021)
BERT	(Ellaky et al., 2021) (Kumar et al., 2021) (Grinberg et al., 2016)
LDA	(Morstatter et al., 2016) (Bello et al., 2018) (Al-Qurishi et al., 2018)
LSA	(Inuwa-Dutse et al., 2018) (Chew et al., 2018)
GloVe	(Benabbou et al., 2022)
Roberta	(Martin-Gutierrez et al., 2021)
Word2vec	(Ellaky et al., 2021)
Fastext	(Ellaky et al., 2021)
Doc2vec	(Ellaky et al., 2021)
Rule Extraction	(Bello et al., 2018)
Hybrid: LR-PCA	(Murugan and Devi, 2019)
Manual	(P. G. Pratama et N. A. Rakhmawati, , 2019)

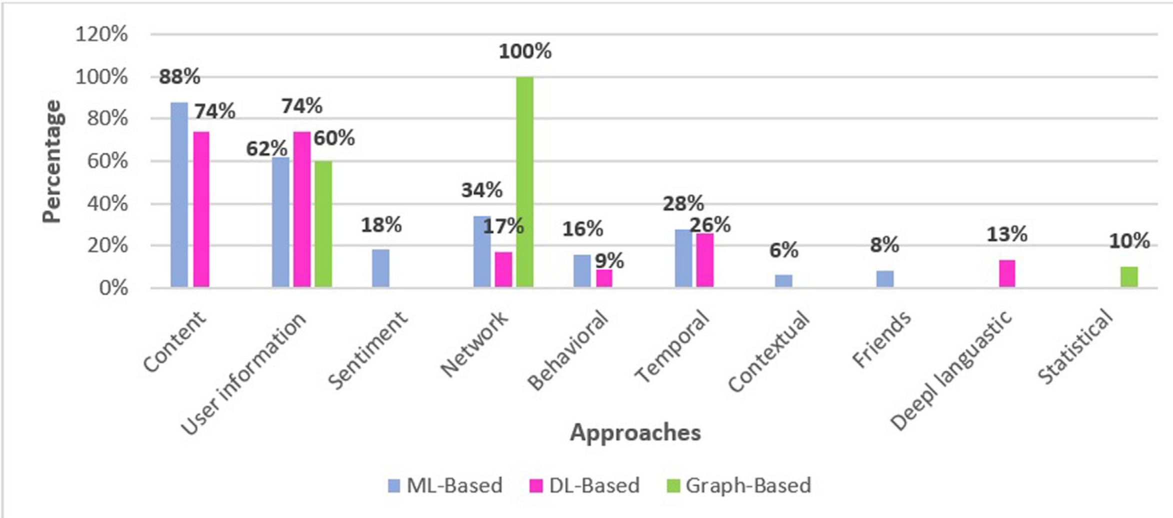

Fig. 10. Features used on the ML-Based, DL-based, and Graph-based algorithms.

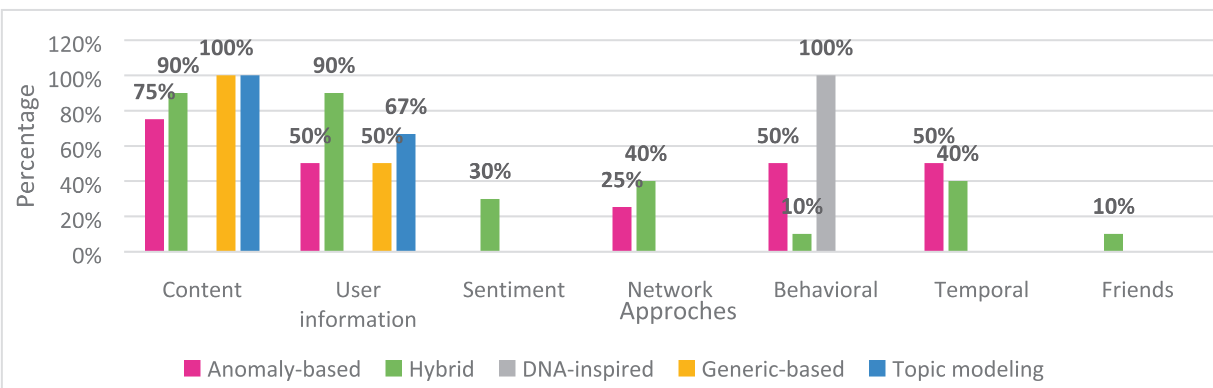

Fig. 11. Features used on the Anomaly-Based, DNA-inspired, Genetic-based, Topic-modeling and Hybrid approaches.

and superfluous characters. Therefore, most of the studied works followed a standard NLP cleaning process. Several cleaning methods have been employed since they reduce noise and enhance NLP comprehension. The elimination of undesirable characters, such as non-Arabic or non-English characters, punctuation, stop words, motion (@), hashtags (#), repeated letters, letter lengthening, diacritical marks, URLs, and usernames (Bello et al., 2018) (Almerekhi and Elsayed, 2015) (Ping and Qin, 2018) (Andriotis and Takasu, 2018). However, some removed symbols or characters can be discriminant for bots' detection because it could result in the loss of essential data. It would be interesting, for example, to calculate the number of URLs, motions, and hashtags, replacing emojis and emoticons with meaningful words and replacing acronyms with their complete form.

- **Stop words**

The majority of authors eliminate stop-words by utilizing libraries that contain words explicitly compiled that are used for text preprocessing or by employing stop words because they are a collection of keywords that are irrelevant for analysis like for the Arabic content that can be automatically preprocessed it requires information retrieval systems (IR) to retrieve the irrelevant words to eliminate during the preprocessing phase. Other techniques, including automated and hybrid (semi-automatic) techniques, have been applied in a few works.

4.4.2. Feature extraction

The datasets used for SMB detection contain several types of features, for example, textual data, numerical data, images, etc. The initial dataset requires a computation phase to calculate the numerical and statistical features and convert the unstructured attributes into a computer-readable format. Feature extraction is a method employed to decrease the dimensionality space of data to a compact dimensionality space to get manageable data. The new features are generated by combining and transforming the original features. Applying feature extraction techniques permits to increase in the training model's performance and decreases the learning process's cost (Safaa et al., 2022). In literature, many techniques were applied for feature extraction, such as statistical methods are used to extract statistical and numerical features from user information, contents, and network features (ex: following/-follower ratio, URL ratio, medium frequency of sharing, etc.) (Al-Rakhami and Al-Amri, 2020); (Balestrucci et al., 2021) , LDA

Table 10
Efficient features for SMBs detection.

Social Bot Type	Features
SMBs	Content: 78%, user information:62%, network features: 40%, temporal features:26%
Sybil, cyborg	Content: 33%, user information:53%, behavioral features: 53%, temporal features:20%
Stegobot	Content: 100%, user information:100%, behavioral features: 50%, network features:50%
Spam bots	Content: 91%, user information: 70%, behavioral features: 13%, network features:40%, sentiment features: 13%, temporal features:13%
Political bots	Content: 60%, user information: 80%, network features:60%, sentiment features: 40%, temporal features: 80%
Game bots	Content: 100%, user information: 100%, behavioral features: 100%

(Bello et al., 2018); (Dutta et al., 2020) ; (Liu et al., 2014) , and manual feature extraction (Pratama and Rakhmawati, 2019). The LSA, and TF-IDF are used to represent text-based content like tweets and comments. Table 8 provides an overview of all feature extraction techniques used in the state-of-the-art works.

From the state-of-the-art studied techniques, the most used techniques for feature extraction are N-gram and TF-IDF. The LDA and LSA methods are typically used for topic modeling the content used in the case of political bots' detection to determine the most important topic discussed in OSNs. However, some embedding techniques have been investigated, such as Bert, GloVe, Wod2vec, Doc2vec, Roberta, and Fastext. In this research work (Ellaky et al., 2021), the efficiency of different embedding methods was compared on ML-based models to detect SMBs. The experiment results were very promising, using RF and DT algorithms and Doc2Vec, which reached a precision score of 100%. %. Certain techniques, such as DistilBERT, XLNet, and sent2vec, have not been applied enough in the literature, which are achieving excellent performance on various NLP tasks, such as question answering, text classification, and sentiment analysis. DL has enabled a boost in performance for NLP applications. It is superior to traditional ML algorithms due to their single end-to-end architecture and ability to handle large training data. However, several works on deep learning and NLP for SMB detection have been published since 2019. Researchers have still not applied many of these techniques to improve SMB detection.

4.4.3. Feature selection

Using selection techniques helps to evade the curse of dimensionality phenomena. It minimizes the dimension of data sets by selecting a limited subset of features that efficiently define the data (Safaa et al., 2022). From the state of the literature, we observed that many feature selection techniques were used. In this work (Pratama and Rakhmawati, 2019), the authors used manual feature selection to detect political bots. In (Rodríguez-Ruiz et al., 2020), the authors manually selected behavioral and numeric features to detect SMBs based on one class classifier. In this work, the authors used (Alothali et al., 2021) information gains ranking filters to avoid overfitting and to reduce the features. And in this work, the authors used Information Gain (IG), Mutual Information (MI), and Chi-Square (Kantepe and Ganiz, 2017). Moreover, in this study (Beğenilmiş and Uskudarli, 2018); the authors used PCA for feature selection to detect organized behavior based on Tweets. The Wrapper methods (Feng et al., 2019) use the model's performance to evaluate the selected feature set. Table 9 presents the feature selection techniques used in the state-of-the-art.

The most common feature selection method used by researchers is PCA. It reduces the number of features by choosing the dimensions of features that exhibit the most significant variation (Murugan and Devi, 2019), thereby enhancing the performance of SVM-NN (Khaled et al., 2018), which can identify fake accounts with 98.3% accuracy and fewer features.

Table 11
Dataset sources usage.

	Twitter	Facebook	Sina Weibo	Renren	Flickr	YouTube	Reddit	VKontakte
Number of Studies	113	7	4	4	3	1	2	1

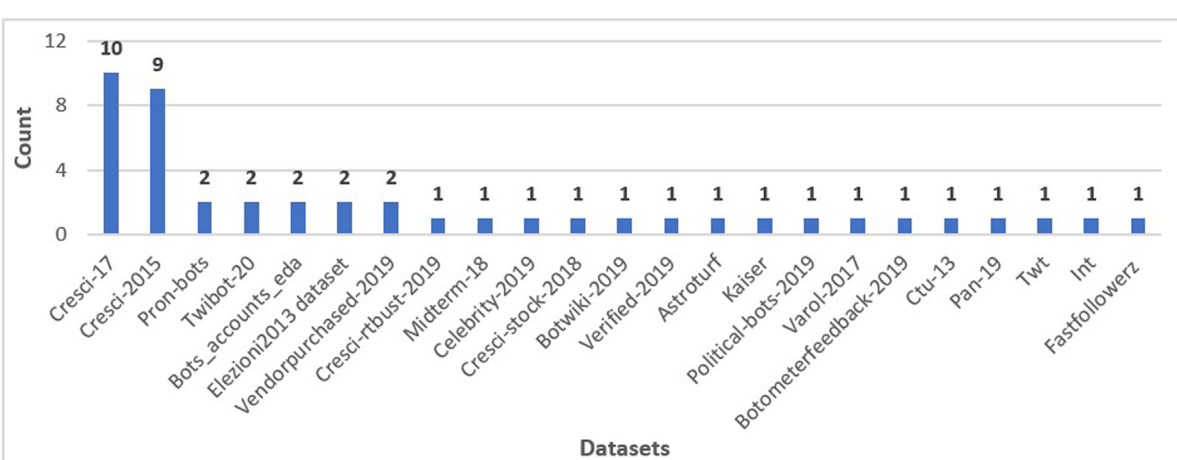

Fig. 12. Public datasets used for social bots' detection.

Table 12
Public datasets provided for SMBs detection.

References	Name	Description	Size	Citation
(Yang et al., 2019)	Pron-bots	It contains Twitter bot accounts and user information.	Bots: 21963	(Feng et al., 2021) (Sayyadiharikandeh et al., 2020)
(Cresci et al., 2017) (Cresci et al., 2017)	Cresci-17	Dataset of genuine, traditional, and social spam bots.	Users: 2764, spam bots: 7049	(Mohammad et al., 2019) (Feng et al., 2021) (Ellaky et al., 2021) (Kumar et al., mai 2021) (Benabbou et al., 2022) (Andriotis and Takasu, 2018) (Cresci et al., 2019) (Loyola-Gonzalez et al., 2019) (Wei and Nguyen, 2019) (Rodríguez-Ruiz et al., 2020)
(Mazza et al., 2019)	Cresci-rtbust-2019	Dataset of the bot and human accounts.	Users: 693	(Mazza et al., 2019)
(Pizarro, 2020)	Midterm-18	Dataset of the politically committed users.	Users: 50538	(Pizarro, 2020)
(Yang et al., 2019)	Celebrity-2019	Dataset of celebrity accounts.	Users: 5971	(Yang et al., 2019)
(Igawa, et al., 2016)	Cresci-stock-2018	Dataset of Twitter accounts posting about fashion.	Users: 13276	(Miller et al., 2014)
(Cresci et al., 2018)	Botwiki-2019	Dataset of bots from https://botwiki.org.	Bots: 698	(Cresci et al., 2018)
(Yang et al., 2020)	Verified-2019	Dataset of Twitter verified accounts.	Users: 1987	(Yang et al., 2020)
(Feng et al., 2021)	Twibot-20	Dataset of the recent generation of Twitter bots and genuine users.	User: 229573, tweets: 33,48 M, followers: 455,958	(Feng et al., 2021) (Feng et al., 2021)
(Sayyadiharikandeh et al., 2020)	Astroturf	Dataset of hyperactive politicals.	Bots: 505	(Ma et al., 2014)
(Rauchfleisch and Kaiser, 2020)	Kaiser	Dataset of German bots' official accounts, German members of parliament, and members of the 115th U.S. Congress.	German bots: 27, official accounts: 532, parliament accounts:516	(Rauchfleisch and Kaiser, 2020)
(Yang et al., 2019)	Political-bots-2019	Dataset of bots run by @rzazula), shared by @josh_emerson.	Bots: 62	(Yang et al., 2019)
(Varol et al., 2017)	Varol-2017	Dataset of human and bot accounts.	Bots: 1525, humans: 451717	(Varol et al., 2017)
(Cresci et al., 2015)	Cresci-2015	Dataset of Fake Twitter accounts.	Users: 3900, tweets: 2.75 M, followers: 1.81 M, friends: 1.78 M	(Rout et al., 2020) (Pham et al., 2022) (Mulamba et al., 2016) (Khaled et al., 2018) (Caruccio et al., 2018) (Lingam et al., 2018) (Cresci et al., 2019) (Lingam et al., 2021) (Zhao et al., 2020)
(Yang et al., 2013)	Vendor-purchased-2019	Dataset of fake follower accounts.	Users: 1088	(Yang et al., 2019) (Sayyadiharikandeh et al., 2020)
(Yang et al., 2019)	Botometer-feedback-2019	Dataset of human and bot accounts.	Users: 386, bots: 143	(Yang et al., 2019)
(García et al., 2014)	CTU-13 dataset	Dataset of botnet traffics.	_	(J. Wang et I. Ch. Paschalidis, « Botnet Detection Based on Anomaly and Community Detection », IEEE Trans. Control Netw. Syst., vol. 4, no 2, p. 392 404, juin, 2017)
(Pizarro, oct. 2020)	Pan-19	Dataset of SMBs and gender profiling.	English: human: 2060, human-tweets: 206000, bots: 2060, bot-tweets: 206000	(Feng et al., 2021)
(Dorri et al., 2018)	Bots_accounts_eda	Twitter accounts of humans and bots.	Users: 25013, bots: 12,425	(Dorri et al., 2018) (Lingam et al., 2018)
(Cresci et al., 2015)	Elezioni2013 dataset	Fake accounts dataset.	Accounts: 1481, tweets: 2.06 M, followers: 1.52 M, friends: 667225	(Mulamba et al., 2016) (Danezis and Mittal, 2009)
(Cresci et al., 2015)	TWT	Fake accounts datasetS	Accounts: 845, tweets: 114192, followers: 28588, friends: 729839	(Mulamba et al., 2016)
(Cresci et al., 2015)	INT	Fake accounts dataset.	Accounts: 1337, tweets:58925, followers: 23173, friends: 517485	(Mulamba et al., 2016)
(Danezis and Mittal, 2009)	Fastfollowerz	Fake accounts dataset.	Accounts: 1169	(Danezis and Mittal, 2009)

4.4.4. Features used in the detection algorithms

This section aims to highlight the features the researchers have exploited to build their detection systems and understand if there is a strong relationship between the selected features and the type of bot detected. We also investigate the impact of feature selection on the design of efficient models for SMB recognition. Fig. 10 illustrates the percentages of the different feature types used to design the ML-Based, DL-Based, and Graph-Based methods.

Fig. 10 shows that the most used features for ML-based systems are content, user information, network, and temporal features. In the case of DL-based algorithms, the most used features are content, user information, network, temporal features, and deep linguistic features. Hence, there is a great deal of similarity between ML and DL-based approaches. Furthermore, DL-based algorithms take advantage of DL's evolution in the NLP domain, which has enabled the use of deep linguistic features such as embedding techniques. Graph-based algorithms' most commonly used features are user information, network, and statistical features. Indeed, they use these features to build the graph, which represents the structure of the investigated accounts' social network and allows the detection of the group of SMBs that engage in malicious activities in OSNs. Fig. 11 plots the percentages of the varying feature types used to implement the other detection approaches.

Table 13
Detail of the Cresci datasets.

Dataset Name	Content	Description	Size	Features
Cresci-17	D1: genuine accounts	Human accounts	Account: 3474, Tweets: 8377, Year: 2011	User information, content, friend, and network features
	D2: Social spam bots1	Re-tweeters of Italian political candidate	Account: 991, Tweets: 1.61 M, Year: 2012	
	D3: Social spam bots2	Spammers of paid applications for mobile	Account: 3457, Tweets: 428542, Year: 2014	
	D4: Social spam bots3	Spammers of products on sale at Amazon.com	Account: 464, Tweets: 1.41 M, Year: 2011	
	D5: Traditional spam bots1	Spammers	Account: 1000, Tweets: 145094, Year: 2009	
	D6: Traditional spam bots2	Spammers using scam URLs	Account: 100, Tweets: 74957, Year: 2014	
	D6: Traditional spam bots3	Spammers of job offer	Account: 433, Tweets: 5.79 M, Year: 2013	
	D7: Traditional spam bots4	Spammers of job offers	Account: 1128, Tweets: 133311, Year: 2009	
	D8: Fake followers	Accounts of fake followers	Account: 3351, Tweets: 196027, Year: 2012	
Cresci-2015	D1: ThefakeProjetct	Verified human accounts	Account: 469, Tweets: 563693, Followers: 258494, Friends: 241,710	User information, content, friend, and network features
	D2: elezioni13	Verified human accounts	Account: 1481, Tweets: 2.06 M, Followers: 1.52 M Friends: 667225, Year: 2013	
	D3: fastfollowerZ	Fake accounts from https://www.fastfollowerz.com	Account: 1169, Tweets: 22910, Followers: 11893, Friends: 253026	
	D4: interTwitter	Fake accounts from https://www.interTwitter.com	Account: 1337, Tweets: 58625, Followers: 23173, Friends: 517485	
	D5: Twittertechnology	Fake accounts from https://www.Twittertechnology.com	Account: 845, Tweets: 114192, Followers: 28588, Friends: 729839	
	D6: human	Verified human accounts	Account: 1950, Tweets: 2,63 M, Followers: 1,78 M, Friends: 908935	
	D7: Fake	Fake Twitter accounts	Account: 1950, Tweets: 118327, Followers: 34553, Friends: 879580	
	D8: Baseline	Union of D6 and D7	Account: 3900, Tweets: 2.75, Followers: 1.81 M, Friends: 1.78 M	

Table 14
Sampling Techniques.

Techniques	References
Random Oversampling, CGAN, SMOTE, ADASYN	(Wu et al., 2020)
SMOTE	(Wu et al., 2020) (Fazil et al., 2021)
Snowball sampling	(Beskow and Carley, 2018)
Entropy-based uncertainty sampling	(Wu et al., 2021)
Synthetic minority oversampling	(Kudugunta and Ferrara, 2018)
Cost-sensitive learning	(Zhao et al., 2020)
Fuzzy-based oversampling, random oversampling, random under sampling	(Liu et al., août 2017)

From Fig. 11, we observe that the most utilized features in anomaly-based systems are content, user information, network, and temporal features. In DNA-based systems, the most frequently employed attributes are behavioral features. Additionally, the most used features for genetic-based and topic-modeling-based systems are content and user information. Conversely, we noticed that the most used features for hybrid models are content, user information, network, temporal, and sentiment features. These results are also confirmed by Table 4, Table 5, and Table 6, which showed that user information, content, network, and temporal features are the attributes used in the ML and DL models, resulting in the best performance. In Table 10, we summarized the most valuable features to detect each type of SMB to give some insights to researchers.

As a result, the content, user information, and network features are the most commonly used attributes to conduct a specific or generic detection of SMBs. Even if they have not been exploited much, sentiment features proved to be effective in detecting political bots that aim to influence the opinion of OSN users by posting positive or negative content. In addition, the temporal features have proven their effectiveness in detecting synchronized massive activities related to events like the presidential elections. Therefore, behavioral features are commonly used to detect Sybil, Cyborg, and game bots.

4.5. Dataset issues (RQ5)

SMB detection systems face several problems related to the dataset characteristics. Indeed, some of these issues have a direct impact on the performance of the approach. The topics addressed in the state-of-the-art include the problem of dataset size, unbalanced data, and data labeling. Furthermore, social bots keep changing their characteristics and behaviors to escape detection systems which require new datasets that follow the evolution of the social

Table 15
Dataset labeling techniques.

Technique	Strengths and Limits	References
Manual	1) Labeling process takes a long time; 2) Not efficient because of human error; 3) Small set of data is produced, the labeling task becomes complex when the size of the data increases; 4) It does not take into consideration the behavior of the bot during the annotation.	(Fazil et al., 2021) (Bello et al., 2018) (Danezis and Mittal, 2009) (Yang et al., 2019) (Cresci et al., 2016) (Gilani et al., f2019) (Cresci et al., 2017) (Shetty et al., 2021) (Chew et al., 2018) (Jeong et al., 2016) (Safaa et al., 2022) (AlRubaian et al., 2015)
Suspended users list	1) Simple technique; 2) Depends on the accuracy of the detection system used by the OSN and criteria used for the suspension.	(Dickerson et al., 2014) (Ping et S. Qin, 2018)
Honeypots	Permit to collect a large set of active bots with considerable reliability.	(Dickerson et al., 2014) (Koggalahewa et al., 2022)
Semi-automatic	Rapid labeling of large datasets enables the quick implementation and adjustment of models to new settings and contexts.	(Beskow and Carley, 2019) (Inuwa-Dutse et al., 2018)
Automatic	1) Simple and rapid labeling; 2) Labeling efficiently depends on the used API. The most commonly used detection tools are Botometer (Davis et al., 2016), Botometer101 (Yang et al., 2022), and Debot (Chavoshi et al., 2016).	(Bruno et al., 2022) (Abu-El-Rub and Mueen, 2019)

bots for effective detection systems. The authors used datasets scraped from OSN by utilizing APIs (Dickerson et al., 2014); (Ji et al., 2016); (Jr et al., 2018) such as Tweepy and Twitter API in some works. On the other hand, many public datasets are suggested by researchers for SMB detection, and they are available for download in Bot Repository (https://botometer.osome.iu.edu/bot-repository/datasets.html).

4.5.1. Dataset sources

Twitter is the most extensively utilized source for data collecting and dataset building for training ML and DL algorithms. Twitter has a data feed API that allows users to collect data in real-time and provides researchers full access to the user's data that need to analyze SMBs and establish detection systems in place. A statistical breakdown of the various existing OSNs' utilization rates is shown in Table 11. As we can see, less research has been done on Facebook, Sina Weibo, Renren, and Flickr than on Twitter.

For spam bots, Sybils, and cyborgs, most researchers used public datasets or combined several datasets to have a dataset corresponding to the model requirements regarding size, diversity of data, features, and data scarcity. For the detection of political bots, in most cases, data was collected using the hashtags of candidates in the electoral elections. Fig. 12 provides the count distribution of public datasets used to detect SMBs.

As shown in Fig. 12, many public datasets address the topic of SMB recognition, and they have been used to train and evaluate numerous state-of-the-art research projects. We identified 23 public datasets devoted to SMB detection. The most used datasets are Cresci-17 and Cresci-2015. For more details, Table 12 gives a presentation of the collected datasets as follows:

- Column References: refers to the article in which the dataset was proposed.
- Name: is the name of the dataset.
- Description: provides a description of the dataset and the type of SMBs it contains.
- Size: is the size of the dataset.
- Citation: refers to the works that used the dataset for their experimentation.

Researchers widely use the Cresci-17 and the Cresci-2015 because they contain various features, which provide more opportunities for researchers to investigate different methods using different characteristics. Since they are labeled datasets, they are used for training supervised models based on ML, DL, genetic, DNA, graph algorithms, etc. The Cresci-17 dataset provides a large set of genuine, traditional, spammer, and fake follower accounts extracted from Twitter, labeled manually by CrowdFlower collaborators by investigating the attributes of each account. The Cresci-2015 dataset includes genuine and fake accounts extracted from Twitter and manually labeled based on the attributes of the scraped data. Table 13 provides more details about these datasets.

It is evident from Table 5 that the Cresci-2015 and the Cresci-17 dataset provide a large set of data and features. They gave good results for approaches based on ML, DL, graph theory, DNA-inspired algorithms, anomaly-based, etc. On the other hand, these datasets contain data that is considerably old. It does not follow the evolution of SMBs and omits the current active SMBs on OSNs. Furthermore, it does not cover the diversity of SMBs each dataset focuses on a specific type of social bot.

4.5.2. Imbalanced data

Another challenge with datasets is the imbalanced data problem. In imbalanced datasets, data may have an asymmetry in the distribution from one class to another. This situation may bias the classification results. In addition, the impact of the choice of sampling techniques on the performance of DL models is significant, and often researchers investigate several methods before making a final choice. For this aim, some techniques were applied, whether under-sampling, over-sampling, or hybrid methods. Table 14 presents the sampling techniques used in the studied articles.

Table 14 shows that only some works have dealt with the problem of imbalanced data, even though the datasets used in the state-of-the-art papers are all imbalanced. However, the proportion of legitimate accounts is superior to the proportion of SMBs. Indeed, training the model with an imbalanced dataset will lead to a biased model that overfills the majority class. Training the model using balanced data has increased the accuracy of spam bot detection in (Fazil et al., 2021) with an accuracy rate of 99.83%, in (Kudugunta and Ferrara, 2018) with an accuracy of 96%, and in the other works referenced in Table 14.

4.5.3. Data labeling

One of the significant challenges facing researchers in the field of SMB detection is to have labeled datasets for training supervised ML and DL models. Labeling a dataset means mastering the techniques to identify whether an account is a bot or not (Orabi et al., 2020). The more the data labels are accurate, the more the model is efficient. Generally, this problem needs to be studied more because most datasets researchers use are labeled. Manual labeling is still the most popular technique; it can be carried out by experts, crowdsourcing, or volunteers (Almerekhi and Elsayed, 2015). This process is time-consuming and inefficient due to the annotation errors that may occur, and only a small set of data can be produced. The second annotation technique is based on the list of suspended accounts (Morstatter et al., 2016), which were detected as bots by checking the status of the account (active, deleted, suspended). The third method is based on honeypots, a group of accounts that behave like bots without interacting with normal users. This is a trap to attract social bots. Thus, any user interacting with this honeypot network will be considered a bot. This technique collects a large set of active bots with a certain reliability. Another annotation technique uses semi-automatic labeling (Teljstedt et al., 2015) by proposing predictive models to aid the labeler. And finally, automatic labeling (Bruno et al., 2022) is based on bot detection tools and APIs like the Botometer API (Yang et al., 2022). Table 15 presents the strengths and limits of each labeling technique used in the literature reviews and the references to the studies that used these techniques.

Unfortunately, there needs to be a clear definition of SMBs, which makes it challenging to develop a strategy for stabilizing the labeling process. Here are some behavioral patterns that are highly reliable for discriminating between social bots and normal users: 1) Absence of original and appropriate content and duplicate contents and posts several times; 2) Retweets are much higher than regular tweets from a genuine account; 3) Retweeting a vast number of tweets in a brief period; 4) The abundant presence of spam or illicit URLs in the posted content that redirects to phishing, malware, or adversaries' links. 5) Display links that do not match the post context; 6) The OSN accounts have a recent creation date; 7) User names and screen names include digits, which might point to an auto-generation of names; 8) Many users publish the same content at the same time, which refers to a synchronization activity launched by the botnet master; 9) Abusively utilizing the trending hashtags to advertise content irrelevant to the context of the used hashtags to gain more visibility and influence.

5. Discussion

This section discusses the results and conclusions reached by the SLR study and gives directions for future research. The main objectives of this paper were to summarize, analyze, and synthesize all the significant elements involved in the social bot detection process, including data source selection, data labeling, feature selection and extraction, and model training, as well as to highlight the limitations of the reviewed studies and to propose openings for improving the quality of SMBs detection systems.

RQ1: From the studied articles, we identified eight types of social profiles that operate in OSN. SMBs include three types: benign, neutral, and malicious. For benign SMBs, there is a category of social bots: promoters, celebrities, informational bots, and chatbots. As well, for malicious SMBs, we have identified six types of social bots: 1) SMBs; 2) spam bots; 3) sybils and cyborgs; 4) political bots; 5) stegobots; 6) gambling bots, as shown in Fig. 3. The new generation of SMBs is provided by cloud services such as Bot-as-a-Service (BaaS), and these SMBs include intelligent functions such as sophisticated content generation similar to humans. Every day, new types of bots may appear that the present systems may not detect because they ignore their behavior. That is why it is necessary to follow their evolution and explore the different characteristics of an OSN account to detect them efficiently. It would be interesting to apply the different detection systems to all current SMBs to detect gaps and predict the evolution of the behavior of social robots over time. Studying emerging social media platforms like TikTok can help discover new types of social media bots. As new platforms emerge, they often introduce new features, technologies, and ways of engaging with users, creating new opportunities for bots to manipulate, influence, and deceive users. TikTok's video-based format and users' short attention spans provide a fertile ground for bots to generate automated videos that mimic human behavior and interactions.

RQ2: ML-based techniques are still the most used for bot detection, followed by DL techniques. The RF, SVM, and AdaBoost are the most employed algorithms for SMB detection, as illustrated in Fig. 7. The DT and RF achieved a precision score equal to 100% with models trained on the Cresci-17 dataset. Therefore, for the ML algorithm, RF was the most used algorithm for spam bots, which achieved an F-measure of 97.9%. The GB algorithm reached an accuracy of 98.97%. DL-based techniques need to be explored more, and the most commonly used DL algorithms are LSTM and CNN. Both the LTSM and CNN algorithms achieved a good precision of 99.8% and an accuracy of 100% (see Fig. 8 and Table 4). However, there is a lack of large datasets to train DL models, which can limit their use. It's crucial to involve social network users in the detection and prevention process of fake accounts through voting or trust techniques, as proposed in some articles. Integrating feature-based and graph-based techniques for fraudulent account identification has also received little attention. The proposed methods are still in their early stages and must be optimized. The supervised techniques are considered early detection, based on the principle of training a model on a dataset of OSN accounts labeled as human or malicious SMBs. The limit of these detection techniques is that evolving SMBs can escape the detection systems because social bots keep developing their characteristics and behaviors over time. Nowadays, social bots use advanced strategies to mimic human behavior by increasing their connections with real friends and followers and owning accounts with cloned names and profile pictures of real accounts to deceive OSN users and detection systems. However, the unsupervised techniques cluster the group of coordinated SMBs based on their synchronized behaviors and activities, which become inaccurate faces of the small groups of poorly coordinated bots that may need more information for their detection, which can lead to a biased detection system.

RQ3: It is evident throughout Table 4, Table 5, and Table 6 that the most commonly used features for specific or generic SMB detection were content, user information, and network features. The sentiment features effectively detect massively synchronized activities associated with elections or political conflicts. Therefore, behavioral features are typically used to detect Sybil, cyborgs, and game bots. To combat malicious SMB evasion, many researchers have used a variety of features to distinguish legitimate accounts from malicious ones. As we have already mentioned, the new generation of SMBs are provided with intelligent functions that allow them to generate and post textual content similar to human-generated text. Our study shows that contextual and deep-linguistic features need to be explored more by researchers to analyze textual content and posts, especially word embedding techniques and deep-NLP. Indeed, for future works, it will be very interesting to experiment with the impact of DL on NLP on the effectiveness of SMB detection systems. Therefore, social bots may exhibit different behaviors, so exploring additional features and models is essential to find the best approach. So, no single feature can accurately identify a social bot, and it's often necessary to use a combination of features and advanced machine learning algorithms to determine the reverent features accurately. Additionally, social bots are constantly evolving and adapting to evade detection, so it's essential to update and refine bot detection algorithms regularly.

RQ4: Finding accurate features is critical to build an effective system to detect SMBs. The feature engineering and preprocessing techniques in the social bot's detection field are discussed in many works, such as data collection, cleaning data, normalization, feature extraction, and selection. As depicted in Tables 8 and 9, feature extraction and feature selection techniques are not used extensively and sufficiently to detect SMBs effectively. Preprocessing social media data for bot detection while preserving the semantic meaning of emojis and the sentiment of the content can be challenging. Several strategies are to consider, such as tokenization, emoji-to-text conversion, word embeddings, and preprocessing pipelines. It is important to experiment with different techniques and pipelines to find the best way to prepare data for the training phase without resulting in a loss of information; some removed symbols or characters can be discriminant for the detection of SMBs. It would be interesting to start by extracting statistical features before cleaning the data, for example, calculating the number of URLs, number of motions, and number of hashtags, replacing emojis and emoticons with their meaningful words, and replacing acronyms with their complete forms. Additionally, it is crucial to analyze the data and adjust the preprocessing accordingly for each type of SMB. It is observable from Table 8 that the majority of work used the traditional bag of words techniques for feature extraction, which cannot capture the semantic and syntactic relationships between terms. The word-embedding has been shown to improve the performance of various NLP tasks, such as sentiment analysis and text classification. By capturing the semantic and syntactic relationships between words, word embedding provides a more nuanced understanding of language that can improve the accuracy of NLP models. Using word embedding can refine and improve detection methods. Based on the works we studied, we found that user information, content, and network features are the most used for SMB detection. While choosing features, accuracy, and speed must be balanced, dimensionality reduction is employed to condense the feature space while taking key features into OSN accounts. It's worth noting that no single feature can identify all bot accounts accurately. Rather, multiple features and techniques should be combined to create a robust bot detection system. Researchers are developing methods to identify each type of SMB separately. However, developing a detection system that can identify all SMB types using a single generic model with the same or comparable features would be interesting.

RQ5: Although the literature has already provided many public datasets for SMB detection. Unfortunately, most represent two generic classes, such as "legitimate" or "fake," or focus on a specific type of SMB. So, they cannot be generalized to all SMBs. In addition, all these datasets depend on the Twittersphere and include limited data that does not allow us to recognize SMBs that exhibit advanced malicious behaviors. The majority of them provide only user information and content features. However, TwiBot-20 is the only dataset that can be used for community-based detection because it provides data on users' friends, followers, and content features in addition to user information and content features. Botometer is a tool developed by researchers to determine the likelihood that a given Twitter account is a bot or a normal account. It analyzes various account characteristics, including its tweeting behavior, metadata, and the content of the tweets. Researchers can use this tool to label Twitter datasets automatically. It also offers a repository of all the public datasets that the researchers provide. In Table 12, we listed all the public datasets provided by researchers for academic works, with detailed descriptions of the types of SMBs and the sizes of the datasets. On the other hand, in some works, the authors used web scraping techniques to study specific cases of SMB detection, such as political and spam bots. Indeed, data was extracted from OSNs based on hashtags, trending topics, or keywords. Therefore, to detect spam bots, fake accounts, Sybils, and cyborgs, most of the researchers used public SMB datasets. From Table 12 and Fig. 12, we can confirm that the most used public datasets by researchers are Cresci-17 and Cresci-2015. Furthermore, for spam bots, with the Cresci-17 dataset, ML and DL models produced good results with a precision that exceeded 95%. Hence, to identify fake accounts. The Cresci-2015 dataset, the most frequently utilized by researchers in various papers, has a precision of over 95%. Hence, Table 5 presents more information about the Cresci-2015 and Cresci-17 datasets. Social media platforms often have restrictions on the data that can be accessed, limiting the data available for bot detection. Additionally, some social bots may use techniques to obscure their activity, making it difficult to detect them. To overcome the limited data availability, missing or confusing ground truth, and obsolescence of existing datasets, researchers must establish reference datasets that contain harmful accounts, such as social bots, cyborgs, and political trolls. The data availability problem requires the most attention from the scientific community and remains largely unsolved. Additionally, the data used for bot detection may be biased toward certain types of social bots, making it difficult to detect others. For example, bots that mimic human behavior may be more difficult to detect than those that exhibit anomalous behavior. Therefore, imbalanced datasets are a common challenge in social media bot detection. In many cases, there may be significantly more data for one class (e.g., human users) than the other (e.g., spambots). Imbalanced datasets in social media bot detection can lead to poor performance for the minority class. Researchers applied many different techniques to handle the issue of data set size and biased class proportions, such as random oversampling, SMOTE, CGAN, SMOTE, and ADASYN (see Table 14). To address this, strategies such as resampling, synthetic data generation, and data augmentation can be used to increase the number of samples and improve the model's performance. However, there is no one-size-fits-all solution for imbalanced datasets, and the optimal approach will depend on the dataset and problem. The issue of dataset labeling for social media bot detection is a challenging problem that requires careful consideration and expertise. The challenge with dataset labeling is that it can be difficult to distinguish between human users and bot accounts. This is because bots can be designed to mimic human behavior, such as tweeting at certain times of the day or using natural language. Additionally, bot detection algorithms can be susceptible to bias, leading to inaccurate results. Self-supervised learning is an improvement over unsupervised learning methods that learn from input data to learn data representations to solve data labeling problems automatically, even for large datasets. Some of the strengths of this technique are its scalability and reduction of time costs by limiting human intervention. However, semi-supervised learning uses both labeled and unlabeled data to train the model, and using these techniques will improve the labeling process and reduce the cost of time. Moreover, we noticed that no public datasets are addressed for stegobots detection. Stegobots use steganographic images to conduct their attacks. So, rather than using only standard features, it would be interesting to investigate how GANs can help to provide realistic stereographic images to refine SMB identification.

6. Conclusion and perspectives

Considering the dangerous impact that malicious SMBs play on OSNs, this has led researchers to dig deeper into this line of research and propose detection systems that help limit their invasion. But this has not stopped the creation of newly evolved SMBs and techniques that can bypass the detection systems. This state-of-the-art aims to identify existing SMB detection systems by applying SLR instructions and guidelines. We reviewed and assessed previous research works in the field of SMBs published in research databases such as ScienceDirect, IEEE, SpringerLink, and Google Scholar. Given the large number of published works that deal with the problem of social bots' detection, we studied only the articles published between 2008 and 2022; we also excluded works that have a small number of citations and those that have not been indexed, which can be considered the limit of this work. We used the PRISMA approach to conduct a bibliographical search of our SLR; nevertheless, only 144 pertinent publications were found, exposing the stages of our search strategy. We first identify the research questions relevant to our study, analyze the relevance of the recovered articles using inclusion and exclusion criteria, and then evaluate the works gathered using quality evaluation methods before attempting to answer the research questions we initially identified. Based on the research retrieved, we classified OSN profiles into three categories: genuine, verified, and cloned or false accounts. We classified social bots into three main groups: benign bots, which automatically post alert tweets, distribute news, engage with users, and attend to their needs; neutral bots, which post or repost jokes; and malicious bots, which are directed by a bot master to carry out automatic tasks and plan attacks. SMBs, spam bots, Sybil and cyborgs, stegobots, political bots, and social game bots are six categories of malicious bots. We conducted related research on each sort of malicious bot based on this classification to complete a pertinent synthesis and analysis of the examined methodologies. We proposed a taxonomy of the SMB's detection technique, which includes eight categories: ML-based, DL-based, graph-based, anomaly-based, topic modeling-based, DNA-inspired, genetic-based, and hybrid approaches. Then, we examined these techniques and showed the strengths and weaknesses of each studied approach by systematizing, evaluating, incorporating, and presenting the outcomes and findings of the reviewed works. The results of our study show that it is so challenging to implement efficient SMB detection systems because social bots keep changing their characteristics over time, and the new generation of SMBs have intelligent functions such as sophisticated content generation similar to humans. To detect them efficiently, it is necessary to track their evolution and investigate the various characteristics of an OSN account. It would be an ideal scenario for the near future. It would also be useful to compare detector performance to that of existing SMBs to simulate how bot characteristics change over time. The RF, SVM, and AdaBoost are the most used ML algorithms for SMBs detection. The most commonly used DL algorithms are LSTM and CNN. However, there is a lack of large datasets to train DL models. Early detection is based on training a model on a dataset of OSN accounts labeled as human or malicious SMBs. Still, these techniques' limits are that evolving SMBs can escape the detection systems. Self-supervised learning is an improvement over unsupervised learning methods that learn from input data to learn data representations to solve data labeling problems automatically, even for large datasets. Existing approaches address these issues, but further enhancements may still be required. There is also a need to implement models that detect different types of SMBs, even those outside the Twittersphere; moreover, certain methods need further exploration, such as leveraging NLP techniques or anomaly-based methods. The most commonly used attributes for conducting specific or generic SMB detection were content, user information, and network characteristics. To combat malicious SMB evasion, researchers have used a variety of features to distinguish legitimate accounts from malicious ones. However, the researchers did not sufficiently explore contextual and deep-linguistic features to analyze textual content deeply, especially word embedding techniques. Deepfakes, an AI-enabled deception method, can now blur the lines between legitimate accounts and the most sophisticated malicious accounts. Feature extraction is essential for ML algorithms to recognize social bots accurately, but no single feature is enough to identify all accounts accurately. Researchers must focus on identifying robust and differentiating traits to create a robust bot detection system. Many public datasets for SMB detection have been published in the literature. Still, most represent two generic classes or are focused on a specific type of SMB. The only dataset that can be used for community-based detection is TwiBot-20. To overcome the limitations of limited data availability, missing or ambiguous ground truth, and the obsolescence of existing datasets, researchers must create reference datasets containing harmful accounts and devise new methods for producing adversarial cases. These issues are largely unsolved and require the most attention from the scientific community. Therefore, the progress of science depends on the policies adopted by decision-makers and operators of OSNs, because if researchers have advanced access to more data, they will be able to respond to new developments and challenges in this field.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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A COMPREHENSIVE SURVEY ON DETECTION OF MALICIOUS SOCIAL BOTS USING LEARNG AUTOMATA.

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Abstract: With the increasing prevalence of social media platforms as communication channels, the risk of malicious activities carried out by social bots has become a significant concern. This survey paper provides a comprehensive overview of the state-of-the-art techniques and methodologies employed in the detection of malicious social bots using machine learning (ML) approaches. exploring the landscape of social bots and their diverse functionalities, emphasizing the need for effective detection mechanisms to mitigate potential threats. Subsequently, a detailed analysis of existing literature is presented, categorizing ML-based approaches into different methodologies such as supervised learning, unsupervised learning, and hybrid models. various ML algorithms employed in social bot detection, including traditional classifiers, deep learning models, and ensemble methods. It evaluates the strengths and limitations of these algorithms in the context of social bot detection, considering factors such as accuracy, scalability, and interpretability.

***Index Terms* - Malicious social bots, Machine learning , Learning Automata.**

INTRODUCTION

In recent years, the proliferation of social media platforms has transformed the landscape of communication, enabling widespread interaction and information dissemination. However, this increased connectivity has also given rise to the proliferation of malicious social bots, automated entities designed to manipulate social discourse, spread misinformation, and engage in other harmful activities. As the threat posed by these malicious actors continues to grow, the development of effective detection mechanisms becomes imperative to safeguard the integrity of online social spaces.

This research paper focuses on the innovative integration of Machine Learning (ML) techniques and Learning Automata to enhance the detection capabilities for malicious social bots. ML, with its ability to analyze vast datasets and identify patterns, has shown promise in addressing the dynamic and evolving nature of social bot behavior. Learning Automata, on the other hand, offers the advantage of adaptive decision-making, allowing the system to dynamically adjust its responses based on the evolving strategies employed by malicious actors. The combination of ML and Learning Automata brings a synergistic approach to social bot detection, leveraging the strengths of both paradigms. This paper aims to explore the theoretical foundations, methodologies, and practical applications of this integrated approach in order to contribute to the ongoing efforts to fortify cybersecurity in the realm of social .The research will delve into the various facets of social bot detection, considering the challenges posed by rapidly changing tactics employed by malicious entities. It will investigate the role of ML algorithms, such as supervised and unsupervised learning, in discerning patterns indicative of malicious behavior. Additionally, the incorporation of Learning Automata will be explored to enhance the adaptability and responsiveness of the detection system in real-time scenarios.

As social bots become increasingly sophisticated in their evasion strategies,the proposed integrated approach seeks to provide a robust defense mechanism capable of not only identifying known malicious patterns but also adapting to novel and evolving tactics. The research also recognizes the ethical considerations associated with the deployment of such technologies, emphasizing the importance of transparency, fairness, and accountability in the development and implementation of social bot detection systems.

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In summary, this research paper endeavors to contribute to the growing bodyof knowledge in the field of cybersecurity by proposing an integrated approach that combines the strengths of Machine Learning and Learning Automata for the effective detection of malicious social bots. Through a multidimensional exploration, the aim is to provide insights that can inform future developments in securing online social spaces against the ever-presentthreat of automated malicious activities.

The emergence of malicious social bots poses a multifaceted challenge thattranscends traditional cybersecurity paradigms. These bots exploit the interconnected nature of social platforms to disseminate misinformation, amplify divisive narratives, and compromise the trustworthiness of online interactions. Detecting and mitigating such threats necessitates innovative approaches that can keep pace with the evolving tactics of these malicious actors.

Machine Learning, with its ability to discern complex patterns from data, hasdemonstrated efficacy in various cybersecurity domains. In the context of social bot detection, ML algorithms offer the potential to learn and adapt to the subtle nuances of bot behavior, making them a valuable tool in the defender's arsenal. However, the dynamic nature of social bot strategies requires not only predictive capabilities but also adaptive responses, which brings Learning Automata into the spotlight.

Learning Automata, rooted in the theory of computational intelligence, provides a framework for decision-making in dynamic and uncertain environments. By incorporating Learning Automata into the detection process, we introduce a dynamic element that enables the system to autonomously adjust its strategies based on the feedback received from the environment. This adaptability is particularly crucial in countering the evolving strategies employed by malicious social bots, which often aim to circumvent static detection mechanisms.

This research paper aims to bridge the gap between theoretical foundations and practical implementations by proposing a cohesive framework that integrates ML and Learning Automata for the detection of malicious social bots. By combining the strengths of pattern recognition and adaptive decision-making, this integrated approach aspires to provide a more comprehensive and resilient defense against the spectrum of social bot activities.

The subsequent sections of this paper will delve into the key components of the proposed framework, exploring the intricacies of feature engineering for ML algorithms, the role of supervised and unsupervised learning in identifying malicious patterns, and the dynamic decision-making processes facilitated by Learning Automata. Additionally, real-world case studies and performance evaluations will be presented to validate the effectiveness and scalability of the integrated approach.

As we embark on this exploration, it is crucial to recognize the collaborative and interdisciplinary nature of addressing the social bot menace. The findings of this research contribute not only to the field of cybersecurity but also to the broader discourse on preserving the authenticity and trustworthiness of online social interactions in an era dominated by automated entities.

The advent of social media has undeniably revolutionized the way we communicate, share information, and connect with others. However, this unprecedented level of connectivity has given rise to a shadow ecosystem of malicious social bots, designed with the intent to exploit vulnerabilities in the social fabric. These bots employ sophisticated tactics, including impersonation, automated content generation, and coordinated campaigns, to manipulate public opinion and sow discord.

This research endeavors to address the escalating threat of malicious social bots through a fusion of two powerful paradigms: Machine Learning and Learning Automata. The integration of these approaches seeks to capitalize on the strengths of both disciplines, offering a holistic solution that not only identifies malicious behavior but also dynamically adapts to the evolving strategies employed by social bots.

Machine Learning, as applied to social bot detection, has the capacity to analyze vast datasets and identify subtle patterns indicative of automated and malicious activity. The supervised learning models can be trained on labeled datasets, allowing them to recognize known patterns of bot behavior. Unsupervised learning, on the other hand, enables the identification of anomalies and deviations from normal user behavior, a crucial capability in detecting previously unseen threats.

Complementing the ML aspect, Learning Automata contribute a dynamic layer to the detection system. Learning Automata, inspired by the principles of adaptive systems, can autonomously adjust their decision-making strategies based on the feedback received from the environment. This inherent adaptability makes them particularly well-suited for addressing the dynamic and evolving nature of social bot tactics.

The unique synergy between ML and Learning Automata proposed in this research aims to provide a robust, adaptable, and scalable solution to the challenge of malicious social bot detection. By understanding and learning from the historical data patterns through ML, and dynamically adjusting to new patterns through Learning Automata, the integrated approach aims to enhance the overall resilience and accuracy of the detection system.

the subsequent sections, we will delve into the intricacies of feature selection, model training, and real-time adaptation processes. Additionally, the paper will explore the ethical considerations surrounding the deployment of such detection systems, emphasizing the importance of transparency, user privacy, and the responsible use of AI technologies in combating the menace of malicious social bots.

In summary, this research paper positions itself at the intersection of cybersecurity, machine learning, and adaptive systems, offering a novel and integrated approach to the detection of malicious social bots. By advancing our understanding and capabilities in this domain, we strive to contribute to the ongoing efforts to safeguard the integrity and trustworthiness of online social interactions. In social media, bot detection is a crucial duty. Automated accounts are a problem on the widely used social media site Twitter According to certain research, 15% or so of Twitter accounts are semi-automated or operate automatically. Twitter's features are one factor that may have contributed to the increase in bot activity. To differentiate between harmful and genuine tweets, this article analyzes the bad behavior of participants by taking into account features that are derived from the posted URLs (in the tweets), such as URL redirection, the frequency of shared URLs, and the presence of spam content in the URL.

LITREATURE SURVEY

2.1. Paper Title: "Detection of Malicious Social Bots Using Learning Automata With URL Features in Twitter Network"

Authors: *Rashmi Ranjan Rout ,Greeshma Lingam, D. V. L. N. Somayajulu.*

Abstract: Malicious social bots generate fake tweets and automate their socialrelationships either by pretending like a follower or by creating multiple fakeaccounts with malicious activities. Moreover, malicious social bots post shortened malicious URLs in the tweet in order to redirect the requests of online social networking participants to some malicious servers. Hence, distinguishing malicious social bots from legitimate users is one of the mostimportant tasks in the Twitter network. To detect malicious social bots, extracting URL-based features (such as URL redirection, frequency of sharedURLs, and spam content in URL) consumes less amount of time in comparison with social graph-based features (which rely on the social interactions of users). Furthermore, malicious social bots cannot easily manipulate URL redirection chains. In this article, a learning automata-based malicious social bot detection (LA-MSBD) algorithm is proposed by integrating a trust computation model with URL-based features for identifying trustworthy participants (users) in the Twitter network. Theproposed trust computation model contains two parameters, namely, direct trust and indirect trust. Moreover, the direct trust is derived from Bayes' theorem, and the indirect trust is derived from the Dempster– Shafer theory (DST) to determine the trustworthiness of each participant accurately algorithm achieves improvement in precision, recall, F-measure, and accuracy compared with existing approaches for MSBD.

2.2.Paper Title: "A Review on Social Bot Detection Techniques andResearch Directions."

Authors: Arzum Karataş, Serap Şahin.

Abstract: The rise of web services and popularity of online social networks(OSN) like Facebook, Twitter, LinkedIn etc. have led to the rise of unwelcome social bots as automated social actors. Those actors can play many malicious roles including infiltrators of human conversations, scammers, impersonators, misinformation disseminators, stock market manipulators, astroturfers, and any content polluter (spammers, malware spreaders) and so on. It is undeniable that social bots have major importanceon social networks. Therefore, this paper reveals the potential hazards of malicious social bots, reviews the detection techniques within a methodological categorization and proposes avenues for future research.

2.3. Paper Title: "Social media bot detection with deep learning methods"

Authors : Susmita Saha,Mohammad Mehedy Masud,Sujith.

Abstract: Social bots are automated social media accounts governed by software and controlled by humans at the backend. Some bots have good purposes, such as automatically posting information about news and even toprovide help during emergencies. Nevertheless, bots have also been used formalicious purposes, such as for posting fake news or rumour spreading or manipulating political campaigns. There are existing mechanisms that allow for detection and removal of malicious bots automatically. However, the botlandscape changes as the bot creators use more sophisticated methods to avoid being detected. Therefore, new mechanisms for discerning between legitimate and bot accounts are much needed. Over the past few years, a fewreview studies contributed to the social media bot detection research by presenting a comprehensive survey on various detection methods including cutting-edge solutions like machine learning (ML)/deep learning (DL) techniques. This paper, to the best of our knowledge, is the first one to only highlight the DL techniques and compare the motivation/effectiveness of these techniques among themselves and over other methods, especially the traditional ML ones. We present here a refined taxonomy of the features usedin DL studies and details about the associated pre-processing strategies required to make suitable training data for a DL model. We summarize the gaps addressed by the review papers that mentioned about DL/ML studies toprovide future directions in this field. Overall, DL techniques turn out to be computation and time efficient techniques for social bot detection with betteror compatible performance as traditional ML techniques.

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PROPOSED METHOD

This study introduces an automatic and highly efficient malicious social botsdetection system. To initiate the process, feature selection is performed on the dataset, preprocessing, Feature extraction, classification. Within theframework of our proposed system, we provide a concise overview of Artificial Neural Networks (ANN), Support Vector Machines (SVM), and decision trees. ANN and SVM are incorporated into the system for the purpose of comparative analysis. The methods are elucidated as follows:

1.Learning automata(LA): By incorporating a trust computation, a Learning Automata-based Malicious Social Bot Detection (LA-MSBD) model is shown.
model with characteristics based on URLs for locating reliable persons in the Twitter network. In this case, the SVM algorithm will receive 13 parameters as input. After processing the inputs, the algorithm will produce a single digit, either 0 or 1.The findings show that when compared to previous methods for MSBD, the suggested algorithm improves precision, recall, F-measure, and accuracy.

2.Machine learning (ML): Machine learning (ML) is a subset of artificial intelligence (AI) that focuses on the development of algorithms and statistical models that enable computer systems to improve their performanceon a specific task through learning from data, without being explicitlyprogrammed. In other words, it is a method of teaching computers to make decisions or predictions based on data, rather than relying on explicit, rule- based programming.

Data Collection: The first step is to gather relevant data that the machine learning model will learn from. This data can be structured or unstructured and should be representative of the problem or task the model will address.

Data Preprocessing: Before feeding the data into model, it needs to be cleaned and preprocessed. It is a technique used in data mining that involvestransforming raw data into an understandable format. The data is cleansed through processes such as filling in missing values, smoothing the noisy data, or resolving the inconsistencies in the data. As it contains some missing value, the dataset is cleaned, and decimal values are converted into proper float values.

Data Splitting: The dataset is typically split into two or more subsets:
Training Data: This is the portion of the data used to train the model. The model learns to make predictions or classifications based on this data.
Validation Data (optional): Sometimes, a separate dataset is used for validation. It helps in tuning hyperparameters and assessing the model's performance during training.
Testing Data: The testing dataset is used to evaluate the model's performance after it has been trained.

Model Selection: The choice of a machine learning algorithm or model is made based on the specific problem at hand, the available data, and the desired outcome. This step is crucial, as different algorithms may be more suitable for different types of problems.

Feature Selection: The data features that used to train machine learning models have a huge influence on the performance of the model. Irrelevant orpartially relevant features can negatively impact model performance.

Classification: The model is trained by fitting the training set to the classifiermodel. The classifier model upon testing, classifies the air quality into goodor bad. The classifications are fairly close to the testing set.

Training: During the learning phase, the model is exposed to the training data. The algorithm uses the training data to adjust its internal parameters, effectively "learning" from the data. The objective is to the difference between the model's predictions and the actual value (for supervised learning) or to discover patterns (for unsupervised learning).

Model Evaluation: After training, the model is tested on the testing dataset to assess its generalization performance. This evaluation helps determine how well the model is likely to perform on new, unseen data.

Iterative Process: The learning phase often involves iterating through steps 4 to 7 multiple times to improve the model's performance.

Feature Selection Algorithms: The quality of the feature set significantly influences the performance evaluation of ML algorithms. Redundant or irrelevant features can detrimentally impact the effectiveness and efficiency of ML algorithms, thereby increasing the computational cost. The feature selection process involves four fundamental steps:

Subset Generation:

This entails the generation of candidate feature subsets for the evaluation phase. Various search strategies, such as Best First, Random Search, and Greedy Search, are employed for this purpose.

Filter Methods:

Filter methods employ statistical tests to select features based on their correlation with the target value. Features are selected or discarded based on their scores in these tests. This approach falls under univariate feature selection, where each feature is examined independently. Notable techniques include Pearson's Correlation, Chi-Square, Signal to Noise Ratio, F-Score, Information Gain, and Analysis of Variance (ANOVA).

Wrapper Methods:

Wrapper methods employ ML algorithms that ultimately influence the choice of features used in the model. The selection or removal of features is based on the impact on the Machine Learning algorithm. Multivariate feature selection is employed here, considering all features simultaneously. This category includes methods like Forward Feature Selection, Backward Feature Elimination, and Recursive Feature Elimination.

Hybrid or Embedded Methods:

Hybrid or embedded methods combine the characteristics of filter and wrapper strategies. These technique sutilize pre-defined feature selection strategies within algorithms. Reduced features can enhance the accuracy of data analysis, including regression or classification.

3.Machine learning techniques:

Supervised Learning*:* In supervised learning, the algorithm is trained using labeled data, which means the data is tagged or categorized. learns to make predictions or classifications based on the patterns it identifies during training. Supervised learning includes classification and regression tasks.

Example algorithms mentioned include Bayesian network, Gaussian process regression, decision trees, K-Nearest Neighbor Algorithm, Support Vector Machines, and more. These algorithms are used for both classification and regression tasks.

Unsupervised Learning:

Unsupervised learning, in contrast to supervised learning, uses unlabeled data and aims to discover patterns, structures, or knowledge within the data. It includes clustering and association tasks.

Algorithms in this category include K-means clustering, Self-organizing maps, Expectation-maximization, and more. These algorithms help identify hidden structures in the data.

Semi-supervised Learning:

Semi-supervised learning combines both labeled and unlabeled data. It leverages the strengths of both supervised and unsupervised learning and is particularly useful when obtaining labeled data is expensive or time- consuming.

Algorithms in this category include Co-training, Generative models, and more, which utilize a combination of labeled and unlabeled data to improve performance.

Reinforcement Learning:
Reinforcement learning involves agents learning by interacting with an environment. The agent takes actions to maximize a cumulative reward or minimize a cumulative cost. It is commonly used in applications where an agent needs to make a sequence of decisions.
Algorithms like Q-Learning, Markov Decision Processes (MDP), and moreare used in reinforcement learning to develop intelligent decision-making agents.

4.Machine learning Algorithms:

Naive Bayes: The naive Bayes model, irrespective of the strong assumptions that it makes, is often used in practice, because of its simplicityand the small number classification of parameters required. The model is generally used for classification — deciding, based on the values of the evidence variables for a given instance, the class to which the instance is most likely to belong.

Random Forest: Random forest (RF) is the ensemble classifier, which collects the results of many decision trees by majority vote. In ensemble learning, the results of multiple classi- fiers are brought together, and a single decision is made on behalf of the community. Each decision tree in the forest is created by selecting different samples from the original data set using the bootstrap technique. Then, the decisions made by many different individual trees are subject to voting and present the class with the highest number of votesas the class estimate of the committee. In the RF method, trees are created by CART (classification and regression trees) algorithms and boot bagging combination method. The data set is divided into training and test data. From the training data set, samples are selected as bootstrap (resampled and sampled) technique, which will form trees (in a bag) and data that will not build trees of the bag).

SYSTEM ARCHITECTURE

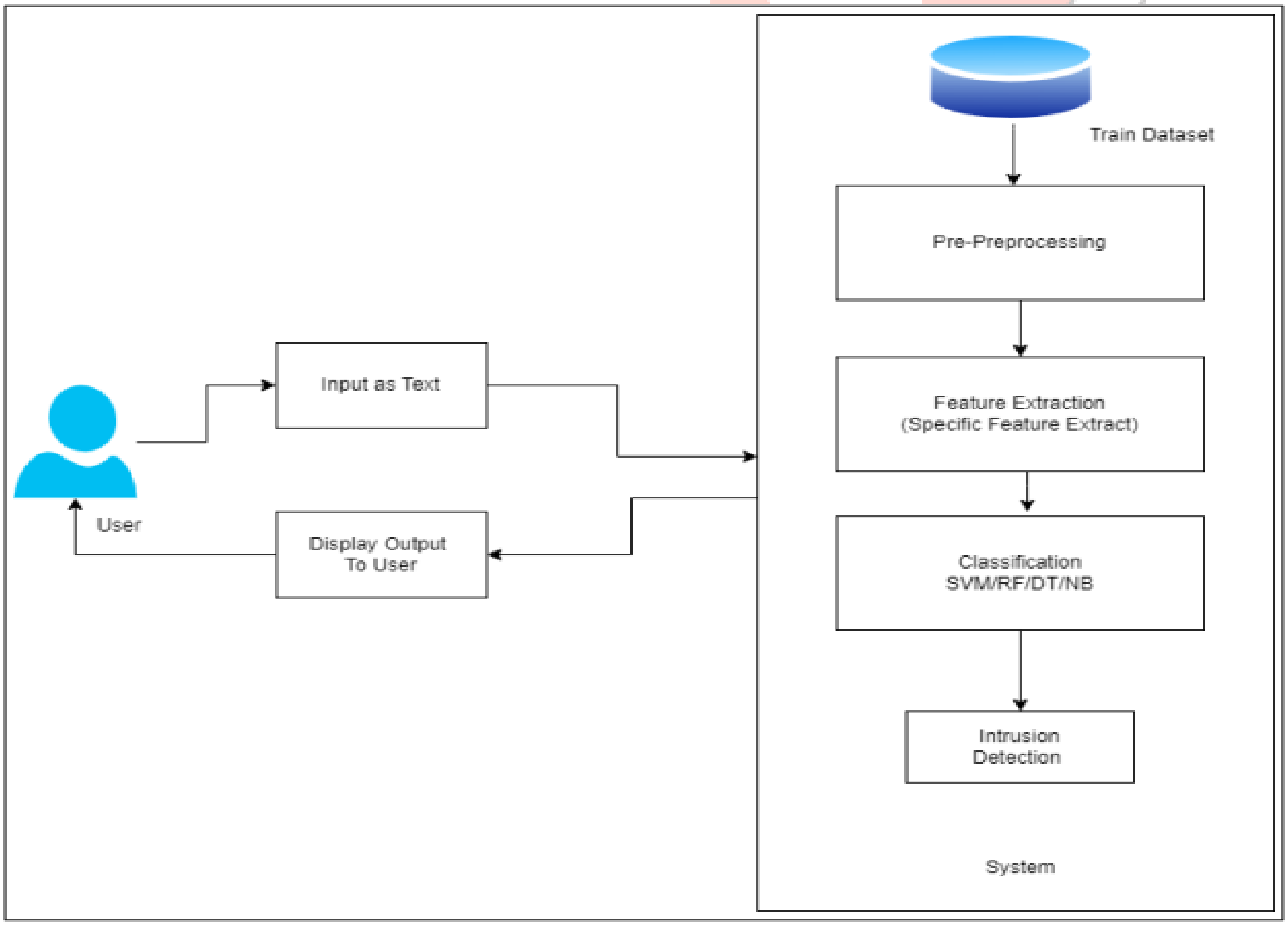

Fig.1. System architecture

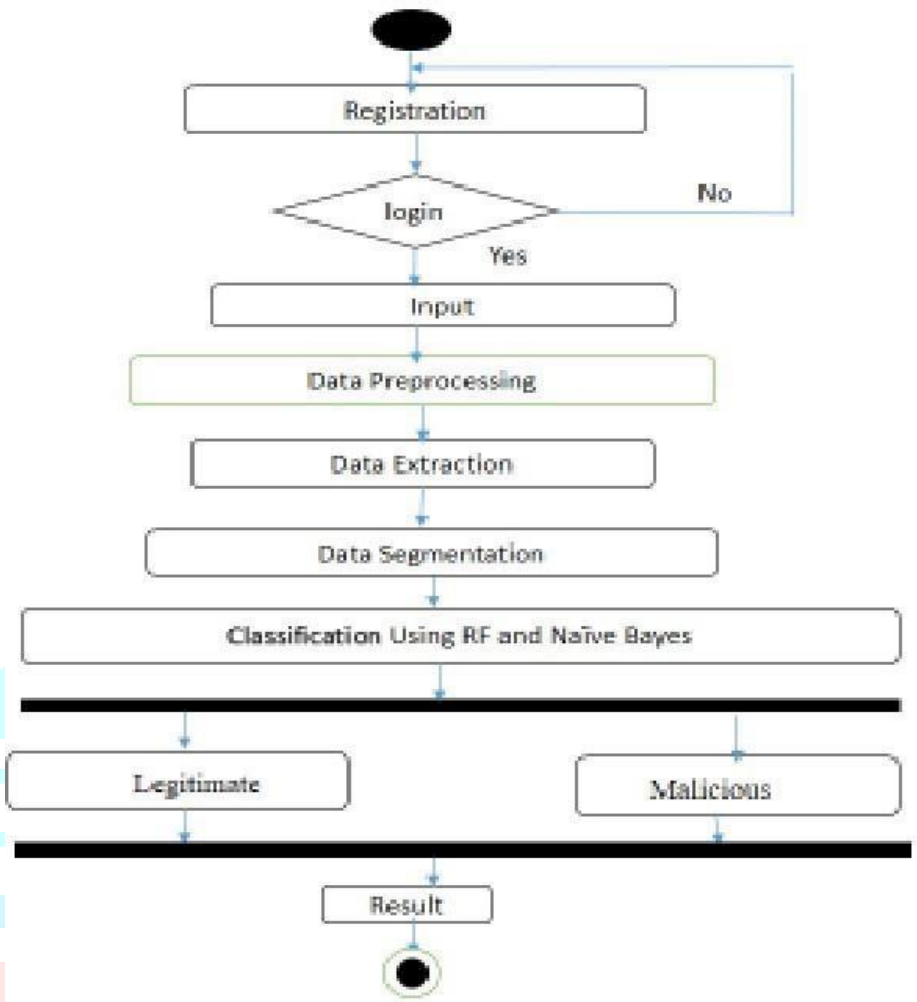

Fig.2. Activity Daigram

CONCLUSION

The prevalence of detecting malicious bots on social media platforms such as Twitter, Facebook, Instagram the need for improved, inexpensive Bot detection methods is apparent. We proposed a Naive Bayes and Random Forest (RF) algorithm allowing us to detect the URL which may be maliciousor harmful for users. In the proposed system till now we have downloaded and installed all the software which are required for system. The dataset hasbeen collected from Kaggle site and preprocessing step have been processed.In next phase the features of preprocessed data will be extracted and the algorithm will be implemented and a model will be saved which can be usedfor classifying the data.

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